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Chapter 1

Holographic entanglement renormalisation for fermionic quantum matter

1.1 Introduction

The manuscript is structured in the following manner. Sec. 1.2 describes the system we work with and defines certain physical ideas and measures of entanglement that are employed in later sections. In Sec. 1.3, we define the scaling transformations mentioned earlier, and discuss its implications on the scaling of entanglement in the system under study. In Sections 1.4, 1.5 and 1.6, we probe the holographic nature of the entanglement, unveiling an emergent spatial dimension in the process and obtain several characteristics of this emergent geometry. Section 1.7 discusses the consequences of a critical Fermi surface on the holographic space, and how the emergent curvature can be linked to a convergence parameter associated with the RG flows. In Section 1.9, we probe the topological aspects of the entanglement, and discuss its implications for the system at hand. We conclude with a discussion of our results in Sec. 4.10 and point out some open questions.

Summary of our main results

For the convenience of the reader, we present below a summary of the main results obtained by us.

1. Quantum information measures (such as entanglement entropy and multipartite information) reveal a hierarchical structure in the nature of entanglement within the system. We relate geometric parameters like distance and curvature of the holographic dimension to an RG beta function of the mass gap, providing thereby an explicit analytic manifestation of the holographic principle. We show that a change in the boundedness of the space corresponds to a change in certain topological winding numbers.

2. We go on to argue that this topological transition and the underlying critical Fermi surface coincides with the formation of a quantum wormhole geometry that connects the UV and the IR of the emergent dimension. The additional (conformal) symmetry at the transition links the emergent metric and the stress-energy tensor.

3. We show that the geometry-independent part of the entanglement is invariant under appropriate scaling transformations for a system of fixed number density, and is related to the Luttinger

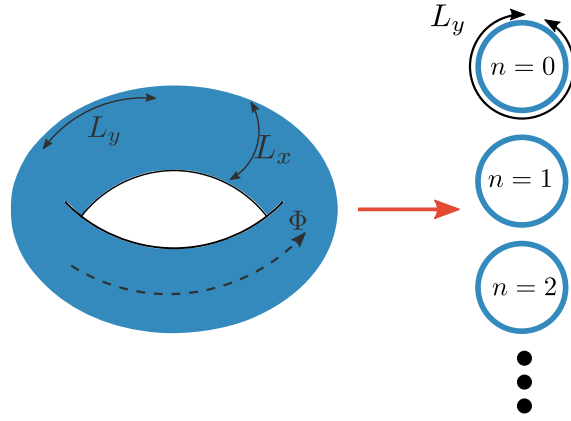


Figure 1.1: The model on the left consists of a system of non-interacting electrons placed on a torus with a flux threading through it. The model on the right consists of a sum of $1 + 1$ -dimensional systems, each consisting of massive electrons placed on a ring; the $1 + 1$ -D systems are decoupled from each other. The mass arises from the flux. These two models can be shown to be equivalent to each other [1–4].

volume of the gapless fermionic system. In the presence of a strong transverse magnetic field, the topological Luttinger volume in the metallic state transforms into the Chern number C of the insulating integer quantum Hall state.

1.2 Preliminaries and definitions

1.2.1 The system

In this work, we focus on the case of Dirac fermions in two spatial dimensions described by the Lagrangian (in natural units $\hbar = c = 1$) $\mathcal{L} = \int dx dy \bar{\psi}(x, y, t) (i\gamma^\mu \partial_\mu - m) \psi(x, y, t)$. The system is placed on a 2-torus of dimensions $L_x \times L_y$ with periodic boundary conditions, such that $\psi(x, y, t) = \psi(x + L_x, y, t) = \psi(x, y + L_y, t)$. Entanglement measures will be calculated in the presence of a gauge field A that transforms the Lagrangian by shifting the momenta: $i\partial_\mu \rightarrow i\partial_\mu + eA$. We choose the vector potential A such that, in any particular setup, it couples to just one component of the momentum (say, the x -component). This results in $\phi = eAL_x/2\pi$ units of flux quantum threading the torus in the y -direction. One of our goals is to express the Luttinger volume V_L in terms of multipartite measures of entanglement $(I_{\{\mathcal{A}_i\}}^g(\phi))$ between multiple subsystems $\{\mathcal{A}_i, i = 1, 2, \dots, g\}$ [5], in the presence of the flux ϕ . The precise definition of I^g and the method of choosing the subsystems will be given later. In order to calculate $I_{\{\mathcal{A}_i\}}^g$, however, we find it convenient to take the thin torus limit: $(L_y/L_x) \rightarrow \infty$, keeping l/L_y fixed. Importantly, this operation does not affect quantities such as V_L as long as we keep the number density unchanged.

In order to calculate entanglement measures, we will express the $2 + 1$ -dimensional system as a sum of an infinite number of massive $1 + 1$ -dimensional systems [1–4]. In the presence of a gauge

field $\vec{A} = A\hat{x}$, the complete Lagrangian (in natural units) is

$$\mathcal{L} = \int dx dy \bar{\psi}(x, y) [\gamma^\mu (i\partial_\mu - eA_\mu) - m] \psi(x, y), \quad (2.1)$$

where $A_\mu = A\delta_{\mu,x}$ and $\phi = eAL_x/2\pi$. The integral ranges over the surface area of the torus, and the time dependence of the fields has been suppressed. Due to the periodic boundary conditions (PBCs) in the x -direction, the momenta k_x are quantised: $k_x^n = \frac{2\pi n}{L_x}, n \in \mathbb{Z}$. We then expand the fields $\psi(x, y)$ in these momenta:

$$\psi(x, y) = \sum_{n=-\infty}^{\infty} e^{ik_x^n x} \psi(k_x^n, y). \quad (2.2)$$

In the absence of the gauge field, the Lagrangian for 2 + 1-dimensional massive non-interacting relativistic electrons (in natural units) is

$$\mathcal{L} = \int dx dy \bar{\psi}(x, y) [\gamma^\mu (i\partial_\mu - eA_\mu) - m] \psi(x, y). \quad (2.3)$$

The integral ranges over the area of the torus on which the electrons reside, and the time dependence of the fields has been suppressed. Each component $\psi^a, a = 0, 1, 2, 3$, of the Dirac spinor $\psi = (\psi^0 \ \psi^1 \ \psi^2 \ \psi^3)$ satisfies the Dirac equation

$$[\gamma^\mu (i\partial_\mu - eA_\mu) - m] \psi^a(x, y) = 0, \quad (2.4)$$

which is equivalent to the form

$$\int dx \bar{\psi}^a(x, y) [\gamma^\nu (i\partial_\nu + eA_\nu) + m] [\gamma^\mu (i\partial_\mu - eA_\mu) - m] \psi^a(x, y) = 0. \quad (2.5)$$

Due to the periodic boundary conditions (PBCs) in the x -direction, the momenta k_x are quantised: $k_x^n = \frac{2\pi n}{L_x}, n \in \mathbb{Z}$. We expand the fields $\psi^a(x, y)$ in these momenta:

$$\psi^a(x, y) = \sum_{n=-\infty}^{\infty} e^{ik_x^n x} \psi^a(k_x^n, y). \quad (2.6)$$

Writing the Dirac equation in terms of the dual fields $\psi^a(k_x^n, y)$ gives

$$\sum_{m,n} \left(\bar{\psi}^a(k_x^m, y) e^{-ik_x^m x} \right) [\gamma^\nu (i\partial_\nu + eA_\nu) + m] [\gamma^\mu (i\partial_\mu - eA_\mu) - m] \left(e^{ik_x^n x} \psi^a(k_x^n, y) \right) = 0. \quad (2.7)$$

Evaluating the right operation gives

$$[\gamma^\mu (i\partial_\mu - eA_\mu) - m] \left(e^{ik_x^n x} \psi^a(k_x^n, y) \right) = e^{ik_x^n x} [\gamma^\mu (i\partial_\mu - eA_\mu) - (\gamma^x(k_x^n + eA_x) + m)] \psi^a. \quad (2.8)$$

Similarly, the left operation gives

$$\begin{aligned} \left(\bar{\psi}^a(k_x^m, y) e^{-ik_x^m x} \right) [\gamma^\nu (i\partial_\nu + eA_\nu) + m] &= \left[[\gamma^\nu (i\partial_\nu - eA_\nu) + m] \left(e^{ik_x^m x} \gamma_0 \psi^a \right) \right]^\dagger \\ &= \bar{\psi}^a e^{-ik_x^m x} [\gamma_\nu (i\partial^\nu + eA_\nu) - (\gamma^x (k_x^n + eA_x) - m)] . \end{aligned} \quad (2.9)$$

Substituting these into the equation eq.(2.7) gives

$$\sum_n \int dy \bar{\psi}^a(k_x^n, y) [-(i\partial_\mu + eA_\mu)(i\partial_\mu - eA_\mu) + (k_x^n + eA_x)^2 + m^2] \psi^a(k_x^n, y) = 0 , \quad (2.10)$$

where we have used $\int dx e^{ix(k_x^m - k_x^n)} = \delta_{m,n}$. In terms of these dual fields, the Lagrangian can be shown to be equivalent to a tower of massive $1 + 1$ -dimensional fermions (see Fig. 1.1):

$$\mathcal{L} = \sum_{k_x^n} \int dy \bar{\psi}(k_x^n, y) (\gamma^\mu (i\partial_\mu - eA_\mu) - M(n, \phi)) \psi(k_x^n, y) , \quad (2.11)$$

with an effective mass $M_{n,\phi}$ given by

$$M_{n,\phi} = \sqrt{m^2 + (k_x^n + eA)^2} = \sqrt{m^2 + \frac{4\pi^2}{L_x^2} (n + \phi)^2} . \quad (2.12)$$

This decoupling ensures that the various modes $n \in \mathbb{Z}$ are disentangled from one another, such that the total density matrix ρ_{2D} of the $2 + 1$ -dimensional system can be written as a product of the density matrices ρ_{1D}^n for each of the modes:

$$\rho_{2D} = \prod_{n=-\infty}^{\infty} \rho_{1D}^n . \quad (2.13)$$

We will focus primarily on the case of massless $2 + 1$ -dimensional electrons by setting $m = 0$, such that the effective mass is given by $M_{n,\phi} = \frac{2\pi}{L_x} |n + \phi|$. Certain results will, however, be generalised to the case of massive fermions ($m \neq 0$). While the massless case corresponds to the gapless excitations of a Fermi liquid or those at the surface of a topological insulator, the massive case corresponds to gapped or insulating states of matter, e.g., the Bogoliubov-de Gennes quasiparticles above an s -wave superconducting gap. Massive Dirac fermions have also been observed experimentally in several materials like ultrathin films [6], topological insulating crystals [7] and van der Waals heterostructures [8]. They are also emergent at low-energies, for instance, in fermionic nonlinear σ -models [9], and can also be realised near the IR fixed point of the kagome Heisenberg quantum antiferromagnets placed in an external magnetic field along the z -direction [10].

1.2.2 Entanglement measures

We will work with two measures of entanglement discussed below. Given the density matrix $\rho = |\Psi\rangle \langle \Psi|$ for the ground state $|\Psi\rangle$ of the complete system, the entanglement entropy (EE) of a set of subsystems $\{\mathcal{A}_i, i = 1, 2, \dots, g\}$ with the rest of the system is given by:

$$S_{\{\mathcal{A}_i\}} = -\text{Tr} [\rho_{\{\mathcal{A}_i\}} \ln \rho_{\{\mathcal{A}_i\}}] ; \quad \rho_{\{\mathcal{A}_i\}} = \text{Tr}_{\{\mathcal{A}_i\}} [\rho] . \quad (2.14)$$

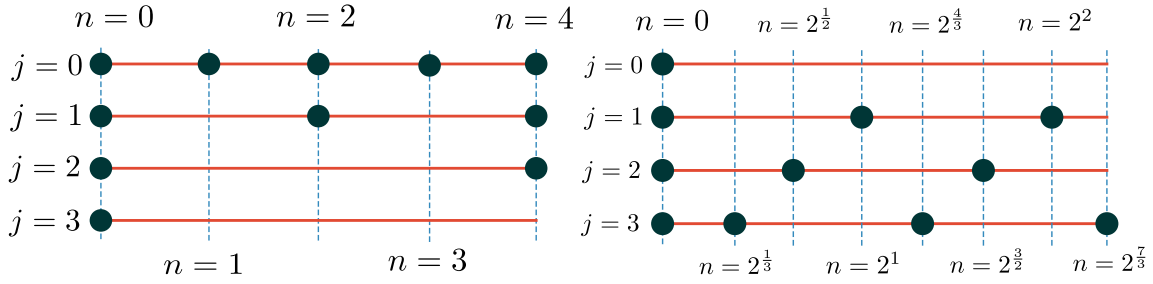


Figure 1.2: Visual depiction of the subsets $\mathcal{A}_z(j)$ employed by us in this work, for $z = 1$ and $z = -1$. As j is changed by 1, the spacing between the modes in the subset increases by a factor of 2^{j^z} . *Top:* $z = 1$. The initial subset $\mathcal{A}_z(0)$ has k -states at the modes $n = 0, 1, 2, 3, 4$. Upon changing j by 1, the new subset obtained has double the spacing between the modes ($2^{j^z} = 1/2^j$) of the previous number of modes. That is, if $j = 0$ has $0, 1, 2, 3, 4$, then $j = 1$ has $0, 2, 4$, $j = 2$ has $0, 4$ and $j = 3$ has 0 . *Bottom:* $z = -1$. In this case, the number of modes get double at each state ($2^{-j^z} = 2^j$). We start with k -modes present only at $n = 0$, but the scaling transformations lead to $j = 1$ having the modes $0, 4$, $j = 2$ having $0, 2, 4$, and so on.

$\text{Tr}[\cdot]$ is the trace operation over all degrees of freedom, while $\text{Tr}_{\{\mathcal{A}_i\}}[\cdot]$ is the partial trace over only the states corresponding to the set of subsystems $\{\mathcal{A}_i\}$. $\rho_{\{\mathcal{A}_i\}}$ is referred to as the reduced density matrix for the rest of the system (complement of the set $\{\mathcal{A}_i\}$). Using EE, one can define the mutual information $I(\{\mathcal{A}_i\} : \{\mathcal{B}_i\})$ between two subsystems A and B :

$$I_2(\{\mathcal{A}_i\} : \{\mathcal{B}_i\}) = S(\{\mathcal{A}_i\}) + S(\{\mathcal{B}_i\}) - S(\{\mathcal{A}_i\} \cup \{\mathcal{B}_i\}), \quad (2.15)$$

where $S(\{\mathcal{A}_i\} \cup \{\mathcal{B}_i\})$ is the EE of $\{\mathcal{A}_i\} \cup \{\mathcal{B}_i\}$ with the rest. Higher order measures of information among N subsystems can also be defined similarly (see, e.g., [5]). Such multi-partite measures of quantum information convey the degree of correlations present within the system - the presence of N -particle correlations ensures all information measures up to the N -partite information will be non-zero.

One final object of interest in this context is the entanglement Hamiltonian $\mathcal{H}(\{\mathcal{A}_i\})$ for a subsystem $\{\mathcal{A}_i\}$, defined through the relation [11, 12]

$$\rho_{\{\mathcal{A}_i\}} = \frac{e^{-\mathcal{H}(\{\mathcal{A}_i\})}}{\text{Tr}(e^{-\mathcal{H}(\{\mathcal{A}_i\})})}. \quad (2.16)$$

The entanglement Hamiltonian is essentially the spectrum of the reduced density matrix, and contains additional information on the nature of correlations beyond the entanglement entropy. For the system we are considering (described in the previous subsection), the total reduced density matrix can be decomposed into decoupled 1D modes (eq. (2.13)). This implies that the total entanglement Hamiltonian is simply a sum of the individual Hamiltonians for the 1D modes:

$$\rho_{2D} = \prod_{n=-\infty}^{\infty} \rho_{1D}^n \sim \prod_{n=-\infty}^{\infty} e^{-\mathcal{H}_{1D}^n(\{\mathcal{A}_i\})} = e^{-\mathcal{H}_{2D}(\{\mathcal{A}_i\})}, \quad (2.17)$$

where $\mathcal{H}_{2D} = \sum_n \mathcal{H}_{1D}^n$.

1.2.3 Luttinger volume

The Luttinger volume V_L is defined as the number of $\vec{k}$ -space points inside the Fermi surface [13, 14] :

$$V_L = \int_{G_{\vec{k}}(\omega=0)>0} \frac{1}{(2\pi)^d} d\vec{k} = \int_0^{k_F} \frac{1}{(2\pi)^d} d\vec{k} . \quad (2.18)$$

The integral (referred to as Luttinger's integral) runs over all $\vec{k}$ -space points where the $\omega = 0$ Greens function is positive, and the Fermi momentum k_F is defined as the set of points in $\vec{k}$ -space where $G_{\vec{k}}(\omega = 0)$ changes sign. Luttinger's integral can be shown to be topological in nature [15, 16], promoting Luttinger's volume to the status of a topological invariant that characterises the system. For systems with long-lived excitations close to the Fermi surface, V_L is equal to the average number density of electrons in the system, leading to Luttinger's theorem [13, 14, 17]. For systems with gapped excitations in the momentum window $k \leq k^*$ above the ground state, the violation ΔV_L of Luttinger's theorem can be quantified through a count of the bound states [18]:

$$\Delta V_L = \int_0^{k_F} \frac{1}{(2\pi)^d} d\vec{k} - \int_{k^*}^{k_F} \frac{1}{(2\pi)^d} d\vec{k} = \int_0^{k^*} \frac{1}{(2\pi)^d} d\vec{k} \quad (2.19)$$

1.3 Entanglement hierarchy in mixed momentum and real space

Before beginning our calculations of entanglement measures, we define the subsystem(s) whose entanglement we want to obtain. The subsystems we will use in this work are of the following kind:

$$\mathcal{A}_z(i) = \tilde{A}_z(i) \times A_z(i) , \quad (3.1)$$

where i takes integral values starting from zero, z can take all integral values apart from 0 and $\times$ represents the Cartesian product between the set of real space points $\tilde{A}_z(i) = \{0 \leq y \leq l\}$ and the set of momentum space points

$$A_z(i) = \left\{ k_x^n \right\}; \quad k_x^n = \frac{2\pi}{L_x} n; \quad \max(\{k_x^n\}) = \frac{2\pi N}{L_x}; \\ n \in \{-N, -N + t_z(i), \dots, -t_z(i), 0, t_z(i), N\}; \quad t_z(i) = 2^{i^z} , \quad (3.2)$$

and where $t_z(i) = 2^{i^z}$ is the difference between consecutive values of n . The subset $\mathcal{A}_z(i)$ is therefore a region in mixed momentum and real spaces, spanned by the variables k_x and y respectively. In y -space, the region is of length l and bounded by the points $y = 0$ and $y = l$. In k_x -space, $\mathcal{A}_z(i)$ comprises the points $\left\{0, \pm 2^{i^z} \frac{2\pi}{L_x}, \pm 2^{i^z} \frac{4\pi}{L_x}, \dots\right\}$ such that $|k_x| \leq \frac{2\pi N}{L_x}$. For example, for the case of $z = 1$ and N being a multiple of 4, we have

$$A_1(0) = \left[-\frac{2\pi}{L_x} N, -\frac{2\pi}{L_x} (N-1), \dots, -\frac{2\pi}{L_x} 2, -\frac{2\pi}{L_x} 1, 0, \frac{2\pi}{L_x} 1, \frac{2\pi}{L_x} 2, \dots, \frac{2\pi}{L_x} (N-1), \frac{2\pi}{L_x} N \right] , \\ A_1(1) = \left[-\frac{2\pi}{L_x} N, -\frac{2\pi}{L_x} (N-2), \dots, -\frac{2\pi}{L_x} 4, -\frac{2\pi}{L_x} 2, 0, \frac{2\pi}{L_x} 2, \frac{2\pi}{L_x} 4, \dots, \frac{2\pi}{L_x} (N-2), \frac{2\pi}{L_x} N \right] , \quad (3.3) \\ A_1(2) = \left[-\frac{2\pi}{L_x} N, -\frac{2\pi}{L_x} (N-4), \dots, -\frac{2\pi}{L_x} 8, -\frac{2\pi}{L_x} 4, 0, \frac{2\pi}{L_x} 4, \frac{2\pi}{L_x} 8, \dots, \frac{2\pi}{L_x} (N-4), \frac{2\pi}{L_x} N \right] .$$

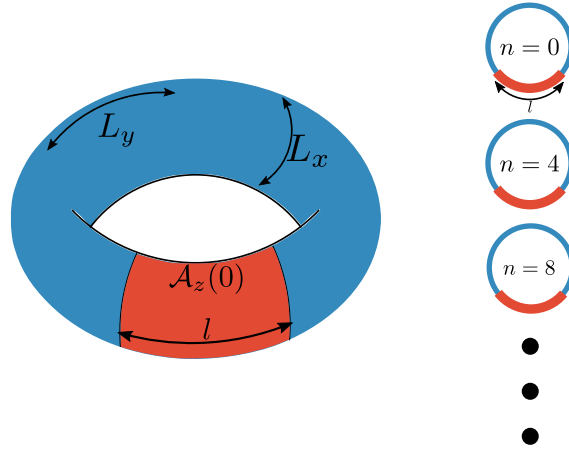


Figure 1.3: *Top*: Depiction of the subset $\mathcal{A}_z(i)$ for the choice $i = 0$. Setting $i = 0$ leads to unchanged length L_x along x and the constraint $0 \leq y \leq l$ in the y -direction. This creates an annulus (red region) of height l and circumference L_x .

Bottom: Typical k -modes present in the subset $\mathcal{A}_z(i)$ for $z = 1$ and $i = 2$. This is because, if $i = 0$ has the modes $n = 0, 1, 2, 3, 4, \dots$, increasing i by 2 increases the spacing between the modes to $2^{2^z} = 4$, so that the mode just after $n = 0$ is $n = 4$, and the one following that is $n = 8$, and so on.

Note that N represents the edge of the Brillouin zone, and will be sent to infinity when we calculate quantities in the continuum limit. In Fig. 1.2, we have shown two such families of subsets $\mathcal{A}_z(0), \mathcal{A}_z(1), \mathcal{A}_z(2), \mathcal{A}_z(3), \dots$, for $z = 1$ and $z = -1$.

We note here that the transformations for $z < 0$ correspond to increasing the density (fine-graining) of k_x -states by adding more momentum states while keeping the bounds fixed. This involves reducing the interval in k_x -space, and hence equivalent to increasing the system size L_x . This is similar to methods such as numerical renormalisation group treatment of the Kondo problem [19] where an increasing number of states are added to the system at increasingly larger distances from the impurity site, such that the system asymptotically goes to the thermodynamic limit. The other case of $z > 0$ is, on the contrary, a decimation process in k_x -space, and equivalent to reducing the system size. Indeed, it will be shown later that the sequence of Hamiltonians generated in the process (for both positive and negative z) are related to one other by renormalisation group transformations.

Within a given set $\mathcal{A}_z(i)$, the distance between adjacent k -space points is $\frac{2\pi}{L_x}t_z(i)$ where $t_z(i) = 2^{i^z}$. Choosing $i = 0$ keeps the set of k_x -space points unchanged, such that we recover the entire Lagrangian of eq. (2.1) in the x -direction for system size L_x , while the y -direction is constrained to length l . This means that $\mathcal{A}_z(0)$ is just an annulus that wraps around the x -direction of the torus and extends to a length l longitudinally in the y -direction (shown in Fig. 1.3). Along with the period $t_z(i)$, one can also define the fraction of maximum points present in the set $\mathcal{A}_z(i)$ as

$$f_z(i) = 1/t_z(i) = 2^{-i^z}. \quad (3.4)$$

For example, for $z > 0$, $f_z(i) = 2^{-i^z}$ expresses the fact that in going from the set at $j = 0$ to the set at $j = i$, the number of elements in the set is scaled down by a factor of 2^{i^z} (see Fig. 1.4).

We now define the shorthand notation $S_z(i)(\phi) \equiv S_{\mathcal{A}_z(i)}(\phi)$ to denote the EE of the subsystem $\mathcal{A}_z(i)$ in the presence of the Aharonov-Bohm flux ϕ . In order to calculate $S_z(i)$, we start with $i = 0$. As mentioned earlier, this is essentially the EE between an annulus of length l and the rest of the 2-torus. Since the density matrices are in product, the total EE is just a sum of the EE arising from each mode:

$$S_{\mathcal{A}_z(0)}(\phi) \equiv S_z(0)(\phi) = \lim_{N \rightarrow \infty} \sum_{n=-N}^N S_{\tilde{A}_z(0)}^n, \quad (3.5)$$

where $\tilde{A}_z(0)$ acts as a projection of the annulus on the $k_x - y$ plane, and S^n indicates that the EE is calculated for the n^{th} mode. Each massive fermionic mode has a correlation length-scale $\xi_n = 1/M_{n,\phi}$. We choose l to be much larger than the largest correlation length: $l \gg \max(\{\xi_n, \forall n\})$. Using the expression for ϕ and the quantisation of the momenta k_x^n , this condition can be expressed as

$$l \gg \frac{L_x}{2\pi} \times \max\left(\left\{\frac{1}{|n + \phi|}\right\}\right) = \frac{L_x}{2\pi \{\phi\}}, \quad (3.6)$$

where $\{\phi\} = \phi - \lfloor \phi \rfloor$ is the fractional part of the flux ϕ in units of the flux quantum. Since we are in the thin torus limit, L_x must remain finite, and the maximum correlation can diverge only if $\{\phi\} = 0$. We ensure that l is larger than the correlation lengths by choosing ϕ to be necessarily non-integral, as this guarantees that the correlation lengths remain finite. Then, the EE of the annulus is given by [2, 3]

$$S_z(0)(\phi) = c \left[\alpha \frac{L_x}{\epsilon} - \ln |2 \sin(\pi \phi)| \right], \quad (3.7)$$

where c is the conformal central charge and ϵ is a UV cutoff; c is $1/3$ for a Dirac fermion. The computation of $S_z(0)(\phi)$ uses the EE $S_{\tilde{A}_z(0)}^n$ of a 1-D chain of fermions with a mass $M_{n,\phi}$ [20, 21]:

$S_{\tilde{A}_z(0)}^n = c \ln \frac{L_x}{2\pi\epsilon} - c \ln |n + \phi|$. On summing over n and regularising the UV cutoff, the first term of eq.(3.7) produces the leading area-law term of $S_0(\phi)$ [3]:

$$\sum_{n=-\infty}^{\infty} c \ln \frac{L_x}{2\pi\epsilon} \rightarrow c\alpha \frac{L_x}{\epsilon}, \quad (3.8)$$

where α is a cutoff-dependent non-universal constant. The flux-dependent term in $S_{\tilde{A}_z(0)}^n$ leads to the sub-leading term of $S_z(0)(\phi)$, and requires a regularisation using the Hurwitz zeta function [3]:

$$\sum_{n=-\infty}^{\infty} \ln |n + \phi| = \sum_{n=-\infty}^{\infty} [\ln |n + \phi| + \ln |n + (1 - \phi)|] = -\frac{\partial \zeta(\phi)}{\partial \phi} - \frac{\partial \zeta(1 - \phi)}{\partial \phi}, \quad (3.9)$$

where $\zeta(\phi) = \sum_{n=-\infty}^{\infty} (n + \phi)^{-1}$ is the Hurwitz zeta function. The derivative converges to $\frac{\partial \zeta(\phi)}{\partial \phi} = \ln \Gamma(\phi) - \frac{1}{2} \ln 2\pi$, as long as the zero mode $n + \phi = 0$ is excluded from the summation. The latter condition is guaranteed as we have chosen ϕ to be a non-integer. Γ is the Gamma function, satisfying $\Gamma(\phi)\Gamma(1 - \phi) = \pi/\sin(\pi\phi)$. Substituting the derivative gives [2, 3, 22]

$$\sum_{n=-\infty}^{\infty} \ln |n + \phi| = \ln 2\pi - \ln [\Gamma(\phi)\Gamma(1 - \phi)] = \ln |2 \sin(\pi \phi)|. \quad (3.10)$$

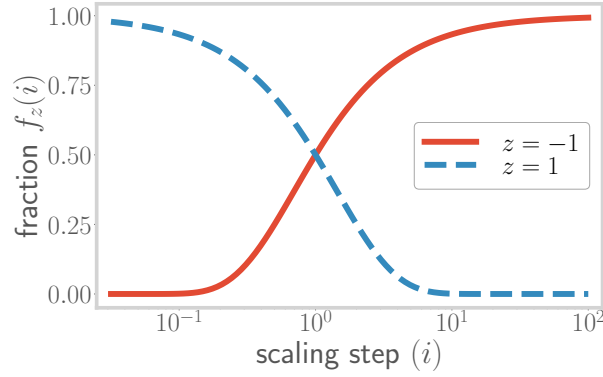


Figure 1.4: A plot of the fraction of states remaining upon performing i number of transformations on the starting Hamiltonian. Obtained by plotting the expression $f = 2^{-i^z}$, for a range of i , and for two values of z (red and blue). This fraction decreases for $z > 0$ (blue curve), and increases for $z < 0$ (red curve).

For a general $\mathcal{A}_z(i)$, there are certain modifications to the expression in eq. (3.7). For the first term in eq.(3.7), we have

$$\lim_{N \rightarrow \infty} \sum_{\substack{n=-N, \\ -N+t_z(i), \dots}}^N c \ln \frac{L_x}{2\pi\epsilon} = \lim_{N \rightarrow \infty} \sum_{\substack{n'=-\frac{N}{t_z(i)}, \\ -\frac{N}{t_z(i)}+1, \dots}}^{\frac{N}{t_z(i)}} c \ln \frac{L_x}{2\pi\epsilon} = t_z^{-1}(i) \lim_{N \rightarrow \infty} \sum_{n=-N}^N c \ln \frac{L_x}{2\pi\epsilon} \rightarrow c f_z(i) \frac{\alpha L_x}{\epsilon}. \quad (3.11)$$

Similarly, for the second term, we have

$$\begin{aligned} \lim_{N \rightarrow \infty} \sum_{\substack{n=-N, \\ -N+t_z(i), \dots}}^N \ln |n + \phi| &= \lim_{N \rightarrow \infty} \sum_{\substack{n'=-\frac{N}{t_z(i)}, \\ -\frac{N}{t_z(i)}+1, \dots}}^{\frac{N}{t_z(i)}} \ln |n' t_z(i) + \phi| = \lim_{N \rightarrow \infty} \sum_{\substack{n'=-\frac{N}{t_z(i)}, \\ -\frac{N}{t_z(i)}+1, \dots}}^{\frac{N}{t_z(i)}} \ln (f_z(i) |n' + f_z(i)\phi|) \\ &= \sum_{n'=-\infty}^{\infty} \ln |n' + f_z(i)\phi|, \end{aligned} \quad (3.12)$$

where we have assumed that $t_z(i)$ is finite, and dropped the constant part $\ln f_z(i)$ (which can be absorbed into the UV cutoff ϵ in eq. (3.11)). Following eq. (3.10), this gives

$$\lim_{N \rightarrow \infty} \sum_{\substack{n=-N, \\ -N+t_z(i), \dots}}^N \ln |n + \phi| = \ln |2 \sin(\pi f_z(i)\phi)|. \quad (3.13)$$

Combining both parts, we obtain the expression for the EE that will be employed in what lies ahead:

$$S_z(i)(\phi) = c f_z(i) \frac{\alpha L_x}{\epsilon} - c \ln |2 \sin(\pi f_z(i)\phi)|. \quad (3.14)$$

It is evident from the expression in eq. (3.14) that, as modes are added into the subsystem, the L_x -dependent geometric part of the EE increases, leading to a *hierarchical structure* in the entanglement. A similar result was previously shown to hold for geometric measures of entanglement of quantum-mechanical systems with eigenstates such as the W and GHZ states and their symmetric superpositions [23]. The variation of the geometric part with the subsystem index j is shown in Fig. 1.4.

The above-mentioned hierarchy in entanglement measures leads to an interesting property for the entanglement: the combined entanglement entropy of any number of systems $\{\mathcal{A}_z(i)\}$ is equal to the entanglement entropy of the most dense system among them. For $z > 0$, the sets become smaller as the RG index increases, while for $z < 0$, the sets become larger as the RG index increases. Thus, we may write

$$S_{\{\mathcal{A}_z(i)\}} = \theta(z)S_{\min\{i\},z} + \theta(-z)S_{\max\{i\},z}, \quad (3.15)$$

The mutual information $I^2(i, j)$ between the systems at two RG steps i and j can then be computed:

$$I^2(i, j) \equiv S_z(i) + S_{j,z} - S_{i \cup j, z} = \theta(-z)S_{\min(i,j),z} + \theta(z)S_{\max(i,j),z}. \quad (3.16)$$

In light of this subsystem-based hierarchical structure, we can also consider the g -partite information $I^g_{\{\mathcal{A}_z(i)\}}$ among g sets $\{\mathcal{A}_z(i)\}$. I^g is defined as [5]

$$I^g_{\{\mathcal{A}_z(i)\}} = \sum_{r=1}^g (-1)^{r+1} \sum_{\substack{\beta \in \mathcal{P}(\{\mathcal{A}_z(i)\}) \\ |\beta|=r}} S_\beta, \quad (3.17)$$

where $\mathcal{P}(\{\mathcal{A}_z(i)\})$ is the power set of $\{\mathcal{A}_z(i)\}$, and $|\beta| = r$ indicates that the sum is carried out only over those sets β that have r number of elements in them. For example, for $g = 3$, we have

$$I^3 = S_{\mathcal{A}_z(0)} + S_{\mathcal{A}_z(1)} + S_{\mathcal{A}_z(2)} - S_{\mathcal{A}_z(0) \cup \mathcal{A}_z(1)} - S_{\mathcal{A}_z(1) \cup \mathcal{A}_z(2)} - S_{\mathcal{A}_z(2) \cup \mathcal{A}_z(0)} + S_{\mathcal{A}_z(0) \cup \mathcal{A}_z(1) \cup \mathcal{A}_z(2)}. \quad (3.18)$$

For our choice of subsystems, the g -partite information can be shown to be equal to the EE of the *intersection set* of the concerned g -sets (details in Appendix A). For the case of $z > 0$, this takes the form

$$I^g_{\{\mathcal{A}_z(i)\}}(\phi) = S_g(\phi) = c f_{g,z} \alpha \frac{L_x}{\epsilon} - c \ln |2 \sin(\pi f_{g,z} \phi)|. \quad (3.19)$$

Since the g -partite information is simply the EE of the g^{th} subsystem, the hierarchy in the EE can thus be seen to lead to multiple levels of correlation in the system, corresponding to an increasing number of parties entering the set.

Finally, we note that the geometric (L_x -dependent) part of the EE in eq. (3.14), as well as any of the information measures I^g (eq. (3.19)), has the same form even if we start with a non-zero mass $m > 0$ [3]. This ensures that the above observations regarding the hierarchical structure of the entanglement hold even for gapped electronic systems.

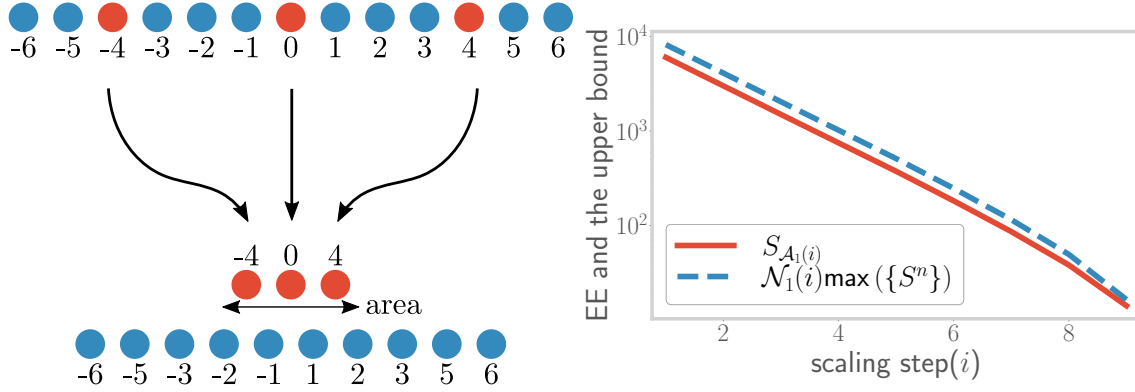


Figure 1.5: *Top:* Modes forming the subsystem $\mathcal{A}_z(i)$ (red circles) form the boundary between $\mathcal{A}_z(i)$ and its complement (blue circles). *Bottom:* Scaling of the subsystem entanglement entropy (red), and its upper bound (blue), as the RG transformations are performed (denoted by increasing index i). The bound is saturated at large i .

1.4 EHM, the Ryu-Takayanagi bound and holographic inequalities

As we will now show, the approach laid out above to the creation of subsystems amounts essentially to performing a sequence of renormalisation group (RG) transformations on the momentum modes of the Hamiltonian in the x -direction. We consider here the general case of massive Dirac electrons $M_{n,\phi} = \sqrt{m^2 + \frac{\pi^2}{L_x^2} (n + \phi)^2}$. Further, we define the superset $\mathcal{A}_z^{(0)}$ of k_x -states: $\mathcal{A}_z^{(0)} = \bigcup_{i=0}^{\infty} \mathcal{A}_z(i)$, such that all k_x -states that can appear at any point of the sequence $\{\mathcal{A}_z(i); i = 0, 1, 2, \dots\}$ are present within it. This superset can now be used to define a parent Hamiltonian $H(0)$ that consists of all the modes in the x -direction:

$$H(0) = \sum_{k_x \in \mathcal{A}_z^{(0)}} H(k_x), \quad (4.1)$$

where $H(k_x)$ is the Hamiltonian for a massive Dirac fermion that extends over a length L_y in the single spatial dimension, and having mass $M_{n,\phi}$. The Hamiltonian at any given step of the RG is then obtained by projecting onto appropriate sets of momenta. The set of momentum at step j is given by $\mathcal{A}_{j,z} = \frac{2\pi}{L_x} \times \{0, \pm t_z(j), \pm, 2t_z(j), \dots\}$ with $t_z(j) = 2^{j_z}$. The projector onto this set is given by

$$P_{j,z} = \sum_{k_x^1=0,1} \sum_{k_x^2=0,1} \dots \sum_{k_x^N=0,1} |\hat{n}_{k_x^1} \hat{n}_{k_x^2} \dots\rangle \langle \hat{n}_{k_x^1} \hat{n}_{k_x^2} \dots|, \quad k_x^j \in \mathcal{A}_{j,z}, \quad (4.2)$$

and the Hamiltonian at the corresponding step is then obtained as

$$H_z(i) = P_z(i) H(0) P_z(i). \quad (4.3)$$

In this formulation of the RG, the index i represents the RG step, while the parameter z indicates the *strength of the scaling* mechanism: for $z > 0$, a higher value of z means that the states of the Hamiltonian become rarer as the index i increases, while for $z < 0$, the states become denser with increasing i . Importantly, we will see subsequently that the parameter z corresponds also to the anomalous dimension of the coupling that leads to a spectral gap in the electronic dispersion, dictating its growth or decay under RG evolution (eq.(5.6)).

Since this is a free theory, the k -modes are decoupled, and the projection operation preserves the spectrum of the remaining degrees of freedom. The renormalised Hamiltonian (eq. (4.3)) can then be said to constitute an exact holographic mapping (EHM) [24, 25] in the direction of the RG transformations. It is also pertinent to note that the sequence of transformations in eq. (4.3) can be thought of as a simple and specific example of a unitary renormalisation group (URG) procedure [18, 26] that first disentangles and then projects out a certain set of degrees of freedom. Because the operation is unitary, the URG preserves the spectrum in the process. In the present case, the disentanglement operation is unity. The URG has already been shown to lead to a holographic renormalisation of the Hilbert space geometry which can be seen through the evolution of the Fubini-Study metric under the URG transformations [18]. Our approach differs from MERA implementations like those in refs. [27, 28] in the fact that while those involve a real space renormalisation scheme, we perform our scaling transformations in momentum space.

More evidence for the holographic nature of the construction comes from the fact that the entanglement entropy for both the massless and massive cases satisfy the bound provided by the Ryu-Takayanagi proposal that relates the entanglement of a conformal field theory (CFT) with the geometry of its gravity dual [29, 30]:

$$S_{\mathcal{A}_z(i)} \leq \mathcal{N}_z(i) \max(\{S^n\}) . \quad (4.4)$$

In the above constraint, $\max(\{S^n\})$ is the maximum single entropy of all the modes n residing in $\mathcal{A}_z(i)$, and $\mathcal{N}$ is the area of the minimal surface that separates the subsystem $\mathcal{A}_z(i)$ from the rest. In order to see how this is satisfied in the presence case, we note that since the subsystem $\mathcal{A}_z(i)$ is interspersed among the rest of the system (see left panel of Fig. 1.5), the minimal surface is as large as the subsystem itself. The area of this surface is therefore given by the number of modes in $\mathcal{A}_z(i)$. Using this information, the right hand side of eq. (4.4) can be written as

$$\mathcal{N}_z(i) \max(\{S^n\}) = \max(\{S^n\}) \mathcal{N}_{f_z(i)} = \max(\{S^n\}) \sum_{n \in \mathcal{A}_z(i)} . \quad (4.5)$$

Following eq. (2.13), the left hand side of eq. (4.4) can, on the other hand, be written as

$$S_{\mathcal{A}_z(i)} = \sum_{n' \in \mathcal{A}_z(i)} S^{n'} \leq \sum_{n' \in \mathcal{A}_z(i)} \max(\{S^n\}) . \quad (4.6)$$

Comparing eqs. (4.5) and (4.6), we recover the Ryu-Takayanagi condition of eq. (4.4). The scaling of both the left and right hand sides of eq. (4.4) has been shown in the right panel of Fig. 1.5.

Entanglement measures like entanglement entropy and mutual information (associated with disjoint intervals) have been shown to satisfy a number of inequalities in quantum field theories with gravity duals. We now consider some of them here, and show that the theory considered by us

here also satisfies these inequalities. One such constraint is the monogamy of mutual information, which says that for three disjoint regions A_1, A_2 and A_3 , we must have $I_2(A_1 : A_2) + I_2(A_1 : A_3) \leq I_2(A_1 : A_2 \cup A_3)$, where $I_2(A_1 : A_2) = S(A_1) + S(A_2) - S(A_1 \cup A_2)$ [31]. For this, we consider three disjoint regions, all of the kind depicted in the left panel of fig. 1.3, but at different positions $y_i \in [l_i, l_i + L], i = 1, 2, 3$. We assume that the distance $l_i - l_{i+1}$ between adjacent regions is much larger than the correlation length. Calabrese and Cardy have shown that the total entanglement entropy of the union of such disjoint regions is obtained by multiplying the entanglement entropy of any one region with the number of disjoint intervals. Putting all this together, we get

$$\begin{aligned} S(A_1) &= S(A_2) = S(A_3), \\ S(A_1 \cup A_2) &= S(A_1 \cup A_3) = 2S(A_1), \\ S(A_1 \cup A_2 \cup A_3) &= 3S(A_1), \end{aligned} \quad (4.7)$$

leading to the above inequality being satisfied:

$$I_2(A_1 : A_2) + I_2(A_1 : A_3) - I_2(A_1 : A_2 \cup A_3) = 0. \quad (4.8)$$

We next consider a cyclic family of inequalities in terms of conditional entropies, introduced and proved in Ref. [32]: $\sum_{i=1}^{2k+1} S(A_i | A_{i+1} \dots A_{i+k}) \geq S(A_1 \dots A_{2k+1})$, where $S(A_i | A_j) = S(A_i \cup A_j) - S(A_j)$. Using the results of eq. (4.7), we get $\sum_{i=1}^{2k+1} S(A_i | A_{i+1} \dots A_{i+k}) = \sum_{i=1}^{2k+1} S(A_i) = (2k + 1)S(A_1)$ and $S(A_1 \dots A_{2k+1}) = (2k + 1)S(A_i)$, leading to $\sum_{i=1}^{2k+1} S(A_i | A_{i+1} \dots A_{i+k}) = S(A_1 \dots A_{2k+1})$, which also satisfies the inequality. In both cases, we find that the inequality is satisfied by saturating the bound; this is a consequence of the fact that the intervals are separated by distances larger than the correlation length and the total entanglement entropies are simply a sum of the individual entropies.

1.5 Holographic geometry of the RG flow

1.5.1 RG evolution as the emergent dimension

To obtain a more specific description of the RG flow as the emergence of a geometric space, we introduce a measure of distance. For this, we will employ another bipartite measure of entanglement - the mutual information I^2 , as defined in eq. (2.15), due to its non-negative nature. We focus on the massless case $m = 0$ for the time-being. Inspired by Refs. [25, 33–35], we adopt the following definition for distance:

$$d_z(i, j) \equiv \ln \frac{I_{\max}^2}{I_z^2(i : j)}, \quad (5.1)$$

where $I_z^2(i : j) \equiv I^2(\mathcal{A}_z(i), \mathcal{A}_{j,z})$ is the mutual information between the two mentioned sets, and $I_{\max}^2$ is the mutual information between two maximally entangled systems. Accordingly, $d_z(j)$ is zero (i.e., the mutual information is maximum) at $j = 0$ if $z > 0$ and $j = \infty$ if $z < 0$, and this distance increases (decreases) with the decrease (increase) in the mutual information as the scaling transformations are performed.

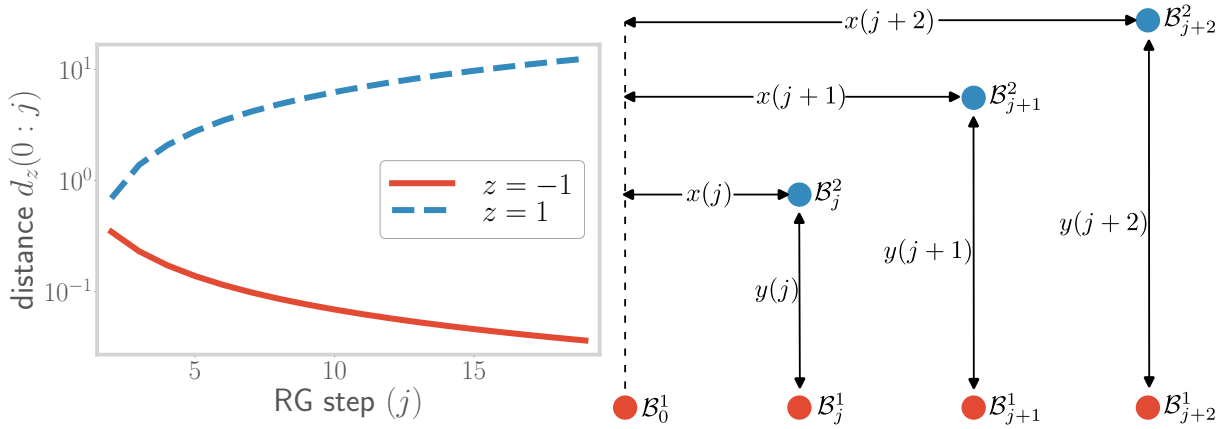


Figure 1.6: *Top*: Variation of the information distance $d_z(0 : j)$ (defined along the RG direction) under the scaling transformations. Obtained by plotting the expression for the distance, as a function of the number of transformations j , for two values of z . For $z = 1$ (blue curve), the distance between two extreme points increases, showing a decrease in entanglement. For $z = -1$ (red curve), the distance decreases, showing an increase in entanglement. *Bottom*: Geometry of subsystems used for calculating the emergent curvature. The red blobs represent the subsystems generated along the emergent direction, through the scaling transformations: $\mathcal{A}_{\epsilon_j} \rightarrow \mathcal{B}_{j+1}$. The blue blobs represent subsystems generated within the 1+1-dimensional field theory, transverse to the emergent direction.

We compute the distance between the points i and j to obtain the scaling of the distance $d_z(j)$. We assume $i < j$ without loss of generality. At each step, we tune the flux ϕ so as to remove the flux-dependent part from the EE. Following eq. (3.16), the mutual information is of the form $\theta(-z)S_i + \theta(z)S_j$, such that the distance is given by

$$d_z(i, j) = \theta(-z) \ln \frac{S_{\max}}{S_z(i)} + \theta(z) \ln \frac{S_{\max}}{S_{j,z}} = \theta(z)j^z \ln 2 + \theta(-z)i^z \ln 2. \quad (5.2)$$

For $z > 0$, the distance increases as j increases (keeping i fixed), as the k -space points are made more coarse-grained under the RG transformations. On the other hand, for $z < 0$, the distance decreases as i is increased (keeping j fixed). The rate of increase or decrease depends on the value of z . This scaling of the distance is shown in the left panel of Fig. 1.6.

At any given RG step j , we define two subsystems $\mathcal{B}_j^1$ and $\mathcal{B}_j^2$. The first subsystem $\mathcal{B}_j^1$ is the entire set of states $\mathcal{A}_{j,z}$ available at the j^{th} RG step, while the second subsystem $\mathcal{B}_j^2$ is the set comprising those modes that remain to be considered at the next RG step if $z > 0$, and the set comprising only the modes that were present in the previous step for $z < 0$. This scheme is shown in the right panel of Fig. 1.6. The mutual information between these two sets defines a measure of distance y_j at the j^{th} RG step:

$$y_z(j) \equiv \ln \frac{I_{\max}^2}{I_z^2(\mathcal{B}_j^1 : \mathcal{B}_j^2)} = \begin{cases} (j+1)^z \ln 2, & z > 0, \\ (j-1)^z \ln 2, & z < 0. \end{cases} \quad (5.3)$$

The distance along the RG direction ($x_z(j)$) is, on the other hand, defined as the distance between

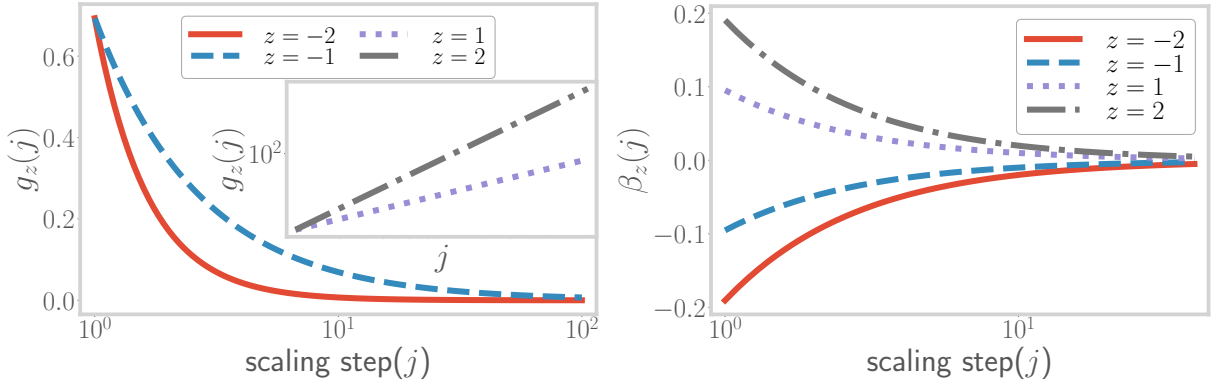


Figure 1.7: *Top:* Depiction of the RG flow of the coupling $g_z(j)$. For $z < 0$ (blue and red curves), the coupling is RG-irrelevant and shows a decrease as j is increased. For $z > 0$ (gray and violet curves), the coupling is RG-relevant and shows an increase as j is increased. *Bottom:* Variation of RG beta function $\beta_z(j)$ for the coupling $g_z(j)$, as a function of step index j . For all values of z , the beta function converges to 0 as $j \rightarrow \infty$, indicating a fixed point of the coupling g .

the current RG step and the one with the maximum entanglement entropy:

$$x_z(j) \equiv \theta(z)d(0, j) + \theta(-z)d(j, \infty) = j^z \ln 2, \quad (5.4)$$

where $d(i, j)$ was defined in eq. (5.1).

The connection of the x - and y -distances to the RG can be made manifest by relating them to the RG beta function. As we have seen above, even though we are working with massless Dirac fermions ($m = 0$), the Aharonov-Bohm flux and the dimensional reduction into $1 + 1$ -dimensional modes generate an effective mass $M_{n,\phi} = \frac{2\pi}{L_x}(n + \phi)$ for $n > 0$. At a specific RG step j , the n^{th} mode is given by $t_z(j) \times n$ on account of the RG transformations in k_x -space. With these points in mind, we define a coupling $g_z(j)$ that acts as a measure of the gap in the single-particle spectrum

$$g_z(j) \equiv \ln \frac{M_{n+1,\phi}(j) - M_{n,\phi}(j)}{2\pi/L_x} = \ln t_z(j) = j^z \ln 2. \quad (5.5)$$

The RG beta-function for the coupling $g_z(j)$ can then be written as

$$\beta_z(j) \equiv \frac{\Delta \log g_z}{\Delta j} = \log \frac{g_z(j+1)}{g_z(j)} = z \ln \left(1 + \frac{1}{j} \right), \quad (5.6)$$

where we have substituted $\Delta j = 1$. The beta function is positive for $z > 0$, indicating that the coupling $g_z(j)$ is RG-relevant in this regime (left panel of Fig. 1.7). On the other hand, the beta function is negative for $z < 0$, leading to irrelevant RG flows in this regime. The RG beta-function $\beta_z(j)$ itself decreases monotonically as j increases under the RG transformations (see Fig. 1.7): for a finite z , a fixed point is reached at $j \rightarrow \infty$ where the beta function goes to zero.

As we will now see, the flow of the RG beta functions can be related to the extremisation of a potential defined in the space of Hamiltonians. This potential is a function of the mutual information

$I_z^2(0 : j)$:

$$\beta_z(j) = -\Delta \left[\ln \left(\ln \frac{I_z^2(0 : j)}{I_{\max}^2} \right) \right]. \quad (5.7)$$

The potential given within the square brackets above corresponds to an RG monotone whose fixed point corresponds with the fixed point of the RG. For relevant flows, the above equation ensures that the function $\ln \left(\ln \frac{I_z^2(0:j)}{I_{\max}^2} \right)$ is always minimised under the RG flow, with the consequence that the trajectories ultimately lead to minimisation of the mutual information. This indicates that the relevance of the mass coupling leads in turn to the decrease of the correlation length and hence a lowering of entanglement between points that are separated in real space. On the other hand, for irrelevant flows, the above-mentioned function is maximised, implying that the RG trajectories now maximise the entanglement. This reflects the fact that the decreasing mass leads to increased correlation lengths and an approach towards criticality.

Given eqs.(5.3), (5.4) and (5.7), we can relate the distances x and y to the beta function β_z :

$$x_z = \left(e^{\frac{\beta_z}{z}} - 1 \right)^{-z} \ln 2, \quad (5.8)$$

$$y_z = \begin{cases} x_z e^{\beta} & z > 0, \\ x_z \left(2 - e^{\frac{\beta}{z}} \right)^z & z < 0. \end{cases} \quad (5.9)$$

The contrasting behaviour of the beta function depending on the sign of z indicates that it tracks the effective number of degrees of freedom that contribute to the entanglement (eq. (3.14)) at any given RG step. Indeed, we find that the effective number of degrees of freedom is the effective c -function (or central charge) $\tilde{c}_z(j)$ of the theory, taking the form $\tilde{c}_z(j) = f_z(i)c$. The beta function is thus related to the c -function as

$$\beta_z = \Delta \left[\ln \left(\ln \frac{c}{\tilde{c}_z(j)} \right) \right]. \quad (5.10)$$

For RG relevant flows, the c -function is found to diminish from the free fermion central charge $c = 1/3$, reaching the value of zero at the fixed point (where only the central $n = 0$ mode remains and the rest have been removed because of the large mass gap). The non-negative, non-increasing nature of the c -function, and its flow towards its minimum at the fixed point, leads to speculate that eq.(5.10) can be thought of as a holographic counterpart of Zamalodchikov's c -theorem [36–39]. For RG irrelevant flows, the behaviour is reversed. The flow of the c -function is shown in the left panel of Fig. 1.8.

1.5.2 Embedding the emergent dimension on a metric space

We have shown above that the renormalisation group flow of the Hamiltonian leads to the emergence of an additional direction, quantified by the parameter x_j . This will now be used to construct a metric, in the following way. We already know that the scaling transformations lead to a graph of theories that are connected via edges that represent the mutual information between any two theories. As a result, this “information graph” can be mapped on to a new “distance graph” where the edges

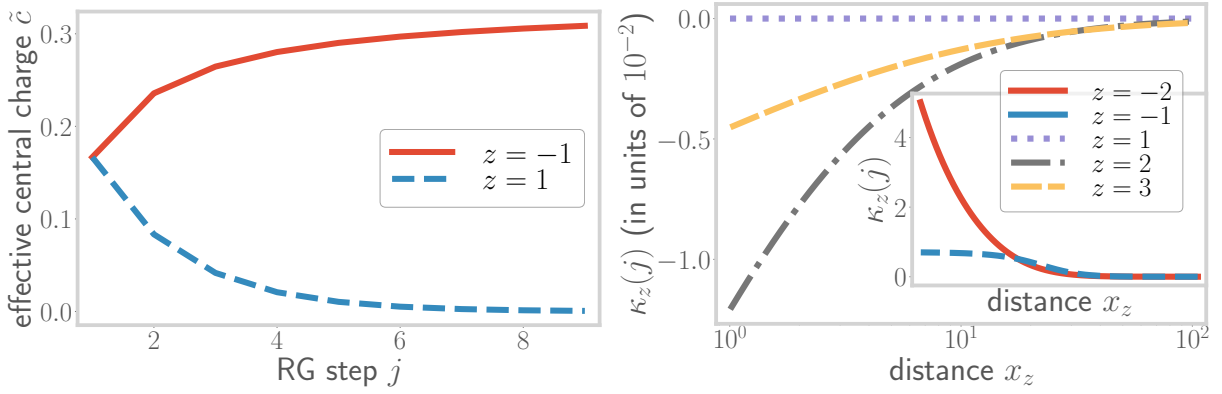


Figure 1.8: *Top*: Variation of the effective central charge defined in eq. (5.10), under the RG transformations for relevant ($z = 1$, blue curve) and irrelevant ($z = -1$, red curve) flows. In the former case, the central charge starts from the fermionic gapless value of $1/3$ and vanishes towards the fixed point, indicating the flow to a gapped massive phase. In the latter case, it starts from the gapped value of 0, and increases to $1/3$ at the fixed point, indicating the flow to a gapless phase. *Bottom*: Variation of the curvature $\kappa_z(j)$ along the RG for multiple values of z . $z = 1$ leads to a flat space with zero curvature, while $z > 1$ and $z < 1$ (inset) lead to negative and positive curvature respectively.

now represent the distance between the theories at the nodes. Finally, these distances can be used to define a metric for the distance graph: the metric between any two nodes will be the distance along the shortest path between these two nodes. This approach as well as some of the terminology are inherited from reference [40].

The RG flow can be represented as an “information graph” $G(V, E)$, where the systems $\mathcal{B}_j^1$ (defined above eq. (5.3)) constitute the set of vertices $V = \{V_j\}$ and the mutual information $I_z^2(j : j')$ between the systems $\mathcal{B}_j^1$ and $\mathcal{B}_{j'}^1$, constitute the edges E_{ij} between the vertices. This information graph is characterised by an adjacency matrix whose elements are defined by the mutual information edge weights $I_z^2(j : j')$ [41]. This graph structure is shown in Fig. (1.6), with the red spheres indicating the vertices. In order to define a metric space, we will map the graph G to another graph $\mathcal{G}$ that has the same vertices as G , but whose edges are weighted according to a well-defined notion of distance.

A path $\mathcal{P}$ in $\mathcal{G}$ is defined as a sequence of m vertices: $\mathcal{P}(V_1, V_2, \dots V_m)$. The distance along the path is defined as the sum of distances along every pair of consecutive vertices: $l(\mathcal{P}) = \sum_{i=1}^{m-1} d(V_i, V_{i+1})$, where $d(V_i, V_{i+1}) = d(i, i+1)$ (see eq. (5.1)). The *geodesic* between two points V_i and V_j is the path $\mathcal{P}$ which minimises the distance $l(\mathcal{P})$, and the distance along the geodesic then defines the *metric* $\tilde{d}(V_i, V_j)$ between two vertices V_i and V_j [40, 41]:

$$\tilde{d}(V_i, V_j) = \min_{\mathcal{P}} \{l(\mathcal{P})\} = \min_{\mathcal{P}} \left\{ \sum_{i=1}^{m-1} d(V_i, V_{i+1}) \right\}, \quad (5.11)$$

where the minimisation is carried out over all possible connected paths $\{\mathcal{P}\}$ between V_i and V_j , and m is the total number of vertices in a particular connected path $\mathcal{P}_m$. Following the expression for

the distance $d(i, j)$ obtained in eq. (5.2), the metric takes the form

$$\tilde{d}(V_i, V_j) = \theta(z)j^z \ln 2 + \theta(-z)i^z \ln 2, \quad (i < j), \quad (5.12)$$

where we have assumed $i < j$, without loss of generality.

Finally, the metric dimension of the emergent space can be determined by finding the set with the lowest possible cardinality that can resolve the metric space [40, 42]. In order for a set to resolve the metric space, we must have $d(i, a) = d(j, a) \implies i = j$ for all points a in the set and for any pair of points i and j in the metric space. In our case, the singleton set $\mathcal{R} = \{0\}$ can resolve the metric space for $z > 0$, as $d(i, 0) = d(j, 0) \implies i^z = j^z \implies i = j$. Similarly, the singleton set $\mathcal{R} = \{\infty\}$ resolves the metric space for $z < 0$. As a result, we conclude that our emergent metric space is one-dimensional in nature.

1.6 Curvature of the emergent space

Further insight on the spatial geometry of the RG flows can be obtained by looking at the curvature generated by the RG transformations. In the plane of the directions x and y generated by the RG flow, the curvature can be written using the ideas of analytic geometry as

$$\kappa_z(j) = \frac{v'_z(j)}{[1 + v_z(j)^2]^{\frac{3}{2}}}, \quad (6.1)$$

where

$$\begin{aligned} v_z(j) &\equiv \frac{\Delta y_z(j)}{\Delta x_z(j)} = \frac{y_z(j+1) - y_z(j)}{x_z(j+1) - x_z(j)} \\ &= \begin{cases} \frac{(j+2)^z - (j+1)^z}{(j+1)^z - j^z} & z > 0, \\ \frac{(j)^z - (j-1)^z}{(j+1)^z - j^z} & z < 0, \end{cases} \end{aligned} \quad (6.2)$$

and

$$v'_z(j) \equiv \frac{v_z(j+1) - v_z(j)}{x_z(j+1) - x_z(j)}. \quad (6.3)$$

The curvature $\kappa_z(j)$ has been plotted against the RG step index j in right panel of Fig. 1.8. For $z = 1$, the first derivative $v_z(j)$ becomes unity, so the second derivative $v'_z(j)$ (and hence the curvature) vanishes, leading to a flat (Minkowski) space. For $z > 1$, the curvature becomes negative, while it is positive for $z < 0$. All the flows lead to an asymptotically flat space (i.e., the curvature vanishes) as $j \rightarrow \infty$.

The different signatures of the emergent curvatures can be attributed to the different ways in which the mass of the theory scales under the decimation procedure. For $z > 0$, the curvature defined in eq. (6.1) can be written in terms of the beta function:

$$\kappa_z(j) = \frac{[e^{\beta(j)} - 1]^{\frac{1}{2}} j^{-z}}{[e^{\beta(j)+\beta(j+1)} - 1]^{\frac{3}{2}}} \left[\frac{e^{\beta(j+2)} - 1}{1 - e^{-\beta(j+1)}} - \frac{e^{\beta(j+1)} - 1}{1 - e^{-\beta(j)}} \right]. \quad (6.4)$$

This equation relates directly a geometric quantity (i.e., the curvature) of the bulk emergent space with the RG beta function for the field theory on the boundary. From Fig. 1.7, as the beta functions flow towards zero, we see from eq. (6.4) that the first fraction inside the square brackets is the first to vanish, followed by the second fraction. The sequential vanishing of these two fractions prior to the vanishing of the denominator $e^{\beta(j)} - 1$ in eq.(6.4) leads to the asymptotic vanishing of the curvature towards the fixed point. Different choices of $z > 0$ can be thought to represent different relevant perturbations that drive the system towards *gapped fixed points*, leading to distinct curvatures of the bulk. By employing the relation $v_z(j) = \Delta S_{j+1,z} / \Delta S_{j,z}$, eq. (6.1) can also be cast in terms of the entanglement entropy $S_{j,z}$, $S_{j+1,z}$ and $S_{j+2,z}$. The resulting non-linear relationship between $\kappa_z(j)$ and $S_{j,z}$ obtained is, however, in contrast to the linear dependence obtained by Cao et al. [40] between entanglement perturbations on the Hilbert space and the curvature of the resultant holographic space.

1.6.1 Topological nature of the sign of curvature

We will now argue that the signature of the curvature κ corresponds to a topological quantum number, and that a change in the signature as z is tuned through 1 amounts to a topological translation. For this, we first point out (following eq. (6.1)) that the sign of the curvature is determined purely by the sign of $\gamma_z(j) \equiv 1 - v_z(j+1)/v_z(j)$ and $\alpha_z(j) = y_z(j+1) - y_z(j)$:

$$\begin{aligned} \kappa_z(j) &= \frac{v'_z(j)}{[1 + v_z(j)^2]^{\frac{3}{2}}} = -\frac{\alpha_z(j) \gamma_z(j)}{(\Delta x_z(j))^2 [1 + v_z(j)^2]^{\frac{3}{2}}} \\ \implies \text{sign} [\kappa_z(j)] &= -\text{sign} [\alpha_z(j)] \text{sign} [\gamma_z(j)] . \end{aligned} \quad (6.5)$$

We now promote j to a continuous complex variable with the real part being greater than or equal to 1: $\text{Re} [j] \geq 1$. The quantities $\alpha_z(j)$ and $\gamma_z(j)$ can be written as

$$\alpha_z(j) = \begin{cases} [(j+2)^z - (j+1)^z] \log 2, & z > 0 \\ [j^z - (j-1)^z] \log 2, & z < 0 \end{cases} , \quad (6.6)$$

$$\gamma_z(j) = \begin{cases} 1 - \frac{[(j+3)^z - (j+2)^z][(j+1)^z - j^z]}{[(j+2)^z - (j+1)^z]^2}, & z > 0 \\ 1 - \frac{[(j+1)^z - j^z]^2}{[j^z - (j-1)^z][(j+2)^z - (j+1)^z]}, & z < 0 . \end{cases} \quad (6.7)$$

As shown in Fig.1.9(left panel), the function γ_z is positive and monotonically decreasing for $z > 1$ or $z \leq -1$, and vanishes identically for $z = 1$. On the other hand, $\alpha_z(j)$ is positive for $z > 1$, zero at $z = 1$ and negative for $z \leq -1$ (see Fig.1.9(right panel)). Combining this, we find that the sign of the curvature can be written as:

$$\text{sign} [\kappa_z] = \begin{cases} -1, & z \geq 1 \\ 1, & z \leq -1 \end{cases} = \begin{cases} -\text{sign} [\gamma_z(j)], & z \geq 1 \\ -\text{sign} [\alpha_z(j)], & z \leq -1 . \end{cases} \quad (6.8)$$

We start by considering the region $z \geq 1$. Since $z = 0$ is excluded by definition, the denominator is always non-zero. This indicates that $\gamma_z(j)$ is continuous and monotonically decreasing in this

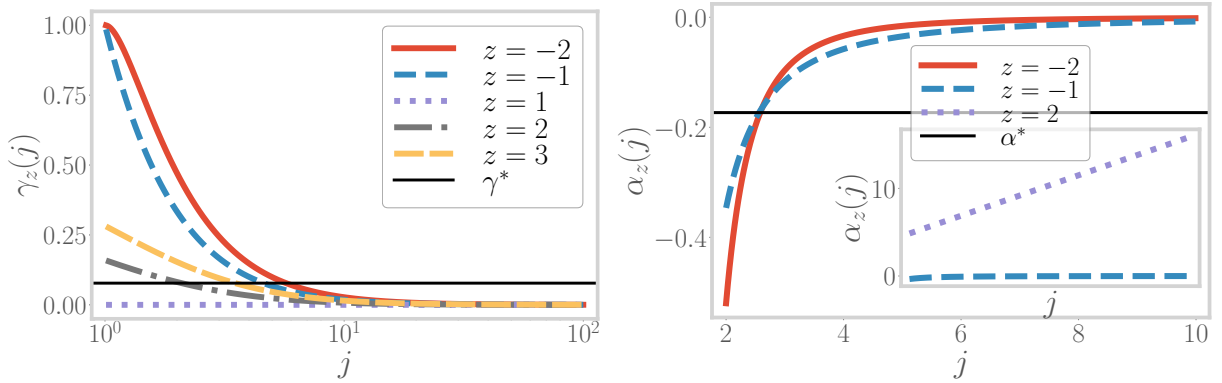


Figure 1.9: *Top*: Variation of γ_z along the RG for multiple values of $z > 0$. All trajectories for $z > 1$ intersect the line $y = \gamma^*$. *Bottom*: Variation of α_z along the RG for multiple values of $z < 0$. All trajectories intersect the line $y = \alpha^*$

range: the maximum value of $\{\gamma_z(j), j \in [1, \infty]\}$ occurs at $j = 1$. Further, the maximum value $\gamma_z(1)$ increases with the value of z , ranging between $\gamma_2(2)$ at $z = 2$ and unity at $z \rightarrow \infty$. These two features are demonstrated in the left panel of Fig. 1.9, and imply that all curves $\{\gamma_z(j) : j\}$ will take values both greater than and less than the value $\gamma^* \equiv \gamma_2(2)$ (dotted line in left panel of Fig. 1.9). Thus, the plot shows that there always exists a real value j_z^* for any given integer value $z > 1$ such that $\gamma_z(j_z^*) = \gamma^* = \gamma_2(2)$. This is precisely what ceases to happen for $z = 1$, as that is the only curve in the left panel of Fig. 1.9 that does not pass through γ^* .

Indeed, the difference between the curvature for the $z = 1$ and $z > 1$ classes can be captured by expressing the presence of the solution j_z^* in the form of a winding number. For this, we use the fact that the sign of $\gamma_j(z)$ is given by the following integral:

$$\text{sign} [\gamma_z(j)] \Big|_{z \geq 1} = \frac{1}{2\pi i} \oint_C dj \frac{\partial}{\partial j} \ln [\gamma_z(j) - \gamma^*] , \quad (6.9)$$

where the contour C extends from $j = 1$ to $j = \infty$, and is shown in the left panel of Fig. 1.12. As the value of z is tuned from 2 to 1, the pole at j_z^* disappears and the integral reduces to zero. As shown in Fig. 1.10, the integral can thus be expressed as the winding number of the curve $\gamma_z(C(j))$ that counts the number of times the contour $\gamma_z(C)$ winds around the singularity at γ^* :

$$\text{sign} [\gamma_z(j)] = \frac{1}{2\pi i} \oint_C dj \frac{\partial(\gamma_z(j) - \gamma^*)}{\gamma_z(j) - \gamma^*} = \frac{1}{2\pi i} \oint_{\gamma_z(C)} \frac{d\gamma_z}{\gamma_z - \gamma^*} . \quad (6.10)$$

This is easily seen from the following argument. If we write a general complex variable y in the form $y = r e^{i\theta}$ and consider a general contour C that starts from $r = r_0, \theta = 0$ and ends at $r = r_0, \theta = 2w\pi$ (such that the winding number of the curve is w), integrals of the form given above simplify and return the winding number itself:

$$\frac{1}{2\pi i} \oint_C \frac{dy}{y} = \frac{1}{2\pi i} \int_{r_0}^{r_0} \frac{dr}{r} + \frac{1}{2\pi} \int_0^{2w\pi} d\theta = w , \quad (6.11)$$

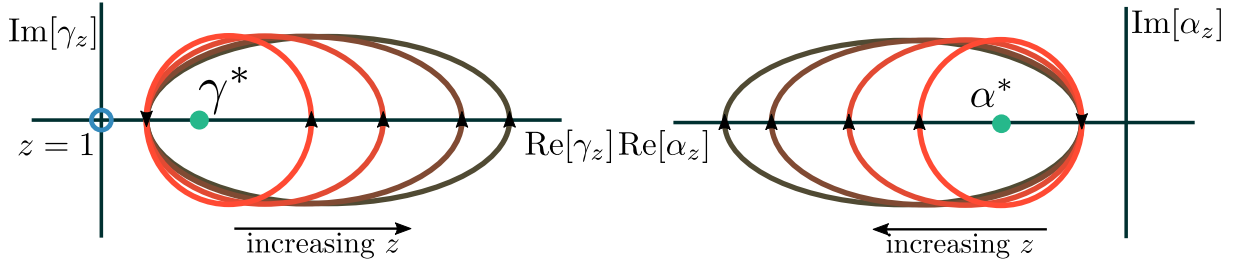


Figure 1.10: *Top*: Contours $\gamma_z(C)$ for multiple values of $z > 0$. As z increases, the contours extend in the positive $\text{Re}[\gamma_z]$ direction. *Bottom*: Contours $\alpha_z(C)$ for multiple values of $z > 0$. As z increases, the contours extend in the negative $\text{Re}[\gamma_z]$ direction.

where we used $dy = y \left(\frac{dr}{r} + d\theta \right)$.

The winding numbers take integer values, and are topological in nature: they are robust against geometric deformations, and change only when the curve crosses the singularity at $\gamma = \gamma^*$. Thus, for $z > 1$, the contour $\gamma_z(C)$ encloses the singularity at γ^* exactly once, leading to a winding number of unity. As z is lowered towards $z = 2$, the contours becomes smaller such that, in order to go from $z = 2$ to $z = 1$, the contour has to be moved across the singularity. This constituting a change in the topology of the contour: for $z = 1$, the contour no longer encloses the pole and the winding number is zero. In fact, at $z = 1$, the contour itself (depicted in blue) is of vanishing radius because $\gamma_1(j)$ has a constant value of zero. To complete the connection, we use eq. (6.5) to relate this topological winding number to the sign of the curvature:

$$\text{sign}[\kappa_z] \Big|_{z \geq 1} = -\mathcal{W}_z(\gamma^*), \quad (6.12)$$

where $\mathcal{W}_z(\gamma^*)$ is the winding number for $\gamma_z(C)$ around γ^* .

One can make a very similar argument for the case of $z \leq -1$. Here, as shown in the right panel of Fig. 3.10, the relevant quantity is $\alpha_z(j)$. For each curve α_z , the minimum value is at $j = 1$. For all z , this minimum starting value is largest for $z = -1$, and all subsequent curves $\alpha_z(j)$ start from even lower values. If we define a value $\alpha^* \equiv \alpha_z(j = 1)/2$, it is then clear that all curves $\alpha_z(j)$ will pass through the value α^* (dotted line in right panel of Fig. 3.10) at some value of j . This is very similar to the situation discussed just above for the case of $z \geq 1$ (with γ^* playing the role of α^*); we, therefore, employ the same technique here. Taking the same contour C as defined above, we write the sign of α_z as the winding number of the curve $\alpha_z(C)$ around the point $\alpha_z = \alpha^*$

$$\text{sign}[\alpha_z(j)] \Big|_{z \leq -1} = -\mathcal{W}'_z(\alpha^*) = \frac{1}{2\pi i} \oint_{\alpha_z(C)} \frac{d\alpha_z}{\alpha_z - \alpha^*}. \quad (6.13)$$

For $z > 0$, α_z is positive, and the curve $\alpha_z(C)$ encloses points in the positive half of the real axis such that the winding number in the right panel of Fig. 1.10 is then zero. This indicates that the function $1 - 2\mathcal{W}'_z(\alpha^*)$ is unity for $z > 0$, leaving any other function in product with it unchanged. On the other hand, it is -1 for $z < 0$, returning the correct sign of $\alpha_z(j)$ in that range. Combining with eq. (6.12), the sign in the entire range can be expressed as

$$\text{sign}[\kappa_z] = \mathcal{W}_z(\gamma^*) \times [2\mathcal{W}'_z(\alpha^*) - 1]. \quad (6.14)$$

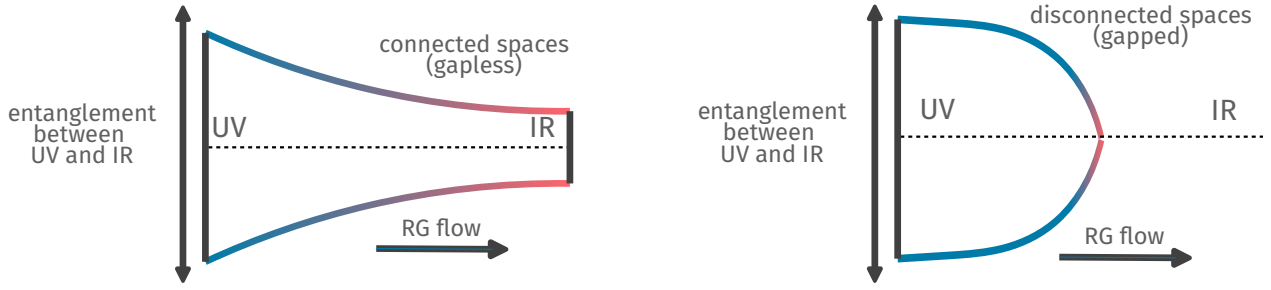


Figure 1.11: Schematic depiction of the emergent geometries in the cases of gapless (top) and gapped (bottom) RG flows. While the gapless RG flow involves a growth in the entanglement between the UV and IR spaces, the latter leads to the vanishing of UV-IR entanglement. This results in the emergence of connected and disconnected geometries, respectively.

We can now summarise the topological nature of the sign of the curvature. For $z > 0$, the second winding number $\mathcal{W}$ is 0. As z varies from 2 to 1, the other winding number $\mathcal{W}$ changes from 1 to 0, constituting the first topological change. In contrast, upon varying z from 1 to -1 , both winding numbers change: $\mathcal{W}$ from 0 to 1, and $\mathcal{W}$ from 0 to 1. This constitutes the second topological change. Together, the two changes show that going from an open geometry (negative curvature) to a closed geometry (positive curvature) involves two topological transformations.

1.7 Entanglement Holography and Fermionic Criticality

1.7.1 Equivalence between a critical Fermi surface and a holographic worm-hole

As mentioned previously under eq. (4.3), the quantity z corresponds to the anomalous dimension of the spectral gap g_z in the effective field theory: a change in the sign of z corresponds to a change in the nature of the RG flow from relevance to irrelevance and vice-versa. The holographic nature of the RG flow dictates that such a change arises by a variation in some quantity in the parent interacting theory from which the effective field theory of non-interacting Dirac fermions can be thought to have originated. Indeed, such a change in the RG relevance has drastic consequences: a relevant spectral distance g_z implies a gap in the excitations at the fixed point, while an irrelevant g_z indicates the presence of gapless excitations. The presence or absence of a gap in the low-energy excitations determines the topology of the electronic Fermi surface through the change of a topological invariant, the Luttinger volume of the system [13, 15, 16, 18, 26] (also see eq. (2.19)).

At the transition between the gapless (metallic) and gapped (insulating) states, the Fermi surface becomes critical, signalled by the vanishing lifetime of the gapless excitations close to it. Our formulation shows that such a gap-inducing transition and its associated quantum critical Fermi surface has consequences for the space generated along the RG flow. For the gapless case ($z < 0$), the degrees of freedom in the UV and the IR have finite entanglement, indicating that they are part of the same connected holographic space (see Figs. 1.6 and 1.7), while for the gapped case ($z > 0$), the entanglement vanishes into the IR, leading to a decoupling of these two regions and their separation

into disconnected spaces [33]. These two scenarios are shown schematically in Fig. 1.11. It is therefore tempting to conclude that at the transition between the gapped and gapless states, a *wormhole geometry* connects the UV and the IR via some minimal entanglement [33, 40]. Our results also suggest that the presence of such a wormhole would then coincide with two topological transitions, the first being the change in the sign of the curvature (as shown in the previous subsection), and the second being the change in the topological Luttinger volume of the underlying quantum field theory (as discussed in the previous paragraph). We end by noting that we have provided, in a later section of this work, additional support for the topological nature of Luttinger's volume and its connection to the holographically generated space by relating it to multipartite entanglement (see eq. (9.17)).

1.7.2 Connecting modular energy with the metric

The notion of an entanglement Hamiltonian was introduced in eq. (2.16). At the transition between the gapless (metallic) and gapped (insulating) states, the system becomes critical, and the Dirac electrons are described by a CFT. It turns out that for such a CFT, the entanglement Hamiltonian takes a particularly simple form. In this subsection, we will relate the emergent metric to the entanglement Hamiltonian obtained at the IR fixed point, focusing on the theory at the transition.

For a 1D CFT, the entanglement Hamiltonian corresponding to a partition of length l is given by [43, 44]

$$\mathcal{H}_{1D} = \frac{2\pi L}{c} \int_0^l dx \frac{L^2 - x^2}{2L^2} T_{00}(x), \quad (7.1)$$

where $T_{\mu\nu}$ is the stress-energy tensor of the CFT and c is the speed of light. We have already shown in eq. (2.17) that the entanglement Hamiltonian for the 2D theory is simply a sum of the 1D counterparts, owing to the decoupling of 1D modes in the Lagrangian. Working in a region where T_{00} is approximately uniform, the 2D entanglement Hamiltonian can be expressed as

$$\mathcal{H}_{2D} \sim \sum_n \mathcal{H}_{1D} = \frac{2\pi L^2}{3c} \sum_n T_{00}^n, \quad (7.2)$$

where T_{00}^n is the average value of T_{00} within the partition for the CFT associated with the n^{th} mode.

In order to make the connection with the emergent geometry, we use the entanglement first law [45], which states that at first order, small variations ΔS of the entanglement entropy can be equated to the associated change in the expectation value of the entanglement Hamiltonian:

$$\Delta S(A) = \Delta I_2(A_0 : A) = \langle \Delta \mathcal{H} \rangle = \frac{2\pi L^2}{3c} \sum_n \langle T_{00}^n \rangle, \quad (7.3)$$

where $\langle \mathcal{H} \rangle = \text{Tr}(\rho \mathcal{H})$ is often referred to as the modular energy, and we have equated $\Delta S(A)$ and $\Delta I_2(A_0 : A)$ following eq. (3.16). Further, since the entanglement at the critical point is expected to be large, we can approximate the expression for the metric between points (i, j) obtained in subsection 1.5.2 as $\tilde{d}(i, j) = -\ln I^2(i : j)/I_{\text{max}}^2 \simeq |I^2(i : j)/I_{\text{max}}^2 - 1|$. Combining this equation with eq. (7.3) leads to the following interesting relation:

$$\Delta \tilde{d}(i, j) \simeq \frac{2\pi L^2}{3c} \sum_n \langle T_{00}^n \rangle. \quad (7.4)$$

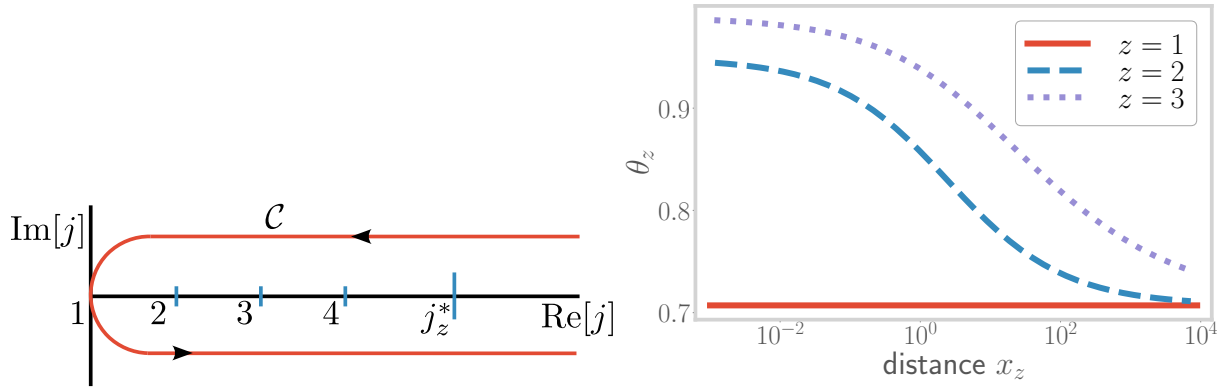


Figure 1.12: *Top*: The contour C used in eq. (6.9). It extends across all values of j , and the integral counts the number of poles of the function $1/(\gamma - \gamma^*)$ encircled by the contour. *Bottom*: Evolution of expansion parameter θ_z with distance x_z , leading to the fixed point at $\theta_z^* = 1/\sqrt{2}$.

The equation relates the change in a geometric measure (the emergent metric) with a mass/energy term. An analogous relation was obtained for states for which the entropy obeys an area law by Cao et al [40], connecting the spatial curvature and the stress tensor, ultimately leading to a semi-classical Einstein's field equation for gravity. Our formulation does not however lead to a relation involving the curvature, and we speculate that this is due to the violation of the area law in entanglement of fermionic critical systems. Instead, we show in the next section that the curvature in our formulation is related to the rate of convergence of the RG flows.

1.7.3 Kinematics of the RG flows

Further statements on the RG flow and the emergent holographic dimension can be made by studying quantities that can serve as candidates for the convergence parameter θ for the RG flows of g_z [46,47]. We consider the following function as a candidate for the convergence parameter for the case of $z > 0$:

$$\theta_z(j) = \frac{1}{\sqrt{v_z^{-2} + 1}} = \left[1 + \left\{ \frac{1 - e^{-\beta_z(j)}}{e^{\beta_z(j+1)-1}} \right\}^2 \right]^{-\frac{1}{2}}. \quad (7.5)$$

The parameter θ_z has been plotted in Fig. 1.12, and the behaviour is observed to be very similar to that of the beta function shown in Fig. 1.7. Indeed, this similarity describes the holographic nature of the RG flow: the emergent geometry (as seen through θ_z) is dictated by the RG flow described by β .

The quantity θ can be interpreted as an estimate of the rate of change of area enclosed by a fixed number of RG trajectories. To see that this is the case, we note that by using $v_j = \Delta y / \Delta x$, the expansion parameter can be written as $\theta = \Delta y(j) / \sqrt{\Delta x^2 + \Delta y^2} \sim \Delta g(j+1) / \sqrt{\Delta x^2 + \Delta y^2}$. Considering RG flows for $g_z(j)$ corresponding to multiple z , the area enclosed by the curves is given by the maximum g_z , and Δg_z is then the change in the area under the RG transformation. The denominator $\sqrt{\Delta x^2 + \Delta y^2}$ is the distance covered in the emergent spacetime, rendering θ as the rate of change of the area.

The equation of motion for this expansion parameter can then be obtained as:

$$\frac{d\theta_z}{dx} = \frac{v_z^{-3}}{(1 + v_z^{-2})^{\frac{3}{2}}} \frac{dv_z}{dx} = \kappa_z. \quad (7.6)$$

The simple form of this relation makes it clear that for relevant flows (corresponding to a negative curvature $\kappa_z < 0$), θ_z flows to a minimum. Following Fig. 1.8, this is accompanied by the minimisation of the c -function to zero. It is, therefore, tempting to conclude that eq. (7.6) for the spatial evolution of the convergence parameter θ is dual to eq. (5.10) that controls the flow of the c -function of the quantum theory along the RG flow.

1.8 Extension to the massive case

We will now generalise several of the main results obtained above to the case of two-dimensional electrons with a single-particle mass-gap $M_{n,\phi} = \sqrt{m^2 + \frac{4\pi^2}{L_x^2}(n + \phi)^2}$. Given that they depend only on the geometric part of the entanglement (which remains unchanged on introducing the mass $M_{n,\phi}$), the following set of key qualitative results and observations discussed earlier continue to hold for the case of a dispersion with a mass-gap:

- the hierarchical structure of the entanglement content (e.g., the area-law term in eqns.(3.14) and (3.17)) and the fact that the entanglement satisfies the Ryu-Takayanagi bound (right panel of Fig.1.5),
- the dependence of the emergent distances x and y on the RG flow (Fig.1.6, eqns.(5.2) and (5.3)), and the fact that the RG beta function tracks the central charge (Fig.1.8),
- various qualitative and quantitative properties of the curvature (e.g., eq.(6.1)), including the topological nature of its sign (eq.(6.14)), and
- the existence of a convergence parameter that is related to the RG flow, and the equation that governs the evolution of the convergence parameter (eq.(7.6)).

The functional forms of certain other results do, however, undergo changes. For instance, the coupling $g_z(j)$ is given by the (more general) expression

$$\begin{aligned} g_z(n, j) &= \ln \frac{M_{n+1,\phi}(j) - M_{n,\phi}(j)}{2\pi/L_x} \\ &= \ln \left[\sqrt{\tilde{m}^2 + (t_z(j)(n+1) + \phi)^2} - \sqrt{\tilde{m}^2 + (t_z(j)n + \phi)^2} \right], \end{aligned} \quad (8.1)$$

where $\tilde{m} = mL_x/2\pi$ is a dimensionless parameter. We will work with the coupling for the mode $n = 0$:

$$g_z(0, j) = \ln \left[\sqrt{\tilde{m}^2 + (t_z(j) + \phi)^2} - \sqrt{\tilde{m}^2 + \phi^2} \right]. \quad (8.2)$$

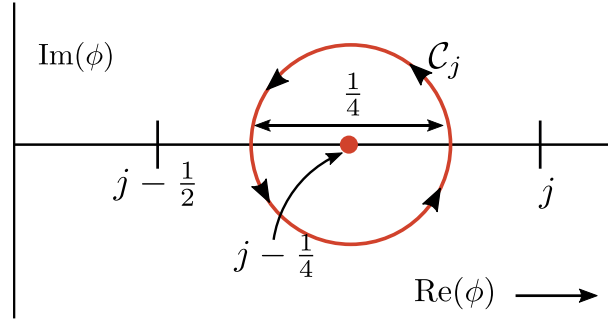


Figure 1.13: Contour C_j used to integrate eq. (9.3). The contour is chosen so as to encircle the simple pole of $1/Q$ at $\phi = j - 1/4$.

In order to simplify the analysis, we work in the large mass limit $\frac{t_z + \phi}{\tilde{m}} \rightarrow 0$. In this limit, upon expanding the square root and retaining only the lowest order term, we obtain

$$g_z(0, j) = \ln \frac{[t_z(j) + \phi]^2 - \phi^2}{2\tilde{m}}. \quad (8.3)$$

The RG beta function $\beta = \ln(g_z(j+1)/g_z(j))$ can, in the large mass limit, then be related to the emergent distances x and y in terms of the coupling $g_z(j)$:

$$x_z(j) = \ln t_z(j) = \ln \left[\sqrt{2\tilde{m} e^{g_z(0,j)} + \phi^2} - \phi \right], \quad (8.4)$$

$$y_z(j) = \begin{cases} \ln \left[\sqrt{2\tilde{m} e^{g_z(0,j+1)} + \phi^2} - \phi \right], & z > 0, \\ \ln \left[\sqrt{2\tilde{m} e^{g_z(0,j-1)} + \phi^2} - \phi \right], & z < 0. \end{cases} \quad (8.5)$$

The quantitative dependence of the c -function on the RG beta function can also be expressed more generally as

$$e^{\beta_z(j)} = \left[\ln \frac{\left(\frac{\tilde{c}(j+1)}{c} + \phi \right)^2 - \phi^2}{2\tilde{m}} \right] \left[\ln \frac{\left(\frac{\tilde{c}(j)}{c} + \phi \right)^2 - \phi^2}{2\tilde{m}} \right]^{-1}. \quad (8.6)$$

Quantities such as the curvature κ and the convergence parameter θ can similarly be expressed in terms of the coupling $g_z(j)$.

1.9 Topological content of the entanglement spectrum

1.9.1 Luttinger's volume as a winding number: the massless case

We will now show how to obtain Luttinger's volume (eq.(2.18)) from the g -partite information entanglement measure I^g of a system of massless 2D electrons. The flux-dependent part of the g -partite information I^g can be used to probe the k -space volume V_L within the Fermi surface. We

define a quantity $Q_g(\phi)$ that is periodic in the flux ϕ and independent of the geometry:

$$Q_g(\phi) = f_{g,z} \left[\frac{1}{\sqrt{2}} - \frac{1}{2} e^{-\frac{1}{c} (I_{\{\mathcal{A}_z(i)\}}^g(\phi) - I_{\{\mathcal{A}_z(i)\}}^g(\phi^{(0)} f_{g,z}))} \right] = f_{g,z} \left[\frac{1}{\sqrt{2}} - |\sin(\pi \phi f_{g,z})| \right], \quad (9.1)$$

where $\phi^{(0)} = \frac{1}{6}$ is the value at which the flux-dependent part of $S_0(\phi)$ vanishes. Further, from eq.(3.19), we have

$$I_{\{\mathcal{A}_z(i)\}}^g(\phi_g) - I_{\{\mathcal{A}_z(i)\}}^g(\phi_g^{(0)} f_{g,z}) = -c \ln(2 |\sin(\pi \phi f_{g,z})|). \quad (9.2)$$

Note that the non-differentiability of $Q_g(\phi)$ at $\phi f_{g,z} = 0, 1, 2, \dots$ is not a concern as ϕ is chosen to be non-integral, ensuring that $\phi f_{g,z}$ is also non-integral.

In order to express the number N_L^x of positive k_x -states available inside the Fermi surface in the x -direction in terms of $Q_g(\phi)$, we need to establish a relation between the flux ϕ and the Fermi surface. The k_x -states of interest to us are given by $k_x^n = \frac{2\pi n}{L_x}$, $n \in \mathbb{Z}$, $0 \leq k_x^n \leq k_x^F$, where k_x^F is the x -component of the Fermi momentum $\vec{k}_F$. Assuming the flux-free Hamiltonian has time-reversal symmetry, there should be an equal number of states with the opposite momenta. Let n^* be the integer for which $k_x^{n^*} \leq k_x^F$ and $k_x^{n^*+1} > k_x^F$. Since the flux couples to the k_x -states as $k_x^n \rightarrow k_x^n + eA = \frac{2\pi n + 2\pi \phi}{L_x}$, tuning ϕ by 1 leads to the entire spectrum of k_x -states being shifted by one quantum number: $k_x^n \rightarrow \frac{2\pi(n+1)}{L_x} = k_x^{n+1}$. This means that the state $k_x^{n^*}$ moves to $k_x^{n^*+1}$, and hence out of the Fermi volume by crossing k_x^F . This flow of the spectrum is shown in Fig. 1.15.

If we now connect the system to a particle reservoir with the chemical potential set to zero, any k_x -state that moves out of the Fermi volume becomes unoccupied. We continue tuning the flux to a value ϕ^* , where the system becomes entirely devoid of any particles. The number of k_x -states that are below k_x^F can then be said to be given by the number of times ϕ goes through an integer value while changing from 0^+ to $\phi^* + 0^+$. Since we do not set the flux to an integer value, we can equivalently count the number of times ϕ goes through, say, three-quarters of an integer - $j - \frac{1}{4} : j = 1, 2, \dots$ - in changing from 0^+ to $\phi^* - 0^+$. We will now show that this count can be expressed in terms of Q_g . We begin by considering $Q_0(\phi) = \frac{1}{\sqrt{2}} - |\sin \pi \phi|$. At every three-quarter value of the flux, the quantity $\phi_j = j - \frac{1}{4} : j = 1, 2, \dots$, $|\sin \pi \phi_j|$ attains the value of $\sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$, such that $Q_0^{-1}(\phi)$ goes through a simple pole. Each such pole ϕ_j can be picked up via a complex integral around a contour C_j that encloses the pole:

$$-f_j \frac{1}{2\sqrt{2}i} \oint_{C_j} \frac{d\phi}{Q_0(\phi)} = 1, \quad (9.3)$$

where f_j is either +1 or -1 and is defined through the equation $|\sin(\pi(j - \frac{1}{4}))| = f_j \sin(\pi(j - \frac{1}{4}))$, and C_j is a circle of diameter 1/4 centered at $\phi = j - \frac{1}{4}$ (shown in Fig. 1.13). The evaluation of the integral eq.(9.3) is shown in Appendix B. In order to drop the sign prefactor f_j , we reorient those contours for which f_j is +1, and such that they wind around in the clockwise direction. The contours are shown in the left panel of Fig. 1.14, and defined as

$$C_j(\phi) = \left\{ \phi : \phi = j - \frac{1}{4} + \frac{1}{8} e^{i\theta}, \theta \in [0, -2\pi f_j] \right\}, \quad (9.4)$$

such that they host a singularity at their center. With this redefinition of the contours, we have

$$\frac{1}{2\sqrt{2}i} \oint_{C_j} \frac{d\phi}{Q_0(\phi)} = 1. \quad (9.5)$$

Counting the number of three-quarter values then reduces to counting the number of poles of Q_0^{-1} in the above-mentioned range $\phi \in (0, \phi^*)$. In this way, we can now express the count N_L^x in terms of Q_0 :

$$N_L^x = \frac{1}{2} \sum_{j=1}^{\phi^*} \frac{1}{2\sqrt{2}i} \oint_{C_j(\phi)} \frac{d\phi}{Q_0(\phi)} = \frac{1}{2\sqrt{2}} \sum_{j=1}^{\phi^*} \frac{1}{2i} \oint_{C_j(\phi)} \frac{d\phi}{Q_0(\phi)}. \quad (9.6)$$

The factor of half in eq.(9.6) is required as we want to count only the number of positive momentum states (and which is equal to the number of negative momentum states).

In going from $Q_0(\phi)$ to a general $Q_g(\phi)$, the only change required is to redefine $\phi \rightarrow \phi_g = f_{g,z}\phi$. This implies that while the states in k -space have become more coarse-grained, the flux ϕ has become more fine-tuned by the same scaling factor. This is a reflection of the overall conservation of particle number density: keeping the chemical potential fixed, the increased spacing between the k_x -states implies that a smaller number of particles are now occupying an equally smaller volume. As the number density is conserved, so is the Luttinger volume (eq.(2.18)), and this shows that our approach is consistent with the holographic Luttinger theorem [48]. Thus, given that $\phi_g^* = \phi^*$, the expression for N_L^x remains practically unchanged:

$$N_L^x = \frac{1}{2\sqrt{2}} \sum_{j=1}^{\phi^*} \frac{1}{2i} \oint_{C_j(\phi_g)} \frac{d\phi_g}{\frac{1}{\sqrt{2}} - |\sin \pi \phi_g|}. \quad (9.7)$$

In order to rewrite N_L^x in terms of the flux ϕ , we start by making the substitution $\phi_g \rightarrow f_{g,z}\phi$:

$$N_L^x = \frac{1}{2\sqrt{2}} \sum_{j=1}^{\phi^*} \frac{1}{2i} \oint_{C_j(f_{g,z}\phi)} \frac{f_{g,z}d\phi}{\frac{1}{\sqrt{2}} - |\sin(\pi f_{g,z}\phi)|}. \quad (9.8)$$

In the above equation, we can recognise $\frac{f_{g,z}}{\frac{1}{\sqrt{2}} - |\sin(\pi f_{g,z}\phi)|}$ as $Q_g^{-1}(\phi)$. Also note that the contour can be written as

$$C_j(f_{g,z}\phi) \equiv C_j^g(\phi) = \left\{ \phi : \phi = t_{g,z} \left(j - \frac{1}{4} + \frac{e^{i\theta}}{8} \right), \theta \in [0, -2\pi f_j] \right\}. \quad (9.9)$$

These curves are again circles of radius $1/8$, but centered at $t_{g,z} \times \left(1 - \frac{1}{4}\right)$, $t_{g,z} \times \left(2 - \frac{1}{4}\right)$, $t_{g,z} \times \left(3 - \frac{1}{4}\right)$ and so on. These are simply a generalisation of the curves defined in eq. (9.4); the curves for $t_{g,z} = 2$ are shown in the right panel of Fig. 1.14. Substituting this in eq.(9.8) gives

$$N_L^x = \frac{1}{2\sqrt{2}} \sum_{j=1}^{\phi^*} \frac{1}{2i} \oint_{C_j^g(\phi)} \frac{d\phi}{Q_g}. \quad (9.10)$$

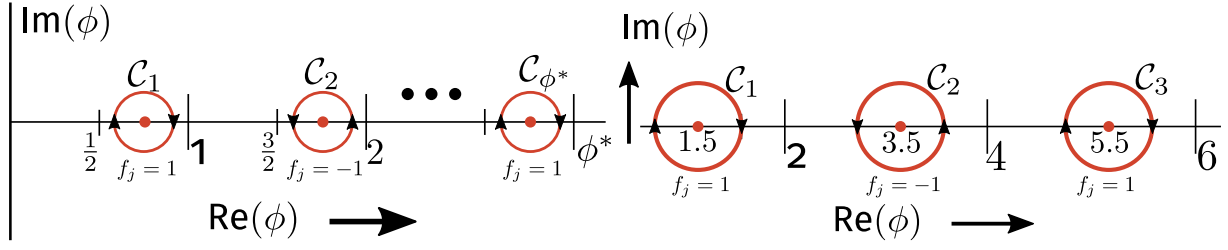


Figure 1.14: *Top:* Contours C_j used to integrate over Q_0^{-1} in eq. (9.6). Tall vertical lines represent integer values of $\text{Re}(\phi)$. All contour start and end within two adjacent integer values. The final contour C_{ϕ^*} ends just short of $\text{Re}(\phi) = \phi^*$. Note the alternating direction of the contours brought about by the alternation of f_j in going from $j = 1$ to $j = 2$, and so on. *Bottom:* Contours C_j^g for $t_{g,z} = 2$, centered at $2\left(1 - \frac{1}{4}\right), 2\left(2 - \frac{1}{4}\right)$ and $2\left(3 - \frac{1}{4}\right)$. Note the alternating directions of the contours.

For a spatially isotropic system, the Fermi volume is given by

$$V_L = v_2 \times \pi (N_L^x)^2 = \frac{1}{2} \frac{\pi^3}{V} \left[\sum_{j=1}^{\phi^*} \frac{1}{2i} \oint_{C_j^g(\phi)} \frac{d\phi}{Q_g(\phi)} \right]^2, \quad (9.11)$$

where V is the real space volume of the system, and $v_d = (2\pi)^d / V$ is the volume of the smallest unit in k -space. For a generally anisotropic Fermi volume, each direction in the d -dimensional k -space can be completely specified by an angular coordinate θ with respect to some predefined coordinate system. For each direction $\hat{\theta}$, we can insert a gauge field $\vec{A} = A\hat{\theta}$ and take the thin torus limit in this direction, obtaining thereby the g -partite information $I^g(\phi_\theta)$ for length l in the direction perpendicular to $\hat{\theta}$. With the θ -dependent I^g , one can then define the quantity $Q_g(\phi_\theta)$ which counts the number of states N_L^θ in the direction $\hat{\theta}$ inside the Fermi volume. In this way, the total Fermi volume can be obtained by integrating over θ :

$$V_L = \frac{\pi^2}{4V} \int_0^{2\pi} d\theta \left[\sum_{j=1}^{\phi_\theta^*} \frac{1}{2\pi i} \oint_{C_j^g(\phi)} \frac{d\phi}{Q_g(\phi)} \right]^2. \quad (9.12)$$

It is important to note that the quantity V_L is independent of the RG index g , and is thus identical in value for all g members of the hierarchy at the g th step of the coarse-graining process.

The topological nature of Luttinger volume can be made further manifest by expressing it in terms of winding numbers, following Ref. [15] for the case of interacting electrons. To do so, we point out that the integrand $Q_g^{-1}(\phi)$ can be written as a derivative:

$$\frac{1}{Q_g(\phi)} = \frac{\sqrt{2}}{\pi} \frac{\partial}{\partial \phi} \ln Y_j^g = \frac{\sqrt{2}}{\pi} \frac{1}{Y_j^g} \frac{\partial Y_j^g}{\partial \phi}, \quad (9.13)$$

where

$$\begin{aligned}
 Y_j^g(\phi) &\equiv \left[\frac{f_j \tan 3\pi/8 - \tan(\pi f_{g,z}\phi/2)}{\tan(\pi f_{g,z}\phi/2) - \tan(\pi/8)} \right]^{f_j} \\
 &= \begin{cases} \frac{\tan 3\pi/8 - \tan(\pi f_{g,z}\phi/2)}{\tan(\pi f_{g,z}\phi/2) - \tan(\pi/8)}, & \text{when } j = \text{odd, and} \\ \frac{\tan(\pi f_{g,z}\phi/2) + \tan(\pi/8)}{-\tan 3\pi/8 - \tan(\pi f_{g,z}\phi/2)}, & \text{when } j = \text{even.} \end{cases}
 \end{aligned} \tag{9.14}$$

Substituting this into the expression for 2-dimensional isotropic Luttinger volume in eq. (9.11), we obtain

$$V_L = \frac{\pi^3}{V} \left[\sum_{j=1}^{\phi^*} \frac{1}{2\pi i} \oint_{Y_j^g(C_j^g)} \frac{dY_j^g}{Y_j^g} \right]^2. \tag{9.15}$$

As before, the specific integral in the above expression is a winding number that counts the number of times the contour $Y_j^g(C_j^g)$ winds around the singularity at $Y_j^g = 0$. Note that these are the same singularities that exist at the centers of the contours C_j^g and that are picked up in, for example, eq. (9.3). To demonstrate this, we point out that the centers of the contours are given by $\phi = t_{g,z} \left(j - \frac{1}{4} \right)$, $j = 1, 2, \dots, \phi^*$. At these values, we have

$$\tan \left(\frac{\pi f_{g,z}\phi}{2} \right) = \tan \left(\frac{\pi j}{2} - \frac{\pi}{8} \right) = \begin{cases} \tan \frac{3\pi}{8}, & j = 1, 3, \dots \\ -\tan \frac{\pi}{8}, & j = 2, 4, \dots \end{cases} \tag{9.16}$$

Comparing with eq. (9.14), we can see that the numerators of Y_j^g are zero at these values of ϕ for both even and odd values of j . This leads to $Y_j^g = 0$, and hence a singularity in $\frac{dY_j^g}{Y_j^g}$. These values of the flux are therefore the common singularities that are picked up by the poles of $Q_g^{-1}(\phi)$ as well as the winding numbers.

We now define w_g^j as the winding number for the curve $Y_j^g(C_j^g)$ around $Y_j^g = 0$, and $\mathcal{W}(\phi^*) = \sum_{j=1}^{\phi^*} w_j^g$ as the total winding number for all the curves $\{Y_j^g(C_j^g), j = 1, 2, \dots\}$. Then, the isotropic Fermi volume takes the compact form

$$V_L = \frac{\pi^3}{V} \mathcal{W}(\phi^*)^2. \tag{9.17}$$

1.9.2 A relation between the quantum hall Chern numbers and Luttinger's volume

Interestingly, the above method employed in using entanglement measures to extract the Luttinger volume can also be applied (with suitable modifications) to an integer quantum hall system (generated upon placing the system in a strong transverse magnetic field) to obtain the number of filled Landau levels (a Chern number). To begin, we imagine a system of free non-relativistic electrons in a cylinder geometry (fig. 1.16), and placed in a magnetic field $\nabla \times \vec{A} = \vec{B}$ that points radially outwards. The field $\vec{B}$ generates Φ flux quanta through the surface of the cylinder. An Aharonov-Bohm

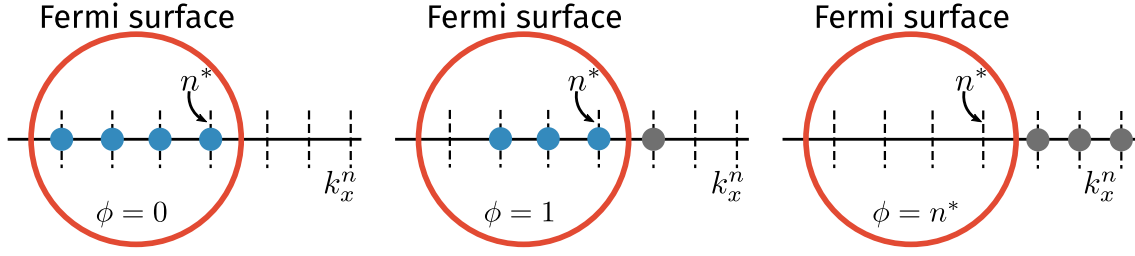


Figure 1.15: Spectral flow of the states within the Fermi volume. The red curve represents the Fermi surface. Upon tuning the flux ϕ , the k -space states (indicated in the form of dotted lines along the k_x -axis) are shifted towards the Fermi surface. The blue circles represent electrons with momenta below k_F , and are part of the ground state. On the other hand, the grey circles represent particles with momenta above k_F : they have moved out of the system, and into the reservoir connected to it (see discussion in main text). Each time ϕ changes by 1, the state nearest to the Fermi surface moves out of the Fermi volume. Tuning ϕ by n^* empties the entire Fermi volume.

flux $\vec{A}' = -2\pi\phi/L_x\hat{x}$ is also placed in the x -direction (i.e., along the circumference of the cylinder). We divide the cylinder into two subsystems A and B (fig. 1.16), such that tuning the dimensionless flux ϕ by 1 transfers a charge equal to the first Chern number, from A to B . The translation invariance in the x -direction ensures that the momentum k_x are good quantum numbers; for each value of k_x , we can define the matrix elements of the single-particle correlation matrix $G(k_y)$ for the subsystem A as [49, 50]

$$G_{ij}^A(k_x) = \langle \psi_{\text{gs}} | c_{k_x,i}^\dagger c_{k_x,j} | \psi_{\text{gs}} \rangle, \quad (9.18)$$

where $|\psi_{\text{gs}}\rangle$ is the ground state and i, j represent the degrees of freedom other than k_x . The total matrix G^A is constructed by joining the smaller blocks $G^A(k_x)$. As ϕ is varied, the variation of the trace of the flux-dependent matrix $G^A(\phi)$ is linear in the flux ϕ : $\text{Tr}[G^A(\phi)] - \text{Tr}[G^A(0)] = -\frac{1}{e}\sigma_H\phi$. This implies that as long as there are extended states at sufficiently small energies, we will have spectral flow that can be captured through the integral:

$$-\frac{\sigma_H}{2\pi ie} \int_{C_H} \frac{d\phi}{\text{Tr}[G^A(\phi)] - \text{Tr}[G^A(0)]}, \quad (9.19)$$

where C_H is a circle of radius $1/2$ in the complex plane centered at $\phi = 0$. This integral is zero if there is no spectral flow, i.e., the effect of the gauge field can be removed by a gauge transformation and the trace is insensitive to the flux. In order to count the number of filled Landau levels (essentially, the filling factor ν), we can now vary the flux quanta Φ (due to $\vec{B}$) so as to change the filling from ν to 0:

$$\nu = -\frac{\sigma_H}{2\pi ie} \sum_{\Phi'=\Phi}^{\Phi^{\max}} \int_{C_H} \frac{d\phi}{\text{Tr}[G_{\Phi'}^A(\phi)] - \text{Tr}[G_{\Phi'}^A(0)]}. \quad (9.20)$$

The subscript Φ' on G^A indicates that the correlations depend on the variable magnetic field $\vec{B}'$. The upper limit $\Phi^{\max}$ is given by the lowest value of the magnetic field $\vec{B}'$ for which only one Landau level is filled. The magnetic field is varied such that exactly an integer number of Landau levels are filled at any given value. Importantly, the filling factor ν is known to be a Chern number $C(\Phi)$ [51, 52].

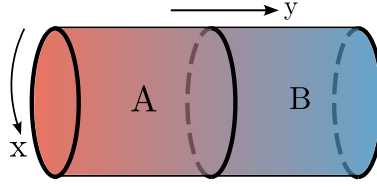


Figure 1.16: Cylindrical geometry for the integer quantum Hall system, with a magnetic field $\vec{B}$ pointing radially outwards through the surface of the cylinder. An Aharonov-Bohm vector potential acts in the x -direction (i.e., along the circumference). The entanglement is calculated between the subsystem A and its complement B .

We can also relate eq. (9.20) to the entanglement of the system by introducing the entanglement spectrum [11, 53], which refers to the set of eigenvalues $\{\lambda_\chi^A(k_x)\}$ of the reduced density matrix $\rho^A(k_x)$ for subsystem A corresponding to the momentum mode k_x in the x -direction and χ is a particular configuration of the degrees of freedom $\{i\}$ in the y -direction. We have the relation [50]

$$\lambda_\chi^A(k_x) = \text{Det} [P_\chi G^A(k_x) (1 - P_\chi) G^A(k_x)] , \quad (9.21)$$

where P_χ is an operator that projects on to the subset of states $\{j\}$ in $\{i\}$ that are occupied in the configuration χ . This relation between G^A and the eigenvalues λ then leads to the same conclusion as in the case without the magnetic field: the pole structure of the entanglement spectrum encodes the topological information of the system revealed in eq.(9.20). In the case of zero magnetic field, the topological quantity revealed from the pole structure was a winding number ($\mathcal{W}$), while in the present case it is a Chern number.

Finally, we reveal the connection between the winding numbers $\mathcal{W}$ in the metallic case and the Chern numbers in the integer quantum Hall system. The degeneracy of each Landau level is equal to the number of flux quanta Φ generated by the magnetic field B . Since the total number of particles in the system is fixed, the electronic states of the filled Fermi sea that form the Luttinger volume (in the absence of the external B-field) are redistributed into the Landau levels with the appropriate degeneracy as a strong transverse magnetic field is turned on. Thus, we have the constraint that the total number of occupied states $C \times \Phi$ in the quantum hall system (C being the Chern number and equal to the number of filled Landau levels) must be equal to $\pi (N_L^x)^2$, the total number of occupied states in the metallic system. Combining eqs. 9.10 and 9.17 with this discussion then allows us to relate C to the winding number $\mathcal{W}(\phi^*)$:

$$C \times \Phi = \left\lfloor \frac{\pi}{4} \mathcal{W}(\phi^*)^2 \right\rfloor = \frac{V_L}{v_2} , \quad (9.22)$$

where $v_2 = 4\pi^2/V$ is the volume of the k -space unit cell (defined below eq. (9.10)) and $\lfloor \cdot \rfloor$ is the floor function and returns the largest integer less than or equal to the argument.

1.10 Conclusions and Outlook

In summary, we have shown that considering the renormalisation group flows of the spectral gap coupling of a system of non-interacting electrons in $2 + 1$ -dimensional theory leads to hierarchical

arrangement of the entanglement with scale, and that the geometrical part of the entanglement can be described as the emergence of an additional spatial dimension, leading to holographic expressions for geometric parameters like distance and curvature. We show that changing the boundedness of the holographic space involves a topological transition with the Fermi surface of the boundary theory turning critical, and involves the generation of a geometric wormhole between the UV and IR degrees of freedom. For the massless case, the presence of an Aharonov-Bohm flux generates a geometry-independent term in the entanglement entropy which can be linked to the Luttinger volume of the gapless fermionic system. By placing the system of $2 + 1$ dimensional electrons in a strong magnetic field, we argue that the Chern number topological invariant for the ground states of the integer quantum Hall ground system are constrained by the value of the Luttinger volume in the underlying metallic system (i.e., prior to the insertion of the field).

How robust are our conclusions towards the inclusion of electronic correlations? Given the adiabatic continuity that links Fermi liquids to the non-interacting Fermi gas, i.e., the preservation of the k -space quantum numbers under adiabatic inclusion of (screened) inter-particle interactions, we expect our results to be equally valid for the many-particle entanglement of the excitations described by the fixed point effective theory for the Fermi liquid. This also suggests that our results for the wormhole geometry and the stress-energy tensor relation at the critical Fermi surface apply very generally to phase transitions that destabilise and gap out the Fermi surface. Specifically, this has profound implications for metal-insulator transitions driven by strong correlations. Non-Fermi liquids are, however, not adiabatically continuous to the Fermi gas. Thus, it should be interesting to adapt our approach to the study of interacting models of electrons that lead to gapless non-Fermi liquids (including those that are known to appear at quantum critical points related to the breakdown of a Fermi liquid). For instance, Ref. [54] reports that the second Renyi entropy of a thin k -space shell proximate to the Fermi surface is related to the quasiparticle residue; how would this change in a non-Fermi liquid? It also appears interesting to identify the effects of electronic correlations on various aspects of the emergent space. The renormalisation group techniques applied in Refs. [54–58] could also be worth investigating towards answering these questions.

It is interesting to note that our results point towards a broader similarity between the gapless fermionic systems considered here and gapped topologically ordered quantum liquids. Recall that the latter are known to possess a ground state degeneracy that is dependent on the topological nature of the interacting electronic system, leading to a subsystem entanglement entropy that has a geometry-independent subleading piece (S_{topo}) referred to as the topological entanglement entropy. It was shown by some of us recently [5] that for a collection of N subsystems (CSS) that form a closed annular structure, the n -partite quantum entanglement information measures I^n ($3 \leq n \leq N$) are a topological invariant equal to the product of S_{topo} and the Euler characteristic of the planar manifold on which the CSS is embedded. This appears to be analogous to our finding of the hierarchical sequence of quantities $\{Q_g\}$ defined at a particular RG step (and arising out of the topological flux-dependent piece of the entanglement entropy), all of whom encode the same Luttinger volume for the gapless systems we have studied (eq. (9.12)). Further, the robustness of the n -partite information I^n against small deformations of the CSS in a topologically ordered system can be compared to the invariance of N_L^x to small deformations of the contours $C_1, C_2, \dots$ (see right panel of Fig. 1.14). It appears tempting to speculate on a possible universal origin underlying these similarities (see also [58] for a recent investigation of the entanglement properties of various quantum liquids).

Our results lead to a number of interesting questions. Firstly, the convergence parameter θ is reminiscent of the expansion rate of congruence flows on Riemannian manifolds in the context of the Raychaudhuri equation [46]. While the evolution equation for θ (eq. (7.6)) obtained by us is not of the same form (as there is no time variable, neither is there a θ^2 term in the expansion rate), we do expect such an equation to hold for an appropriately defined expansion parameter [47, 59]. This likely requires the introduction of temporal dynamics of the emergent spatial dimension, and lies well beyond the purview of the present work. Secondly, it remains to be seen how the geometry-independent part of the entanglement behaves in the presence of a mass ($m > 0$) for the Dirac fermions: a spectral gap is expected to have consequences for Luttinger's theorem [18] (see also eq. (2.19)), as well as for the many-particle entanglement. For instance, Ref. [54] reports that the second Renyi entropy shows the signatures of the condensation of Cooper pairs in the BCS problem.

Finally, we would like to point out an interesting conclusion that emerges from our analysis. As observed by us, the system considered here shows a hierarchy of multipartite entanglement measures beyond just the mutual information. Further, we have shown that the mutual information leads to the emergence of the holographic dimension. However, whether the higher-party measures also have a holographic interpretation remains an open question. We note that studies of the time-variation of tripartite information during an equilibrium process after the injection of energy suggest that it remains negative throughout the process [60–62], indicating that care must be taken in defining geometric measures using it. Nevertheless, our study raises the possibility of obtaining a generalisation of the gauge-gravity duality from the perspective of many-particle entanglement. We speculate that the higher order information measures will lead to a modification of the universal entanglement Hamiltonian obtained at criticality. Further, that these modifications can be visualised as a *hypergraph*: just as a graph contains edges that connect any two nodes, a hypergraph contains *hyperedges* that can connect more than two nodes. The eigenvalues $\epsilon^{(n)}$ (with $n > 2$) of the entanglement Hamiltonian would act as hyperedges connecting the degrees of freedom $\{i, j, \dots\}$. We leave an investigation of these aspects to future works.

Appendix A: Evaluation of multi-partite information

We define the multipartite information among g sets $\{\mathcal{A}_z(i)\}$ as [5]

$$I_{\{\mathcal{A}_z(i)\}}^g = \sum_{r=1}^g (-1)^{r-1} \sum_{\beta \in \mathcal{P}(\{\mathcal{A}_z(i)\})}^{| \beta | = r} S_{\beta} , \quad (10.1)$$

where $\mathcal{P}(\{\mathcal{A}_z(i)\})$ is the power set of $\{\mathcal{A}_z(i)\}$, and $| \beta | = r$ implies that we sum only over those sets β that have r number of elements in them. We will first evaluate the internal sum σ_r over the set β for a general r :

$$\sigma_r = \sum_{\beta \in \mathcal{P}(\{\mathcal{A}_z(i)\})}^{| \beta | = r} S_{\beta} . \quad (10.2)$$

For this, we first note that following eq. (3.15), the entanglement entropy (EE) of the union of a given sequence of sets $\{i\}$ is given by the EE of the set with the highest (lowest) index for a positive

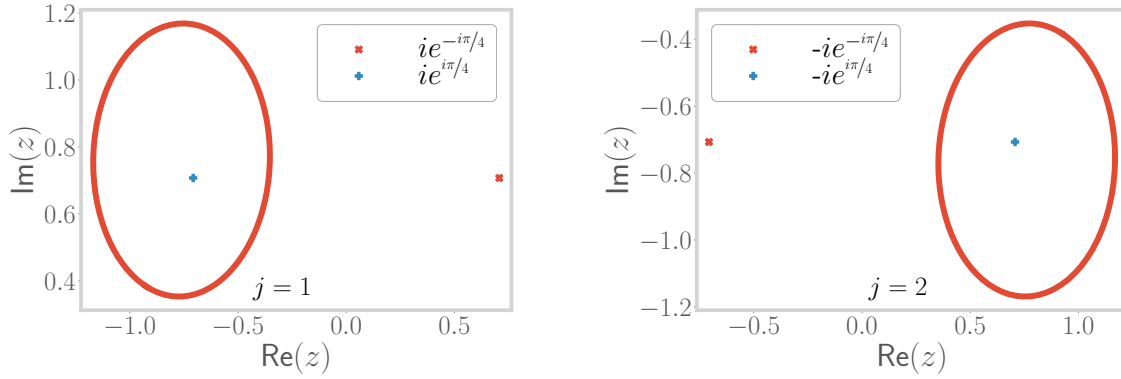


Figure 1.17: Contours $z(C_j)$ for two values of j employed for evaluating the integral eq.(10.8), and the locations of the poles $z_{\pm}^{\pm}$. For each j , only one pole (shown in blue) contributes, as the other (shown in red) is not enclosed by the contour.

(negative) z . As a result, the EE S_{β} for the set β is simply $S_{\beta} = \theta(z)S_{\min(\beta)} + \theta(-z)S_{\max(\beta)}$. If $n_j^{(r,+)}$ ($n_j^{(r,-)}$) is the number of sets of cardinality r that have the element $\mathcal{A}_{j,z}$ as the element with the smallest (largest) index j , the internal sum can be written as

$$\sigma_r = \sum_{j=1}^{g-(r-1)} \theta(z) n_j^{(r,+)} S_{j,z} + \sum_{j=r}^g \theta(-z) n_j^{(r,-)} S_{j,z}. \quad (10.3)$$

The number $n_j^{(r,+)}$ corresponds to the number of ways one can pick $r-1$ elements out of a set of $g-j$ elements: $n_j^{(r,+)} = \binom{g-j}{r-1}$. Similarly, $n_j^{(r,-)}$ corresponds to the number of ways one can pick $r-1$ elements out of a set of j elements: $n_j^{(r,-)} = \binom{j}{r-1}$. We can also extend the limits on the two summations by defining $\binom{n}{m} = 0$ for $n < m$. With these considerations, eq. (10.3) takes the form

$$\sigma_r = \sum_{j=1}^g \left[\theta(z) \binom{g-j}{r-1} + \theta(-z) \binom{j}{r-1} \right] S_{j,z}. \quad (10.4)$$

This can now be substituted into the full expression for the multipartite equation, giving

$$\begin{aligned} I_{\{\mathcal{A}_z(i)\}}^g &= \sum_{r=1}^g (-1)^{r-1} \sum_{j=1}^g \left[\theta(z) \binom{g-j}{r-1} + \theta(-z) \binom{j}{r-1} \right] S_{j,z} \\ &= \sum_{j=1}^g S_{j,z} \left[\theta(z) \sum_{r=0}^{g-j} (-1)^r \binom{g-j}{r} + \theta(-z) \sum_{r=0}^j (-1)^r \binom{j}{r} \right]. \end{aligned} \quad (10.5)$$

From the binomial theorem, we can now recognise each internal sum as over r as a polynomial expansion. That is, upon comparing with $(x-1)^n = \sum_{m=0}^n \binom{n}{m} (-1)^m x^{n-m}$, we obtain $\sum_{r=0}^{g-j} (-1)^r \binom{g-j}{r} =$

$\lim_{x \rightarrow 1} (x-1)^{g-j} = \delta_{g,j}$. Similarly, the other sum evaluates to $\delta_{j,0}$. Substituting these into the expression for $I_{\{\mathcal{A}_z(i)\}}^g$ gives

$$\begin{aligned} I_{\{\mathcal{A}_z(i)\}}^g &= \sum_{j=1}^g S_{j,z} [\theta(z)\delta_{g,j} + \theta(-z)\delta_{j,0}] \\ &= \theta(z)S_g + \theta(-z)S_0. \end{aligned} \quad (10.6)$$

By noting that S_g and S_0 are, respectively, the EE of the intersection set of $\{S_z(i)\}$, one can combine the two terms in the previous equation into

$$I_{\{\mathcal{A}_z(i)\}}^g = \theta(z)S_g + \theta(-z)S_0 = S_{\cap\{\mathcal{A}_z(i)\}}. \quad (10.7)$$

Appendix B: Evaluation of the contour integral that counts poles

We prove eq. (9.3) in this appendix. The integral in question is given by

$$\mathcal{I} = \oint_{C_j} \frac{d\phi}{\mathcal{Q}_0(\phi)} = \oint_{C_j} \frac{d\phi}{\frac{1}{\sqrt{2}} - |\sin \pi \phi|}, \quad (10.8)$$

where C_j is the contour shown in Fig. 1.13 and the variable ϕ is given by $\phi = \{\phi_j + \frac{1}{8}e^{i\theta} : \theta \in [0, 2\pi]\}$, where $\phi_j = j - \frac{1}{4}$ is the center of the contour. In order to evaluate the integral, we make a variable change: $z = e^{i\pi\phi}$, giving $dz = i\pi z d\phi$ and $\sin \pi \phi = \frac{1}{2i} \left(z - \frac{1}{z}\right)$. Depending on the value of j , $\sin \pi \phi$ can be either positive or negative. We denote either case by a subscript $\mathcal{I}_{\pm}$ corresponding to the contour $C_{\pm}$, giving $|\sin(\pi\phi)| = \pm \sin(\pi\phi)$. Casting the integral in terms of z , we obtain

$$\mathcal{I}_{\pm} = \frac{1}{i\pi} \oint_{z(C_{\pm})} \frac{dz/z}{\frac{1}{\sqrt{2}} \mp \frac{1}{2i} \left(z - \frac{1}{z}\right)} = \oint_{z(C_{\pm})} \frac{\mp \frac{2}{\pi} dz}{\left(z \mp \frac{i}{\sqrt{2}}\right)^2 - \frac{1}{2}}. \quad (10.9)$$

The integrand in $\mathcal{I}_{+}$ has two simple poles at $z_{\pm}^{\pm} = \pm i e^{\mp i\pi/4}$. Similarly, the integrand in $\mathcal{I}_{-}$ has two simple poles at $z_{\pm}^{\pm} = \mp i e^{\mp i\pi/4}$. We plot the contour and the two poles for two contours of opposite signature in Fig. 1.17; it is clear that only the pole at $z_{\pm}^{+}$ contributes to the integral. Using the residue theorem, the integral evaluates to

$$\begin{aligned} \mathcal{I}_{\pm} &= \oint_{z(C_{\pm})} \frac{\mp \frac{2}{\pi} dz}{(z - z_{\pm}^{+})(z - z_{\pm}^{-})} \\ &= \mp \frac{2}{\pi} 2\pi i \lim_{z \rightarrow z_{\pm}^{+}} \frac{1}{z - z_{\pm}^{-}} \\ &= \mp 2\sqrt{2}i. \end{aligned} \quad (10.10)$$

This proves eq. (9.3).

Chapter 2

Kondo frustration via charge fluctuations: A route to Mott localisation

2.1 Introduction

The rich physics of metal-insulator transitions in strongly correlated systems has been an active subject of study for quite some time [63–65], yet much still remains to be understood. It involves diverse aspects such as spin and charge fluctuations, quasiparticle renormalisation effects, anomalous metallic phases and unconventional superconductivity, and has been studied using an equally diverse array of methods like mean-field theory, renormalisation group approaches, numerical techniques like exact diagonalisation, quantum Monte Carlo and dynamical mean-field theory, and many others. In particular, dynamical mean-field theory (DMFT) [66–74] obtains an exact solution of the Mott metal-insulator transition (MIT) for the 1/2-filled Hubbard model [75–79] on the Bethe lattice with infinite coordination number, in terms of an Anderson impurity model with a self-consistently determined bath obtained by requiring, in an iterative manner, that its local Greens function be equal to that of the impurity site. The above-mentioned transition can be captured by the local spectral function through (i) the continuous appearance of a Mott gap, followed by (ii) the sharpening and vanishing of the central Kondo resonance. These two features are often referred to, respectively, as the Mott-Hubbard [78] and Brinkman-Rice [79] scenarios of the Mott MIT. The exact nature of the solution arises from the fact that all non-local contributions to the lattice self-energy are observed to vanish upon taking the limit of an infinite coordination number for the lattice model. Thus, the simplification is that the dynamics of any local site on the lattice is determined completely by a quantum impurity problem [70, 80]. It must also be noted that this exact solution precludes any long-range order in the system, and corresponds to the case of a maximally frustrated Hubbard model involving long-range and frustrating inter-site hopping such that both the metallic and insulating phases remain paramagnetic [81]. Due to the exact and non-perturbative nature of the DMFT solution for the MIT in $d = \infty$, the method has been extended to models of strongly correlated electrons in finite spatial dimensions [72, 82, 83], as well as the study of the electronic properties of various correlated materials [73, 84, 85].

Despite this progress, a key aspect of the DMFT solution for the Hubbard model in $d = \infty$ remains to be understood. During the search for a self-consistent impurity model, the conduction

bath is modified drastically in order to become correlated [86]. The numerical implementation of self-consistency, however, precludes a deeper understanding of the precise nature of the correlations present in the conduction bath of the impurity model, and its implications for the electron dynamics of the associated bulk lattice (Hubbard) model. Indeed, as we demonstrate below, our work paves the way for a more insightful exploration of the physics of the Mott Hubbard transition in strongly correlated electronic systems. Further, it has considerable consequences for the development of functional quantum materials and next-generation quantum technologies. Below, we lay out the specific questions addressed by us, and summarise our results at the end of this section.

- i. Is there a minimal but effective quantum impurity model Hamiltonian that describes the Mott MIT of the $1/2$ -filled Hubbard model on the Bethe lattice in $d = \infty$? Finding such an impurity model would also reduce the considerable computational effort that is presently required in self-consistent approaches.
- ii. What are the fluctuations that destroy the metal and lead to the insulating phase? Can we obtain a universal theory for these competing tendencies?
- iii. The coexistence of metallic and insulating phases at $T = 0$ within DMFT shows that the insulating solution is present within the many-body spectrum of the metallic phase. Can an associated impurity model Hamiltonian display the emergence of the insulating state prior to the transition? Analytic insight of this kind is particularly important because approaching a coexistence region numerically is often fairly tricky.
- iv. Is it possible to obtain a low-energy theory for the local gapless excitations precisely at the MIT, where the metal is on the brink of destruction? How do these excitations compare with those of the local Fermi liquid, e.g., in terms of self-energies and two-particle correlation functions?

Obtaining answers to these questions would go a long way towards understanding the dominant fluctuations at the heart of the Mott transition and the physical processes that become important as one approaches it. It would also help in understanding the mechanism of DMFT's search for the self-consistent impurity model within the space of Anderson impurity models.

The essence of our approach is to model phenomenologically the lattice self-energy obtained from DMFT in the form of additional bath correlations within an extended Anderson impurity model. In addition to the usual on-site repulsion (U) and single-particle hybridisation (V) between the impurity and the conduction bath of the Anderson impurity model (eq. (2.1)), we introduce (i) an additional on-site correlation (U_b) on the bath site with which the impurity couples, and (ii) an antiferromagnetic Kondo coupling (J) between the impurity and the conduction bath (eq. (2.2)). We note that a similar impurity model-based approach was taken towards understanding the physics of the heavy fermions several years ago by Si and Kotliar [87, 88]. We postpone a comparison of our work with theirs to the discussions section. The rest of the work is structured as follows. Sec. (2.2.1) describes the extended model that we will study, and the unitary renormalisation group (URG) method that we employ to study it is presented in Sec. (2.2.2). In Sec. (2.3) and (2.4), we describe the phase diagram and various characteristics of the impurity phase transition. In Sec. (2.6), we use our extended model to explain various features of the coexistence region observed in DMFT.

In Sec. (2.7), we describe the effect of the impurity on the low-lying excitations of the bath, near and at the transition. We conclude in Sec. (2.8) with some discussions and possible future directions. For the convenience of readers, we first present below a brief summary of our main results.

Summary of our main results

- *Presence of a local metal-insulator transition:* At a critical value of the parameter $r = -U_b/J$, the effective impurity model shows a transition from a Kondo screened phase into an un-screened local moment phase. The quantum critical point (QCP) involves a degeneration of the Kondo singlet and the local moment states.
- *The physics of Kondo screening and local pairing drives the transition:* The transition involves the frustration of the Kondo screening of the impurity by enhanced local pairing fluctuations in the bath and can be described by a universal theory written in terms of J and U_b .
- *Emergence of insulating solutions in the metallic phase:* Our analysis reveals that at a certain value of the parameter r prior to the transition, the single-particle hybridisation parameter (V) turns irrelevant (in the RG sense), and this leads to the emergence of the local moment solutions within the many-particle spectrum through an excited state quantum phase transition (ESQPT).
- *Critical fluctuations and the coexistence region:* We observe the appearance of long-ranged quantum fluctuations extending into the conduction bath in the vicinity of both the ESQPT at $r = r_{c1}$ and QPT at $r = r_{c2}$. We believe that these are the likely origin of the critical fluctuations observed above the finite temperature second-order critical point in DMFT [81, 89]. The two-step process at $T = 0$ also provides a natural explanation for the coexistence of metallic and insulating features in the phase diagram, in the regime $r_{c1} < r < r_{c2}$.
- *Emergence of non-Fermi liquid excitations at the QCP:* Precisely at the QCP, the local Fermi liquid is replaced by a quasi-local non-Fermi liquid (NFL) that spans the impurity, zeroth and first sites of the conduction bath. The NFL results from a degeneracy between the local moment and singlet states and leads to (i) “Andreev scattering” of incoming states into orthogonal outgoing states, (ii) anomalous power-law behaviour in the self-energies and two-particle correlations with universal exponents, and (iii) a fractional entanglement entropy of the impurity.
- *Correlated Fermi liquid excitations in the Hubbard sidebands:* A many-body perturbation theoretic treatment of the Hubbard sidebands reveals that they are comprised of the holon-doublon excitations created by the hybridisation of the impurity site with the conduction bath. These excitations consist of decoupled local Fermi liquids for the holons and doublons at the lowest order, which, at higher orders, become coupled via correlated holon-doublon scattering between impurity and bath.

2.2 Model Hamiltonian and method

2.2.1 The extended Anderson impurity model

The single-impurity Anderson model (SIAM) [90, 91] consists of a single impurity site with local repulsive correlation U hybridising with a non-interacting fermionic conduction bath through a (momentum-independent) single-particle transfer whose coupling is V . For the case of a half-filled impurity site, the Hamiltonian of the SIAM is given by

$$\mathcal{H}_A = -\frac{U}{2} (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2 + \sum_{\vec{k}, \sigma} \epsilon_{\vec{k}} \tau_{\vec{k}, \sigma} + V \sum_{\sigma} \left(c_{d\sigma}^\dagger c_{0\sigma} + \text{h.c.} \right), \quad (2.1)$$

where $\tau_{\vec{k}, \sigma} \equiv c_{\vec{k}, \sigma}^\dagger c_{\vec{k}, \sigma} - 1/2$ indicate the occupancy of the single-particle momentum state $|\vec{k}\rangle$. Also, $c_{d\sigma}$ and $c_{0\sigma} = \sum_k c_{k\sigma}$ are the fermionic annihilation operators of spin σ for the impurity and conduction bath site to which it couples (henceforth referred to as the *zeroth site*) respectively. The conduction bath is typically considered to possess a constant (i.e., energy-independent) density of states.

The SIAM (along with its $U \rightarrow \infty$ limit, the Kondo model) has been studied using several analytical and numerical techniques [19, 92–106]. The general conclusion for the positive U case at $T = 0$ is that on the particle-hole symmetric (that is, half-filled) line, the impurity local moment is always screened by the conduction electrons (referred to as the Kondo cloud [107–118]). Enhanced spin-flip scattering at low-energies leads to the formation of a macroscopic singlet ground state and local Fermi liquid gapless excitations [119, 120]. In order to enhance the SIAM, we introduce two extra two-particle interaction terms into the Hamiltonian:

- a spin-exchange term $J \vec{S}_d \cdot \vec{S}_0$ between the impurity spin $\vec{S}_d$ and the spin $\vec{S}_0$ of the zeroth site, and
- a local particle-hole symmetric correlation term $-U_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2$ on the bath zeroth site.

With these additional terms, the Hamiltonian of the *extended single-impurity Anderson model* (henceforth referred to as the e-SIAM) is, at particle-hole symmetry, given by

$$\mathcal{H}_{\text{E-A}} = \mathcal{H}_A + J \vec{S}_d \cdot \vec{S}_0 - \frac{1}{2} U_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2. \quad (2.2)$$

All the terms in the Hamiltonian have been depicted schematically in the left panel of Fig. (2.1). The additional interaction terms J and U_b enjoy particle-hole, SU(2)-spin and U(1)-charge symmetries. The e-SIAM Hamiltonian (eq. (2.2)) therefore preserves all the local symmetries of the half-filled Hubbard model on a lattice, and can potentially serve as an effective auxiliary quantum impurity model (with a correlated bath) describing the local physics of the latter.

2.2.2 The unitary renormalisation group method

In order to obtain the various low-energy phases of the e-SIAM, we perform a scaling analysis of the associated Hamiltonian (eq. (2.2)) using the recently developed unitary renormalisation group

(URG) method [18,26]. The method has been applied successfully on a wide variety of problems of correlated fermions [10, 18, 26, 106, 121–125]. The method proceeds by resolving quantum fluctuations in high-energy degrees of freedom, leading to a low-energy Hamiltonian with renormalised couplings and new emergent degrees of freedom. Typically, for a system with Fermi energy ϵ_F and bandwidth D_0 , the sequence of isoenergetic shells $\{D_{(j)}\}$, $D_{(j)} \in [\epsilon_F, D_0]$ define the states whose quantum fluctuations we sequentially resolve. The momentum states lying on shells $D_{(j)}$ that are far away from the Fermi surface comprise the UV states, while those on shells near the Fermi surface comprise the IR states. This scheme is shown in the right panel of Fig. (2.1).



Figure 2.1: *Left*: Schematic 1D representation of the extended SIAM Hamiltonian. The red sphere is the impurity site with on-site Hubbard term U . It is connected via two couplings V and J to the bath zeroth site (large blue sphere) that has its own on-site Hubbard term U_b . The small blue spheres make up the rest of the bath, connected through the single-particle hopping t . *Right*: High energy - low energy scheme defined and used in the URG method. The states away from the Fermi surface form the UV subspace and are decoupled first, leading to a Hamiltonian which is more block-diagonal and comprised of only the IR states near the Fermi surface.

As a result of the URG transformations, the Hamiltonian $H_{(j)}$ at a given RG step j involves scattering processes between the k -states that have energies lower than $D_{(j+1)}$. The unitary transformation $U_{(j)}$ is then defined so as to remove the number fluctuations of the currently most energetic set of states $D_{(j)}$ [18,26]:

$$H_{(j-1)} = U_{(j)} H_{(j)} U_{(j)}^\dagger, \text{ such that } [H_{(j-1)}, \hat{n}_j] = 0. \quad (2.3)$$

The eigenvalue of $\hat{n}_j$ has, thus, been rendered an integral of motion (IOM) under the RG transformation.

The unitary transformations can be expressed in terms of a generator $\eta_{(j)}$ that has fermionic algebra [18,26]:

$$U_{(j)} = \frac{1}{\sqrt{2}} \left(1 + \eta_{(j)} - \eta_{(j)}^\dagger \right), \quad \left\{ \eta_{(j)}, \eta_{(j)}^\dagger \right\} = 1, \quad (2.4)$$

where $\{\cdot\}$ is the anticommutator. The unitary operator $U_{(j)}$ that appears in Eq. (2.7) can be cast into the well-known general form $U = e^{\mathcal{S}}$, $\mathcal{S} = \frac{\pi}{4} (\eta_{(j)}^\dagger - \eta_{(j)})$ that a unitary operator can take, defined by an anti-Hermitian operator $\mathcal{S}$. The generator $\eta_{(j)}$ is given by the expression [18,26]

$$\eta_{(j)}^\dagger = \frac{1}{\hat{\omega}_{(j)} - \text{Tr}(H_{(j)} \hat{n}_j)} c_j^\dagger \text{Tr}(H_{(j)} c_j). \quad (2.5)$$

The operators $\eta_{(j)}, \eta_{(j)}^\dagger$ behave as the many-particle analogues of the single-particle field operators $c_j, c_j^\dagger$ - they change the occupation number of the single-particle Fock space $|n_j\rangle$. The important operator $\hat{\omega}_{(j)}$ originates from the quantum fluctuations that exist in the problem because of the non-commutation of the kinetic energy terms and the interaction terms in the Hamiltonian:

$$\hat{\omega}_{(j)} = H_{(j-1)} - H_{(j)}^i. \quad (2.6)$$

$H_{(j)}^i$ is the part of $H_{(j)}$ that commutes with $\hat{n}_j$ but does *not* commute with at least one $\hat{n}_l$ for $l < j$. The RG flow continues up to energy D^* , where a fixed point is reached from the vanishing of the RG function. Detailed comparisons of the URG with other methods (e.g., the functional RG, spectrum bifurcation RG etc.) can be found in Refs. [18, 123]. More information on the unitary RG method is provided in Section 1 of the Supplementary Materials [126].

2.3 $T = 0$ scaling theory for the e-SIAM

2.3.1 URG equations for the e-SIAM

The derivation of the RG equations for the extended Anderson impurity model (e-SIAM) Hamiltonian is shown in Section 2 of the Supplementary Materials [126]. The bath coupling U_b is marginal. We provide below the RG equations of the remaining couplings for a given quantum fluctuation scale ω :

$$\begin{aligned} \Delta U &= 4V^2 n_j \left(\frac{1}{d_1} - \frac{1}{d_0} \right) - n_j \frac{J^2}{d_2}, \\ \Delta V &= -\frac{3n_j V}{8} \left[J \left(\frac{1}{d_2} + \frac{1}{d_1} \right) + \frac{4U_b}{3} \sum_{i=1}^4 \frac{1}{d_i} \right], \\ \Delta J &= -\frac{n_j J (J + 4U_b)}{d_2}, \end{aligned} \quad (3.1)$$

where the denominators d_i are given by

$$d_0 = \omega - \frac{D}{2} + \frac{U_b}{2} - \frac{U}{2}, \quad d_1 = \omega - \frac{D}{2} + \frac{U_b}{2} + \frac{U}{2} + \frac{J}{4}, \quad (3.2)$$

$$d_2 = \omega - \frac{D}{2} + \frac{U_b}{2} + \frac{J}{4}, \quad d_3 = \omega - \frac{D}{2} + \frac{U_b}{2}. \quad (3.3)$$

The symbols used in the RG equations have the following meanings: ΔU represents the renormalisation of the coupling U in going from the j^{th} Hamiltonian to the $(j-1)^{\text{th}}$ Hamiltonian by decoupling the isoenergetic shell at energy $D_{(j)}$ (see right panel of Fig.(2.1)). n_j is the number of electronic states on the shell $D_{(j)}$. We note that the labels U_0, J_0, V_0 that appear in various figures (and elsewhere in the text) represent the bare values of the associated couplings U, J and V . The RG equations reduce, in the perturbative regime of the couplings U, V and J , to the well-known “poor man’s” scaling forms obtained for the SIAM [95] and the single-channel Kondo model [94] respectively.

The RG fixed point Hamiltonian describes the low-energy phase of the system. In general, if the RG fixed point is reached at an energy scale D^* , the fixed point Hamiltonian $\mathcal{H}^*$ is obtained simply from the fixed point values of the couplings (and by noting that the states above D^* are now part of the IOMs):

$$\mathcal{H}^* = \sum_{\sigma, \vec{k}}^{\|\epsilon_{\vec{k}}\| < D^*} \epsilon_{\vec{k}} \tau_{\vec{k}, \sigma} + V^* \sum_{\sigma} \left(c_{d\sigma}^\dagger c_{0\sigma} + \text{h.c.} \right) + J^* \vec{S}_d \cdot \vec{S}_0 - \frac{1}{2} U^* (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2 - \frac{1}{2} U_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2 .$$

The fixed point values of the couplings are obtained by solving the RG equations numerically.

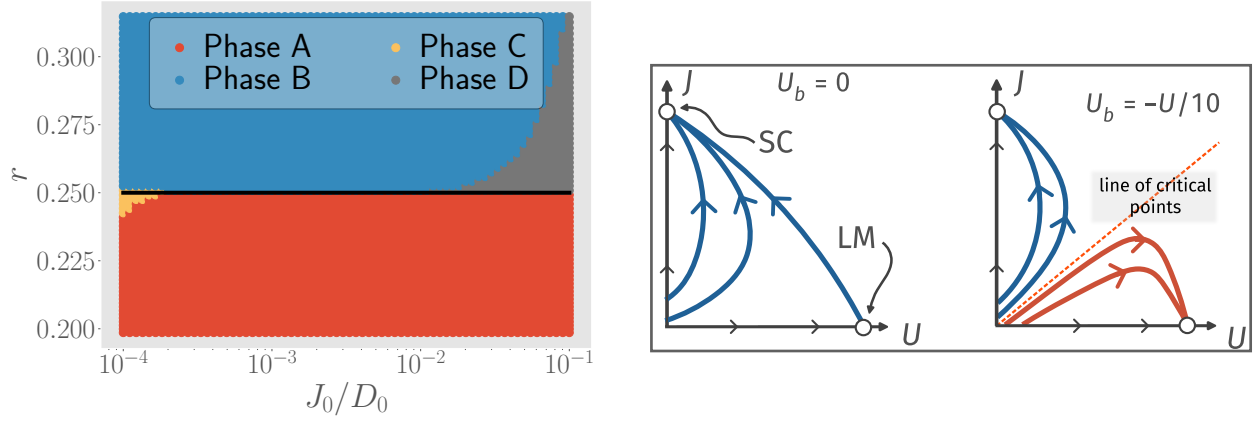


Figure 2.2: *Left*: Phase diagram of the e-SIAM, in the space of r and J_0/D_0 . U_b is set to $-U_0/10$. The various phases are described in the text. *Right*: Schematic nature of RG flows for $U_b = 0$ and $U_b = -U/10$. Blue curves represent RG flows towards the strong-coupling (SC) fixed point, while red curves represent RG flows towards the local moment (LM) fixed point.

2.3.2 Phase diagram

We will work in the low-energy regime where the quantum fluctuation scale ω is such that all the denominators are negative: $d_i < 0 \forall i$. Moreover, we constrain the impurity and local bath correlations (U and U_b respectively) through the relation $U_b = -U/10$. While the precise value of the factor of $1/10$ is unimportant, we have chosen a factor considerably smaller than 1 in order to demonstrate that an interesting body of results can be obtained with a value of $|U_b|$ that is much smaller than that of U . Further, the negative sign in the above relation is significant, as we shall see below that the MIT is obtained for the case of U_b being negative (i.e., attractive on-site correlations on the zeroth site of the bath). The relation between U_b and U is motivated on phenomenological grounds such that the MIT occurring in the bulk lattice model (obtained upon increasing the on-site Hubbard repulsion to large values) corresponds, in the auxiliary model mapping within DMFT, to the local MIT observed upon tuning the impurity correlation U (and the related bath correlation U_b) within the proposed effective impurity model Hamiltonian. The physical significance of attractive on-site correlations in the bath lies in providing a mechanism for the frustration of Kondo screening

(in the RG equation for an antiferromagnetic Kondo coupling $J(> 0)$, eq. (9.28)) within an Anderson impurity coupled to a single channel of conduction electrons. We provide a detailed discussion of this point in the concluding section of our work.

Regime	Low-energy effective Hamiltonian	impurity ground-state
1. $0 < r < r_{c1}$	$\text{K.E}^* + V^* \sum_{\sigma} \left(c_{d\sigma}^{\dagger} c_{0\sigma} + \text{h.c.} \right) + J^* \vec{S}_d \cdot \vec{S}_0$	$\frac{1}{\sqrt{2}} (\uparrow_d\rangle \downarrow_0\rangle - \downarrow_d\rangle \uparrow_0\rangle + 2_d\rangle 0_0\rangle + 0_d\rangle 2_0\rangle)$
2. $r_{c1} < r < r_{c2}$	$\text{K.E}^* + J^* \vec{S}_d \cdot \vec{S}_0 - \frac{1}{2} U_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2$	$\frac{1}{\sqrt{2}} (\uparrow_d\rangle \downarrow_0\rangle - \downarrow_d\rangle \uparrow_0\rangle)$
3. $r_{c2} < r$	$\text{K.E}^* - \frac{U^*}{2} (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2 - \frac{U_b}{2} (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2$	$\{ \uparrow_d\rangle, \downarrow_d\rangle\}$

Table 2.1: Effective Hamiltonians and ground-states of the three important parts of the phase diagram in Fig. (2.2). The ground-state is simplified to capture only the configuration of the impurity site and at most the bath zeroth site. K.E.* represents the kinetic energy of the conduction electron states residing within the fixed point window D^* .

The phase diagram is shown in the left panel of Fig. (2.2) in terms of the parameter $r = |U_b|/J$ (y-axis) and the ratio of the bare Kondo coupling (J_0) to the bare conduction bath bandwidth (D_0) (x-axis), we first define two important points in the space of couplings. These are values of the parameter r where there is a qualitative change in the nature of RG flows, and hence in the low-energy physics of the model

- $r = r_{c1} \left(= - \left(\frac{U_b}{J} \right)_{c1} = \frac{3}{20} = \left(\frac{U}{10J} \right)_{c1} > 0 \right)$: At this point, the coupling V becomes irrelevant,
- $r = r_{c2} \left(= - \left(\frac{U_b}{J} \right)_{c2} = \frac{1}{4} = \left(\frac{U}{10J} \right)_{c2} \right)$: At this point, the coupling J also turns irrelevant. Note that $r_{c2} > r_{c1}$.

We will use these two values of the parameter r as checkpoints around which we can describe the low-energy physics. There are three important parts in Fig. (2.2):

- red region, $0 < r < r_{c1}$: the $J - V$ model; V and J are both relevant, but U is irrelevant; spin and charge delocalisation on the impurity; spin-charge mixing in ground-state
- blue region, $r_{c1} < r < r_{c2}$: the $J - U_b$ model; J is relevant, but V and U are both irrelevant; charge localisation and spin delocalisation on the impurity; singlet ground-state
- violet region, $r_{c2} < r$: the $U - U_b$ model; U is relevant, but V and J are both irrelevant; spin and charge localisation on the impurity; local moment ground-state. This phase describes a local Mott insulator on the impurity site, characterised by vanishing local double occupancy in ground state (black curve in left panel of Fig. 2.4).

Typical RG flows that lead to these phases are shown in Fig. (2.3). We also note that the grey region shown in the top right corner of Fig.(2.2) corresponds to a model in which all three couplings (J , U and V) are RG irrelevant. We find, however, that this phase is an artefact of solving the RG equations for an impurity coupled to a finite-sized conduction bath, and it gradually disappears upon increasing the bath size.

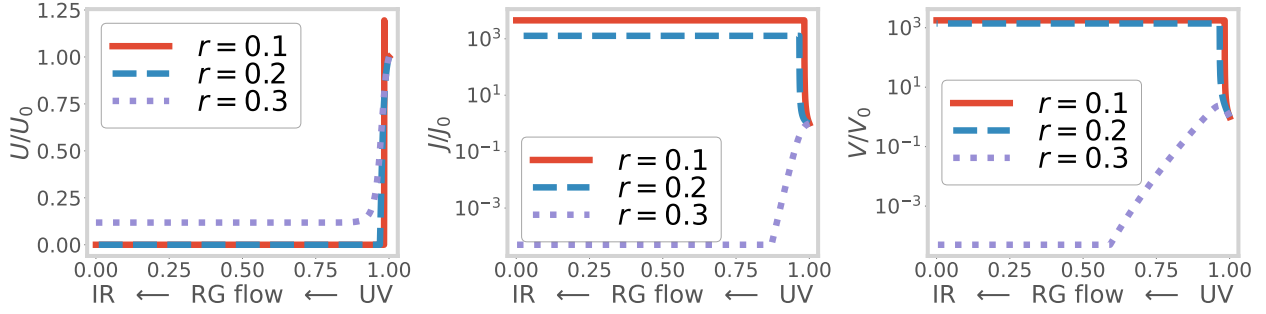


Figure 2.3: Variation of couplings U , V and J along the RG transformations, for three values of the transition-tuning ratio $r = -U_b/J_0$. The x-axis represents the distance of the running cutoff from the Fermi surface; the rightmost point is the first RG step (UV) and the leftmost point is the final RG step (IR). The red curves represent the flows for $r < r_{c1}$, where both V and J are relevant. The blue curves represent the RG flows for $r_{c1} < r < r_{c2}$, where J is relevant but V is irrelevant. The violet curves represent RG flows for $r > r_{c2}$, where both V and J are irrelevant but U flows to a finite value.

2.3.3 Phase transition at r_{c2}

The effective Hamiltonians and corresponding impurity ground-states for various phases have been listed in table (2.1). The variation in the ground state has been checked by numerically solving the fixed point Hamiltonian for various bare values of the couplings and is shown in the left panel of Fig. (2.4). A sharp change in the ground state from spin-singlet to local moment shows that the blue and violet phases are separated by an *impurity delocalisation-localisation transition* (black line in Fig. (2.2)). The transition occurs at finite values of the correlations: $r_{c2} = -\left(\frac{U_b}{J}\right)_{c2} = \frac{1}{4} = \left(\frac{U}{10J}\right)_{c2}$. This is a stark contrast from the standard SIAM, where the transition can happen only at on-site correlation $U \rightarrow \infty$. Therefore, the presence of the critical point at finite values of the various couplings transforms the landscape of RG phase diagram shown schematically in the right panel of Fig. (2.2) (right panel): the RG flows split into two classes - those that flow towards the strong-coupling Kondo screened fixed point and those that flow towards the local moment fixed point.

By a simple rewriting of the RG equation for J (eq. (9.28)), the impurity transition can be seen to arise from a competition between the Kondo screening physics of J and the local pairing physics of U_b :

$$\Delta J = \frac{\overbrace{(J + 2U_b)^2 n_j}^{\text{usual Kondo physics}}}{|d_2|} - \frac{\overbrace{(2U_b)^2 n_j}^{\text{competing pairing physics}}}{|d_2|}, \quad (3.4)$$

where $d_2 = \omega - D/2 + U_b/2 + J/4$. The competition between the effective Kondo term $(J + 2U_b)^2$ and the competing pairing term $-4U_b^2$ leads to the presence of two stable phases - one that is Kondo screened and one that remains unscreened.

In the DMFT treatment of the 1/2-filled Hubbard model on the Bethe lattice in infinite dimensions, the vanishing of non-local contributions to the lattice self-energy means that the lattice Greens

function can be computed self-consistently by solving a local quantum impurity problem [70]. In the rest of the work, we provide extensive evidence that the local transition observed in the e-SIAM is very similar to the Mott MIT observed in DMFT. We conclude thereby that the e-SIAM models faithfully the round-trip excursions of an electron on the Bethe lattice, and that the impurity phase transition observed in the e-SIAM offers a local description of the Mott MIT in the bulk Hubbard model. With this evidence in mind, we will henceforth refer to the transition at r_{c2} as a *local metal-insulator transition*.

2.3.4 The case of repulsive U_b

Henceforth, we will only consider the case of attractive U_b ($U_b > 0$), because that is the regime in which an impurity phase transition is realised at finite values of the couplings. However, before moving on, we will comment briefly on the effects of a repulsive U_b ($U_b > 0$). From the RG equation for J in eq. 9.28, we see that ΔJ is always positive for $U_b > 0$. Moreover, something similar happens in the RG equation for V , where the two terms in the square brackets now enforce each other (J and U_b now have the same sign). Together, they indicate that there will not be any Kondo destruction in the case of $U_b > 0$, and the only fixed points are the weak-coupling one ($J^* = 0$) and the strong coupling one ($J^* = \infty$); there is no local moment phase. This case is therefore adiabatically connected to the case of $U_b = 0$.

2.4 Descriptors of the local MIT

The present section gives more clarity on the nature of the impurity phase transition in the form of additional descriptors of the transition such as impurity spectral function and measures of entanglement, computed from the e-SIAM Hamiltonian. Throughout the rest of the work, we have set the value of the coupling V equal to the Kondo coupling J while generating the quantitative plots.

2.4.1 Evolution of the impurity spectral function

The impurity spectral function of the standard SIAM (eq. (2.1)) always displays a central peak at finite values of U (along with Hubbard sidebands at sufficiently large U), indicating the presence of gapless local Fermi liquid excitations on the impurity site [19, 127, 128]. To demonstrate the impurity localisation transition, we compute the impurity local spectral function of the e-SIAM (eq. (2.2)). This involves numerically diagonalising the effective Hamiltonian at various energy scales along the RG flow and computing the spectral weight at a range of frequencies from UV to IR. We find that the spectral function (shown in the right panel of Fig. (2.4)) displays three notable features, in agreement with DMFT results [70]:

- The appearance of a *preformed gap* (flattening of spectral function between the central peak and sidebands) as r crosses r_{c1} ($= -\left(\frac{U_b}{J}\right)_{c1} = \frac{3}{20} = \left(\frac{U}{10J}\right)_{c1}$): The preformed optical gap simply indicates the separation of the spin (central peak) and charge (sidebands) degrees of freedom beyond r_{c1} . This separation is brought about by the irrelevance of V . We note that

the consequences of an irrelevant hybridisation V can be accounted for in many-body perturbation theory, leading to “in-gap” processes within the optical gap of the impurity spectral function [129]. However, we expect that such contributions will be small, and cannot fill in the optical gap observed in Fig. 2.4 completely.

- The sharpening of the central peak, and concomitant increase in the preformed gap, as r is increased in the range $r_{c1} < r < r_{c2}$.
- The vanishing of the central peak and appearance of a *hard gap* as r crosses r_{c2} : The irrelevance of J beyond r_{c2} destroys the Kondo screening and localises the impurity moment. This involves the destabilisation of the singlet ground state, and the stabilisation of the local moment states in its place.

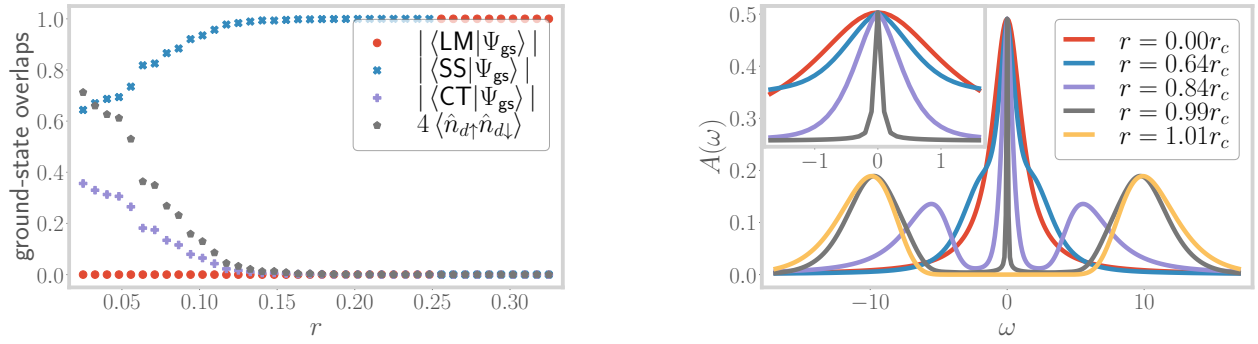


Figure 2.4: *Left:* Overlap of the RG fixed point ground state $|\Psi\rangle_{\text{gs}}$ with the spin-singlet state $|\text{SS}\rangle$, the zero charge member of the charge triplet states $|\text{CT}\rangle$ and the local moment states $|\text{LM}\rangle$, for all three regimes of the model. The black curve corresponds to the double occupancy on the impurity site and is observed to be strongly suppressed beyond r_{c1} . Beyond the critical point $r_{c2} = 0.25$, the overlaps with the entangled states vanish, indicating the transition to a decoupled local moment. *Right:* Variation of the impurity spectral function from $r = 0$ to $r > r_{c2}$. At small r , the central peak is broad, but at larger r , it sharpens, and the difference in spectral weight is used in creating the Hubbard sidebands. For $r > r_{c2}$, the central peak vanishes.

These features are also reflected in ground-state correlation measures like spin-flip and charge isospin-flip correlations (see left panel of Fig. (2.5)), which are defined as follows:

$$\frac{1}{2} \left(\langle S_i^+ S_j^- \rangle + \text{h.c.} \right) = \frac{1}{2} \left(\langle c_{i\uparrow}^\dagger c_{i\downarrow} c_{j\downarrow}^\dagger c_{j\uparrow} \rangle + \text{h.c.} \right), \quad \frac{1}{2} \left(\langle C_i^+ C_j^- \rangle + \text{h.c.} \right) = \frac{1}{2} \left(\langle c_{i\uparrow}^\dagger c_{i\downarrow}^\dagger c_{j\downarrow} c_{j\uparrow} \rangle + \text{h.c.} \right). \quad (4.1)$$

At small values of r , both the correlations are large as the ground-state has both spin and charge content (row 1 of table (2.1)). The subsequent decrease in the impurity-bath charge flip correlation (violet points of Fig. (2.5)), and the simultaneous increase in the impurity-bath spin-flip correlation (red points of Fig. (2.5)), can be attributed to the irrelevance of V at $r \simeq r_{c1}$ and the appearance of the preformed gap in the spectral function. Beyond r_{c2} , the spin-flip correlation sharply drops to zero due to the irrelevance of J . They are replaced by intra-bath correlations like the charge

isospin-flip correlation between the zeroth site and the first site (blue points of Fig. (2.5)). Such intra-bath correlations are promoted by the bath on-site term U_b . The sudden change in the nature of the ground state, and resulting correlations, at r_{c2} indicates the presence of a quantum critical point whose nature will be analysed further in subsequent sections.

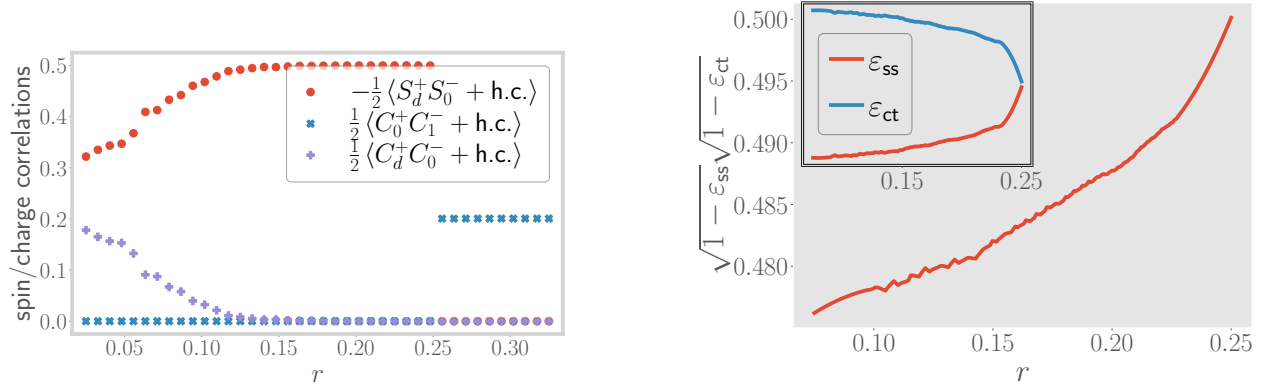


Figure 2.5: *Left:* Variation of impurity-bath spin-flip correlation (red) and charge isospin-flip correlation (violet), as well as intra-bath charge isospin-flip correlation (blue), from $r \sim 0$ to $r > r_{c2}$. The imp.-bath isospin correlation vanishes at r_{c1} , indicating the change in RG relevance of V . The imp.-bath spin correlation vanishes at r_{c2} , indicating the marginality of J at that point. The intra-bath correlation picks up after the transition, showing the decoupling of the impurity from the bath. *Right:* Variation of the geometric entanglement ϵ_{ss} (blue) w.r.t. the singlet state and that w.r.t the charge triplet zero state, ϵ_{ct} (red), with r . The former becomes maximum (unity) at r_{c1} , showing that the true ground state has no charge content. The latter discontinuously jumps to unity at r_{c2} , therefore acting as an order parameter for the transition.

2.4.2 Using entanglement to track correlations across the transition

We will now show that a certain measure of entanglement behaves as an order parameter for the transition. We can define a geometric measure of entanglement in terms of wavefunctions $|\psi_1\rangle$ and $|\psi_2\rangle$ [130–132]:

$$\epsilon(\psi_1, \psi_2) = 1 - |\langle \psi_1 | \psi_2 \rangle|^2. \quad (4.2)$$

From this definition, if $|\psi_1\rangle$ corresponds to a separable state, the entanglement content of the state $|\psi_2\rangle$ is small if it's overlap with $|\psi_1\rangle$ is large. For brevity, we will use the notation $\epsilon_{ss} \equiv \epsilon(\psi_{ss}, \psi_{gs}^{(2)})$, $\epsilon_{ct} \equiv \epsilon(\psi_{ct}, \psi_{gs}^{(2)})$ to represent the geometric entanglement between the e-SIAM ground-state $|\psi_{gs}\rangle$ and the singlet state $|\psi_{ss}\rangle$ or the charge triplet zero state $|\psi_{ct}\rangle$. The latter two states are shown in Table (2.1). The entanglement measures ϵ_{ss} and ϵ_{ct} can be related to the impurity Greens function through the following equation (details can be found in Sec. 3 of the Supplementary

Materials [126]):

$$G_d(\omega) = \sum_n \left[(1 - \varepsilon_{ss}) G_{\Phi_{ss}, \Phi_{ss}}(\omega, n) + (1 - \varepsilon_{ct}) G_{\Phi_{ct}, \Phi_{ct}}(\omega, n) + 2\sqrt{(1 - \varepsilon_{ss})}\sqrt{(1 - \varepsilon_{ct})} G_{\Phi_{ss}, \Phi_{ct}}(\omega, n) \right], \quad (4.3)$$

where

$$G_{\psi_1, \psi_2}(\omega, n) = \frac{1}{2} \frac{\langle \psi_1 | c_{d\sigma} | \Psi_n \rangle \langle \Psi_n | c_{d\sigma}^\dagger | \psi_2 \rangle + \text{h.c.}}{\omega + E_{gs} - E_n} + \frac{1}{2} \frac{\langle \psi_1 | c_{d\sigma}^\dagger | \Psi_n \rangle \langle \Psi_n | c_{d\sigma} | \psi_2 \rangle + \text{h.c.}}{\omega - E_{gs} + E_n}. \quad (4.4)$$

This expression displays that the evolution of the impurity Greens function G_d with r (and related correlation functions shown in the left panel of Fig.(2.5)) is dependent on that of the entanglement measures ε_{ss} and ε_{ct} (shown in the right panel of Fig. (2.5)). Indeed, we find that the geometric entanglement ε_{ct} with the charge sector increases towards unity as the transition at r_{c2} is approached, and remains unity after the transition. A more sensitive measure is the singlet entanglement ε_{ss} : it initially decreases to zero at r_{c1} owing to the irrelevance of V , but rises discontinuously to unity at the transition and becomes equal to the charge entanglement in the local moment phase. Therefore, ε_{ss} acts as an order parameter for the local MIT at r_{c2} . We find that the cross-term $\sqrt{(1 - \varepsilon_{ss})}\sqrt{(1 - \varepsilon_{ct})}$ decreases monotonically to zero with increasing r (not shown), displaying a continual decrease in the mixing of the spin and charge sectors and the *immobilisation of the doublons and holons on the impurity site*.

In general, any one-particle or two-particle fluctuation $\langle O_1 O_2^\dagger \rangle$ that acts on the combined Hilbert space of the impurity and the zeroth site can be expressed in terms of these entanglement measures $\varepsilon_{ss}, \varepsilon_{ct}$. The detailed derivations and expressions are given in Section 3 of the Supplementary Materials [126]. As a demonstration, consider the spin-spin correlation $\langle S_d^+ S_0^- \rangle$ between the impurity and the zeroth site shown in the left panel of Fig. (2.5). The general expression given in Section 3 of the Supplementary Materials [126] is of the form

$$\begin{aligned} \langle S_d^+ S_0^- \rangle = & (1 - \varepsilon_{ss}) \langle \Phi_{ss} | S_d^+ S_0^- | \Phi_{ss} \rangle + (1 - \varepsilon_{ct}) \langle \Phi_{ct} | S_d^+ S_0^- | \Phi_{ct} \rangle \\ & + \sqrt{1 - \varepsilon_{ss}} \sqrt{1 - \varepsilon_{ct}} \left(\langle \Phi_{ss} | S_d^+ S_0^- | \Phi_{ct} \rangle + \langle \Phi_{ct} | S_d^+ S_0^- | \Phi_{ss} \rangle \right). \end{aligned} \quad (4.5)$$

From the expression given, we find that only the singlet overlap is non-zero, such that $\langle S_d^+ S_0^- \rangle$ is directly proportional to the quantity $1 - \varepsilon_{ss}$. As the entanglement measure ε_{ss} increases towards the transition (right panel of Fig. (2.5)), the quantity $1 - \varepsilon_{ss}$ decreases, in turn leading to the decrease in the magnitude of the correlation observed in the left panel of Fig. (2.5). For another explicit connection between correlations and entanglement measures, we present relations between the quantum Fisher information (QFI) [133] and many-particle Greens functions in Section 3 of the Supplementary Materials [126]. There, we also plot the QFI for a number of two-particle operators as a function of r , notably the ones corresponding to the degree of compensation for the impurity ($\langle \vec{S}_d \cdot \vec{S}_0 \rangle$) and the impurity magnetisation ($\langle S_d^z \rangle$). These two quantities (and hence the corresponding QFI) are important because they track the local MIT and act as order parameters for the transition, and the QFI corresponding to these two operators quantify the quantum fluctuations present in the system

corresponding to these order parameters. We show in Sec. 3 of the Supplementary Materials that the two phases on either side of the transition are characterised by distinct values of this pair of QFI: while the QFI corresponding to the degree of compensation is zero in the Kondo screened phase, it becomes non-zero in the local moment phase, and the opposite is true for the QFI arising from the impurity magnetisation. The phase precisely at the transition is distinct from those on either side because it displays a non-zero value for both of the QFI. While it is expected that a critical point would show enhanced fluctuations of multiple kinds (giving rise to universality), it is enlightening to find that this is also reflected in a measure of many-particle entanglement.

2.5 Universal theory for the local metal-insulator transition

In order to identify the competing tendencies near the transition at r_{c2} , we will now obtain the minimal effective Hamiltonian that displays the same transition. This involves integrating out the degrees of freedom that do not affect the low-energy physics near r_{c2} . We note that, due to the irrelevance of V , there is no scattering between the spin and charge states on the impurity at low energies. Further, the impurity charge states $|0\rangle$ and $|\uparrow_d\downarrow_d\rangle$ have been pushed to the high energy Hubbard sidebands because of the large value of U close to r_{c2} . As a result, the impurity charge states can be safely decoupled from the spin states through a Schrieffer-Wolff transformation. Up to second order in V^2/U , this transformation accounts for the effects of U and V by generating an additional (Kondo) spin-exchange term $\delta J \left(\sim \frac{V^2}{U+U_b} \right)$, as well as an additional on-site correlation $\delta U_b \left(\sim \frac{2V^2}{U+U_b+J/2} - \frac{8V^2}{U-U_b} \right)$ on the zeroth site of the conduction bath. In this way, we obtain the following renormalised effective Hamiltonian:

$$H_{\text{MIT}} = \mathcal{J} \vec{S}_d \cdot \vec{S}_0 - \frac{1}{2} \mathcal{U}_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2 + H_{\text{K.E.}}, \quad (5.1)$$

where $\mathcal{J} = J + \delta J$ is the renormalised s-d interaction and $\mathcal{U}_b = U_b + \delta U_b$ is the renormalised local correlation on the bath zeroth site. A schematic 1D construction of the effective Hamiltonian is shown in the left panel of Fig. (2.6). We note that a similar approach towards extracting an effective theory for the Mott MIT has been employed in the past ([134, 135]). However, those works essentially led to renormalised Kondo models that do not possess any frustration of the Kondo screening. Thus, they cannot display an impurity transition in the absence of the requirement of self-consistency and can describe only the physics of the metallic regime. Importantly, the effective $\mathcal{J} - \mathcal{U}_b$ model (eq. (5.1)) is consistent with the IR fixed point Hamiltonian obtained in the appropriate regime $r_{c1} < r < r_{c2}$ (row 2 of table (2.1)).

The RG equations for $\mathcal{J}$ and $\mathcal{U}_b$ of the simplified effective impurity model eq. (5.1) can be obtained by setting $U = V = 0$ in the RG equations (9.28):

$$\Delta \mathcal{J} = -\frac{n_j \mathcal{J} (\mathcal{J} + 4\mathcal{U}_b)}{d_2}, \quad \Delta \mathcal{U}_b = 0. \quad (5.2)$$

For $\mathcal{J} + 4\mathcal{U}_b > 0$, the Kondo coupling is relevant and the low-energy phase is a paramagnetic local Fermi liquid with gapless excitations. But for $\mathcal{J} + 4\mathcal{U}_b < 0$, the Kondo coupling becomes

irrelevant and the ground state is a decoupled local moment that is isolated from the bath. Such an effective picture of the transition can be understood if one notes that a straightforward way to destroy the Kondo screening is to inhibit the coordinated spin fluctuations between the impurity and the bath. Such frustration of Kondo screening is precisely the effect of the $\mathcal{U}_b$ term: it promotes charge fluctuations on the bath site coupled directly to the impurity, reducing thereby the spectral weight for spin-flip scatterings between the impurity and bath.



Figure 2.6: *Left*: Schematic 1D construction of universal theory for the metal-insulator transition, obtained by integrating out the charge states of the impurity near the transition. *Right*: Zero bandwidth limit of the low-energy effective Hamiltonian for r between r_{c1} and r_{c2} (second row in table (2.1)). The bath is reduced to just a single degree of freedom - the zeroth site that is directly coupled to the impurity site.

Indeed, the local correlation $\mathcal{U}_b$ encourages entanglement between the sites of the bath and makes the formation of the impurity-bath singlet difficult. In other words, the $\mathcal{J}-\mathcal{U}_b$ model displays the destabilisation of the singlet by redistributing the entanglement from the impurity+bath system to purely within the bath (see left panel of Fig. (2.7)). Given the simplicity of these arguments, our analysis makes the case that local pairing fluctuations of the bath in eq. (5.1) offer a universal mechanism by which to frustrate the Kondo effect and lead thereby to an impurity transition that is the local counterpart of the Mott MIT obtained by auxiliary model approaches such as DMFT. We note that similar conclusions were reached in Refs. [87, 88, 102] for the emergence of non-Fermi liquid phases at critical points in the mixed valence regime of the periodic Anderson model.

Further insight into the destabilisation of the singlet can be obtained from a zero-bandwidth approximation of the bath. Under such an approximation, the IR effective Hamiltonian obtained from the RG flow (row 2 of Fig. (2.1) takes the form of a two-site model with modified couplings (shown in the right panel of Fig. (2.6)):

$$\tilde{\mathcal{J}} \vec{S}_d \cdot \vec{S}_0 - \frac{1}{2} \tilde{\mathcal{U}}_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2. \quad (5.3)$$

The spectrum of this model is shown in the right panel of Fig. (2.7). For $|\tilde{\mathcal{U}}_b|/\tilde{\mathcal{J}} < 3/2$, the singlet ground-state (red) is separated from the excited local moment states (blue) by a gap of $\frac{-3\tilde{\mathcal{J}}}{4} - \frac{\tilde{\mathcal{U}}_b}{2}$. As we tune r towards r_{c2} , $|\tilde{\mathcal{U}}_b|$ increases and the fixed point value $\tilde{\mathcal{J}}$ decreases, leading to an overall reduction in the gap. At the critical point, the gap closes and the states become degenerate at zero energy; this is the equivalent of the quantum critical point at r_{c2} of the full impurity model that was discussed earlier. We note that the vanishing of the energy of the metallic state was also observed by Brinkman and Rice from a Gutzwiller-type variational calculation of the Hubbard model at half-filling [79].

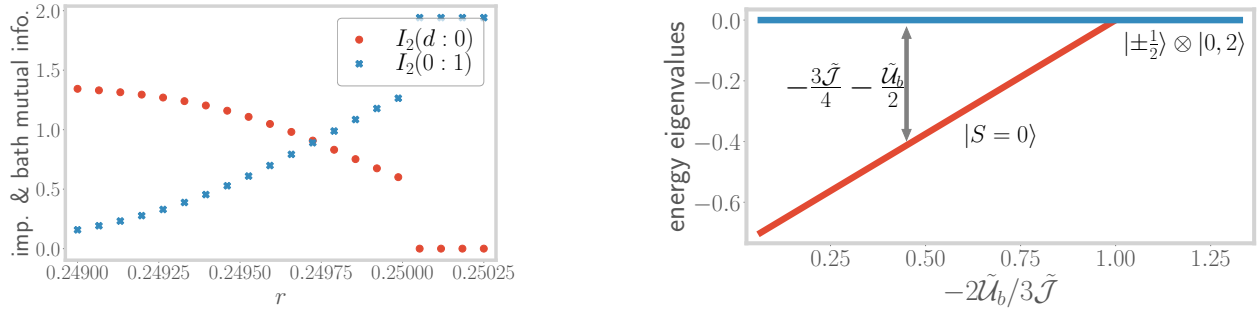


Figure 2.7: *Left:* Very close to the transition, the mutual information $I_2(d : 0)$ (red curve) between impurity and the zeroth site reduces, while that between the zeroth and the first site ($I_2(0 : 1)$, blue curve) increases, showing the redistribution of entanglement. *Right:* Spectrum of the zero bandwidth Hamiltonian of the right panel of Fig. (2.6). The blue line represents the local moment states at zero energy, while the red line represents the singlet state at an energy that continuously increases upon increasing r .

2.5.1 Local pairing and the destruction of Kondo screening

Very close to the transition at r_{c2} , we find signatures of the breakdown of the Kondo cloud in terms of decaying impurity-zeroth site correlations (blue curve in the left panel of Fig. (2.8)) and enhanced intra-bath correlations. Remarkably, we find an *increase in pairing correlations* between the bath zeroth site and first site (red curve in the left panel of Fig. (2.8)). As discussed above, these are observed to be responsible for the destruction of the Kondo screening: spin and charge degrees of freedom are mutually exclusive, and the increased charge fluctuations on the bath zeroth site suppress the spin-flip scattering processes between the impurity and the zeroth site. We will show later that these fluctuations lead to the destruction of the local Fermi liquid excitations in our model, and replace them with non-Fermi local excitations in the neighbourhood of r_{c2} . We believe that the observed growth in non-local pairing correlations near the transition at r_{c2} is likely tied to a putative low-energy divergence of the local pairing susceptibility. A similar observation was made in Ref. [87] from an auxiliary model-based analysis of a hole-doped extended Hubbard model in infinite dimensions that is pertinent to the physics of the heavy fermions. Similar to the conclusions of Ref. [87], we expect the pairing fluctuations of the bath to become dominant upon tuning the e-SIAM away from half-filling, signalling an instability towards a superconducting state.

2.5.2 Emergent self-consistency in the e-SIAM

As is well-established, the DMFT self-consistency equation is equivalent to requiring that the impurity Greens function become equal to the local Greens function in the bath [70]. Such a condition is also used in the projective self-consistent technique of Moeller et al. [134] for the states within the central Kondo resonance. We will now show that our model displays a qualitatively similar emergent feature. To proceed, by employing a one-step URG transformation, we integrate out the impurity site from the rest of the fixed-point Hamiltonian of eq. (3.4). The details are shown in Sec. 4 of the Supplementary Materials [126]. The essential idea is similar to that of the Schrieffer-Wolff

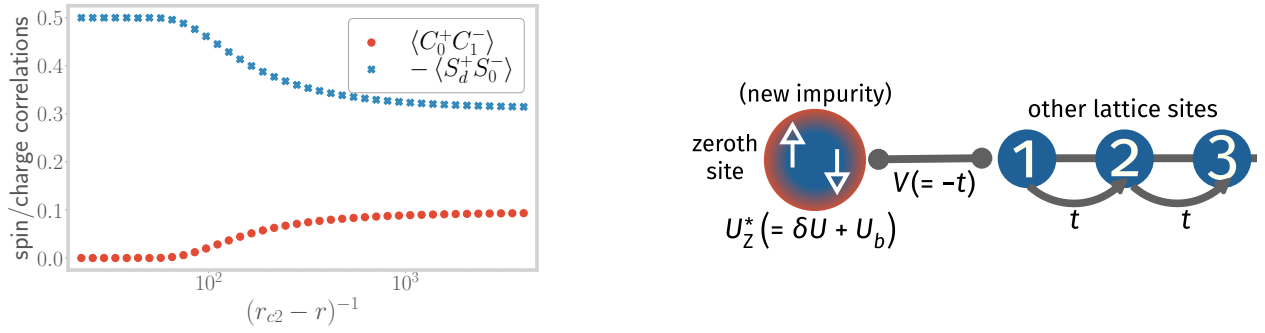


Figure 2.8: *Left:* Spin-flip correlation between the impurity and the bath (blue), and charge isospin-flip correlation between the bath zeroth and first sites (red), both as a function of the tuning parameter r , very close to the transition. The operators are defined as $S_i^+ = c_{i\uparrow}^\dagger c_{i\downarrow}$ and $C_i^+ = c_{i\uparrow}^\dagger c_{i\downarrow}^\dagger$. *Right:* New effective SIAM Hamiltonian generated upon integrating out the coupling between the impurity and the bath zeroth site. In the new SIAM, the zeroth site acts as the new impurity site with a renormalised on-site correlation $U_Z^* = U_b + \delta U_Z^*$.

transformation: removing the impurity-bath couplings J and V generates an additional repulsive correlation δU_Z^* on the zeroth site. The new low-energy model H_Z is therefore an Anderson impurity model with a net local correlation $U_Z^* = \delta U_Z^* + U_b$ on the zeroth site (which is now the “new impurity site”), and a single-particle hybridisation $V_Z^* = -t$ coupling with a conduction bath formed by the remaining sites. This new impurity model is depicted schematically in the right panel of Fig. (2.8).

$$H_Z^* = \underbrace{-\frac{1}{2}U_Z^* (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2}_{\text{new impurity} = 0^{\text{th}} \text{ site}} + \underbrace{V_Z^* \sum_{\langle j,0 \rangle} (c_{0\sigma}^\dagger c_{j\sigma} + \text{h.c.})}_{\text{hopping between } 0^{\text{th}} \text{ site \& new bath}} + \underbrace{(-t) \sum_{\langle i,j \rangle} (c_{i\sigma}^\dagger c_{j\sigma} + \text{h.c.})}_{\text{K.E. of new bath}}. \quad (5.4)$$

Since, for $r < r_{c2}$, the model always flows to strong coupling at low energies, the largest energy scales are J^* and V^* . Using this, the effective correlation U_Z^* on the zeroth site can be expressed to leading order as

$$U_Z^* \simeq \frac{J^*}{4} \frac{1}{1 + 2\gamma} - V^* \frac{\gamma}{\gamma^2 - \frac{1}{4}}, \quad \text{where } \gamma \equiv V^*/J^*. \quad (5.5)$$

From $r = 0$ to $r = r_{c1}$, the factor γ decreases due to the gradual removal of single-particle hopping from the impurity site (irrelevance of V), leading to an increase in the correlation U_Z^* . This is simply a restatement of the fact that the scattering processes that create the central impurity resonance induce a repulsive correlation on the bath zeroth site. The fact that we end up with a standard SIAM on the zeroth site once the impurity site has been integrated out means that the spectral function of this new impurity will again go through a sharpening of the central peak (and the appearance of the Hubbard sidebands) upon increasing the parameter r of the original e-SIAM. This ensures that the impurity and zeroth site spectral functions look qualitatively similar up to r_{c1} . One can now repeat iteratively this process - decoupling the zeroth site generates a standard SIAM with a

repulsive correlation on the first site, and so on. The fact that excursions starting from any point along the bath can be described by a positive U SIAM is, therefore, the emergent self-consistency in our model. Similar indications of a correlated spectral function on lattice sites far away from the impurity were also observed in Ref. [111] from finite- U slave boson calculations of the SIAM. Beyond r_{c1} , the irrelevance of V means that U_Z^* reduces to just $J^*/4$. As r is now increased towards r_{c2} , the correlation decreases because J is moving towards its critical point, indicating that the bath zeroth site is moving away from its local moment regime. This is another reflection of the increase in pairing fluctuations of the bath, as well as the lowering of spin-flip fluctuations between the impurity and the bath.

For $r > r_{c2}$, the impurity site decouples from the bath in the impurity model. This impurity model can be promoted to a bulk model as follows. Recall that in the auxiliary model mapping, any lattice site $\vec{r}_i$ of the bulk lattice can act as the impurity, and one can think of the impurity model with the impurity at $\vec{r}_i$ as a representation of the excursion of an electron from any such site $\vec{r}_i$ into the rest of the bath. When all such impurity models undergo the transition at $r = r_{c2}$ simultaneously, the result is the decoupling of all sites from their respective baths and a paramagnetic bulk insulator is obtained.

2.6 Coexistence of local metallic and insulating phases in the e-SIAM

The DMFT solution of the Hubbard model on the Bethe lattice exhibits a coexistence region of metallic and insulating solutions between two spinodal lines $U_{c1}(T)$ and $U_{c2}(T)$ [70], with the metallic solution having lower internal energy and the insulating solution being a metastable state at a higher energy [70, 134, 136–138]. The $T = 0$ transition is observed to be second order in nature and occurs through a merging of the metallic and insulating solutions at U_{c2} [134, 139]. On the other hand, at $T > 0$, the MIT happens along the first-order line $U_c(T)$ ($U_{c1} < U_c < U_{c2}$) where the free energies of the two solutions become equal, and the two spinodal lines merge into a second order critical point at a sufficiently high temperature T_c . We will now show that our Hamiltonian-based approach gives a clear zero temperature picture of various aspects that can lead to the emergence of the metal-insulator coexistence at $T > 0$.

2.6.1 Excited state quantum phase transition at r_{c1} : the Mott-Hubbard scenario

We have already discussed in detail the nature of the phase transition at r_{c2} : the zero-bandwidth picture provided (around eq. (5.3)) shows the merging of the metallic and insulating solutions at that point, enabling identification of r_{c2} with the $T = 0$ continuous phase transition observed at U_{c2} in the DMFT phase diagram. The focus of this subsection is the physics of the other important point r_{c1} . As has been mentioned before, this point marks the RG irrelevance of the single-particle hybridisation amplitude V , and leads to the exclusion of the charge states from the ground-state (see left panel of Fig. (2.4)). This exclusion means that there is now one fewer scattering channel by

which the impurity electron can hybridise with the bath. In turn, this leads to a *partial localisation of the impurity*, and can be thought of as the first step towards the more complete localisation that occurs at r_{c2} .

Apart from the change in the ground state, the physics at r_{c1} also involves a phase transition in certain excited states of the spectrum. To expose this, we consider the following states in the zero-bandwidth spectrum of the impurity model given in eq. (2.2):

$$\begin{aligned}
 |1, \sigma, \pm\rangle &= \alpha_{\pm} |\sigma_d\rangle |0_0\rangle \mp \sqrt{1 - \alpha_{\pm}^2} |0_d\rangle |\sigma_0\rangle, \quad |3, \sigma, \pm\rangle = \alpha_{\pm} |\sigma_d\rangle |2_0\rangle \mp \sqrt{1 - \alpha_{\pm}^2} |2_d\rangle |\sigma_0\rangle, \\
 E_+ &= -\frac{U_0}{4} + \sqrt{V_0^2 + \frac{U_0^2}{16}}, \quad E_- = -\frac{U_{\text{im}}}{4} - \sqrt{V_{\text{im}}^2 + \frac{U_{\text{im}}^2}{16}}, \\
 \alpha_+(U_0, V_0) &= \frac{V_0}{\sqrt{V_0^2 + \left(E_+ + \frac{U_0}{2}\right)^2}}, \quad \alpha_- = \frac{E_- + \frac{U_{\text{im}}}{2}}{\sqrt{V_{\text{im}}^2 + \left(E_- + \frac{U_{\text{im}}}{2}\right)^2}}.
 \end{aligned} \tag{6.1}$$

where $(\sigma = \uparrow, \downarrow)$ and $E_{\pm}$ is the energy of the states $|1(3), \sigma, \pm\rangle$ and the subscripts d and 0 refer to the impurity and bath zeroth site respectively. The states $|1(3), \sigma, +\rangle$ are high-energy states, so their energy E_+ and coefficient α_+ involve the bare single-particle hybridisation V_0 and impurity on-site repulsion U_0 . On the other hand, the other states $|1(3), \sigma, -\rangle$ are closer to the IR energy scale and involve renormalised intermediate-scale couplings V_{im} and U_{im} . Both E_+ and E_- are four-fold degenerate because of the SU(2) spin ($\uparrow \leftrightarrow \downarrow$) and particle-hole ($|0\rangle \leftrightarrow |2\rangle$) symmetries. For example, the degenerate subspace corresponding to E_+ is the set of states $\{|1, \uparrow, +\rangle, |1, \downarrow, +\rangle, |3, \uparrow, +\rangle, |3, \downarrow, +\rangle\}$.

All these states are in general delocalised - they involve the impurity hybridising with the bath via V . At $r = r_{c1}$, however, the coupling V becomes irrelevant and $V_{\text{im}} \rightarrow 0$, so that the coefficient α_- of the low-energy state becomes unity beyond that point. The low-energy states $|1(3), \sigma, -\rangle$ thus become localised at r_{c1} and give rise to a set of excited and *degenerate local moment states*, i.e., $\alpha_- \rightarrow 0$ as $r \rightarrow r_{c1}$ leads to

$$|1(3), \sigma, -\rangle \rightarrow \begin{cases} |\uparrow_d\rangle |0_0\rangle, |\downarrow_d\rangle |0_0\rangle \\ |\uparrow_d\rangle |2_0\rangle, |\downarrow_d\rangle |2_0\rangle \end{cases}. \tag{6.2}$$

The other coefficient α_+ remains non-zero because, as mentioned earlier, V is non-zero in the UV scales of the RG flow. As a result, the high-energy states $|1(3), \sigma, +\rangle$ remain delocalised, and are separated from the localised states by an energy-scale

$$\lim_{r \rightarrow r_{c1}^-} (E_+ - E_-) \simeq \sqrt{V_0^2 + \left(\frac{U_0}{4}\right)^2} + \sqrt{V_{\text{im}}^2 + \left(\frac{U_{\text{im}}}{4}\right)^2} + \frac{U_{\text{im}} - U_0}{4}. \tag{6.3}$$

This is shown schematically in the left panel of Fig. (2.9), and corresponds to the preformed gap in the zero mode spectrum. The point r_{c1} therefore represents a *delocalisation-localisation excited state quantum phase transition* (ESQPT) where degenerate local moment states are emergent as excited states in the many-body spectrum. This localisation of the impurity at r_{c1} is shown in the form of excited state mutual information in the right panel of Fig. (2.9). This ESQPT acts as a

precursor to the QPT at r_{c2} , where the local moment states become degenerate with the spin-singlet ground state. The local moment states are stabilised as ground states in the insulating phase for $r > r_{c2}$.

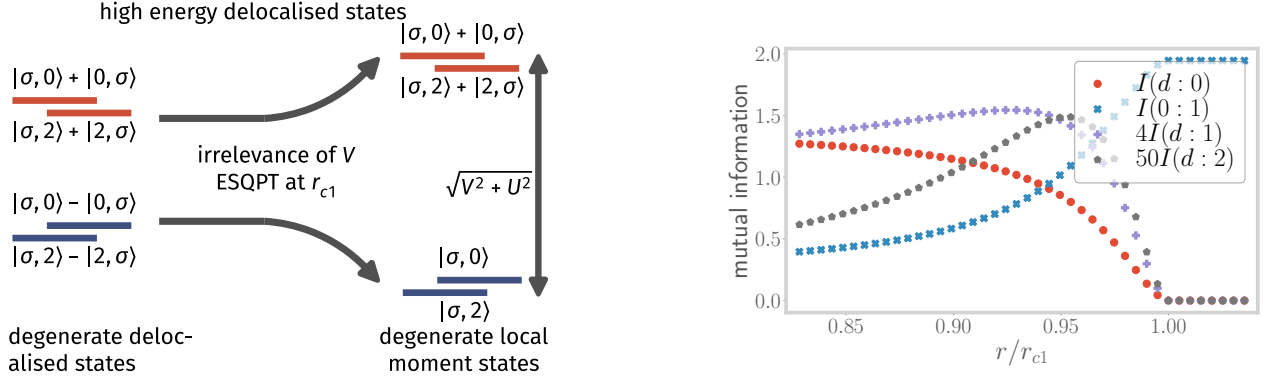


Figure 2.9: *Left*: Evolution of the excited states mentioned in Eqs. (6.1) as r is tuned through r_{c1} . The high-energy states (red) retain their impurity-bath coupling, while their low-energy counterparts (blue) become disentangled and form a degenerate set of local moment states. The high-energy states allow delocalisation into the bath and lead to the broad Hubbard sidebands. *Right*: Variation of mutual information (MI) between various parties, across the ESQPT at r_{c1} . The MI (red) between the impurity and the zeroth site vanishes at r_{c1} , showing the localisation of the impurity. The same between the bath zeroth site and the next site becomes maximum at r_{c1} . The MI between the impurity and the other sites (purple and gray) of the bath show an initial rise near r_{c1} , indicating that entanglement is becoming long-ranged near the ESQPT.

2.6.2 Theory for the charge excitations in the Hubbard sidebands

Beyond r_{c1} , excitations into the *local moment states* $|1, \sigma, -\rangle$ and $|3, \sigma, -\rangle$ reside at the edge of the central peak in the impurity spectral function (purple curve in the right panel of Fig.(2.4)) but provide no spectral weight due to the lack of electron mobility. This explains the development of a preformed gap in the impurity spectral function. That these local moment states have to reside at the edge of the central peak becomes clear when we note that as the width of the central peak shrinks continuously, the local moment states must also recede towards zero frequency and finally replace the zero frequency peak at $r = r_{c2}$ in order to give rise to the insulating local moment phase for $r > r_{c2}$. On the other hand, the *still-delocalised high-energy states* $|1, \sigma, +\rangle$ and $|3, \sigma, +\rangle$ are pushed into the Hubbard sidebands, and their hybridisation with the bath through V and t is responsible for the broadening of the sidebands.

This isolation of the delocalised states into the sidebands means that charge delocalisation processes are now excluded from the physics at low energies. As accessing the sidebands involves excitations at exorbitantly high energy scales, such processes can only happen virtually and involve very short time scales. The central Kondo resonance observed at low energies, therefore, does not support any charge delocalisation, and metallic excitations propagate only through spin-flip scattering processes of the impurity. This reveals that the Mott insulator comes about through a local

binding of doublons and holons [75, 140, 141]. Closely related to this is the Mott-Hubbard scenario of the MIT, which is equivalent to our observation of the appearance of an optical gap in the spectrum after r_{c1} .

The fact that the sidebands are broad indicates that there are gapless excitations propagating from the impurity site into the conduction bath, but whose energy lies outside the Mott gap. These can also be viewed as metallic excitations of the bath that are induced by the coupled impurity. In order to expose the nature of these excitations, we perform a calculation similar to the local Fermi liquid calculation of Nozières [127]. This involves considering the states at $\omega \sim \pm U/2$ as the ground-states of the Hubbard sidebands, and then computing how excitations into the bath renormalise the ground-state subspace. Up to the second order in the hopping strength t , the effective Hamiltonian for the excitations of the bath can be expressed as

$$H_{\text{eff}}^{(2)} = \frac{4t^2(1 - \alpha_+^2)}{E_+ - E_{\text{gs}}} C_{\text{tot}}^z C_1^z - t \sum_{i>0, \sigma} \left(c_{i\sigma}^\dagger c_{i+1\sigma} + \text{h.c.} \right), \quad (6.4)$$

where E_+ and α_+ have been defined in eq. (6.1), $C_i^z |2(0)\rangle = +(-)\frac{1}{2} |2(0)\rangle$ is the z-component of the charge isospin operator at a particular site i , and $\vec{C}_{\text{tot}} = \vec{C}_d + \vec{C}_0$ is the total isospin for the impurity and zeroth sites. $E_{\text{gs}} = U/2$ is the two-site energy of the ground-state subspace within the sideband. These excitations are of the local Fermi liquid kind - the absence of any isospin-flip scattering term promotes the independent delocalisation of the doublon and holon states into the bath. The second-order effective Hamiltonian can therefore be written as the sum of two decoupled parts corresponding to holon and doublon propagation respectively, and suggesting adiabatic continuity with the $U = 0$ Hubbard model [142].

However, fourth-order corrections lead to the appearance of scattering processes that convert holon and doublons into one another, resulting in the local Fermi liquid becoming correlated

$$H_{\text{eff}}^{(4)} = \frac{\gamma^4 \alpha_+^2 \beta^2}{(1 - \alpha_+^2)(E'_+ - E_{\text{gs}})} \left[-C_{\text{tot}}^z C_{\text{tot}}^2 C_1^z + \sqrt{2} \mathcal{P}_{\text{tot}}^4 (C_0^+ - C_d^+) C_1^- + \text{h.c.} \right], \quad (6.5)$$

where $E'_+ = \frac{U}{4} - \frac{3J}{8} + \sqrt{4V^2 + \left(\frac{U}{4} + \frac{3J}{8}\right)^2}$ is the energy of the state containing the charge isospin triplet zero, and $\beta = \frac{2V}{\sqrt{4V^2 + (U+3J/4)^2}}$. $\mathcal{P}_{\text{tot}}^4$ projects on to the $n_d + n_0 = 4$ subspace. The scattering processes in eq. (6.5) are clearly non-Fermi liquid in nature as they allow the inter-conversion of the charge isospin eigenstates, and therefore reduce the lifetime of the quasiparticle excitations of the local Fermi liquid obtained in eq. (6.5). Additional details pertaining to the calculation of this effective Hamiltonian are present in Sec. 5 of the Supplementary Materials [126].

2.6.3 Zero temperature origin of critical fluctuations above the second order point

Above the second order point $r_c(T_c)$, the DMFT phase diagram shows a rapid crossover from the paramagnetic metallic solution to the paramagnetic insulating solution [70, 143]. Remarkably, signatures of quantum critical scaling in this crossover region have been recently predicted from theoretical analyses [81, 89], as well as detected experimentally in transport measurements of several

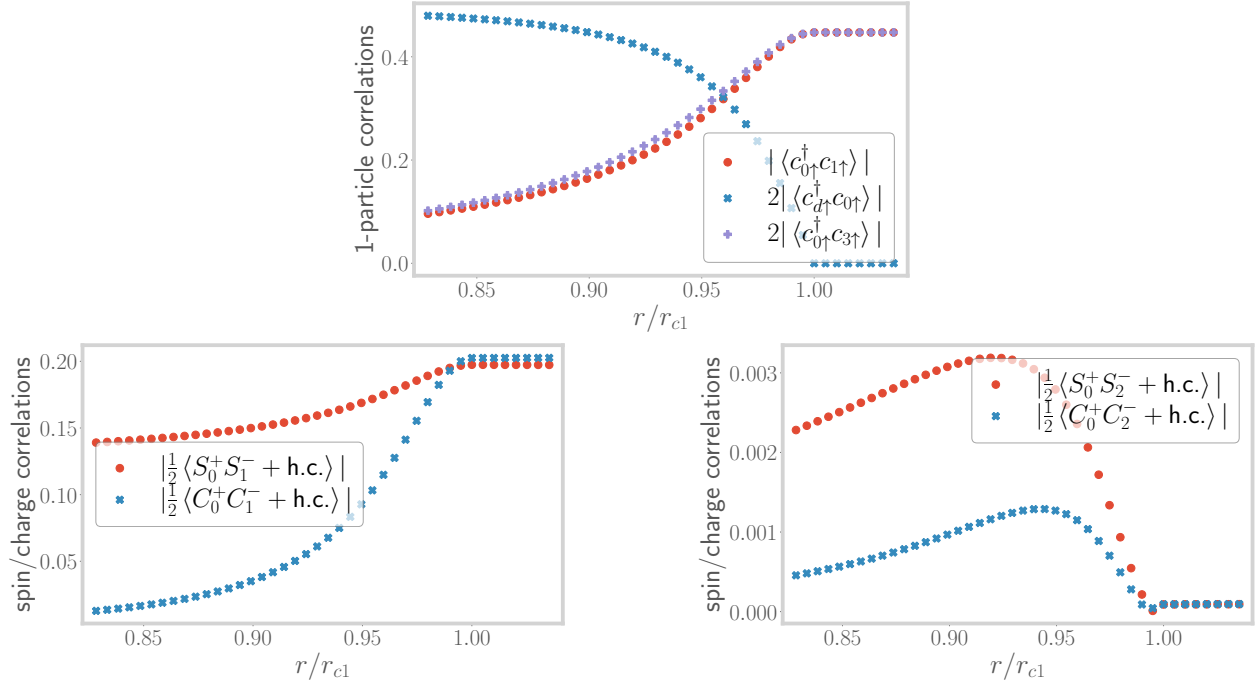


Figure 2.10: *Top:* Variation of the one particle correlations in the state $|1, \sigma, -\rangle$ between impurity and bath zeroth site (blue), as well as within the bath (red and violet). The former vanishes, depicting the excited state localisation transition at r_{c1} . *Left:* Variation of the spin-flip (red) and charge isospin-flip (blue) correlations between impurity and bath, close to r_{c1} . They both increase, indicating that the impurity is now less strongly coupled with the bath. *Right:* Variation of the spin-flip (red) and charge isospin-flip (blue) correlations between the zeroth and second sites of the bath, close to r_{c1} . Both show an initial increase, leading to the propagation of relatively long-range correlations into the bath.

organic compounds [144, 145]. In Refs. [81, 89], this has been ascribed to the existence of a *hidden quantum criticality* in the maximally frustrated 1/2-filled Hubbard model on the Bethe lattice with infinite coordination number. In order to locate the zero temperature origin of these signatures of quantum criticality, we inspect carefully the quantum-mechanical fluctuations near the two important points r_{c1} and r_{c2} in our model.

Close to r_{c1} , we compute correlations in the state $|1, \sigma, -\rangle$. As shown in the top panel of Fig. (2.10), we find that while the one-particle correlation between the impurity and the bath vanishes, the same between the zeroth and first sites of the bath picks up. Importantly, these correlations extend beyond the immediate neighbourhood of the impurity. This is observed in, for example, the one-particle correlation between the zeroth site and the second site (purple curve in the top panel of Fig. (2.10)). Some other signatures of correlations in the bath are shown in Fig. (2.10): while the left panel displays increased spin-flip and charge isospin-flip correlations between the zeroth and first sites of the bath, a similar phenomenon is observed between the zeroth and second sites of the bath in the right panel. Our findings are consistent with the presence of scale-invariant solutions obtained from recent NRG-DMFT calculations [146] that correspond to the metastable insulating solutions in the coexistence region.

The growth of similar long-ranged correlations within the bath (and leading away from the immediate neighbourhood of the impurity) near r_{c2} as well. We recall that correlations between the impurity and the zeroth site are observed to decrease (see left panel of Fig.(2.5)). Instead, as shown in Fig. (2.11), non-trivial two-particle correlations arise between the impurity and bath zeroth sites with bath sites that are farther away. The spreading of the spin-spin correlations shown in the left panel of Fig. (2.11) indicates a “stretching” of the Kondo singlet state prior to its destruction. We will show later that these enhanced correlations also result in a diverging quasiparticle mass at r_{c2} , indicating a breakdown of the local Fermi liquid metal. Further, the right panel of Fig. (2.11) indicates that longer-ranged pairing correlations develop between the bath zeroth site and bath sites farther away upon approaching the transition.

In general, within the metallic phase and away from r_{c1} or r_{c2} , the impurity is strongly coupled to the bath zeroth site and the impurity-bath entanglement follows an area law characteristic of the local nature of the impurity problem. However, near the excited state and ground state transitions at r_{c1} or r_{c2} respectively, these results indicate a spreading of entanglement into the bath: more and more bath sites beyond the zeroth site get correlated with the impurity site as the ESQPT or the QCP is approached. This is corroborated in Fig. 3 of the Supplementary Materials [126]. This suggests a significant enhancement of the entanglement beyond the area law and is consistent with the critical quantum fluctuations observed in the various two-particle correlations.

The results of this subsection indicate that it might be possible to experimentally detect the signatures of such critical quantum fluctuations around the spinodals of the DMFT phase diagram even at $T > 0$. We note that signatures of second-order criticality have been indeed observed recently in vanadium sesquioxide, in the form of critical slowing down and critical opalescence [147]. It is, therefore, tempting to speculate that these signatures arise from the presence of critical quantum fluctuations.

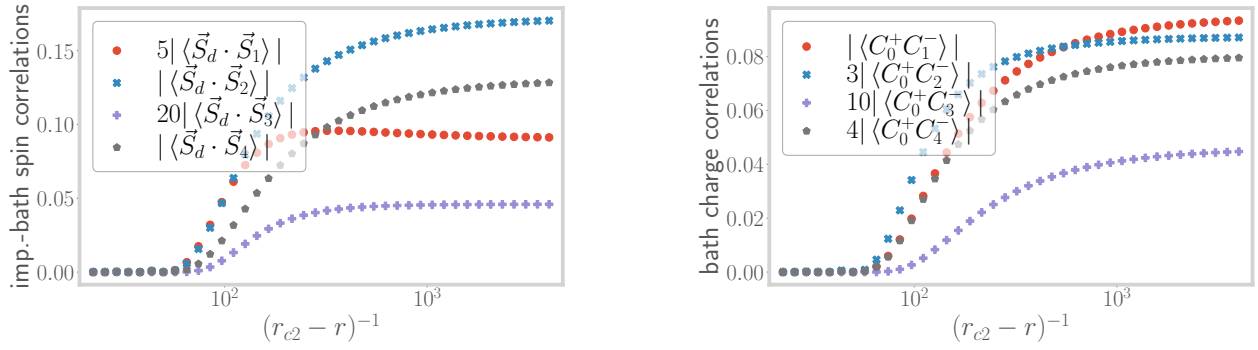


Figure 2.11: *Left:* Variation of the spin-spin correlations between the impurity and bath sites 1 through 4, close to r_{c2} . They show an increase, revealing the distribution of entanglement into the bath. *Right:* Variation of pairing correlations within the bath. These correlations show an increase, because of the weakening of the Kondo singlet.

2.7 Non-Fermi liquid signatures at the MIT

2.7.1 Death of the local Fermi liquid: the Brinkman-Rice scenario

The low-lying metallic excitations of the bath can be obtained by considering the singlet ground-state of the model (of energy $\sim -3\tilde{\mathcal{J}}/4$) in the metallic phase, and then studying the effect of an electron hopping term (with coupling t) between the singlet and the rest of the bath as a perturbation (see left panel of Fig. (2.12)). Such a strong-coupling expansion in powers of $t^2/\tilde{\mathcal{J}}$ leads to the usual local Fermi liquid effective Hamiltonian in the Kondo model [19, 106, 127, 148]:

$$H_{\text{LFL}} = \mathcal{F} \hat{n}_{1\uparrow} \hat{n}_{1\downarrow} + H_{\text{KE}}, \quad \mathcal{F} \sim t^4 / \tilde{\mathcal{J}}^3, \quad (7.1)$$

where $\hat{n}_{1\sigma}$ are the number operators for the first site (site adjacent to the zeroth site of the conduction bath), H_{KE} is the kinetic energy arising from the nearest-neighbour hopping among all sites in the bath apart from the zeroth site, and $\mathcal{F}$ is the local Fermi liquid correlation strength. The Kondo singlet (formed between the impurity spin and the bath zeroth site) has decoupled from the rest of the lattice, and eq. (7.1) describes the effective Hamiltonian for the rest of the bath beyond the zeroth site.

Following an identical approach, we find that the effective Hamiltonian for the low-lying excitations in the metallic phase is given by

$$H_{\text{LFL}} = \mathcal{F} [\hat{n}_{1\uparrow} \hat{n}_{1\downarrow} + (1 - \hat{n}_{1\uparrow})(1 - \hat{n}_{1\downarrow})] + H_{\text{KE}}, \quad \text{where } \mathcal{F} = \frac{2t^4}{\tilde{\mathcal{J}} (3\tilde{\mathcal{J}}/4 + \tilde{\mathcal{U}}_b)^2}. \quad (7.2)$$

We note that for our extended SIAM, the local Fermi liquid correlation $\mathcal{F}$ diverges as the transition is approached at $3\tilde{\mathcal{J}}/4 + \tilde{\mathcal{U}}_b \rightarrow 0$. The divergence of $\mathcal{F}$ leads to the divergence of the renormalised mass of the quasiparticles, and can also be obtained from a Gutzwiller variational calculation as shown by Brinkman and Rice [79]. This shows the breakdown of perturbation theory and is indicative of the fact that the ground state is about to change at the metal-insulator transition.

The death of the local Fermi liquid is also seen from the vanishing of the dynamically generated low-energy Kondo screening scale T_K . Towards obtaining T_K close to the transition, we follow the approach used by Moeller et al. 1995 [134] and Held et al. 2013 [135]. Near the transition, they obtained a Kondo model from the SIAM by applying a Schrieffer-Wolff transition that removes the charge fluctuations of the impurity site and retains the physics of only the Kondo coupling J . This amounts to removing the side peaks from the impurity spectral function and focusing on the low-energy central peak (right panel of Fig. (2.12)). Held et al. then integrated the RG equation for this Kondo model by using a Lorentzian DOS $\rho(D)$ in the bath: $\rho(D) = \frac{\rho_0 \Gamma^2}{D^2 + \Gamma^2}$. This is motivated by



Figure 2.12: *Left*: Setup for obtaining the low-energy excitations above the strong-coupling ground state. The effects of the rest of the bath is included by treating the hopping between the zeroth site (blue sphere on the left) and the first site (sphere labelled 1) as a perturbation on top of the singlet ground state formed by the impurity (red sphere) and the zeroth site. *Right*: At r_{c1} , the Hubbard sidebands have become isolated from the central peak in the local spectral function because of the finite optical gap. Removing the UV excitations that couple the low-energy and high-energy bands then leads to a theory of the Kondo model with a Lorentzian DOS.

the fact that the central peak of the impurity spectral function is a Lorentzian, and the bath becomes equivalent to the impurity site under self-consistency.

We implement the same approach on the e-SIAM, such that the Schrieffer-Wolff transformation leads to a $J - U_b$ model with a Lorentzian electronic DOS in the bath. The expression of the Kondo temperature for such a system has been derived in Sec. 6 of the Supplementary Materials [126]. Very close to the transition, we have $r \rightarrow 0.25^-$, and the Kondo temperature is given by

$$T_K = \frac{D_0}{k_B} \exp \left[-\frac{\ln(r_{c2} - r)}{4U_b \rho_0} \right], \quad (7.3)$$

where $r_{c2} = 0.25$ and D_0 is the bare bandwidth. Note that the pre-factor of the logarithm is positive as U_b is negative: $-4U_b \rho_0 = |4U_b \rho_0|$. As we approach the transition, the parameter r takes the limit $r \rightarrow r_{c2}^-$, and the Kondo temperature scale vanishes:

$$\lim_{r \rightarrow r_{c2}^-} T_K = \frac{D_0}{k_B} \lim_{r \rightarrow r_{c2}^-} (r_{c2} - r)^{|4U_b \rho_0|} \rightarrow 0. \quad (7.4)$$

Following the renormalised perturbation theory approach of Hewson [149, 150], the imaginary part of the self-energy of the local Fermi liquid quasiparticles is given by

$$\text{Im} [\Sigma(\omega)] \sim \frac{\mathcal{F}^2 \omega^2}{D_0 T_K^2}, \quad (7.5)$$

while the quasiparticle residue is given by $Z \sim T_K$. The divergence of $\text{Im} [\Sigma(\omega)]$ at the transition and the vanishing of T_K and the quasiparticle residue are important indicators of the loss of the local Fermi liquid excitations and the breakdown of Kondo screening.

The vanishing of the Kondo temperature scale also leads to the divergence of thermodynamic quantities such as the impurity contribution to local spin susceptibility χ_{imp} and the specific heat coefficient γ_{imp} . As the low-energy theory is a local Fermi liquid, χ_{imp} and γ_{imp} retain their Fermi liquid forms but involve the highly renormalised Kondo temperature scale [19, 151]:

$$\begin{aligned} \chi_{\text{imp}} &= \frac{w}{4k_B T_K}, \quad \lim_{r \rightarrow r_{c2}^-} \chi_{\text{imp}} = \frac{w}{4k_B} (r_{c2} - r)^{-4|U_b \rho_0|}, \\ \gamma_{\text{imp}} &= \frac{\pi^2 k_B^2 w}{6T_K}, \quad \lim_{r \rightarrow r_{c2}^-} \gamma_{\text{imp}} = \frac{\pi^2 k_B^2 w}{6} (r_{c2} - r)^{-4|U_b \rho_0|}, \end{aligned} \quad (7.6)$$

where $w \sim 0.4128$ is the Wilson number for the Kondo model [19, 106]. These results also show that the ratio of χ_{imp} and γ_{imp} , referred to as the Wilson ratio $R \equiv \frac{4\pi^2 k_B^2 \chi_{\text{imp}}}{3\gamma_{\text{imp}}}$, remains pinned at the single-channel Kondo value of $R_{\text{LFL}} = 2$ for $r \rightarrow r_{c2}^-$. This shows that the low-frequency Landau quasiparticles are able to survive until very close to the QCP. As we will show in the next subsection, the metallic excitations precisely at the QCP acquire non-Fermi liquid character and involve additional correlations between holons and doublons. These additional correlations are expected to lead to an enhancement of the Wilson ratio. Evidence for this can be found in the DMFT calculation of a local Wilson ratio very close to the Brinkman-Rice transition, $R_{\text{loc}} = 2.9 \pm 0.2$ [70].

2.7.2 Emergence of non-Fermi liquid excitations at the transition

In order to obtain an effective Hamiltonian for the excitations precisely at the critical point $r = 1/4$, we first note that the ground-state subspace of the zero bandwidth $J - U_b$ effective model (eq. (5.3)) becomes degenerate at this point:

$$\frac{1}{\sqrt{2}} (|\uparrow\rangle_d |\downarrow\rangle_0 - |\downarrow\rangle_d |\uparrow\rangle_0), \quad |\sigma\rangle_d |0\rangle_0, \quad |\sigma\rangle_d |2\rangle_0, \quad (\sigma = \uparrow, \downarrow), \quad (7.7)$$

with all the states lying at zero energy. Here, the first ket (with subscript d) represents the impurity spin configuration while the second ket (with subscript 0) represents the configuration of the bath zeroth site). We now diagonalise these states in the presence of the electron hopping t into the rest of the conduction bath, and obtain the effective Hamiltonian for the low-lying excitations mediated by t . This is described in detail in Sec. 7 of the Supplementary Materials [126]. We note that the ground state is four-fold degenerate at energy $-t$ in the presence of the hopping, and comprises the following states:

$$|N_{\text{tot}} = 2, S_{\text{tot}}^z = 0\rangle, \quad |N_{\text{tot}} = 4, S_{\text{tot}}^z = 0\rangle, \quad |N_{\text{tot}} = 3, S_{\text{tot}}^z = \frac{1}{2}\rangle, \quad |N_{\text{tot}} = 3, S_{\text{tot}}^z = -\frac{1}{2}\rangle, \quad (7.8)$$

where $N_{\text{tot}} = \hat{n}_d + \hat{n}_0 + \hat{n}_1$ and $S_{\text{tot}}^z = S_d^z + S_0^z + S_1^z$ are the total number operator and total magnetisation operator respectively for the impurity, bath zeroth and first sites taken together. It is worth observing that

- the presence of the unscreened states $|N_{\text{tot}} = 3, S_{\text{tot}}^z = \pm \frac{1}{2}\rangle$ in the ground-state subspace is a result of the inexact screening at the critical point, and
- the presence of a degenerate ground-state manifold indicates that these excitations of the bath will likely be of the non-Fermi liquid (NFL) kind [87, 152, 153].

The precise form of the effective Hamiltonian for this non-Fermi liquid, as well as the behaviour of various correlation functions, is obtained by considering the ground-state manifold in conjunction with the excited states at energy $-t$.

Since the total Hamiltonian conserves the total spin S_{tot}^z , the effective Hamiltonian separates into two sectors, $S_{\text{tot}}^z = 0$ and $|S_{\text{tot}}^z| = \frac{1}{2}$. Detailed calculations on obtaining this effective Hamiltonian from the RG fixed point theory are shown in Sec. 7 of the Supplementary Materials [126]. In order to highlight certain distinct features, we present simplified effective Hamiltonians only for the $S_{\text{tot}}^z = 0$ and $S_{\text{tot}}^z = \frac{1}{2}$ sectors. We first take a look at the $S_{\text{tot}}^z = 0$ sector:

$$H_{\text{eff}}^{S_{\text{tot}}^z=0} = t \vec{S}_d \cdot (\vec{S}_{1,-} + \vec{S}_{3,-}) . \quad (7.9)$$

The effective spin-1/2 ladder operators $\vec{S}_{1,\pm}$ act on the positive and negative parity sectors (labelled by $\pm$) of the $\hat{n}_0 + \hat{n}_1 = 1$ subspace spanned by the states

$$|\uparrow\uparrow\rangle_{1,\pm} = |\uparrow\rangle_0 |0\rangle_1 \pm |0\rangle_0 |\uparrow\rangle_1, \quad |\downarrow\downarrow\rangle_{1,\pm} = |\downarrow\rangle_0 |0\rangle_1 \pm |0\rangle_0 |\downarrow\rangle_1, \quad (7.10)$$

leading to a Pauli matrix representation of $\vec{S}_{1,\pm}$ in that basis. The Pauli matrix operators $\vec{S}_{3,\pm}$ similarly act on the $\hat{n}_0 + \hat{n}_1 = 3$ subspace spanned by the states $|\uparrow\uparrow\rangle_{3,\pm} = |\uparrow\rangle_0 |2\rangle_1 \pm |2\rangle_0 |\uparrow\rangle_1$ and $|\downarrow\downarrow\rangle_{3,\pm} = |\downarrow\rangle_0 |2\rangle_1 \pm |2\rangle_0 |\downarrow\rangle_1$, obtained by applying the transformation $|0\rangle \rightarrow 2$ on the $\hat{n}_0 + \hat{n}_1 = 1$ counterparts. We now point out two interesting features of this simplified effective Hamiltonian in eq. (7.9).

- The fact that the impurity is now interacting with emergent spins $\vec{S}_1$ and $\vec{S}_3$ that span over two lattice sites (0 and 1) instead of just the zeroth site indicates that the entanglement between the impurity and bath is now extended beyond the bath zeroth site, and that *the singlet is being stretched*.
- The presence of two spins that are trying to simultaneously screen the impurity spin introduces frustration in the dynamics of the impurity. This is reminiscent of the 2-channel Kondo effect, and the connection is made more precise in the concluding section of our work.

Both these features are precursors to the ultimate and complete destruction of screening in the local moment regime, and reflect the fact that the impurity-bath system *is now in an over-screened state*.

We now focus on the effective Hamiltonian of the $S_{\text{tot}}^z \neq 0$ subspace:

$$H_{\text{eff}}^{S_{\text{tot}}^z \neq 0} = -\frac{1}{2}t \left(S_d^- \mathcal{B}_\uparrow^+ + \text{h.c.} \right) . \quad (7.11)$$

The Pauli matrix operators $\mathcal{B}_\uparrow^+$ and $\mathcal{B}_\uparrow^-$ flip between the doublet of states $|\uparrow\rangle_0 |\uparrow\rangle_1$ and $\frac{1}{\sqrt{2}} (|2\rangle_0 |0\rangle_1 + |0\rangle_0 |2\rangle_1)$:

$$\begin{aligned}\mathcal{B}_\uparrow^- |\uparrow\rangle_0 |\uparrow\rangle_1 &= \frac{1}{\sqrt{2}} (|2\rangle_0 |0\rangle_1 + |0\rangle_0 |2\rangle_1), \\ \mathcal{B}_\uparrow^+ \frac{1}{\sqrt{2}} (|2\rangle_0 |0\rangle_1 + |0\rangle_0 |2\rangle_1) &= |\uparrow\rangle_0 |\uparrow\rangle_1.\end{aligned}\tag{7.12}$$

The structure of the effective Hamiltonian in eq. (7.11) leads to another drastic difference from the behaviour of the local Fermi liquid: it allows for the electrons to be “Andreev scattered” into an orthogonal state by hopping from the first site into the zeroth site. This is easily seen by considering the scattering processes in eq. (7.12): a state $|\uparrow\rangle_0 |\uparrow\rangle_1$ (i.e., with site 1 in $\uparrow$ configuration) has a finite probability of being flipped into the $|\downarrow\rangle_0 |2\rangle_1$ state (i.e., with site 1 in the doublon configuration). *An incident electron $c_{1\uparrow}^\dagger$ can therefore emerge as a doublon $c_{1\uparrow}^\dagger c_{1\downarrow}^\dagger$ or a hole $c_{1\uparrow}$ upon scattering from the zeroth site.* This is a direct consequence of the entanglement between the singlet and the first site at the critical point that was absent in the metallic regime of the e-SIAM. We note that similar orthogonal scattering processes also occur in a two-channel Kondo problem between effective pseudo-particles and pseudo-holes [154].

At this point, it is worth noting that the polarised ground-state $|S_{\text{tot}}^z = \frac{1}{2}\rangle$ can be written as an equal superposition of the singlet state $|\text{SS}\rangle_{d0} \otimes |\uparrow\rangle_1$ and the local moment states $\frac{1}{\sqrt{2}} (|\uparrow, 0, 2\rangle - |\uparrow, 2, 0\rangle)$.

This symmetry-broken ground state, along with its counterpart $|S_{\text{tot}}^z = -\frac{1}{2}\rangle$, act as a bridge between the ground states of the metallic and insulating phases. They are thus important in displaying the breakdown of the Kondo cloud at the QCP, and the consequences arising from it. We will hence consider these states in the next subsection and demonstrate some exotic non-Fermi liquid properties of these states, e.g., inexact screening of the impurity, fractional impurity magnetisation and fractional impurity entanglement entropy.

Finally, by mapping the quantum impurity problem of the e-SIAM onto that of a classical Coulomb gas [87, 92], it can be shown that the local self-energies $\Sigma_{dd}(\omega)$ and $\Sigma_{00}(\omega)$ of the impurity and zeroth sites respectively and certain two-particle correlation functions have power-law behaviours in the frequency domain:

$$\text{Re} [\Sigma_{dd}] (\omega) \sim |\omega|^{\gamma_{dd}}, \quad \text{Re} [\Sigma_{00}] (\omega) \sim |\omega|^{\gamma_{00}},\tag{7.13}$$

$$\langle S_d^+ \rangle \sim |\omega|^{(\alpha_1-1)/2}, \quad \langle c_{0\uparrow}^\dagger c_{0\downarrow}^\dagger \rangle \sim |\omega|^{(\alpha_3-1)/2}.\tag{7.14}$$

The precise forms of the exponents in terms of the conduction electron scattering phase shifts, as well as several other technical details of the calculation, are provided in Sec. 8 of the Supplementary Materials [126]. While the various exponents are found to be non-universal functions of the fixed point value of J and the coupling U_b for $r \rightarrow r_{c2}-$, we argue in subsection (2.7.4) that they assume universal values precisely at the QCP ($r = r_{c2}$). The algebraic behaviour of these local correlations is reminiscent of critical behaviour ascribed to the class of local quantum criticality [155, 156].

2.7.3 Impurity magnetisation and entanglement entropy

More indications of non-Fermi liquid behaviour at the QCP is obtained from a calculation of the impurity magnetisation and the impurity entanglement entropy ($S_{\text{EE}}(d)$) (shown in Sec. 9 of the

Supplementary Materials [126]) for the symmetry-broken states $|S_{\text{tot}}^z = \sigma/2\rangle$. This leads to a fractional entanglement entropy (in units of $\log 2$) for each of the two states:

$$\begin{aligned} \rho_{\text{imp}} &= \begin{pmatrix} \frac{1}{2} + m_{\text{imp}}^z & 0 \\ 0 & \frac{1}{2} - m_{\text{imp}}^z \end{pmatrix} = \begin{pmatrix} \frac{3}{4} & 0 \\ 0 & \frac{1}{4} \end{pmatrix}, \\ S_{\text{EE}}(d) &= -\text{Tr}(\rho_{\text{imp}} \ln \rho_{\text{imp}}) \simeq 0.81 \log 2, \end{aligned} \quad (7.15)$$

where the impurity magnetisation m_{imp}^z takes the value $m_{\text{imp}}^z = 1/4$ at the QCP (i.e., half the value for a local moment). Further, $S_{\text{EE}}(d)$ can be written in terms of an effective impurity degeneracy g_{imp} , which we define using the impurity magnetisation: $g_{\text{imp}} \equiv 1 + 2|m_{\text{imp}}^z|$, such that it takes the expected values of 1 and 2 in the local Fermi liquid and local moment phases (with m_{imp}^z having values 0 and $\frac{1}{2}$ respectively). The effective degeneracy in the polarised subspace $S_{\text{tot}}^z = \pm 1/2$ at the QCP then turns out to be $3/2$, which is half-way between a unique state and a doublet. This corresponds to partial screening of the impurity degrees of freedom at the QCP, in contrast to complete screening in the metallic phase ($g_{\text{imp}} = 1$) and the absence of screening in the insulating phase ($g_{\text{imp}} = 2$).

In Sec. 9 of the Supplementary Materials [126], we show that the incomplete magnetisation (and hence the fractional value of S_{EE}) arises from the mixing of the local Fermi liquid ground-state $|\phi\rangle = \otimes_{k < k_F} |k \uparrow\rangle |k \downarrow\rangle$ and the gapless excitations above it, $|e\rangle_\sigma = e_\sigma^\dagger |\phi\rangle$, $e \equiv \sum_{k \in FS} c_{k\sigma}^\dagger$, leading to the decay of the local Fermi liquid quasiparticles [152]. This is captured by the modified ground-state $|- \rangle$ and lowest-lying excited states $|+ \rangle$ at the QCP:

$$|\pm\rangle = \frac{1}{\sqrt{2}} (|SS\rangle \otimes |e\rangle_\sigma \pm |LM\rangle_\sigma \otimes |\phi\rangle). \quad (7.16)$$

The spin-singlet state $|SS\rangle \sim \sum_\sigma \sigma |\sigma\rangle_d |\bar{\sigma}\rangle_0$ and the local moment state $|LM\rangle_\sigma = |\sigma\rangle_d |2\rangle_0$ represent the configurations of the impurity and the bath zeroth sites. Such a ground-state $|- \rangle$ should be contrasted with the local Fermi liquid ground-state $|SS\rangle \otimes |\phi\rangle$ that is stable in the metallic phase for $r_{c1} < r < r_{c2}$. This change in the ground-state manifests in the vanishing of the quasiparticle residue of the local Fermi liquid excitations: $Z = |\langle + | e_\sigma^\dagger | - \rangle|^2 = |\langle + | (|LM\rangle_\sigma \otimes |e\rangle_\sigma) |^2 = 0$, consistent with the orthogonality catastrophe [157] described below eq. (7.5). As the local Fermi liquid quasiparticles are rendered unstable, they are replaced by stable composite three-site excitations of the form, $S_d^- c_{0\bar{\sigma}} e_\sigma^\dagger$. This is further reflected in the fact that the corresponding residue for such three-site composite excitation is non-zero at the QCP: $|\langle + | S_d^- c_{0\bar{\sigma}} e_\sigma^\dagger | - \rangle|^2 > 0$.

2.7.4 Friedel scattering phase shift and fractional excess charge

The orthogonality catastrophe faced by the infrared excitations of the conduction bath at the QCP points towards the presence of an “unrenormalised” phase shift [158] of the scattering k -states. Indeed, using the Friedel sum rule [159–161], we obtain a $\pi/2$ phase shift for the extended states at the Fermi surface for the QCP, which is only half the unitary limit. This can be seen by recalling that the ground state in eq. (7.16) is an equal admixture of a decoupled local moment state and a singlet state. Through the singlet state, the impurity contributes an excess charge of unity to the Fermi surface of the conduction bath (because of the presence of gapless spin-flip fluctuations). The local

moment state does not contribute any excess charge, because the impurity is decoupled from the bath. The net excess charge then comes out to be $n_{\text{exc}} = \frac{1}{2} (1 + 0) = \frac{1}{2}$. From the Friedel sum rule, the phase shift at the fixed point is $\delta^* = \pi n_{\text{exc}} = \pi/2$, as mentioned above. Expectedly, n_{exc} is zero in the local moment phase. Also, as the exponents of the algebraic form for the local quantities shown in eq. (7.13) are functions of the scattering phase shift, we find that these exponents take universal values precisely at the QCP.

The values of the excess charge at the QCP and in the local moment phase lead to important corollaries. Firstly, the jump in the excess charge by unity in going from $r < r_{c2}$ to $r > r_{c2}$ is in fact a reduction in the Luttinger volume V_L of the conduction bath [13, 162]. V_L counts the number of extended states present at and below the Fermi volume, and corresponds to a topological quantum number [15, 16, 18, 163]. Since the excess charge n_{exc} calculated above is simply the impurity contribution to V_L at the Fermi surface, the difference in the excess charge across the transition implies that the two phases acquire different values for the invariant V_L . Thus, n_{exc} tracks the change (ΔV_L) in the Luttinger volume, and the transition is topological in nature. Secondly, in a recent work, Sen et al. [164] have shown that the Mott MIT of the infinite-dimensional Hubbard model proceeds through the dissociation of domain walls in a fictitious Su-Schrieffer-Heeger chain connected to the physical lattice sites. If one couples their observation with the fact that the two dissociated domain walls at the ends of the SSH chain are together known to host a single charge [165, 166], it becomes evident that the fractional excess charge we obtain at the QCP corresponds to the state that is localised in the SSH chain near the physical lattice site as obtained by Sen et al. The well-known bulk-boundary correspondence of the SSH model (see, e.g., [167]) suggests that the excess charge n_{exc} and the change ΔV_L in the Luttinger volume are the respective topological invariants of the boundary and bulk in the e-SIAM that are tied to one another. Further, the appearance of a half-quantised n_{exc} at the MIT of the e-SIAM corresponds, in the SSH model, to the dissociation of a domain wall-anti domain wall pair with each carrying a half charge [165, 166].

We show in Sec. 9 of the Supplementary Materials [126] that it is in fact possible to connect the scattering phase shift (δ^*), impurity magnetisation (m_{imp}^z), effective degeneracy ($\tilde{g}$) and excess charge (n_{exc}) in a single relation:

$$\delta^* = \pi n_{\text{exc}} = \left[1 + \left(\frac{2 - \tilde{g}}{\tilde{g} - 1} \right)^2 \right]^{-1} = \frac{4m_{\text{imp}}^z{}^2}{1 - 4|m_{\text{imp}}^z| + 8m_{\text{imp}}^z{}^2}. \quad (7.17)$$

As we now discuss, this provides a unified picture of the effect of frustration on the impurity spin. As we have seen earlier, the presence of U_b in the Hamiltonian of the e-SIAM introduces local moment states into the spectrum and leads to states with non-vanishing impurity magnetisation at the QCP. This non-zero magnetisation can be interpreted as a partial screening of the impurity spin, in turn resulting in an effective impurity degeneracy ($1 < \tilde{g} < 2$ between that of a unique screened state and an unscreened local moment. The partial screening also manifests in only half the excess charge being contributed to the conduction bath. This reduction in the excess charge acts as a change in the boundary conditions felt by the conduction electrons coupled to the impurity and manifests as a phase shift that is less than the unitarity limit of π .

2.8 Discussions and Outlook

In summary, we have shown that an attractive correlation on the bath zeroth site is enough to frustrate the Kondo effect and stabilise the local moment phase. The destruction of the Kondo cloud, and the associated local Fermi liquid, occurs through *pairing fluctuations in the bath and proximate to the impurity* (left panel of Fig. (2.8)). This is reminiscent of a subdominant superconducting tendency that was observed in the half-filled Hubbard model at $T = 0$ from a unitary RG treatment in Refs. [123, 168]. We also note the finding of non-local attractive effective interactions in a recent theoretical study of the 2D Hubbard model [169]. We find that the critical point displays non-Fermi liquid behaviour with a vanishing quasiparticle residue Z . The strong agreement of our results with several aspects of DMFT suggests that the local self-energy obtained self-consistently in that method can be represented quite faithfully through the e-SIAM. It will, thus, be important to test the predictions offered by our approach directly within the DMFT method. We note that the non-Fermi liquid at the critical point displays a *partial correlation of doublons and holons* that is distinct from not only the correlated Fermi liquid metal (unbound holons and doublons), but also the paramagnetic insulator (comprised of bound holons and doublons). The excitations that propagate through the two unscreened states in the ground-state subspace (Eq. (7.8)) involve the simultaneous creation of a doublon and a holon: in these channels, the doublons and holons cannot propagate in the absence of each other. Such an incomplete correlation is an intermediate step towards complete confinement in the local moment phase. It appears interesting to experimentally test some of these ideas on a mesoscopic quantum dot [170, 171] which is, apart from the usual electron tunnel coupling to an electronic reservoir, additionally coupled through the proximity effect to a superconducting lead.

We stress here that the e-SIAM analysed in this work represents an impurity model that (i) has a single correlated channel of conduction electrons, (ii) is consistent with the symmetries of a half-filled Hubbard model (SU(2)-spin, U(1)-charge and particle-hole), and (iii) shows a local metal-insulator transition. The model, therefore, provides a minimal route towards obtaining a Hamiltonian-based understanding of DMFT. Further, at the level of the renormalisation group flows that involve the frustration of Kondo screening, the effect of the attractive on-site interaction U_b introduced by us is equivalent to that of additional bath correlations that have been introduced in related impurity models. These include a single impurity Anderson model with multiple anisotropic conduction channels studied by Giamarchi et al. [172], and a periodic Anderson model enhanced with explicit s-d coupling (as well as a density-density interaction) between the conduction and impurity bands, studied by Si and Kotliar [87, 102, 173] as well as by Ruckenstein et al., [174]. In the former K -channel model of Ref. [172], the z -component (V_l , $l \in [1, K - 1]$) of the Kondo interaction term in the $K - 1$ additional conduction channels acts as a source of frustration for the spin-flip term γ_0 of the Kondo interaction in the original $l = 0$ channel, leading to a breakdown of Kondo screening. The U_b coupling in the e-SIAM acts as a similar source of frustration: the equivalence lies at the level of the renormalisation group equations, such that both V_l and U_b oppose the RG relevance of γ_0 and J respectively. This can be made more precise by comparing the RG equations of our work (eqs.(9.28)) and that of Giamarchi et al. [172]: the mapping between the two frustration terms U_b and V_l is found to be of the form $U_b \sim \sum_{l=1}^{K-1} V_l^2 \rho$, where ρ is the conduction bath DOS. A similar relationship can be found between the RG equations obtained by us and those for the extended periodic Anderson model studied by Si and Kotliar [87]. In this sense, our analysis applies

to a wide variety of models where the Kondo effect is stable in a certain regime of parameters but is destroyed in other parameter regimes by some form of quantum-mechanical frustration of the impurity spin degree of freedom.

Extensions of the present work involve analysing the mixed valence regime of the impurity site (i.e., away from half-filling) within the e-SIAM auxiliary model. This allows the possibility that the finite-temperature critical end-point of the DMFT first-order transition could turn into a quantum critical point from the merging of the $T = 0$ ESQPT and QPT observed in the present work. Recent DMFT calculations [175] show a shrinking of the coexistence region with hole doping but fall short of revealing a QCP. If such a QCP does exist, can it harbour pair fluctuations between the impurity and zeroth bath sites that become dominant upon doping, signalling thereby a putative superconducting instability of a related bulk model? We recall that a superconducting state of matter was indeed observed to be emergent from a QCP in the hole-doped Hubbard model at $T = 0$ via a unitary RG treatment in Refs. [122, 168]. The phenomenon of electronic differentiation in k -space can perhaps be captured by considering cluster variants of the e-SIAM, i.e., multiple impurities connected to one another through single-particle hopping and/or RKKY-like interactions [176, 177]. In general, the presence of multiple and varied classes of correlations in the auxiliary model - localisation from Mott physics, delocalisation from spin and charge fluctuations, and pairing from local attractive correlations - makes this a strong candidate for an auxiliary model that carries the potential for describing the emergence of a variety of novel phases of correlated quantum matter.

2.9 Appendix: Derivation of renormalisation group equations for the extended-SIAM

The Hamiltonian we are working with is the extended Anderson impurity model (described in Section III of the main manuscript):

$$\mathcal{H} = -\frac{1}{2}U (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2 + \sum_{\vec{k}, \sigma} \epsilon_{\vec{k}} \tau_{\vec{k}, \sigma} + J \vec{S}_d \cdot \vec{S}_0 + V \sum_{\sigma} \left(c_{d\sigma}^{\dagger} c_{0\sigma} + \text{h.c.} \right) - \frac{1}{2}U_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2. \quad (9.1)$$

The renormalisation in the Hamiltonian $H_{(j)}$ at the j^{th} RG step, upon decoupling an electronic state $q\beta$, is given by the expression:

$$(\Delta H_{(j)})_{\vec{q}, \beta} = H_{(j-1)} - H_{(j)} = c_{\vec{q}, \beta}^{\dagger} T_{\vec{q}, \beta} \frac{1}{\omega_e - H_D} T_{\vec{q}, \beta}^{\dagger} c_{\vec{q}, \beta} + T_{\vec{q}, \beta}^{\dagger} c_{\vec{q}, \beta} \frac{1}{\omega_h - H_D} c_{\vec{q}, \beta}^{\dagger} T_{\vec{q}, \beta}, \quad (9.2)$$

where $\omega_{e,h}$ are the quantum fluctuation scales for the electron and hole scattering channels, H_D is the part of the Hamiltonian that is diagonal in k -space, and $T_{\vec{q}, \beta} + T_{\vec{q}, \beta}^{\dagger}$ is the part of the Hamiltonian that does not commute with $\hat{n}_{\vec{q}, \beta}$ (in short, it is the part that is off-diagonal with respect to a particular state $\vec{q}, \beta$). This off-diagonal part is made up of contributions from V, J and U_b : $T_{\vec{q}\uparrow} = \left[V c_{d\uparrow}^{\dagger} + J \sum_{\vec{k}} S_d^+ c_{\vec{k}\downarrow}^{\dagger} + U_b \sum_{\vec{k}_1, \vec{k}_2, \vec{k}_3} c_{\vec{k}_1\uparrow}^{\dagger} c_{\vec{k}_3\downarrow}^{\dagger} c_{\vec{k}_4\downarrow} \right] c_{\vec{q}\uparrow}$, while the diagonal part is given by $H_D = \epsilon_{\vec{q}} \tau_{q\uparrow} + \frac{1}{4} J S_d^z (\hat{n}_{q\uparrow} - \hat{n}_{q\downarrow}) - \frac{1}{2} U_b (\tau_{q\uparrow} - \tau_{q\downarrow})^2$. The contribution from U_b is obtained by taking the

Hartree contributions corresponding to the states $|q\sigma\rangle$ from the full interacting term in the Hamiltonian. Note that we have ignored potential scattering terms in the off-diagonal part $T_{q\beta}$. In order to allow both spin-flip and charge transfer scattering processes, we start from initial states with $\tau_{q\uparrow} = -\tau_{q\downarrow}$. As a result, the contribution to H_D from the kinetic energy is $\epsilon_q \tau_{q\uparrow} = (\pm D) \times \left(\pm \frac{1}{2}\right) = D/2$, while the contribution from U_b is $-\frac{1}{2}U_b \left(\pm \frac{1}{2} - \left(\mp \frac{1}{2}\right)\right)^2 = -U_b/2$.

2.9.1 Renormalisation of U_b

U_b can renormalise only via itself. The relevant renormalisation term in the particle sector is

$$U_b^2 \sum_{q\beta} \sum_{\substack{k_1, k_2, k_3, \\ k'_1, k'_2, k'_3}} c_{q\beta}^\dagger c_{k_1\beta} c_{k_3\beta}^\dagger c_{k'_1\bar{\beta}} \frac{1}{\omega - H_D} c_{k'_2\bar{\beta}}^\dagger c_{k'_3\bar{\beta}} c_{k_2\beta} c_{q\beta} . \quad (9.3)$$

In order to renormalise U_b , we need to contract one more pair of momenta. There are two choices. The first is by setting $k_3 = k'_3 = q$. The two internal states, then, are $q\beta$ and $q\bar{\beta}$. As discussed above, the intermediate state energy is $-U_b/4$. We therefore have

$$\frac{U_b^2 n_j}{\omega - D/2 + \frac{U_b}{2}} \sum_{\beta} \sum_{k_1, k_2, k'_1, k'_2} c_{k_1\beta} c_{k'_1\bar{\beta}} c_{k'_2\bar{\beta}}^\dagger c_{k_2\beta}^\dagger = \frac{U_b^2 n_j}{\omega - D/2 + \frac{U_b}{2}} \sum_{\beta} \sum_{k_1, k_2, k'_1, k'_2} c_{k'_2\bar{\beta}}^\dagger c_{k'_1\bar{\beta}} c_{k_2\beta} c_{k_1\beta} \quad (9.4)$$

Another way to contract the momenta is by setting $k'_1 = k'_2 = q$, which gives a renormalisation of

$$\frac{U_b^2 n_j}{\omega - D/2 + \frac{U_b}{2}} \sum_{\beta} \sum_{k_1, k_2, k_3, k'_3} c_{k_1\beta} c_{k_3\bar{\beta}}^\dagger c_{k'_3\bar{\beta}} c_{k_2\beta}^\dagger = -\frac{U_b^2 n_j}{\omega - D/2} \sum_{\beta} \sum_{k_1, k_2, k_3, k'_3} c_{k_3\bar{\beta}}^\dagger c_{k'_3\bar{\beta}} c_{k_2\beta} c_{k_1\beta} . \quad (9.5)$$

The two contributions cancel each other. The same cancellation happens in the hole sector as well.

2.9.2 Renormalisation of the impurity correlation U

The coupling U is renormalised by two kinds of vertices: V^2 and J^2 . We will consider these processes one at a time. For convenience, we define $\epsilon_d = -U/2$.

The renormalisation arising from the first kind of terms, in the particle sector, is

$$\begin{aligned} \sum_{q\beta} c_{q\beta}^\dagger c_{d\beta} \frac{V^2}{\omega - H_D} c_{d\beta}^\dagger c_{q\beta} &= \sum_{q\beta} V^2 \hat{n}_{q\beta} (1 - \hat{n}_{d\beta}) \left(\frac{1 - \hat{n}_{d\bar{\beta}}}{\omega_0 - E_0} + \frac{\hat{n}_{d\bar{\beta}}}{\omega_1 - E_1} \right) \\ &= V^2 n_j \sum_{\beta} (1 - \hat{n}_{d\beta}) \left(\frac{1 - \hat{n}_{d\bar{\beta}}}{\omega_0 - E_0} + \frac{\hat{n}_{d\bar{\beta}}}{\omega_1 - E_1} \right) . \end{aligned} \quad (9.6)$$

q runs over the momentum states that are being decoupled at this RG step: $|q| = \Lambda_j$. $E_{1,0}$ are the diagonal parts of the Hamiltonian at $\hat{n}_{d\bar{\beta}} = 1, 0$ respectively: $E_1 = \frac{D}{2}$ and $E_0 = \frac{D}{2} + \epsilon_d - \frac{J}{4}$. In order to relate ω_0 and ω_1 with the common fluctuation scale ω for the conduction electrons, we

will replace these quantum fluctuation scales with the current renormalised values of the single-particle self-energy for the initial state from which we started scattering. For $\hat{n}_{d\bar{\beta}} = 0$, there is no additional self-energy because the impurity does not have any spin: $\omega_0 = \omega$. For $\hat{n}_{d\bar{\beta}} = 1$, we have an additional self-energy of ϵ_d arising from the correlation on the impurity: $\omega_1 = \omega + \epsilon_d$. Substituting the values of $E_{0,1}$ and $\omega_{0,1}$, we get

$$V^2 n_j \sum_{\beta} (1 - \hat{n}_{d\beta}) \left(\frac{1 - \hat{n}_{d\bar{\beta}}}{\omega - \frac{D}{2} - \epsilon_d + \frac{J}{4}} + \frac{\hat{n}_{d\bar{\beta}}}{\omega - \frac{D}{2} + \epsilon_d} \right). \quad (9.7)$$

Performing a similar calculation for the hole sector gives the contribution:

$$V^2 n_j \sum_{\beta} \hat{n}_{d\beta} \left(\frac{1 - \hat{n}_{d\bar{\beta}}}{\omega - \frac{D}{2} + \epsilon_d} + \frac{\hat{n}_{d\bar{\beta}}}{\omega - \frac{D}{2} - \epsilon_d + \frac{J}{4}} \right). \quad (9.8)$$

We now come to the second class of terms: spin-spin. We first look at the particle sector:

$$\frac{J^2}{4} \sum_{q\beta} c_{d\bar{\beta}}^{\dagger} c_{d\beta} c_{q\beta}^{\dagger} c_{-q\bar{\beta}} \frac{1}{\omega - H_D} c_{d\beta}^{\dagger} c_{d\bar{\beta}} c_{q\beta}^{\dagger} c_{q\beta} = \frac{J^2}{4} n_j \frac{1}{\omega - \frac{D}{2} + \frac{J}{4}} \sum_{\beta} \hat{n}_{d\bar{\beta}} (1 - \hat{n}_{d\beta}). \quad (9.9)$$

The diagonal part in the denominator was simple to deduce in this case because the nature of the scattering requires the spins S_d^z and $\frac{\beta}{2} (\hat{n}_{q\beta} - \hat{n}_{q\bar{\beta}})$ to be anti-parallel. This ensures that the intermediate state has an energy of $E = \frac{D}{2} + \epsilon_d - \frac{J}{4}$, and the quantum fluctuation scale is $\omega' = \omega + \epsilon_d$, such that $\omega' - E = \omega - \frac{D}{2} + \frac{J}{4}$. In the hole sector, we have

$$\frac{J^2}{4} n_j \frac{1}{\omega - \frac{D}{2} + \frac{J}{4}} \sum_{\beta} \hat{n}_{d\beta} (1 - \hat{n}_{d\bar{\beta}}). \quad (9.10)$$

From eqs. (9.7), (9.8), (9.9) and (9.10), we write

$$\begin{aligned} \Delta U &= \Delta\epsilon_2 + \Delta\epsilon_0 - 2\Delta\epsilon_1 \\ &= -\frac{4V^2 n_j}{\omega + \frac{U_b}{2} - \frac{D}{2} - U/2} + \frac{4V^2 n_j}{\omega + \frac{U_b}{2} - \frac{D}{2} + U/2 + \frac{J}{4}} - \frac{J^2 n_j}{\omega + \frac{U_b}{2} - \frac{D}{2} + \frac{J}{4}}, \end{aligned} \quad (9.11)$$

where we have restored the contribution from U_b in the denominator.

2.9.3 Renormalisation of the hybridisation V

Renormalisation of V happens through two kinds of processes: VJ and VU_b . Within the first kind, the scattering can be either via S_d^z or through $S_d^{\pm}$. For the first kind, we have the following contribution in the particle sector:

$$\begin{aligned} &\sum_{q\beta} V c_{q\beta}^{\dagger} c_{d\beta} \frac{1}{\omega - H_D} \frac{1}{4} J \sum_k (\hat{n}_{d\beta} - \hat{n}_{d\bar{\beta}}) c_{k\beta}^{\dagger} c_{q\beta} \\ &= \frac{1}{4} V J n_j \frac{1}{2} \left(\frac{1}{\omega'_1 - E} + \frac{1}{\omega'_2 - E} \right) \sum_{k\beta} (1 - \hat{n}_{d\bar{\beta}}) c_{d\beta} c_{k\beta}^{\dagger}. \end{aligned} \quad (9.12)$$

The transformation from $\frac{1}{\omega - H_D}$ to $\frac{1}{2} \left(\frac{1}{\omega'_1 - E} + \frac{1}{\omega'_2 - E} \right)$ is made so that we can account for both the initial state and the final state energies through the two fluctuation scales ω'_1 and ω'_2 respectively; we calculate the denominators for both the initial and final states and then take the mean of the two (hence the factor of half in front). This was not required previously because, in the earlier scattering processes, the impurity returned to its initial state at the end, at least in terms of $\epsilon_d (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2$, and so we had $\omega'_1 = \omega'_2 = \omega'$. Substituting the energies and the ω -scales, we get

$$- \left(\frac{\frac{n_j}{4} V J \frac{1}{2}}{\omega - \frac{D}{2} + \frac{J}{4}} + \frac{\frac{n_j}{4} V J \frac{1}{2}}{\omega - \frac{D}{2} - \epsilon_d + \frac{J}{4}} \right) \sum_{k\beta} \left(1 - \hat{n}_{d\bar{\beta}} \right) c_{k\beta}^\dagger c_{d\beta} . \quad (9.13)$$

One can generate another such process by exchanging the single-particle process and the spin-exchange process:

$$\sum_{q\beta} \frac{1}{4} J \sum_k \left(\hat{n}_{d\beta} - \hat{n}_{d\bar{\beta}} \right) c_{q\beta}^\dagger c_{k\beta} \frac{1}{\omega - H_D} V c_{d\beta}^\dagger c_{q\beta} . \quad (9.14)$$

This is simply the Hermitian conjugate of the previous contribution. Combining this with the previous then gives

$$- \frac{n_j}{8} V J \left(\frac{1}{\omega - \frac{D}{2} + \frac{J}{4}} + \frac{1}{\omega - \frac{D}{2} - \epsilon_d + \frac{J}{4}} \right) \sum_{k\beta} \left(1 - \hat{n}_{d\bar{\beta}} \right) \times \left(c_{d\beta}^\dagger c_{k\beta} + \text{h.c.} \right) . \quad (9.15)$$

If we similarly calculate the contributions from the spin-exchange processes involving $S_d^\pm$, we get

$$- \frac{1}{4} V J n_j \left(\frac{1}{\omega - \frac{D}{2} + \frac{J}{4}} + \frac{1}{\omega - \frac{D}{2} - \epsilon_d + \frac{J}{4}} \right) \sum_{k\beta} \left(1 - \hat{n}_{d\beta} \right) \left(c_{k\bar{\beta}}^\dagger c_{d\bar{\beta}} + \text{h.c.} \right) . \quad (9.16)$$

The contributions from the hole sector are obtained by making the transformation $\hat{n}_{d\bar{\beta}} \rightarrow 1 - \hat{n}_{d\bar{\beta}}$ on the particle sector contributions. The total renormalisation to V from VJ processes are

$$- \frac{3n_j}{8} V J \left(\frac{1}{\omega + U_b/4 - \frac{D}{2} + \frac{J}{4}} + \frac{1}{\omega + U_b/4 - \frac{D}{2} + U/2 + \frac{J}{4}} \right) \sum_{k\beta} \left(c_{d\beta}^\dagger c_{k\beta} + \text{h.c.} \right) . \quad (9.17)$$

The renormalisation in V from U_b arises through terms of VU_b and U_bV kind. The first term gives

$$- \sum_{k\beta} c_{d\beta}^\dagger c_{k\beta} \left[\frac{\hat{n}_{d\bar{\beta}}}{2} \left(\frac{n_j U_b V}{\omega - \frac{D}{2} - \frac{U}{2} + \frac{U_b}{4}} + \frac{n_j U_b V}{\omega - \frac{D}{2} + \frac{U_b}{4}} \right) + \frac{1 - \hat{n}_{d\bar{\beta}}}{2} \left(\frac{n_j U_b V}{\omega - \frac{D}{2} + \frac{U_b}{4} + \frac{U}{2} + \frac{J}{4}} + \frac{n_j U_b V}{\omega - \frac{D}{2} + \frac{U_b}{4}} \right) \right] . \quad (9.18)$$

The second term is of the form $\sum_{q\beta} \sum_k U_b V c_{q\beta}^\dagger c_{d\beta} \frac{1}{\omega - H_D} \hat{n}_{q\bar{\beta}} c_{k\beta}^\dagger c_{q\beta}$, and this is just the Hermitian conjugate of the previous term, so these two terms together lead to

$$- n_j U_b V \sum_{k\beta} \left(c_{d\beta}^\dagger c_{k\beta} + \text{h.c.} \right) \left[\frac{\hat{n}_{d\bar{\beta}}}{2} \left(\frac{1}{\omega - \frac{D}{2} - \frac{U}{2} + \frac{U_b}{4}} + \frac{1}{\omega - \frac{D}{2} + \frac{U_b}{4}} \right) + \frac{1 - \hat{n}_{d\bar{\beta}}}{2} \left(\frac{1}{\omega - \frac{D}{2} + \frac{U_b}{4} + \frac{U}{2} + \frac{J}{4}} + \frac{1}{\omega - \frac{D}{2} + \frac{U_b}{4}} \right) \right] . \quad (9.19)$$

In the hole sector, we have

$$-n_j U_b V \sum_{k\beta} c_{k\beta}^\dagger \left[\frac{\hat{n}_{d\bar{\beta}}}{2} \left(\frac{1}{\omega_1 - E_1} + \frac{1}{\omega'_1 - E_1} \right) + \frac{1 - \hat{n}_{d\bar{\beta}}}{2} \left(\frac{1}{\omega_0 - E_0} + \frac{1}{\omega'_0 - E_0} \right) \right] c_{d\beta} . \quad (9.20)$$

$E_1 = D/2 - U_b/4 - U/2 - J/4$, $E_0 = D/2 - U_b/4 - K/4$. The fluctuation scales are $\omega_1 = \omega = \omega'_0$, $\omega'_1 = \omega - U/2 = \omega_0$. Substituting these gives

$$- \sum_{k\beta} c_{d\beta}^\dagger c_{k\beta} \left[\frac{1 - \hat{n}_{d\bar{\beta}}}{2} \left(\frac{n_j U_b V}{\omega - \frac{D}{2} - \frac{U}{2} + \frac{U_b}{4}} + \frac{n_j U_b V}{\omega - \frac{D}{2} + \frac{U_b}{4}} \right) + \frac{\hat{n}_{d\bar{\beta}}}{2} \left(\frac{n_j U_b V}{\omega - \frac{D}{2} + \frac{U_b}{4} + \frac{U}{2} + \frac{J}{4}} + \frac{n_j U_b V}{\omega - \frac{D}{2} + \frac{U_b}{4} + \frac{J}{4}} \right) \right] . \quad (9.21)$$

The other term, obtained by exchanging V and U_b , gives the Hermitian conjugate, so the overall contribution from the hole sector is the same as the total contribution from the particle sector, but with $\hat{n}_{d\bar{\beta}} \rightarrow 1 - \hat{n}_{d\bar{\beta}}$. Combining both the sectors, we get

$$- \sum_{k\beta} \left(c_{d\beta}^\dagger c_{k\beta} + \text{h.c.} \right) \left[\left(\frac{n_j U_b V/2}{\omega - \frac{D}{2} - \frac{U}{2} + \frac{U_b}{4}} + \frac{n_j U_b V/2}{\omega - \frac{D}{2} + \frac{U_b}{4}} \right) + \left(\frac{n_j U_b V/2}{\omega - \frac{D}{2} + \frac{U_b}{4} + \frac{U}{2} + \frac{J}{4}} + \frac{n_j U_b V/2}{\omega - \frac{D}{2} + \frac{U_b}{4} + \frac{J}{4}} \right) \right] . \quad (9.22)$$

2.9.4 Renormalisation of the spin-exchange coupling J

The term $J \vec{S}_d \cdot \vec{S}_0$ can be split into three parts: $J^z S_d^z$, $\frac{1}{2} J^+ S_d^+ S_0^-$ and $\frac{1}{2} J^- S_d^- S_0^+$. We will only calculate the renormalisation in J^+ , which will be equal to that of J^- , J^z due to spin-rotation symmetry. The terms that renormalise J^+ are of the form $S_d^+ S_d^z$ and $S_d^z S_d^+$. In the particle sector, we have

$$- \sum_q \sum_{kk'} \frac{1}{4} J^2 S_d^+ c_{q\downarrow}^\dagger c_{k'\uparrow} \frac{1}{\omega - H_D} S_d^z c_{k\downarrow}^\dagger c_{q\downarrow} = n_j \frac{1}{4} J^2 \left(-\frac{1}{2} S_d^+ \right) \sum_{kk'} c_{k\downarrow}^\dagger c_{k'\uparrow} \frac{1}{\omega + \frac{U_b}{2} - \frac{D}{2} + \frac{J}{4}} ,$$

$$\sum_q \sum_{kk'} \frac{1}{4} J^2 S_d^z c_{q\uparrow}^\dagger c_{k'\uparrow} \frac{1}{\omega - H_D} S_d^+ c_{k\downarrow}^\dagger c_{q\uparrow} = -n_j \frac{1}{4} J^2 \left(\frac{1}{2} S_d^+ \right) \sum_{kk'} c_{k\downarrow}^\dagger c_{k'\uparrow} \frac{1}{\omega + \frac{U_b}{2} - \frac{D}{2} + \frac{J}{4}} . \quad (9.23)$$

The denominator is determined using $E = \frac{D}{2} + \epsilon_d - \frac{J}{4}$ and $\omega' = \omega + \epsilon_d$. In the hole sector, we similarly have

$$\sum_q \sum_{kk'} \frac{1}{4} J^2 S_d^+ c_{q\downarrow}^\dagger c_{q\uparrow} \frac{1}{\omega - H_D} S_d^z c_{q\uparrow}^\dagger c_{k'\uparrow} = n_j \frac{1}{4} J^2 \left(-\frac{1}{2} S_d^+ \right) \sum_{kk'} c_{k\downarrow}^\dagger c_{k'\uparrow} \frac{1}{\omega + \frac{U_b}{2} - \frac{D}{2} + \frac{J}{4}} ,$$

$$- \sum_q \sum_{kk'} \frac{1}{4} J^2 S_d^z c_{q\downarrow}^\dagger c_{q\downarrow} \frac{1}{\omega - H_D} S_d^+ c_{q\downarrow}^\dagger c_{k'\uparrow} = -n_j \frac{1}{4} J^2 \left(\frac{1}{2} S_d^+ \right) \sum_{kk'} c_{k\downarrow}^\dagger c_{k'\uparrow} \frac{1}{\omega + \frac{U_b}{2} - \frac{D}{2} + \frac{J}{4}} . \quad (9.24)$$

The renormalisation due to U_b also happens through multiple terms. One of the terms is

$$\frac{1}{2}JU_b \sum_q \sum_{k,k'} S_d^+ c_{q\downarrow}^\dagger c_{k\uparrow} \frac{1}{\omega - H_D} \hat{n}_{q\uparrow} c_{k'\downarrow}^\dagger c_{q\downarrow} = -\frac{1}{2} \frac{JU_b n_j}{\omega - \frac{D}{2} + \frac{U_b}{2} + \frac{J}{4}} \sum_{k,k'} S_d^+ c_{k'\downarrow}^\dagger c_{k\uparrow}. \quad (9.25)$$

The factor of half in front is the same half factor that appears in front of the $S_1^+ S_2^-, S_1^- S_2^+$ terms when we rewrite $\vec{S}_1 \cdot \vec{S}_2$ in terms of $S^z, S^\pm$. Another term is obtained by switching J and U_b :

$$\frac{1}{2}JU_b \sum_q \sum_{k,k'} \hat{n}_{q\downarrow} c_{q\uparrow}^\dagger c_{k\uparrow} \frac{1}{\omega - H_D} S_d^+ c_{k'\downarrow}^\dagger c_{q\uparrow} = -\frac{1}{2} \frac{JU_b n_j}{\omega - \frac{D}{2} + \frac{U_b}{2} + \frac{J}{4}} \sum_{k,k'} S_d^+ c_{k'\downarrow}^\dagger c_{k\uparrow}. \quad (9.26)$$

The corresponding terms in the hole sector are

$$-\frac{1}{2} \frac{JU_b n_j}{\omega - \frac{D}{2} + \frac{U_b}{2} + \frac{J}{4}} \sum_{k,k'} S_d^+ c_{k'\downarrow}^\dagger c_{k\uparrow}, -\frac{1}{2} \frac{JU_b n_j}{\omega - \frac{D}{2} + \frac{U_b}{2} + \frac{J}{4}} \sum_{k,k'} S_d^+ c_{k'\downarrow}^\dagger c_{k\uparrow}. \quad (9.27)$$

2.9.5 Final URG equations

In summary, the renormalisation in the couplings takes the form:

$$\begin{aligned} \Delta U &= 4V^2 n_j \left(\frac{1}{d_1} - \frac{1}{d_0} \right) - n_j \frac{J^2}{d_2}, & \Delta V &= -\frac{3n_j V}{8} \left[J \left(\frac{1}{d_2} + \frac{1}{d_1} \right) + \frac{4U_b}{3} \sum_{i=1}^4 \frac{1}{d_i} \right], \\ \Delta J &= -\frac{n_j J (J + 4U_b)}{d_2}, & \Delta U_b &= 0, \end{aligned} \quad (9.28)$$

where the denominators d_i are given by

$$d_0 = \omega - \frac{D}{2} + \frac{U_b}{2} - \frac{U}{2}, \quad d_1 = \omega - \frac{D}{2} + \frac{U_b}{2} + \frac{U}{2} + \frac{J}{4}, \quad (9.29)$$

$$d_2 = \omega - \frac{D}{2} + \frac{U_b}{2} + \frac{J}{4}, \quad d_3 = \omega - \frac{D}{2} + \frac{U_b}{2}. \quad (9.30)$$

For the sake of completeness, we present the RG equation for a charge isospin Kondo coupling K between the impurity and the bath:

$$\Delta K = -\frac{n_j K (K + 4U_b)}{\omega - \frac{D}{2} + \frac{U_b}{2} + \frac{K}{4}}. \quad (9.31)$$

Indeed, the RG equation for K is observed to be identical in form to that for the spin Kondo coupling J , obtained through the transformation $J \rightarrow K$. This indicates that charge fluctuations between the bath zeroth and first sites (that are incited by an attractive interaction U_b) lead to the RG irrelevance of $K (> 0)$, and thereby a destabilisation of the charge Kondo effect similar to that presented in the main manuscript for J .

2.10 Appendix: Expressing correlation functions in terms of entanglement

We will first relate the impurity Greens function to the geometric entanglement. Given a ground state $|\Psi\rangle_{\text{gs}}$ and a spectrum of energies $\{E_n\}$, the retarded impurity Greens function (in time domain) is defined as $G_{d\sigma}(t) = -i\theta(t) \langle \{c_{d\sigma}(t), c_{d\sigma}^\dagger\} \rangle$. It can be given a spectral representation in terms of the eigenstates $\{|\Psi\rangle_n\}$ of the e-SIAM:

$$G_{d\sigma}(\omega) = \frac{1}{Z} \sum_n \left[\frac{|\langle \Psi_{\text{gs}} | c_{d\sigma} | \Psi_n \rangle|^2}{\omega + E_{\text{gs}} - E_n} + \frac{|\langle \Psi_n | c_{d\sigma} | \Psi_{\text{gs}} \rangle|^2}{\omega - E_{\text{gs}} + E_n} \right], \quad (10.1)$$

where $Z \equiv \sum_n e^{-\beta E_n}$ is the partition function. We will now insert a complete basis into the expression. The basis will be the set of eigenstates of the Hamiltonian $H = H_1 + H_2$, where H_1 is the two-site $J - U_b$ Hamiltonian formed by the impurity and zeroth sites, and H_2 is a tight-binding Hamiltonian formed by the remaining sites. Since the Hamiltonians are decoupled, the eigenstates $|\Psi\rangle_{m,n}$ will be direct product states formed by combining the eigenstates $|\phi\rangle_m, |\psi\rangle_n$ of the individual Hamiltonians H_1 and H_2 respectively: $|\Phi\rangle_{m,n} = |\phi\rangle_m \otimes |\psi\rangle_n$. Inserting this basis leads to the expression:

$$|\Psi\rangle_{\text{gs}} = \sum_{m,n} |\Phi\rangle_{m,n} (\langle \phi_m | \otimes \langle \psi_n |) |\Psi_{\text{gs}}\rangle. \quad (10.2)$$

The ground state of H_1 is the spin-singlet: $|\phi\rangle_0 = |ss\rangle$. We denote the ground state of the tight-binding Hamiltonian H_2 by $|\psi\rangle_0$. Because of the highly renormalised couplings V and J , the impurity site is almost maximally entangled with the zeroth site, such that the ground state $|\Psi\rangle_{\text{gs}}$ in the $r < 0.25$ regime can be thought of as a direct product of the two-site ground state, $c|ss\rangle + \sqrt{1-c^2}|ct\rangle$, in direct product with the tight-binding ground state:

$$|\Psi\rangle_{\text{gs}} \simeq \left(c|ss\rangle + \sqrt{1-c^2}|ct\rangle \right) \otimes |\psi\rangle_0 = |\Phi\rangle_{\text{ss}} + |\Phi\rangle_{\text{ct}}, \quad (10.3)$$

where $|\Phi\rangle_{\text{ss(ct)}} = |ss(ct)\rangle \otimes |\psi\rangle_0$. This suggests that not all terms in the summation of eq. (10.2) contribute; out of all $\{|\psi\rangle_n\}$, only the ground state $n = 0$ contributes, while only $|ss\rangle$ and $|ct\rangle$ contribute from the set $\{|\phi\rangle_m\}$. The summation then simplifies to

$$|\Psi\rangle_{\text{gs}} = |\Phi\rangle_{\text{ss}} \langle ss | \Psi_{\text{gs}}^{(2)} \rangle + |\Phi\rangle_{\text{ct}} \langle ct | \Psi_{\text{gs}}^{(2)} \rangle, \quad (10.4)$$

where $|\Psi_{\text{gs}}^{(2)}\rangle$ is the two-site part of the ground state and can be obtained by starting with the full ground state $|\Psi\rangle_{\text{gs}}$ and tracing over the other lattice sites of the system.

We assume that the global phases of the wavefunctions are real, and since the internal weights of the wavefunctions are real as well, the overlaps $\langle ss | \Psi_{\text{gs}}^{(2)} \rangle, \langle ct | \Psi_{\text{gs}}^{(2)} \rangle$ are also real. We can use these overlaps to define a geometric measure of entanglement [130–132]:

$$\varepsilon(\psi_1, \psi_2) = 1 - |\langle \psi_1 | \psi_2 \rangle|^2. \quad (10.5)$$

If $|\psi_1\rangle$ is thought of as a separable state, then $|\psi_2\rangle$ should be less entangled if its overlap with $|\psi_1\rangle$ is large, which is indeed borne out by the definition. The overlaps then become $\langle ss|\Psi_{gs}^{(2)}\rangle = \sqrt{1 - \varepsilon(ss, \Psi_{gs}^{(2)})}$, and similarly for the state $|ct\rangle$. For brevity, we will use the notation $\varepsilon_{ss} \equiv \varepsilon(ss, \Psi_{gs}^{(2)})$, $\varepsilon_{ct} \equiv \varepsilon(ct, \Psi_{gs}^{(2)})$. The retarded impurity Greens function for spin σ can now be written in terms of these measures:

$$G_{d\sigma}(\omega) = \frac{1}{Z} \sum_n \left[(1 - \varepsilon_{ss}) \left(\frac{|\langle \Phi_{ss} | c_{d\sigma} | \Psi_n \rangle|^2}{\omega + E_{gs} - E_n} + \frac{|\langle \Psi_n | c_{d\sigma} | \Phi_{ss} \rangle|^2}{\omega - E_{gs} + E_n} \right) + (1 - \varepsilon_{ct}) \frac{|\langle \Phi_{ct} | c_{d\sigma} | \Psi_n \rangle|^2}{\omega + E_{gs} - E_n} \right. \\ \left. + (1 - \varepsilon_{ct}) \frac{|\langle \Psi_n | c_{d\sigma} | \Phi_{ct} \rangle|^2}{\omega - E_{gs} + E_n} + \sqrt{(1 - \varepsilon_{ss})} \sqrt{(1 - \varepsilon_{ct})} \frac{\langle \Phi_{ss} | c_{d\sigma} | \Psi_n \rangle \langle \Psi_n | c_{d\sigma}^\dagger | \Phi_{ct} \rangle + \text{h.c.}}{\omega + E_{gs} - E_n} \right. \\ \left. + \sqrt{(1 - \varepsilon_{ss})} \sqrt{(1 - \varepsilon_{ct})} \frac{\langle \Phi_{ct} | c_{d\sigma} | \Psi_n \rangle \langle \Psi_n | c_{d\sigma}^\dagger | \Phi_{ss} \rangle + \text{h.c.}}{\omega - E_{gs} + E_n} \right]. \quad (10.6)$$

We now generalise this to any real-space two-particle correlation involving operators O_1, O_2 that are at most two-particle operators and act on the combined Hilbert space of the impurity site and the zeroth site. The so-called lesser Greens function corresponding to these operators is defined as

$$G_{O_1^\dagger, O_2}^<(t) \equiv i \langle O_1^\dagger O_2(t) \rangle, \quad G_{O_1^\dagger, O_2}^<(t \rightarrow \infty) = i \langle \Psi_{gs} | O_1^\dagger O_2 | \Psi_{gs} \rangle. \quad (10.7)$$

We focus on the static case ($t \rightarrow \infty$) here. Following eq. (10.4), we can rewrite the ground states in terms of the entanglement measures mentioned above.

$$|\Psi\rangle_{gs} = |\Phi\rangle_{ss} \sqrt{1 - \varepsilon_{ss}} + |\Phi\rangle_{ct} \sqrt{1 - \varepsilon_{ss}}, \quad (10.8)$$

where $|\Phi\rangle_{ss(ct)}$ are the zero spin and zero charge isospin eigenstates of the Hamiltonian $H = H_1 + H_2$ defined below eq. (10.1). Substituting this in the static lesser Greens function gives

$$\frac{1}{i} G_{O_1^\dagger, O_2}^<(t \rightarrow \infty) = (1 - \varepsilon_{ss}) \langle \Phi_{ss} | O_2 O_1^\dagger | \Phi_{ss} \rangle + (1 - \varepsilon_{ct}) \langle \Phi_{ct} | O_2 O_1^\dagger | \Phi_{ct} \rangle \\ + \sqrt{1 - \varepsilon_{ss}} \sqrt{1 - \varepsilon_{ct}} \left(\langle \Phi_{ss} | O_2 O_1^\dagger | \Phi_{ct} \rangle + \langle \Phi_{ct} | O_2 O_1^\dagger | \Phi_{ss} \rangle \right). \quad (10.9)$$

This can be extended to generalised retarded Greens functions $G_{O_1^\dagger, O_2}^R(t) \equiv -i\theta(t) \langle \{O_1^\dagger(t), O_2\} \rangle$, whose spectral representation is of the form

$$G_{O_1^\dagger, O_2}^R(\omega) = \frac{1}{Z} \sum_n \left[\frac{\langle \Psi_{gs} | O_2 | \Psi_n \rangle \langle \Psi_n | O_1^\dagger | \Psi_{gs} \rangle}{\omega + E_{gs} - E_n} + \frac{\langle \Psi_n | O_2 | \Psi_{gs} \rangle \langle \Psi_{gs} | O_1^\dagger | \Psi_n \rangle}{\omega - E_{gs} + E_n} \right]. \quad (10.10)$$

Following an approach very similar to the one that led to eq. (10.6), we can cast the generalised Greens function in terms of the geometric entanglement measures:

$$G_{O_1^\dagger, O_2}^R(\omega) = \frac{1}{Z} \sum_n \left[(1 - \varepsilon_{ss}) \left(\frac{(O_1)_{ss,n}^* (O_2)_{ss,n}}{\omega + E_{gs} - E_n} + \frac{(O_1)_{n,ss}^* (O_2)_{n,ss}}{\omega - E_{gs} + E_n} \right) + (1 - \varepsilon_{ct}) \frac{(O_1)_{ct,n}^* (O_2)_{ct,n}}{\omega + E_{gs} - E_n} \right. \\ \left. + (1 - \varepsilon_{ct}) \frac{(O_1)_{n,ct}^* (O_2)_{n,ct}}{\omega - E_{gs} + E_n} + \sqrt{(1 - \varepsilon_{ss})} \sqrt{(1 - \varepsilon_{ct})} \frac{(O_2)_{ss,n} (O_1)_{ct,n}^* + \text{h.c.}}{\omega + E_{gs} - E_n} \right. \\ \left. + \sqrt{(1 - \varepsilon_{ss})} \sqrt{(1 - \varepsilon_{ct})} \frac{(O_2)_{ct,n} (O_1)_{ss,n}^* + \text{h.c.}}{\omega - E_{gs} + E_n} \right], \quad (10.11)$$

where $(O_2)_{ss,n}$ is the matrix element $\langle \Phi_{ss} | O_2 | \Psi_n \rangle$, and similar definitions exist for $(O_1)_{ss,n}$ and its $|\Phi_{ct}\rangle$ counterparts.

Closely related to geometric entanglement is the quantum Fisher information (QFI) F_Q [133, 178], which quantifies how sensitive a state ρ is to unitary transformations generated by an observable $\hat{O}$. For a pure state $\rho = |\psi\rangle \langle \psi|$, $F_Q(\psi, \hat{O})$ is defined in terms of the variance of the operator $\hat{O}$ (i.e., the connected correlation function):

$$F_Q(\psi, \hat{O}) = 4\Delta(\hat{O})^2 = 4 \left(\langle \psi | \hat{O}^2 | \psi \rangle - \langle \psi | \hat{O} | \psi \rangle^2 \right), \quad (10.12)$$

and it provides an upper bound on the precision that can be achieved in measuring the parameter θ that is dual to the observable $\hat{O}$: $\mathcal{N}(\Delta\theta)^2 \geq F_Q^{-1}$, $\mathcal{N}$ being the number of independent measurements [133, 178]. $F_Q(\psi, \hat{O})$ is thus a measure of the entanglement arising from quantum fluctuation content in $|\psi\rangle$ (considered with respect to an eigenstate of $\hat{O}$). This can be made more manifest by considering a traceless operator $\hat{M}$ with eigenstates $\{|m\rangle\}$. Without loss of generality, we can rescale the operator such that $\sum m^2 = 1/2$. There are several examples of such operators in the context of the present work, such as the impurity magnetisation operator (S_d^z), the Kondo spin-flip operator ($S_d^+ S_0^- + \text{h.c.}$) and the Kondo isospin-flip operator ($C_d^+ C_0^- + \text{h.c.}$). If $\varepsilon_m(\psi) \equiv 1 - |\langle m | \psi \rangle|^2$ is the geometric entanglement between a given state $|\psi\rangle$ and the eigenstates of $\hat{M}$, the QFI corresponding to $\hat{M}$ in the state ψ can be written as

$$F_Q(\psi, \hat{M}) = 4 \left[\frac{1}{2} - \sum_m m^2 \varepsilon_m - \left(\sum_m m \varepsilon_m \right)^2 \right]. \quad (10.13)$$

The total geometric entanglement $\sum_m \varepsilon_m$ is constrained to be $d_M - 1$, where d_M is the dimension of the operator $\hat{M}$. Because each eigenvalue m has a magnitude of at most $1/\sqrt{2}$ (all m^2 must add up to $1/2$), the maximum magnitude of the last two terms in eq. (10.13) (and hence the minimum value of the QFI) is attained for the case when some of the ε_m are zero. However, because the eigenstates are orthogonal, only one of them can have perfect overlap with the state $|\psi\rangle$ and only one ε_m can be zero at a time. The case of minimum QFI therefore corresponds to $\varepsilon_{m^*} = 0$, $\varepsilon_{m \neq m^*} = 1$, leading to $F_Q = 0$. The opposite situation arises when all the geometric entanglement measures are non-zero and equal, $\varepsilon_m = 1/d_M$, leading to a maximum QFI value of $F_Q = 2/d_M$. A uniformly spread geometric entanglement distribution, therefore, leads to a larger QFI, while a more focused distribution leads to a smaller QFI. This sums up the relation between the quantum Fisher information and geometric entanglement.

If the operator $\hat{M}$ is such that its eigenstates are separable states from the perspective of a certain party, the vanishing QFI arises when $|\psi\rangle$ has zero geometric entanglement with respect to one of the separable states, and the QFI saturates when $|\psi\rangle$ has a finite geometric entanglement with all of the states. A large QFI can thus be associated with more entanglement. Moreover, if the eigenstates of $\hat{M}$ are symmetry-broken states, a vanishing QFI indicates that $|\psi\rangle$ is very close to such a symmetry-broken state, while a large QFI indicates that $|\psi\rangle$ is a uniform superposition of such symmetry-broken states and therefore preserves the symmetry as a whole. A larger QFI points towards the presence of more quantum fluctuations and the lack of susceptibility towards symmetry-breaking.

In order to demonstrate these ideas in the present problem, we computed the QFI in the ground-state of the e-SIAM for a number of Hermitian operators, as a function of the parameter $r = -U_b/J_0$. These are shown in the left panel of Fig. (2.13) (left panel). In order to explain the behaviour depicted in the figure, we point out again that the ground-states for $r \ll r_{c2}$, $r \lesssim r_{c2}$ and $r > r_{c2}$ are $|SS\rangle + |CT\rangle$, $|SS\rangle$ and $|LM\rangle$ respectively, where $|SS\rangle$, $|CT\rangle$ and $|LM\rangle$ are the spin-singlet state, charge triplet and local moment states respectively. We focus on the spin-flip QFI, corresponding to the operator $\hat{O} = S_d^+ S_0^- + \text{h.c.}$.

- For $r < r_{c1}$, the ground-state involves both $|SS\rangle$ and $|CT\rangle$, meaning that neither of the two entanglement measures ε_{SS} or ε_{CT} will be zero. As mentioned in the preceding paragraph, this leads to the maximal QFI, as can be seen in the figure as well.
- At $r = r_{c1}$, the charge triplet content vanishes and the ground-state is purely a singlet beyond that point. As a result, in the range $r_{c1} < r < r_{c2}$, the geometric entanglement corresponding to the singlet state is zero, leading to a minimal and vanishing QFI. This is again shown in the figure.
- Finally, for $r > r_{c2}$, the local moment states become the ground-states, where the bath zeroth site displays all four configurations as a superposition, because of the hopping into the rest of the bath. This state is again not an eigenstate of the spin-flip operator, and the geometric entanglement will be uniformly distributed across the states, none being zero. This explains the large QFI in the local moment phase.

Similar explanations hold for the single-particle hopping FQI between the impurity and the zeroth site, and the charge isospin FQI between the zeroth site and the first site. In Fig. (2.13) (right panel), we show the QFI for the same three operators very close to the MIT (i.e., for $r \lesssim r_{c2}$). Two important quantities that track the local MIT and act as order parameters for the transition are the degree of compensation for the impurity ($\langle \vec{S}_d \cdot \vec{S}_0 \rangle$) and the impurity magnetisation ($\langle S_d^z \rangle$), and the first two QFI (red and blue curves) in Fig. 2.13 are therefore particularly important because that they quantify the quantum fluctuations present in the system corresponding to these order parameters. Our analysis shows that the two phases on either side of the transition are characterised by distinct values of this pair of QFI: while the QFI corresponding to the degree of compensation zero in the Kondo screened phase, it becomes non-zero in the local moment phase, and the opposite is true for the QFI arising from the impurity magnetisation. The phase precisely at the transition is distinct from those on either side because it displays a non-zero value for both the QFI. While it is expected that a critical point would show enhanced fluctuations of multiple kinds (giving rise to universality), it is enlightening to find that this is also reflected in a measure of many-particle entanglement.

All three QFIs are observed to saturate to finite, non-zero values at the MIT, indicating that the non-Fermi liquid state therein possesses quantum fluctuations of all three varieties.

We will now relate the QFI to some other measures of correlation. The QFI corresponding to an observable can be directly related to the static lesser Greens functions associated with that observable:

$$F_Q(\psi, \hat{O}_2) = 4 \left(\left\langle \hat{O}_2 \hat{O}_1^\dagger \right\rangle \Big|_{\hat{O}_1^\dagger = \hat{O}_2} - \left\langle \hat{O}_2 \hat{O}_1^\dagger \right\rangle^2 \Big|_{\hat{O}_1 = 1} \right) = 4 \left[G_{\hat{O}_2, \hat{O}_2}^<(t \rightarrow \infty) - \left(G_{1, \hat{O}_2}^<(t \rightarrow \infty) \right)^2 \right] \quad (10.14)$$

In the context of the present work, eq. (10.9) allows relating the QFI to the geometric entanglement measured with respect to the spin-singlet and charge triplet states. By combining eqs. (10.9) and (10.14), we get

$$\begin{aligned}
 F_Q(\psi, \hat{O}_2) &= 4 \left[(1 - \varepsilon_{ss}) \langle \Phi_{ss} | \hat{O}_2^2 | \Phi_{ss} \rangle + (1 - \varepsilon_{ct}) \langle \Phi_{ct} | \hat{O}_2^2 | \Phi_{ct} \rangle \right. \\
 &\quad + \sqrt{1 - \varepsilon_{ss}} \sqrt{1 - \varepsilon_{ct}} \left(\langle \Phi_{ss} | \hat{O}_2^2 | \Phi_{ct} \rangle + \langle \Phi_{ct} | \hat{O}_2^2 | \Phi_{ss} \rangle \right) - \left\{ (1 - \varepsilon_{ss}) \langle \Phi_{ss} | \hat{O}_2 | \Phi_{ss} \rangle \right. \\
 &\quad \left. \left. + (1 - \varepsilon_{ct}) \langle \Phi_{ct} | \hat{O}_2 | \Phi_{ct} \rangle + \sqrt{1 - \varepsilon_{ss}} \sqrt{1 - \varepsilon_{ct}} \left(\langle \Phi_{ss} | \hat{O}_2 | \Phi_{ct} \rangle + \langle \Phi_{ct} | \hat{O}_2 | \Phi_{ss} \rangle \right) \right\}^2 \right] \quad (10.15)
 \end{aligned}$$

Further, the retarded Greens function defined above in eq. (4.5) can very generally be related to the QFI. By using the simplified forms $\frac{i}{2} G_{O_2, O_2}^R(t \rightarrow \infty) = \langle (O_2)^2 \rangle$ and $\frac{i}{2} G_{1, O_2}^R(t \rightarrow \infty) = \langle O_2 \rangle$, we get

$$F_Q(\psi, \hat{O}_2) = 2i G_{O_2, O_2}^R(t \rightarrow \infty) + \left(G_{1, O_2}^R(t \rightarrow \infty) \right)^2. \quad (10.16)$$

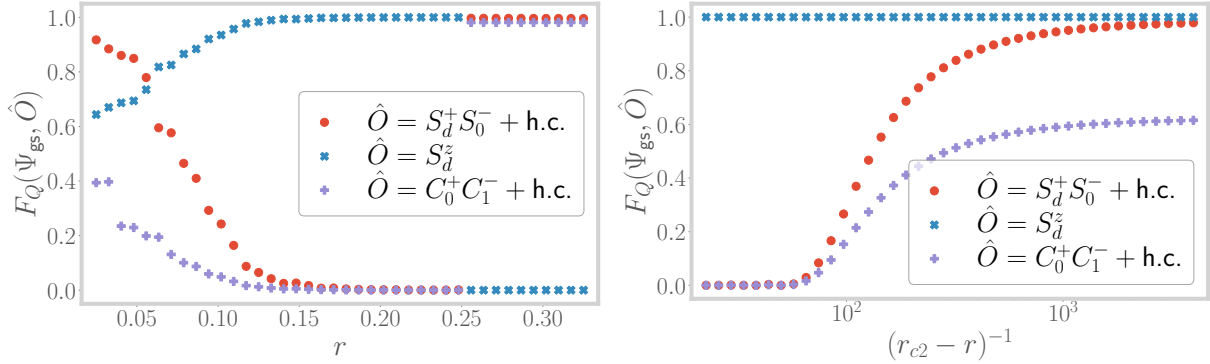


Figure 2.13: *Left:* Variation of the quantum Fisher information for three separate fluctuations, across the entire range of r . *Right:* The QFI corresponding to the same three operators, but very close to the QCP at r_{c2} .

For the sake of completeness, we generalise the above relations to finite temperatures by adopting the approach of Hauke et al. [133]. For a mixed state at inverse temperature β characterised by the density matrix $\rho = \sum_n f_n |n\rangle \langle n|$ where $(E_n, |n\rangle)$ are the eigenvalue-eigenstate pairs and $f_n = e^{-\beta E_n} / \sum_n e^{-\beta E_n}$ are the Boltzmann weights, the FQI can be assigned a spectral representation of the form [133]

$$F_Q(\rho, \hat{O}) = 2 \sum_{m,n} \frac{(f_m - f_n)^2}{f_m + f_n} |\langle m | \hat{O} | n \rangle|^2. \quad (10.17)$$

By using the identity $\frac{f_m - f_n}{f_m + f_n} = \int_{-\infty}^{\infty} d\omega \delta(\omega + E_m - E_n) \tanh(\beta\omega/2)$, the finite temperature FQI can be written in the form

$$F_Q(\rho, \hat{O}) = 2 \int_{-\infty}^{\infty} d\omega \tanh(\beta\omega/2) \sum_{m,n} (f_m - f_n) \delta(\omega + E_m - E_n) |\langle m | \hat{O} | n \rangle|^2. \quad (10.18)$$

In order to relate this with a correlation function, we now point out that the lesser Greens function $G^<$ defined in eq. (10.7), as well as the greater Greens function $G^>$ that we will now define, also admit spectral representations in terms of the eigenstates $|n\rangle$ of the full problem:

$$G_{O^\dagger, O}^<(t) \equiv i \langle \hat{O}^\dagger \hat{O}(t) \rangle \implies G_{O^\dagger, O}^<(\omega) = 2\pi i \sum_{m,n} f_n \delta(\omega + E_m - E_n) |\langle m | \hat{O}^\dagger | n \rangle|^2 \quad (10.19)$$

$$G_{O^\dagger, O}^>(t) \equiv -i \langle \hat{O}(t) \hat{O}^\dagger \rangle \implies G_{O^\dagger, O}^>(\omega) = -2\pi i \sum_{m,n} f_m \delta(\omega + E_m - E_n) |\langle m | \hat{O} | n \rangle|^2 \quad (10.20)$$

If $\hat{O}$ is a Hermitian operator, eqs. (10.19) can be used to rewrite the QFI in terms of $G_{O^\dagger, O}^\lessgtr(\omega)$. Indeed, by combining eqs. (10.19) and eq. (10.18) with the assumption $\hat{O} = \hat{O}^\dagger$, we get

$$F_Q(\rho, \hat{O}) = -\frac{1}{\pi i} \int_{-\infty}^{\infty} d\omega \tanh(\beta\omega/2) \left(G_{O, O}^>(\omega) + G_{O, O}^<(\omega) \right). \quad (10.21)$$

This constitutes the general relation between the finite temperature QFI for a general Hermitian many-particle operator $\hat{O}$ and the corresponding many-particle correlations at zero and finite frequencies.

The Kubo linear response function associated with the operator $\hat{O}(t)$ due to a perturbation from the same operator is associated with a correlation function/susceptibility $C_{\hat{O}}(t, t') = i\theta(t - t') \langle [\hat{O}(t), \hat{O}(t')] \rangle$. By going to the frequency domain, the imaginary part of such a function can be related to the sum of the lesser and greater Greens functions:

$$\begin{aligned} C_{\hat{O}}(\omega) &= i \int_0^\infty d(t-t') e^{i\omega(t-t')} \left(\langle \hat{O}(t) \hat{O}(t') \rangle - \langle \hat{O}(t') \hat{O}(t) \rangle \right) \\ \implies C_{\hat{O}}(\omega) - C_{\hat{O}}(\omega)^* &= i \int_0^\infty d(t-t') \left(e^{i\omega(t-t')} - e^{-i\omega(t-t')} \right) \left(\langle \hat{O}(t) \hat{O}(t') \rangle - \langle \hat{O}(t') \hat{O}(t) \rangle \right) \\ &= i \int_{-\infty}^\infty d(t-t') e^{i\omega(t-t')} \left(\langle \hat{O}(t) \hat{O}(t') \rangle - \langle \hat{O}(t') \hat{O}(t) \rangle \right) \\ &= - \left(G_{O, O}^>(\omega) + G_{O, O}^<(\omega) \right) \end{aligned} \quad (10.22)$$

This allows us to connect the QFI with the imaginary part of the Kubo correlation function/susceptibility obtained in Ref. [133]:

$$F_Q(\rho, \hat{O}) = \frac{1}{\pi i} \int_{-\infty}^{\infty} d\omega \tanh(\beta\omega/2) (C_{\hat{O}}(\omega) - C_{\hat{O}}(\omega)^*). \quad (10.23)$$

2.11 Appendix: One-shot decoupling of impurity from zeroth site

We will decouple the impurity states from the fixed point Hamiltonian using a single URG transformation. This will, of course, generate correlations on the zeroth site. The resulting Hamiltonian will be an AIM with the hopping between the zeroth site and the rest of the chain acting as the effective hybridisation. Schematically, we will have

$$\begin{aligned} H^* = H_{\text{imp}} + V_{\text{imp-0}} + H_0 + H_{0-1} + H_{\text{rest}} &\xrightarrow{\text{decouple imp.}} E_{\text{imp}} + \tilde{H}_0 + H_{0-1} + H_{\text{rest}} \\ &= H_{\text{new imp}} + V_{\text{new imp - rest}} + H_{\text{rest}}. \end{aligned} \quad (11.1)$$

Since the impurity is not coupled with any site beyond the zeroth site, the parts of the Hamiltonian that involve "rest" will not change in the process. This also means that the decoupling can be performed by looking at the smaller Hamiltonian

$$H_{\text{imp}+0} = H_{\text{imp}} + H_0 + V_{\text{imp}-0} = -\frac{U^*}{2} (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2 - U_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2 + J^* S_d^z S_0^z + V^* \sum_{\sigma} \left(c_{d\sigma}^\dagger c_{0\sigma} + \text{h.c.} \right) + \frac{J^*}{2} \left(c_{d\uparrow}^\dagger c_{d\downarrow} c_{0\downarrow}^\dagger c_{0\uparrow} + \text{h.c.} \right). \quad (11.2)$$

2.11.1 Renormalisation from V^*

From the off-diagonal term involving V^* , we generate the following term, in the particle sector for 0:

$$\sum_{\sigma} c_{0\sigma}^\dagger c_{d\sigma} \frac{V^{*2}}{\tilde{\omega} - H_d} c_{d\sigma}^\dagger c_{0\sigma} = \sum_{\sigma} c_{0\sigma}^\dagger c_{d\sigma} \frac{V^{*2}}{\tilde{\omega} + \frac{U^*}{2} (1 - \hat{n}_{d\bar{\sigma}})^2 + U_b \hat{n}_{0\bar{\sigma}} + \frac{J^*}{4} (1 - \hat{n}_{d\bar{\sigma}}) \hat{n}_{0\bar{\sigma}}} c_{d\sigma}^\dagger c_{0\sigma} \quad (11.3)$$

where $\tilde{\omega}$ is the quantum fluctuation operator for the impurity site and $H_d = -\frac{U^*}{2} (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2 - U_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2 + J^* S_d^z S_0^z$, is the diagonal part of the Hamiltonian for the imp+0 system. In order to resolve the operators in the denominator, we expand the unity in the numerator using the identity $1 = \hat{n}_{d\bar{\sigma}} \hat{n}_{0\bar{\sigma}} + \hat{n}_{d\bar{\sigma}} \hat{h}_{0\bar{\sigma}} + \hat{h}_{d\bar{\sigma}} \hat{n}_{0\bar{\sigma}} + \hat{h}_{d\bar{\sigma}} \hat{h}_{0\bar{\sigma}}$, where $\hat{h} = 1 - \hat{n}$ is hole operator. On Substituting this, we get

$$\begin{aligned} \sum_{\sigma} c_{0\sigma}^\dagger c_{d\sigma} \frac{V^{*2}}{\tilde{\omega} - H_d} c_{d\sigma}^\dagger c_{0\sigma} &= V^{*2} \sum_{\sigma} c_{0\sigma}^\dagger c_{d\sigma} \frac{\hat{h}_{d\bar{\sigma}} \hat{h}_{0\bar{\sigma}} + \hat{h}_{d\bar{\sigma}} \hat{n}_{0\bar{\sigma}} + \hat{n}_{d\bar{\sigma}} \hat{h}_{0\bar{\sigma}} + \hat{n}_{d\bar{\sigma}} \hat{n}_{0\bar{\sigma}}}{\tilde{\omega} + \frac{U^*}{2} \hat{h}_{d\bar{\sigma}} + U_b \hat{n}_{0\bar{\sigma}} + \frac{J^*}{4} \hat{h}_{d\bar{\sigma}} \hat{n}_{0\bar{\sigma}}} c_{d\sigma}^\dagger c_{0\sigma} \\ &= V^{*2} \sum_{\sigma} \hat{h}_{d\bar{\sigma}} \hat{n}_{0\bar{\sigma}} \left[\frac{\hat{h}_{d\bar{\sigma}} \hat{h}_{0\bar{\sigma}}}{\tilde{\omega}_{00} + \frac{U^*}{2}} + \frac{\hat{h}_{d\bar{\sigma}} \hat{n}_{0\bar{\sigma}}}{\tilde{\omega}_{01} + \frac{U^*}{2} + U_b + \frac{J^*}{4}} + \frac{\hat{n}_{d\bar{\sigma}} \hat{h}_{0\bar{\sigma}}}{\tilde{\omega}_{10}} + \frac{\hat{n}_{d\bar{\sigma}} \hat{n}_{0\bar{\sigma}}}{\tilde{\omega}_{11} + U_b} \right]. \quad (11.4) \end{aligned}$$

$\tilde{\omega}_{(0,1),(0,1)}$ represents the quantum fluctuation scale corresponding to the configuration in the numerator.

In order to "freeze" the impurity dynamics, we will substitute $\hat{n}_{d\sigma} = \hat{n}_{d\bar{\sigma}} = \frac{1}{2}$, because of the Z_2 symmetry and the particle-hole symmetry of the impurity levels. This gives

$$\frac{1}{4} V^{*2} \sum_{\sigma} \hat{n}_{0\sigma} \left[\frac{\hat{h}_{0\bar{\sigma}}}{\tilde{\omega}_{00} + \frac{U^*}{2}} + \frac{\hat{n}_{0\bar{\sigma}}}{\tilde{\omega}_{01} + \frac{U^*}{2} + U_b + \frac{J^*}{4}} + \frac{\hat{h}_{0\bar{\sigma}}}{\tilde{\omega}_{10}} + \frac{\hat{n}_{0\bar{\sigma}}}{\tilde{\omega}_{11} + U_b} \right]. \quad (11.5)$$

The state that most closely represents the metallic ground state is $\tilde{\omega}_{10}$, we take that as the reference quantum fluctuation scale $\bar{\omega}$. It is of the order of $\bar{\omega} \sim -\frac{J^*}{4} - \frac{U^*}{2} - U_b$. We will now relate the other scales to $\bar{\omega}$ by expressing them in terms of the energy of the initial state:

$$\tilde{\omega}_{00} \sim -U_b = \bar{\omega} + \frac{J^*}{4} + \frac{U^*}{2}, \quad \tilde{\omega}_{01} \sim 0 = \bar{\omega} + \frac{J^*}{4} + \frac{U^*}{2} + U_b, \quad \tilde{\omega}_{11} \sim -\frac{U^*}{2} = \bar{\omega} + \frac{J^*}{4} + U_b. \quad (11.6)$$

Substituting these gives

$$\frac{1}{4} V^{*2} \sum_{\sigma} \hat{n}_{0\sigma} [\hat{h}_{0\bar{\sigma}} \alpha_1 + \hat{n}_{0\bar{\sigma}} \alpha_2], \quad (11.7)$$

where $\alpha_1 = \left(\bar{\omega} + U^* + \frac{J^*}{4}\right)^{-1} + (\bar{\omega})^{-1}$ and $\alpha_2 = \left(\bar{\omega} + U^* + 2U_b + \frac{J^*}{2}\right)^{-1} + \left(\bar{\omega} + 2U_b + \frac{J^*}{4}\right)^{-1}$.

Because of the particle-hole symmetry on the impurity as well as in the bath, the renormalisation from the hole sector is obtained simply by transforming $\hat{n} \leftrightarrow \hat{h}$:

$$\frac{1}{4}V^{*2} \sum_{\sigma} \hat{h}_{0\sigma} [\hat{n}_{0\bar{\sigma}}\alpha_1 + \hat{h}_{0\bar{\sigma}}\alpha_2] . \quad (11.8)$$

The total renormalisation arising from V is therefore

$$\frac{1}{4}V^{*2} \sum_{\sigma} \left[\alpha_1 (\hat{n}_{0\sigma}\hat{h}_{0\bar{\sigma}} + \hat{h}_{0\sigma}\hat{n}_{0\bar{\sigma}}) + \alpha_2 (\hat{n}_{0\sigma}\hat{h}_{0\bar{\sigma}} + \hat{h}_{0\sigma}\hat{h}_{0\bar{\sigma}}) \right] = \frac{1}{2}V^{*2} (\alpha_1 - \alpha_2) (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2 + \text{constant} . \quad (11.9)$$

2.11.2 Renormalisation from J^*

The renormalisation arising from decoupling the Kondo coupling has two terms. The first term arises when the spin of the zeroth site is initially up:

$$\begin{aligned} \frac{J^{*2}}{4} c_{d\downarrow}^{\dagger} c_{d\uparrow} c_{0\uparrow}^{\dagger} c_{0\downarrow} \frac{1}{\bar{\omega} - H_d} c_{0\downarrow}^{\dagger} c_{0\uparrow} c_{d\uparrow}^{\dagger} c_{d\downarrow} &= c_{d\downarrow}^{\dagger} c_{d\uparrow} c_{0\uparrow}^{\dagger} c_{0\downarrow} \frac{J^{*2}/4}{\bar{\omega} + \frac{U^*}{2} + U_b + \frac{J^*}{4}} c_{0\downarrow}^{\dagger} c_{0\uparrow} c_{d\uparrow}^{\dagger} c_{d\downarrow} \\ &= \frac{J^{*2}}{4} \frac{\hat{n}_{d\downarrow} \hat{h}_{d\uparrow} \hat{n}_{0\uparrow} \hat{h}_{0\downarrow}}{\bar{\omega} + \frac{U^*}{2} + U_b + \frac{J^*}{4}} . \end{aligned} \quad (11.10)$$

The quantum fluctuation scale $\tilde{\omega}$ for this process can be similarly related to $\bar{\omega}$: $\tilde{\omega} \sim -\frac{U^*}{2} - U_b - \frac{J^*}{4} = \bar{\omega}$. This gives

$$\frac{J^{*2}}{4} \frac{\hat{n}_{d\downarrow} \hat{h}_{d\uparrow} \hat{n}_{0\uparrow} \hat{h}_{0\downarrow}}{\bar{\omega} + \frac{U^*}{2} + U_b + \frac{3J^*}{4}} = \frac{J^{*2}}{16} \frac{\hat{n}_{0\uparrow} \hat{h}_{0\downarrow}}{\bar{\omega} + \frac{U^*}{2} + U_b + \frac{J^*}{4}} , \quad (11.11)$$

as the renormalisation for this configuration. At the final step, we substituted $\hat{n}_{d\sigma} = \hat{h}_{d\sigma} = \frac{1}{2}$. For the other configuration where the spin of the zeroth site is down, we get

$$\frac{J^{*2}}{16} \frac{\hat{n}_{0\downarrow} \hat{h}_{0\uparrow}}{\bar{\omega} + \frac{U^*}{2} + U_b + \frac{J^*}{4}} . \quad (11.12)$$

Adding both sectors, we get

$$\frac{J^{*2}}{16} \frac{1}{\bar{\omega} + \frac{U^*}{2} + U_b + \frac{J^*}{4}} (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2 . \quad (11.13)$$

2.11.3 Total renormalisation

Adding the contributions from both V^* and J^* , the net renormalisation is the generation of a local correlation term $-U' (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2$ on the zeroth site, where U' is given by

$$U' = \frac{-J^{*2}/16}{\bar{\omega} + \frac{U^*}{2} + U_b + \frac{J^*}{4}} - \frac{1}{2}V^{*2} \left(\frac{1}{\bar{\omega} + U^* + \frac{J^*}{4}} + \frac{1}{\bar{\omega}} - \frac{1}{\bar{\omega} + U^* + 2U_b + \frac{J^*}{2}} - \frac{1}{\bar{\omega} + 2U_b + \frac{J^*}{4}} \right) \quad (11.14)$$

The effective Hamiltonian for the zeroth site is therefore

$$H_{0+\text{rest}} = \underbrace{-(U' + U_b) (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2}_{\text{new correlated impurity}} - t \underbrace{\sum_{\substack{j \in \text{n.n. of } 0, \\ \sigma}} (c_{0\sigma}^\dagger c_{j\sigma} + \text{h.c.})}_{\text{hopping between new impurity \& new bath}} - t \underbrace{\sum_{\langle i, j \rangle} (c_{i\sigma}^\dagger c_{j\sigma} + \text{h.c.})}_{\text{K.E. of new bath}} \quad (11.15)$$

2.12 Appendix: Effective Hamiltonian for the excitations within the Hubbard sidebands

Beyond r_{c1} , the Hubbard sidebands can be seen to separate from the central Kondo resonance in the impurity spectral function. These sidebands are broad, indicating that the impurity site is hybridising with the bath from within the sidebands, at high energies $\omega \sim U/2$. These processes lead to the presence of gapless excitations within the rest of the bath, outside the Mott gap. To find the effective Hamiltonian for the rest of the bath (sites 1 and onwards), we will take the eigenstates of the two-site Hamiltonian H that reside within the sideband as our ground-states, and treat the hopping from the zeroth site to the rest of the bath as a perturbation. The two-site Hamiltonian is

$$H_{\text{two site}} = V \sum_{\sigma} (c_{d\sigma}^\dagger c_{0\sigma} + \text{h.c.}) + J \vec{S}_d \cdot \vec{S}_0 - \frac{U}{2} (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2 + \frac{U}{2}. \quad (12.1)$$

In order to align the doubly-occupied impurity level at the more conventional value of $U/2$, we have shifted the Hamiltonian by the same constant value. The effect of U_b has been ignored because it is small compared to U at the frequencies pertinent to the physics of the sidebands. The two-site states that reside within the Hubbard sidebands are

$$\begin{aligned} |\Psi_{cc}\rangle &\equiv |c\rangle_d |c\rangle_0; \quad c \in \{0, 2\}; \quad |\Psi_{CS}\rangle \equiv \frac{1}{\sqrt{2}} (|2\rangle_d |0\rangle_0 - |0\rangle_d |2\rangle_0); \\ |\Psi_{\sigma,c}\rangle &= \alpha |\sigma\rangle_d |c\rangle_0 + \sqrt{1 - \alpha^2} |c\rangle_d |\sigma\rangle_0; \quad \sigma \in \{\uparrow, \downarrow\}; \\ |\Psi_{CT}\rangle &= \beta \frac{1}{\sqrt{2}} (|\uparrow\rangle_d |\downarrow\rangle_0 - |\downarrow\rangle_d |\uparrow\rangle_0) + \sqrt{1 - \beta^2} \frac{1}{\sqrt{2}} (|2\rangle_d |0\rangle_0 + |0\rangle_d |2\rangle_0), \end{aligned} \quad (12.2)$$

where the coefficients of the eigenstates are given by

$$\alpha \simeq \frac{V}{\sqrt{V^2 + (E_{X1} + \frac{U}{2})^2}}, \quad \beta = \frac{2V}{\sqrt{4V^2 + (E_{CS} + \frac{U}{2} + \frac{3J}{4})^2}}. \quad (12.3)$$

The energies of the states are

$$\begin{aligned} E_{\text{gs}} \equiv E_{00} = E_{22} = E_{CS} &= \frac{U}{2}, \quad E_{X1} \equiv E_{\sigma,c} = \frac{U}{4} + \sqrt{V^2 + \frac{U^2}{16}}, \\ E_{X2} \equiv E_{CS} &= \frac{U}{4} - \frac{3J}{8} + \sqrt{4V^2 + \left(\frac{U}{4} + \frac{3J}{8}\right)^2}. \end{aligned} \quad (12.4)$$

The ground state (comprising the states shown in eq. (12.2)) is triply degenerate. In order to study the dynamics of the bath, we now introduce the perturbation $H_I = -t \sum_{\sigma} (c_{0\sigma}^{\dagger} c_{1\sigma} + \text{h.c.})$. We also take an expanded set of eigenstates $\{|\Psi_n\rangle \otimes |a\rangle\}$, where $\{|\Psi_n\rangle\}$ is the set of eigenstates of $H_{\text{two site}}$, and $|a\rangle \in \{|0\rangle, |\uparrow\rangle, |\downarrow\rangle, |2\rangle\}$ represents the configuration of the first site. The eigenstates of $H_{\text{two site}}$ are also eigenstates the total number operator $\hat{n}_d + \hat{n}_0 + \hat{n}_1$, and since H_I does not conserve the total number, all odd order shifts are zero: $\langle \Psi_n | H_I^b | \Psi_n \rangle$, $b = 1, 3, \dots$, is zero for all the eigenstates.

For the second order shift, we need to calculate the overlaps $\langle \Psi_m | H_I | \Psi_n \rangle$, $m \neq n$ for n in the ground state. Since all three ground-states involve the impurity and zeroth sites in charge configuration (0 or 2), $H_I | \Psi_n \rangle$ will result in states of the form $|0(2)\rangle_d |\sigma\rangle_0 |a\rangle_1$. This resulting state has non-zero overlap only with the part $\sqrt{1 - \alpha^2} |c\rangle_d |\sigma\rangle_0 |a\rangle$ of the expanded state $|\Psi_{\sigma,c}\rangle |a\rangle$. The square of each such overlap is, therefore, $t^2 (1 - \alpha^2)$.

- Excitations from the ground-states $|\Psi_{00}\rangle |0\rangle$ and $|\Psi_{22}\rangle |2\rangle$ have zero overlap with any other state, such that the second order shifts in these states is zero.
- Excitations from the ground-states $|\Psi_{00}\rangle |\sigma\rangle$ and $|\Psi_{22}\rangle |\sigma\rangle$ have non zero overlap with one excited state each, such that the second order shift in these states are $\gamma^2 \equiv t^2 (1 - \alpha^2) / (E_{X1} - E_{\text{gs}})$.
- Excitations from the remaining ground states have non-zero overlaps with two excited states each, such that the second order shift in these states are $2\gamma^2$.

Upon dropping the zeroth order part (because it has no dynamics for the rest of the bath), the effective Hamiltonian at second order in t takes the form

$$H_{\text{eff}}^{(2)} = \gamma^2 \left[(|\Psi_{00}\rangle \langle \Psi_{00}| + |\Psi_{22}\rangle \langle \Psi_{22}| + 2 |\Psi_{\text{CS}}\rangle \langle \Psi_{\text{CS}}|) (\sum_{\sigma} |\sigma\rangle_1 \langle \sigma|_1) + 2 (|\Psi_{22}\rangle \langle \Psi_{22}| + |\Psi_{\text{CS}}\rangle \langle \Psi_{\text{CS}}|) |0\rangle_1 \langle 0|_1 + 2 (|\Psi_{00}\rangle \langle \Psi_{00}| + |\Psi_{\text{CS}}\rangle \langle \Psi_{\text{CS}}|) |2\rangle_1 \langle 2|_1 \right] \quad (12.5)$$

By using the completeness relation $1 = \sum_{\sigma} |\sigma\rangle_1 \langle \sigma|_1 + |0\rangle_1 \langle 0|_1 + |2\rangle_1 \langle 2|_1$ of the Hilbert space of the first site, this evaluates to

$$H_{\text{eff}}^{(2)} = \text{constant} + \gamma^2 (|\Psi_{22}\rangle \langle \Psi_{22}| - |\Psi_{00}\rangle \langle \Psi_{00}|) (|2\rangle_1 \langle 2|_1 - |0\rangle_1 \langle 0|_1) , \quad (12.6)$$

where the "constant" term refers to operators that do not depend on the dynamics of the rest of the bath. By defining the charge isospin operator $C^z |2(0)\rangle = +(-)\frac{1}{2} |2(0)\rangle$, the effective Hamiltonian can be written as (dropping the non-dynamical part)

$$H_{\text{eff}}^{(2)} = 4\gamma^2 C_{\text{tot}}^z C_1^z , \quad (12.7)$$

where C_i^z is the charge isospin for site i and $C_{\text{tot}}^z \equiv C_d^z + C_0^z$ is the z-component of the total charge isospin for the impurity and the zeroth sites.

The fourth order shift in the energy of the eigenstate $|\Psi_n\rangle$ will be given by

$$E_n^{(4)} = \sum_{j,k,l} \frac{(H_I)_{nl} (H_I)_{lk} (H_I)_{kj} (H_I)_{jn}}{(E_l - E_n) (E_k - E_n) (E_j - E_n)} - E_n^{(2)} \left[\frac{(H_I)_{kn}}{E_k - E_n} \right]^2 , \quad (12.8)$$

where the matrix elements $(H_I)_{in}$ are defined as $\langle i|H_I|n\rangle$. This fourth-order shift vanishes for almost all the ground-states. For the remaining states, it becomes important to treat their degeneracy. The "appropriate" combinations of states that provide non-zero fourth order contribution are

$$|+\rangle = \sqrt{\frac{2}{3}} |\Psi_{22}\rangle |0\rangle_1 + \sqrt{\frac{1}{3}} |\Psi_{CS}\rangle |2\rangle_1, \quad |-\rangle = -\sqrt{\frac{2}{3}} |\Psi_{00}\rangle |2\rangle_1 + \sqrt{\frac{1}{3}} |\Psi_{CS}\rangle |0\rangle_1. \quad (12.9)$$

Corresponding to these ground states, the fourth-order shift in the eigenvalue is

$$E^{(4)} = \frac{3\gamma^4\alpha^2\beta^2}{(1-\alpha^2)(E_{X2} - E_{gs})}, \quad (12.10)$$

such that the fourth-order contribution to the effective Hamiltonian is

$$\begin{aligned} H_{\text{eff}}^{(4)} &= E^{(4)} (|+\rangle \langle +| + |-\rangle \langle -|) \\ &= \frac{\gamma^4\alpha^2\beta^2}{(1-\alpha^2)(E_{X2}-E_{gs})} \left[-C_{\text{tot}}^z C_{\text{tot}}^2 C_1^z + \sqrt{2} \mathcal{P}_{\text{tot}}^4 \left(C_0^+ - C_d^+ \right) C_1^- + \text{h.c.} \right]. \end{aligned} \quad (12.11)$$

The operator $\mathcal{P}_{\text{tot}}^4$ acts on states belonging to the combined $d + 0$ Hilbert space and projects on to the $\hat{n}_d + \hat{n}_0 = 4$ subspace.

2.13 Appendix: Behaviour of the Kondo scale T_K close to the transition

Towards obtaining the emergent Kondo temperature scale T_K close to the transition, we follow the approach used by Moeller et al. 1995 [134] and Held et al. 2013 [135]. Near the transition, they obtained a Kondo model from the full AIM by applying a Schrieffer-Wolff transition that removes the charge fluctuations of the impurity site and retains the physics of only the s-d coupling J . This amounts to removing the side peaks from the impurity spectral function and focusing on the low-energy central peak. Held et al. then integrated the RG equation for this Kondo model by using a Lorentzian DOS in the bath. This is motivated by the fact that the central peak of the impurity spectral function is a Lorentzian, and the bath becomes equivalent to the impurity site under self-consistency. We implement the same approach but on our extended impurity model, such that the transformation then leads to a $J - U_b$ model. The relevant RG equations are mentioned in Section IV of the main manuscript:

$$\Delta J = -\frac{n_j J (J + 4U_b)}{\omega - D/2 + U_b/2 + J/4}, \quad \Delta U_b = 0, \quad (13.1)$$

where $n_j = \rho(D)|\Delta D|$ is proportional to the bath density of states. In Section VII of the main manuscript, we show that the spectral function of the zeroth site is tracked by that on the impurity site through the coherent spin-flip fluctuations, and this shows that the appropriate density of states for the bath in this regime is of the Lorentzian kind (same as the shape of the central peak of the impurity spectral function). We therefore set

$$\rho(D) = \frac{\rho_0 \Gamma^2}{D^2 + \Gamma^2}, \quad (13.2)$$

where Γ represents the half-width of the Lorentzian. A similar line of arguments was presented by Held et al. [135]. Close to the transition, the RG flow of the Kondo coupling J is stunted by the competing pairing term, and the RG equation can be simplified into the Poor man's scaling 1-loop form:

$$\frac{dJ}{dD} \simeq -2\rho_0\Gamma^2 \frac{J(J+4U_b)}{(\Gamma^2 + D^2)D}. \quad (13.3)$$

We now integrate this RG equation over the range $D \in [D_0, D^*]$, where D_0 is the bare bandwidth, and D^* is the fixed-point bandwidth. D^* represents the energy scale for the conduction electrons in the IR.

$$\begin{aligned} \int_{J_0}^{J^*} \frac{dJ}{J(J+4U_b)} &= - \int_{D_0}^{D^*} \frac{2\rho_0\Gamma^2 dD}{(\Gamma^2 + D^2)D} \\ \Rightarrow \frac{1}{4U_b\rho_0} \left(\ln \frac{J^*}{J_0} - \ln \frac{J^*+4U_b}{J_0+4U_b} \right) &= \ln \frac{D^* + \Gamma^2}{D_0 + \Gamma^2} - \ln \frac{D^*}{D_0}. \end{aligned} \quad (13.4)$$

We will now take the limit of $J_0 + 4U_b \rightarrow 0^-$ and *equate the singular terms on both sides* of the equation:

$$\ln \frac{D^*}{D_0} = -\frac{1}{4U_b\rho_0} \ln \frac{J_0 + 4U_b}{J_0} = -\frac{1}{4U_b\rho_0} \ln (r_{c2} - r) + \text{non-singular part}. \quad (13.5)$$

We have replaced the couplings J and U_b with the dimensionless parameters $r = -U_b/J_0$ and $r_{c2} = 1/4$. Dropping the non-singular part and solving for D^* then gives the IR energy scale:

$$D^* = D_0 \exp \left[-\frac{\ln (r_{c2} - r)}{4U_b\rho_0} \right]. \quad (13.6)$$

The dynamically generated Kondo temperature scale is then obtained as

$$T_K = \frac{D^*}{k_B} = \frac{D_0}{k_B} \exp \left[-\frac{\ln (r_{c2} - r)}{4U_b\rho_0} \right]. \quad (13.7)$$

Note that the prefactor of the logarithm is positive because U_b is negative: $-4U_b\rho_0 = |4U_b\rho_0|$. As we approach the transition, the parameter r takes the limit $r \rightarrow r_{c2}^-$, and the Kondo temperature scale vanishes:

$$\lim_{r \rightarrow r_{c2}^-} T_K = \frac{D_0}{k_B} \lim_{r \rightarrow r_{c2}^-} (r_{c2} - r)^{|4U_b\rho_0|} \rightarrow 0. \quad (13.8)$$

2.14 Appendix: Effective Hamiltonian for the low-lying excitations at the critical point

In order to obtain an effective Hamiltonian for the low-lying excitations in the bath, we will introduce a hopping term V between the zeroth site and the first site into the two-site Hamiltonian H_0 , and

treat that term perturbatively.

$$H = \underbrace{J\vec{S}_d \cdot \vec{S}_0 - U_b (\hat{n}_{0\uparrow} - \hat{n}_{0\downarrow})^2}_{H_0} - t \underbrace{\sum_{\sigma} (c_{0\sigma}^{\dagger} c_{1\sigma} + \text{h.c.})}_{V}. \quad (14.1)$$

As mentioned before, H_0 is our zeroth Hamiltonian, while V is the perturbation. At the critical point, the zeroth Hamiltonian has a 20-fold degenerate ground-state manifold, and we will show later that it is this degeneracy that leads to NFL excitations. The ground-states are

$$\{|S = S^z = 0\rangle, |\uparrow, 0\rangle, |\uparrow, 2\rangle, |\downarrow, 0\rangle, |\downarrow, 2\rangle\} \otimes \{|0\rangle, |\uparrow\rangle, |\downarrow\rangle, |\uparrow\downarrow\rangle\}. \quad (14.2)$$

The ket to the left of $\otimes$ describes the configuration of the impurity and zeroth sites; $S = S^z = 0$ represents the singlet, while the remaining four states $|\sigma, 0\rangle, |\sigma, 2\rangle$ are local moment states. The ket to the right of $\otimes$ represents the configuration of the first site. Eight of these states have zero matrix elements within the ground-state subspace, even in the presence of the perturbation, so we drop these states. Since the full Hamiltonian conserves the total number of particles $N_{\text{tot}} = \sum_{\sigma} (\hat{n}_{d\sigma} + \hat{n}_{0\sigma} + \hat{n}_{1\sigma})$ and the total spin $S_{\text{tot}}^z = S_d^z + S_0^z + S_1^z$, the remaining 12 states can be split into decoupled blocks based on the values of these two operators:

$$\begin{aligned} N_{\text{tot}} = c + 2, S_{\text{tot}}^z = 0 & : |S = S^z = 0\rangle |c\rangle, |\uparrow, c, \downarrow\rangle, |\downarrow, c, \uparrow\rangle; c = 0, 2, \\ N_{\text{tot}} = 3, S_{\text{tot}}^z = \pm \frac{1}{2} & : |S = S^z = 0\rangle |\sigma\rangle, |\sigma, 0, 2\rangle, |\sigma, 2, 0\rangle; \sigma = \uparrow, \downarrow. \end{aligned} \quad (14.3)$$

The perturbation V will lift the degeneracy in each block; to calculate the modified spectrum, we diagonalise each block in the presence of V . We will first write down the action of the perturbation on the singlet ground-states: this will make it easier to identify the true eigenstates.

$$\begin{aligned} N_{\text{tot}} = c + 2, S_{\text{tot}}^z = 0 & : V |SS_c\rangle = \frac{t}{\sqrt{2}} (|\downarrow, c, \uparrow\rangle - |\uparrow, c, \downarrow\rangle), c = 0, 2, \\ N_{\text{tot}} = 3, S_{\text{tot}}^z = \frac{\sigma}{2} & : V |SS_{\sigma}\rangle = \frac{t}{\sqrt{2}} |\sigma\rangle \otimes (|0, 2\rangle + |2, 0\rangle), \sigma = \uparrow (1), \downarrow (-1), \end{aligned} \quad (14.4)$$

where we have introduced the notation $|SS_{\alpha}\rangle = |S = S^z = 0\rangle \otimes |\alpha\rangle, \alpha = 0, \uparrow, \downarrow, 2$. This shows that the action of the perturbation on the singlet state is to create linear combinations of the local moment states in the respective blocks. We therefore define the two possible linear combinations $|\text{LM}_{\pm}^{N_{\text{tot}}, S_{\text{tot}}^z}\rangle$ within each block:

$$\begin{aligned} N_{\text{tot}} = c + 2, S_{\text{tot}}^z = 0 & : |\text{LM}_{\pm}^{c+2,0}\rangle = \frac{1}{\sqrt{2}} (|\uparrow, c, \downarrow\rangle \pm |\downarrow, c, \uparrow\rangle), c = 0, 2, \\ N_{\text{tot}} = 3, S_{\text{tot}}^z = \frac{\sigma}{2} & : |\text{LM}_{\sigma}^{3, \frac{1}{2}}\rangle = \frac{1}{\sqrt{2}} (|\sigma, 0, 2\rangle \pm |\sigma, 2, 0\rangle), \sigma = \uparrow (1), \downarrow (-1). \end{aligned} \quad (14.5)$$

In terms of these linear combinations, we then get

$$\begin{aligned} V |SS_0\rangle &= t |\text{LM}_{-}^{2,0}\rangle, V |\text{LM}_{-}^{2,0}\rangle = t |SS_0\rangle, V |\text{LM}_{+}^{2,0}\rangle = 0; V |SS_{\uparrow}\rangle = t |\text{LM}_{+}^{3,\uparrow}\rangle, \\ V |\text{LM}_{+}^{3,\uparrow}\rangle &= t |SS_{\uparrow}\rangle, V |\text{LM}_{-}^{3,\uparrow}\rangle = 0, V |SS_{\downarrow}\rangle = t |\text{LM}_{+}^{3,\downarrow}\rangle, V |\text{LM}_{+}^{3,\downarrow}\rangle = t |SS_{\downarrow}\rangle, \\ V |\text{LM}_{-}^{3,\downarrow}\rangle &= 0; V |SS_2\rangle = t |\text{LM}_{-}^{4,0}\rangle, V |\text{LM}_{-}^{4,0}\rangle = t |SS_2\rangle, V |\text{LM}_{+}^{4,0}\rangle = 0. \end{aligned} \quad (14.6)$$

The zero matrix elements at the far right of each line should be understood as the fact that V takes those states out of the ground-state subspace. For e.g., the state $V |LM_-^{2,0}\rangle$ is not really zero, but the overlap of $V |LM_-^{2,0}\rangle$ with any other state in the ground-state subspace spanned by the 20 degenerate states is zero. We now see that each of the 3×3 blocks decouples into a 2×2 block $\begin{pmatrix} 0 & t \\ t & 0 \end{pmatrix}$ and a 1×1 block (0) when written in terms of the linear combinations $|LM_\pm\rangle$. The state in the final block is therefore already an eigenstate, but with zero energy, so that will drop out from the effective Hamiltonian. The remaining 2×2 block has eigenstates $|\pm^{N_{\text{tot}}, S_{\text{tot}}^z}\rangle$ that are linear combinations of the singlet states and the surviving local moment combination, with eigenvalues $\pm t$:

$$\begin{aligned} N_{\text{tot}} = 2, S_{\text{tot}}^z = 0 : |\pm^{2,0}\rangle &= \frac{1}{\sqrt{2}} \left(|SS_0\rangle \pm |LM_-^{2,0}\rangle \right), \\ N_{\text{tot}} = 3, S_{\text{tot}}^z = \frac{1}{2} : |\pm^{3,\frac{1}{2}}\rangle &= \frac{1}{\sqrt{2}} \left(|SS_\uparrow\rangle \pm |LM_+^{3,\frac{1}{2}}\rangle \right), \\ N_{\text{tot}} = 3, S_{\text{tot}}^z = -\frac{1}{2} : |\pm^{3,-\frac{1}{2}}\rangle &= \frac{1}{\sqrt{2}} \left(|SS_\downarrow\rangle \pm |LM_+^{3,-\frac{1}{2}}\rangle \right), \\ N_{\text{tot}} = 4, S_{\text{tot}}^z = 0 : |\pm^{4,0}\rangle &= \frac{1}{\sqrt{2}} \left(|SS_2\rangle \pm |LM_-^{4,0}\rangle \right). \end{aligned} \quad (14.7)$$

At this point, it is worth noting that the polarised ground-states $|S_{\text{tot}}^z = \sigma \frac{1}{2}\rangle$, $\sigma = \pm 1$ can be written as an equal superposition of the singlet states $|SS\rangle_{d0} \otimes |\sigma\rangle_1$ and the local moment states $\frac{1}{\sqrt{2}} (|\sigma, 0, 2\rangle - |\sigma, 2, 0\rangle)$. The polarised symmetry-broken states therefore act as a bridge between the ground states of the two phases and are important in displaying the breakdown of the Kondo cloud at the QCP and the consequences arising from it (like inexact screening of the impurity and fractional magnetisation and entanglement entropy, which will be discussed in the next section).

By combining the two eigenstates $|\pm\rangle$ for each block, one can then reconstruct the effective Hamiltonian $\mathcal{H}_{\text{eff}}^{N_{\text{tot}}, S_{\text{tot}}^z}$ that describes the dynamics of the first site from within the ground-state subspace of the zeroth Hamiltonian. Formally, this is written as

$$\begin{aligned} \mathcal{H}_{\text{eff}}^{N_{\text{tot}}, S_{\text{tot}}^z} &= t |+\rangle^{N_{\text{tot}}, S_{\text{tot}}^z} \langle +|^{N_{\text{tot}}, S_{\text{tot}}^z} - t |-\rangle^{N_{\text{tot}}, S_{\text{tot}}^z} \langle -|^{N_{\text{tot}}, S_{\text{tot}}^z}, \\ &= \begin{cases} t \left[|SS_\alpha\rangle \langle LM_-^{N_{\text{tot}}, S_{\text{tot}}^z}| + |LM_-^{N_{\text{tot}}, S_{\text{tot}}^z}\rangle \langle SS_\alpha| \right], & \alpha = 0, 2, \\ t \left[|SS_\alpha\rangle \langle LM_+^{N_{\text{tot}}, S_{\text{tot}}^z}| + |LM_+^{N_{\text{tot}}, S_{\text{tot}}^z}\rangle \langle SS_\alpha| \right], & \alpha = \uparrow, \downarrow. \end{cases} \end{aligned} \quad (14.8)$$

We first write down the effective Hamiltonian for $\alpha = 0$ which corresponds to $N_{\text{tot}} = 2, S_{\text{tot}}^z = 0$:

$$\mathcal{H}_{\text{eff}}^{2,0} = t \left[|SS_0\rangle \langle LM_-^{2,0}| + |LM_-^{2,0}\rangle \langle SS_0| \right]. \quad (14.9)$$

In order to convert this into a second-quantised form, we will use the following identities:

$$|SS_0\rangle = \frac{1}{\sqrt{2}} \left(P_d^\uparrow c_{0\downarrow}^\dagger - P_d^\downarrow c_{0\uparrow}^\dagger \right) P_0^{(0)} P_1^{(0)} |0\rangle; \quad |LM_-^{2,0}\rangle = \frac{1}{\sqrt{2}} \left(P_d^\uparrow c_{1\downarrow}^\dagger - P_d^\downarrow c_{1\uparrow}^\dagger \right) P_0^{(0)} P_1^{(0)} |0\rangle. \quad (14.10)$$

where $P_i^{(m)}$ ($m = 0, 1, 2$) projects onto the $\hat{n}_i = m$ subspace on the i^{th} site, and P_d^σ projects the impurity spin into $|\sigma\rangle$; $\sigma = \uparrow, \downarrow$ configuration. We then have

$$\begin{aligned} |\text{SS}_0\rangle \langle \text{LM}_-^{2,0}| &= \frac{1}{2} \left(P_d^\uparrow c_{0\downarrow}^\dagger - P_d^\downarrow c_{0\uparrow}^\dagger \right) P_0^{(0)} P_1^{(0)} \left(P_d^\uparrow c_{1\downarrow} - P_d^\downarrow c_{1\uparrow} \right) \\ &= P_1^{(0)} \frac{1}{2} \left(P_d^\uparrow c_{0\downarrow}^\dagger c_{1\downarrow} + P_d^\downarrow c_{0\uparrow}^\dagger c_{1\uparrow} - S_d^+ c_{0\downarrow}^\dagger c_{1\uparrow} - S_d^- c_{0\uparrow}^\dagger c_{1\downarrow} \right) P_0^{(0)}, \end{aligned} \quad (14.11)$$

where we have used $P^2 = P$ if P is a projector, and that $P_d^\uparrow P_d^\downarrow = S_d^+$. Substituting these into the effective Hamiltonian and recognising that $P_d^\uparrow = \frac{1}{2} + S_d^z$, $P_d^\downarrow = \frac{1}{2} - S_d^z$ gives

$$\begin{aligned} \mathcal{H}_{\text{eff}}^{2,0} &= \frac{t}{2} \left[\left(\frac{1}{2} + S_d^z \right) c_{0\downarrow}^\dagger P_0^{(0)} P_1^{(0)} c_{1\downarrow} - S_d^+ c_{0\downarrow}^\dagger P_0^{(0)} P_1^{(0)} c_{1\uparrow} - S_d^- c_{0\uparrow}^\dagger P_0^{(0)} P_1^{(0)} c_{1\downarrow} \right. \\ &\quad \left. + \left(\frac{1}{2} - S_d^z \right) c_{0\uparrow}^\dagger P_0^{(0)} P_1^{(0)} c_{1\uparrow} \right] + \text{h.c.} . \end{aligned} \quad (14.12)$$

In order to make this more illuminating, we define the holon excitation operators $h_{i\sigma}^\dagger = c_{i\sigma}^\dagger P_i^{(0)}$. In terms of these operators, the effective Hamiltonian becomes

$$\begin{aligned} \mathcal{H}_{\text{eff}}^{2,0} &= \frac{t}{2} \left[\left(\frac{1}{2} + S_d^z \right) \left(h_{0\downarrow}^\dagger h_{1\downarrow} + h_{1\downarrow}^\dagger h_{0\downarrow} \right) - S_d^+ \left(h_{0\downarrow}^\dagger h_{1\uparrow} + h_{1\downarrow}^\dagger h_{0\uparrow} \right) - S_d^- \left(h_{0\uparrow}^\dagger h_{1\downarrow} + h_{1\uparrow}^\dagger h_{0\downarrow} \right) \right. \\ &\quad \left. + \left(\frac{1}{2} - S_d^z \right) \left(h_{0\uparrow}^\dagger h_{1\uparrow} + h_{1\uparrow}^\dagger h_{0\uparrow} \right) \right] \\ &= -t \left[S_d^z \frac{1}{2} \left(h_{0\uparrow}^\dagger h_{1\uparrow} + h_{1\uparrow}^\dagger h_{0\uparrow} - h_{0\downarrow}^\dagger h_{1\downarrow} - h_{1\downarrow}^\dagger h_{0\downarrow} \right) + \frac{1}{2} S_d^+ \left(h_{0\downarrow}^\dagger h_{1\uparrow} + h_{1\downarrow}^\dagger h_{0\uparrow} \right) \right. \\ &\quad \left. + \frac{1}{2} S_d^- \left(h_{0\uparrow}^\dagger h_{1\downarrow} + h_{1\uparrow}^\dagger h_{0\downarrow} \right) \right] + \frac{t}{4} \left(h_{0\uparrow}^\dagger h_{1\uparrow} + h_{1\uparrow}^\dagger h_{0\uparrow} + h_{0\downarrow}^\dagger h_{1\downarrow} + h_{1\downarrow}^\dagger h_{0\downarrow} \right) . \end{aligned} \quad (14.13)$$

The first part of the Hamiltonian (with a $-t$ factor in front) is reminiscent of a two-spin Heisenberg Hamiltonian. To investigate this, we write down the potential second spin:

$$S_1^z \equiv \frac{1}{2} \left(h_{0\uparrow}^\dagger h_{1\uparrow} + h_{1\uparrow}^\dagger h_{0\uparrow} - h_{0\downarrow}^\dagger h_{1\downarrow} - h_{1\downarrow}^\dagger h_{0\downarrow} \right) . \quad (14.14)$$

Note that each term that comprises the full operator S_1^z has a combined projector $P_{01}^{(0)} = P_0^{(0)} P_1^{(0)} = P_1^{(0)} P_0^{(0)}$ in the middle with a single annihilation operator to the right. This means that S^z projects onto the smaller set of only those states that satisfy $\hat{n}_0 + \hat{n}_1 = 1$. That is also the meaning of the subscript 1 in S_1^z . There are four such states: $|\uparrow, 0\rangle, |\downarrow, 0\rangle, |0, \uparrow\rangle, |0, \downarrow\rangle$. In fact, it is easy to see that the eigenstates of S^z are

$$|\uparrow\rangle_{1,\pm} = \frac{1}{\sqrt{2}} (|\uparrow, 0\rangle \pm |0, \uparrow\rangle), \quad S_1^z |\uparrow\rangle_{1,\pm} = \pm \frac{1}{2} |\uparrow\rangle_{1,\pm}, \quad (14.15)$$

$$|\downarrow\rangle_{1,\pm} = \frac{1}{\sqrt{2}} (|\downarrow, 0\rangle \pm |0, \downarrow\rangle), \quad S_1^z |\downarrow\rangle_{1,\pm} = \mp \frac{1}{2} |\downarrow\rangle_{1,\pm} . \quad (14.16)$$

The label $\pm$ on top of the eigenstates indicates that these are also eigenstates of the parity operator $\mathcal{P}_{01}$ that switches the site labels 0 and 1:

$$\mathcal{P}_{01} |\alpha, \alpha'\rangle = |\alpha', \alpha\rangle; \quad \alpha, \alpha' \in \{0, \uparrow, \downarrow, 2\} . \quad (14.17)$$

In fact, the eigenstates $|\uparrow\rangle_{1,+1}$ and $|\downarrow\rangle_{1,-1}$ both have $S^z = \frac{1}{2}$, but are orthogonal because they have different parities. Analogous to the S_1^z operator, we can also read off the potential $S_1^\pm$ ladder operators from the effective Hamiltonian:

$$S_1^+ = h_{0\uparrow}^\dagger h_{1\downarrow} + h_{1\uparrow}^\dagger h_{0\downarrow}, \quad S_1^- = (S_1^+)^\dagger. \quad (14.18)$$

Applying these ladder operators on the eigenstates of S_1^z gives

$$S_1^+ |\downarrow\rangle_{1,\pm} = \pm |\uparrow\rangle_{1,\pm}; \quad S_1^- |\uparrow\rangle_{1,\pm} = \pm |\downarrow\rangle_{1,\pm}; \quad S_1^+ |\uparrow\rangle_{1,\pm} = S_1^- |\downarrow\rangle_{1,\pm} = 0. \quad (14.19)$$

This makes it clear that the pair of states $|\uparrow\rangle_1^+$ and $|\downarrow\rangle_1^+$ form a spin-half doublet, as does the other pair $|\uparrow\rangle_1^-$ and $|\downarrow\rangle_1^-$. We can define new spin operators that act only on a single parity subspace:

$$S_{1,\pm}^a = \pm S^a \frac{1}{2} (1 \pm \mathcal{P}_{01}); \quad a = x, y, z. \quad (14.20)$$

The operator $\frac{1}{2} (1 \pm \mathcal{P}_{01})$ projects on to the subspace with ± 1 parity. In the notation $S_{1,\pm}^a$, the label $\pm$ indicates the parity sector to which the operator applies, while 1 indicates that we are operating in the subspace with $\hat{n}_0 + \hat{n}_1 = 1$. Two operators corresponding to different parity subspaces will commute because they act on disjoint Hilbert spaces: $[S_{+1}^a, S_{-1}^b] = 0$. In terms of these projected spin operators, we recover the usual spin-half SU(2) algebra:

$$\begin{aligned} S_{1,\pm}^z |\uparrow\rangle_{1,\pm} &= \frac{1}{2} |\uparrow\rangle_{1,\pm}; & S_{1,\pm}^z |\downarrow\rangle_{1,\pm} &= -\frac{1}{2} |\downarrow\rangle_{1,\pm}; \\ S_{1,\pm}^+ |\downarrow\rangle_{1,\pm} &= |\uparrow\rangle_{1,\pm}; & S_{1,\pm}^- |\uparrow\rangle_{1,\pm} &= |\downarrow\rangle_{1,\pm}. \end{aligned} \quad (14.21)$$

The part of the effective Hamiltonian that involves the impurity can now be written in terms of these new spin operators:

$$t \left(\vec{S}_d \cdot \vec{S}_{-,1} - \vec{S}_d \cdot \vec{S}_{+,1} \right); \quad \vec{S}_\pm = (S_\pm^x \ S_\pm^y \ S_\pm^z). \quad (14.22)$$

The last term in eq. (14.13) (with a prefactor of $\frac{t}{4}$) can now be identified as the parity operator $\mathcal{P}_{01}$ acting within the subspace $\hat{n}_0 + \hat{n}_1 = 1$, because it has eigenstates $|\uparrow, 0\rangle \pm |0, \uparrow\rangle$ and $|\downarrow, 0\rangle \pm |0, \downarrow\rangle$ with eigenvalues ± 1 . The full effective Hamiltonian in the $N_{\text{tot}} = 2$ subspace then takes the compact form

$$\mathcal{H}_{\text{eff}}^{2,0} = t \left(\vec{S}_d \cdot \vec{S}_{1,-1} - \vec{S}_d \cdot \vec{S}_{1,+1} \right) + \frac{t}{4} \mathcal{P}_{01} P_{01}^{(1)}, \quad (14.23)$$

where $P_{01}^{(1)} = P_0^{(1)} P_1^{(0)} + P_0^{(0)} P_1^{(1)}$ projects on to the $\hat{n}_0 + \hat{n}_1 = 1$ subspace. In light of these new spin operators, the original eigenstates $|\pm^{2,0}\rangle$ can be written in terms of these emergent spins $|\uparrow\rangle_{1,\pm}$ and $|\downarrow\rangle_{1,\pm}$:

$$\begin{aligned} |\pm^{2,0}\rangle &= \frac{1}{2} (|\uparrow, \downarrow, 0\rangle - |\downarrow, \uparrow, 0\rangle \pm |\uparrow, 0, \downarrow\rangle \mp |\downarrow, 0, \uparrow\rangle) = \frac{1}{2} [|\uparrow\rangle (|\downarrow, 0\rangle \pm |0, \downarrow\rangle) - |\downarrow\rangle (|\uparrow, 0\rangle \pm |0, \uparrow\rangle)] \\ &= \frac{1}{\sqrt{2}} (|\uparrow\rangle |\downarrow\rangle_{1,\pm} - |\downarrow\rangle |\uparrow\rangle_{1,\pm}). \end{aligned} \quad (14.24)$$

Both the ground and excited states are therefore spin-singlet states, but they belong to the negative and positive parity sectors respectively.

We now proceed to the $N_{\text{tot}} = 4$ sector, which is equivalent to $\alpha = 2$ and $\hat{n}_0 + \hat{n}_1 = 3$. From eq. (14.8), we note that the effective Hamiltonian of the $\alpha = 2$ sector is obtained from that of the $\alpha = 0$ sector by replacing the holons with doublons, $|0\rangle \rightarrow |2\rangle$. This allows us to use the results of the $N_{\text{tot}} = 2$ sector simply by making the above-mentioned holon-doublon transformation. Accordingly, the composite up and down spins of the 0 + 1-site system become

$$|\uparrow\rangle_{3,\pm} = \frac{1}{\sqrt{2}} (|\uparrow, 2\rangle \pm |2, \uparrow\rangle), \quad |\downarrow\rangle_{3,\pm} = \frac{1}{\sqrt{2}} (|\downarrow, 2\rangle \pm |2, \downarrow\rangle), \quad (14.25)$$

which then allow us to define spin operators for the positive and negative parity sectors:

$$\mathcal{S}_{\pm}^z = \frac{1}{2} (|\uparrow\rangle_{3,\pm} - |\downarrow\rangle_{3,\pm}), \quad \mathcal{S}_{\pm}^+ = |\uparrow\rangle_{3,\pm} \langle\downarrow|_{3,\pm}, \quad (14.26)$$

analogous to those of the $N_{\text{tot}} = 2$ sector in eqs. (14.15) and (14.18). These definitions then allow us to rewrite the effective Hamiltonian in terms of the eigenstates of $\mathcal{S}_{\pm}^z$, and hence in terms of the spin operators $\vec{\mathcal{S}}_{3,\pm}$. The eigenstates $|\pm^{4,0}\rangle$ and the effective Hamiltonian $H_{\text{eff}}^{4,0}$ for this sector can be written as

$$|\pm^{4,0}\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle_d |\downarrow\rangle_{3,\pm} - |\downarrow\rangle_d |\uparrow\rangle_{3,\pm}). \quad (14.27)$$

The effective Hamiltonian can then be written in terms of these emergent spins

$$H_{\text{eff}}^{4,0} = \frac{t}{2} \left[(|\uparrow\rangle_d |\downarrow\rangle_{3,+} - |\downarrow\rangle_d |\uparrow\rangle_{3,+}) (\langle\uparrow|_d \langle\downarrow|_{3,+} - \langle\downarrow|_d \langle\uparrow|_{3,+}) \right. \\ \left. - (|\uparrow\rangle_d |\downarrow\rangle_{3,-} - |\downarrow\rangle_d |\uparrow\rangle_{3,-}) (\langle\uparrow|_d \langle\downarrow|_{3,-} - \langle\downarrow|_d \langle\uparrow|_{3,-}) \right]. \quad (14.28)$$

By replacing the emergent spins with the spin operators $\vec{\mathcal{S}}_{3,\pm}$, we get the final form of the Hamiltonian

$$H_{\text{eff}}^{4,0} = t \left(\vec{\mathcal{S}}_d \cdot \vec{\mathcal{S}}_{3,-} - \vec{\mathcal{S}}_d \cdot \vec{\mathcal{S}}_{3,+} \right) + \frac{t}{4} \mathcal{P}_{01} P_{01}^{(3)}, \quad (14.29)$$

where $\mathcal{P}_{01}$ is the parity transformation operator defined in eq. (14.17) and $P_{01}^{(3)}$ projects on to the $\hat{n}_0 + \hat{n}_1 = 3$ subspace. This completes the effective Hamiltonian for the $S_{\text{tot}}^z = 0$ sector.

We now come to the $S_{\text{tot}}^z = +1/2$ sector. From eq. (14.8), the effective Hamiltonian for this sector is

$$H_{\text{eff}}^{3,\frac{1}{2}} = t \left[|\text{SS}\uparrow\rangle \langle\text{LM}^{3,\frac{1}{2}}| + |\text{LM}^{3,\frac{1}{2}}\rangle \langle\text{SS}\uparrow| \right], \quad (14.30)$$

where

$$|\text{SS}\uparrow\rangle = \frac{1}{\sqrt{2}} \left(P_d^\uparrow c_{0\downarrow}^\dagger - P_d^\downarrow c_{0\uparrow}^\dagger \right) c_{1\uparrow}^\dagger |0\rangle; \quad |\text{LM}^{3,\frac{1}{2}}\rangle = \frac{1}{\sqrt{2}} P_d^\uparrow \left(c_{0\uparrow}^\dagger c_{0\downarrow}^\dagger + c_{1\uparrow}^\dagger c_{1\downarrow}^\dagger \right) |0\rangle, \quad (14.31)$$

such that

$$|\text{SS}\uparrow\rangle \langle \text{LAM}^{3,\frac{1}{2}}_-| = \frac{1}{2} \left(P_d^\uparrow c_{0\downarrow}^\dagger - P_d^\downarrow c_{0\uparrow}^\dagger \right) c_{1\uparrow}^\dagger (c_{0\downarrow} c_{0\uparrow} + c_{1\downarrow} c_{1\uparrow}) P_d^\uparrow \quad (14.32)$$

$$= -\frac{1}{2} \left[P_d^\uparrow \left(c_{1\uparrow}^\dagger c_{0\uparrow} \hat{n}_{0\downarrow} + c_{0\downarrow}^\dagger c_{1\downarrow} \hat{n}_{1\uparrow} \right) + S_d^- \left(c_{1\uparrow}^\dagger c_{0\downarrow} \hat{n}_{0\uparrow} - c_{0\uparrow}^\dagger c_{1\downarrow} \hat{n}_{1\uparrow} \right) \right] \quad (14.33)$$

The full effective Hamiltonian then becomes

$$\begin{aligned} H_{\text{eff}}^{3,\frac{1}{2}} &= -\frac{t}{2} \left[P_d^\uparrow \left\{ \left(c_{1\uparrow}^\dagger c_{0\uparrow} + c_{0\downarrow}^\dagger c_{1\downarrow} \right) \left(P_0^{(2)} P_1^{(0)} + P_0^{(0)} P_1^{(2)} \right) + \left(c_{0\uparrow}^\dagger c_{1\uparrow} + c_{1\downarrow}^\dagger c_{0\downarrow} \right) P_0^{(1)} P_1^{(1)} \right\} \right. \\ &\quad \left. + S_d^- \left(c_{1\uparrow}^\dagger c_{0\downarrow} - c_{0\uparrow}^\dagger c_{1\downarrow} \right) \left(P_0^{(2)} P_1^{(0)} + P_0^{(0)} P_1^{(2)} \right) + S_d^+ \left(c_{0\downarrow}^\dagger c_{1\uparrow} - c_{1\downarrow}^\dagger c_{0\uparrow} \right) P_0^{(1)} P_1^{(1)} \right] \\ &= t \left[P_d^\uparrow \sqrt{2} \mathcal{A}_{\frac{1}{2}}^z - \frac{1}{2} S_d^+ \mathcal{B}_{\frac{1}{2}}^- - \frac{1}{2} S_d^- \mathcal{B}_{\frac{1}{2}}^+ \right], \end{aligned} \quad (14.34)$$

where we have defined

$$\mathcal{A}_{\frac{1}{2}}^z = -\frac{1}{2\sqrt{2}} \left[\left(c_{1\uparrow}^\dagger c_{0\uparrow} + c_{0\downarrow}^\dagger c_{1\downarrow} \right) \left(P_0^{(2)} P_1^{(0)} + P_0^{(0)} P_1^{(2)} \right) + \left(c_{0\uparrow}^\dagger c_{1\uparrow} + c_{1\downarrow}^\dagger c_{0\downarrow} \right) P_0^{(1)} P_1^{(1)} \right], \quad (14.35)$$

and

$$\mathcal{B}_{\frac{1}{2}}^+ = \left(c_{1\uparrow}^\dagger c_{0\downarrow} - c_{0\uparrow}^\dagger c_{1\downarrow} \right) \left(P_0^{(2)} P_1^{(0)} + P_0^{(0)} P_1^{(2)} \right); \quad \mathcal{B}_{\frac{1}{2}}^- = \left(c_{0\downarrow}^\dagger c_{1\uparrow} - c_{1\downarrow}^\dagger c_{0\uparrow} \right) P_0^{(1)} P_1^{(1)} \quad (14.36)$$

The effective Hamiltonian for the $S_{\text{tot}}^z = -1/2$ sector is obtained by applying the transformations $P_d^\sigma \rightarrow P_d^{\bar{\sigma}}, c_{i\sigma} \rightarrow c_{i\bar{\sigma}}$ on the Hamiltonian. This is motivated by the fact that applying the same transformations on the $S_{\text{tot}}^z = 1/2$ eigenstates give the $S_{\text{tot}}^z = -1/2$ eigenstates.

$$H_{\text{eff}}^{3,-\frac{1}{2}} = t \left[P_d^\uparrow \sqrt{2} \mathcal{A}_{-\frac{1}{2}}^z - \frac{1}{2} S_d^+ \mathcal{B}_{-\frac{1}{2}}^- - \frac{1}{2} S_d^- \mathcal{B}_{-\frac{1}{2}}^+ \right]. \quad (14.37)$$

The operators $\mathcal{A}_{-\frac{1}{2}}^z$ and $\mathcal{B}_{-\frac{1}{2}}^\pm$ are obtained by applying the above mentioned transformation on their $S_{\text{tot}}^z = +1/2$ counterparts.

2.15 Appendix: Non-Fermi liquid exponents in self-energy and two-particle correlations

The one-particle self-energies relating to the impurity and zeroth sites, as well as some relevant two-particle correlation functions, can be shown to follow power-law behaviour close to the Brinkman-Rice transition at r_{c2} , by mapping the impurity model to a classical Coulomb gas, similar to the work of Anderson, Yuval and Hamman for the Kondo model [92] and that of Si and Kotliar for a periodic Anderson model [87]. Indeed, the latter work uses an auxiliary that is similar to our extended Anderson impurity model. In the following, we adapt their results to our model. These non-universal power-laws are signatures of the non-Fermi liquid excitations that emerge at r_{c2} .

Near r_{c2} , extended SIAM can be reduced to a $J - U_b$ model (shown in subsection V.A of the main manuscript). The conduction electrons scattering off the impurity suffer phase shifts because of the presence of the potentials J and U_b . These phase shifts are given by $\delta_{\text{sp}} = \arctan(\pi\rho_0 J/2)$ and $\delta_{\text{ch}} = \arctan(\pi\rho_0 U_b/2)$ [87, 179], where ρ_0 is the electronic density of states in the conduction bath. The low-energy behaviour of the local impurity and bath self-energies can be expressed in terms of these phase shifts [87]:

$$\begin{aligned} \Sigma_{dd}(\omega) &\sim \omega^{\gamma_{dd}}, \quad \Sigma_{d0}(\omega) \sim \omega^{\gamma_{d0}}, \quad \Sigma_{00}(\omega) \sim \omega^{\gamma_{00}}, \\ \gamma_{dd} &= \frac{5}{4} - \left(\frac{\delta_{\text{tot}}^*}{\pi}\right)^2 - \left(\frac{\delta_{\text{ch}}^*}{\pi}\right)^2, \quad \gamma_{d0} = 2 \left[\left(\frac{\delta_{\text{ch}}^*}{\pi}\right)^2 + \left(\frac{\delta_{\text{tot}}^*}{\pi}\right)^2 - \frac{1}{4} \right], \\ \gamma_{00} &= \left(\frac{\delta_{\text{tot}}^*}{\pi} - \frac{1}{2}\right)^2 + \left(\frac{\delta_{\text{ch}}^*}{\pi}\right)^2 - 2, \end{aligned} \quad (15.1)$$

where we have defined a total phase shift parameter $\frac{\delta_{\text{tot}}}{\pi} \equiv \frac{\delta_{\text{sp}} + \delta_{\text{ch}}}{\pi} - \frac{1}{2}$. This can be extended to certain correlation functions as well. The impurity spin-flip correlation, the impurity-bath excitonic correlation and the zeroth-site pairing correlation [87]:

$$\begin{aligned} \langle S_d^+(\tau) S_d^-(0) \rangle &\sim \tau^{-\alpha_1} \implies \langle S_d^+ \rangle(\omega) \sim \omega^{(\alpha_1-1)/2}, \\ \langle (c_{d\uparrow}^\dagger c_{0\uparrow})(\tau) (c_{0\uparrow}^\dagger c_{d\uparrow})(0) \rangle &\sim \tau^{-\alpha_2} \implies \langle c_{d\uparrow}^\dagger c_{0\uparrow} c_{0\uparrow}^\dagger c_{d\uparrow} \rangle(\omega) \sim \omega^{(\alpha_2-1)/2}, \\ \langle (c_{0\uparrow}^\dagger c_{0\downarrow})(\tau) (c_{0\downarrow} c_{0\uparrow})(0) \rangle &\sim \tau^{-\alpha_3} \implies \langle c_{0\uparrow}^\dagger c_{0\downarrow} \rangle(\omega) \sim \omega^{(\alpha_3-1)/2}, \end{aligned} \quad (15.2)$$

where the exponents are given by

$$\alpha_1 = 2 \left(1 - \frac{\delta_{\text{sp}}^*}{\pi}\right)^2, \quad \alpha_2 = \left(\frac{\delta_{\text{tot}}^*}{\pi} - \frac{1}{2}\right)^2 + \left(\frac{\delta_{\text{ch}}^*}{\pi}\right)^2, \quad \alpha_3 = \left(\frac{\delta_{\text{tot}}^*}{\pi} + \frac{1}{2}\right)^2 + \left(1 + \frac{\delta_{\text{ch}}^*}{\pi}\right)^2. \quad (15.3)$$

2.16 Appendix: Entanglement entropy, magnetisation and phase shift, at the QCP

As mentioned in the previous section, the polarised ground-states $|- \rangle^{3, \pm \frac{1}{2}}$ of eq. (14.7) are special because they involve the strong-coupling ground-state as well as the local moment ground-state. The overall ground states, therefore, contain both entangled and non-entangled impurity contributions, leading to a departure from maximal entanglement. We will now calculate the impurity entanglement entropy $S_{\text{EE}}(d)$ in this polarised ground-state(s) that emerges exactly at the QCP, and show that $S_{\text{EE}}(d)$ has a fractional value (in units of $\log 2$). This turns out to be linked to the fact that the impurity is screened partially in these states, leading to an impurity magnetisation that is non-vanishing but only half of that in the local moment phase.

The fractional entropy arises because of the singular scattering of the gapless excitations at the Fermi surface leading to the destruction of the local Fermi liquid. In order to access these processes, we will work with a *semi-continuous* conduction bath beyond the zeroth site (total number of lattice

sites $N \rightarrow \infty$): $H_{\text{bath}} = \sum_{k,\sigma} \epsilon(k) \psi_{\sigma}^{\dagger}(k) \psi_{\sigma}(k)$. This conduction bath that is composed of the k -states formed from the Hilbert space of the lattice sites beyond the zeroth site will be referred to as the reduced conduction bath. This is in addition to the Hamiltonian for the impurity and zeroth sites: $H_{d0} = J \vec{S}_d \cdot \vec{S}_0 - \frac{U_b}{2} \left(c_{0\uparrow}^{\dagger} c_{0\uparrow} - c_{0\downarrow}^{\dagger} c_{0\downarrow} \right)^2$, where $\vec{S}_d$ is the spin operator for the impurity site, and $c_{0\sigma}^{\dagger}$ creates an electron of spin σ at the zeroth site. The full Hamiltonian also involves a single-particle hopping term that couples these two Hamiltonians: $H = H_{d0} + H_{\text{bath}} + H_{\text{int}}$, where

$$H_{\text{int}} = -t \sum_{\sigma} \left(c_{0\sigma}^{\dagger} c_{1\sigma} + \text{h.c.} \right) = -\frac{t}{N} \sum_{k,\sigma} \left[c_{0\sigma}^{\dagger} \psi_{\sigma}(k) + \text{h.c.} \right]. \quad (16.1)$$

$c_{1\sigma}^{\dagger} = \frac{1}{N} \sum_k \psi_{\sigma}^{\dagger}(k)$ is the creation operator for the first site of the conduction bath of N sites (but effectively the zeroth site for the reduced conduction bath defined by H_{bath}). We will treat H_{int} perturbatively in t/J , in order to obtain the set of renormalised g -fold degenerate ground-states $\left\{ |\psi_{\text{gs}}^{(n)}\rangle; n = 1, \dots, g \right\}$.

As mentioned before, the intention here is to calculate magnetisation and entanglement entropy in the non-Fermi liquid ground states. In the case of a degenerate subspace with $S_{\text{tot}}^z = \pm 1/2$ where $S_{\text{tot}}^z = S_d^z + S_0^z + S_1^z$ is the total spin operator, the magnetisation m_d^z must be calculated in any *one* of the degenerate states for our problem. This is accomplished by solving the problem in the presence of an infinitesimal magnetic field term $-h S_{\text{tot}}^z$ that lifts the degeneracy and chooses one of the symmetry-broken ground-states. This is similar to the "two self-energy description" introduced by David Logan for computing quantities in the local moment phase/Mott insulating phase in terms of different self-energies obtained by inserting positive and negative magnetic fields [80, 180, 181]. Due to the symmetry of the Hamiltonian under $S_{\text{tot}}^z \rightarrow -S_{\text{tot}}^z$, both the limits $h = 0^+$ and $h = 0^-$ give the same results for the density matrix and the entanglement entropy. We will work with $h = 0^+$. Since the total Hamiltonian H and the perturbation H_{int} preserve the total spin $S_{\text{tot}}^z = S_d^z + S_0^z + \int dk S_k^z$, it is sufficient to solve only for the sector with the most positive value of S_{tot}^z , since those are the states that will be selected by the positive magnetic field $h = 0^+$.

In the absence of H_{int} , the set of ground-states with a positive S_{tot}^z will contain, to begin with, the local moment states $|\uparrow\rangle_d |0\rangle_0 |\phi\rangle$ and $|\uparrow\rangle_d |2\rangle_0 |\phi\rangle$, where $|\phi\rangle$ represents the filled Fermi sea of the reduced conduction bath, acting as its spinless ground-state. These states have $S_{\text{tot}}^z = 1/2$. We can label these states as $|n_{\text{tot}}, \mu_d^z\rangle$, using the total number of particles $n_{\text{tot}} = n_d + n_0 + n_{\text{bath}}$ (which is also conserved, and where we set $n_{\text{bath}} = 1$ for the set $|\phi\rangle$), and the average magnetisation μ_d^z of the impurity state: $|2, \frac{1}{2}\rangle \equiv |\uparrow\rangle_d |0\rangle_0 |\phi\rangle$, $|4, \frac{1}{2}\rangle \equiv |\uparrow\rangle_d |2\rangle_0 |\phi\rangle$. Since the singlet state $|\text{SS}\rangle_{d0}$ (on the d and zeroth sites) is degenerate with the local moment states $|\uparrow\rangle_d |0(2)\rangle_0$, one can construct other states with $S_{\text{tot}}^z = 1/2$ by creating magnetic excitations at the Fermi surface. The fact that we are working with a thermodynamically large bath ($N \rightarrow \infty$) means that there will be gapless excitations vanishingly close to the Fermi surface. Such states are of the form $|\text{SS}\rangle_{d0} |e_{\uparrow}\rangle$ (labeled as $|4, 0\rangle$ in the $|n_{\text{tot}}, \mu_d^z\rangle$ notation) and $|\text{SS}\rangle_{d0} |h_{\downarrow}\rangle$ (labeled as $|2, 0\rangle$), where $|e_{\sigma}\rangle$ and $|h_{\downarrow}\rangle$ are electron and hole states respectively:

$$|e_{\sigma}\rangle = \sum_k^{\text{FS}} \psi_{\sigma}^{\dagger}(k) |\phi\rangle, \quad |h_{\sigma}\rangle = \sum_k^{\text{FS}} \psi_{\sigma}(k) |\phi\rangle. \quad (16.2)$$

The sum is over k -states on the Fermi surface. The singlet state was set to have zero spin on the impurity site in the state notation, because $\langle S_d^z \rangle$ vanishes in the singlet state. The full set of relevant degenerate ground-states, classified by the total number of particles, is

$$n_{\text{tot}} = 4 : |4, 0\rangle, |4, \frac{1}{2}\rangle; \quad n_{\text{tot}} = 2 : |2, 0\rangle, |2, \frac{1}{2}\rangle. \quad (16.3)$$

Within each 2×2 block, the interaction matrix H_{int} takes the form (in the basis of the states in eq. (16.3))

$$(H_{\text{int}})_{\text{gs}}^{n_{\text{tot}}=2} = \frac{t}{N} \times \begin{pmatrix} 0 & 1/\sqrt{2} \\ 1/\sqrt{2} & 0 \end{pmatrix} = - (H_{\text{int}})_{\text{gs}}^{n_{\text{tot}}=4}, \quad (16.4)$$

and the $S_{\text{tot}}^z = +\frac{1}{2}$ ground-states are then easily obtained to be of the form $|\psi_{\text{gs}}\rangle^{(n_{\text{tot}})} = \frac{1}{\sqrt{2}} \left(|n_{\text{tot}}, 0\rangle - |n_{\text{tot}}, \frac{1}{2}\rangle \right)$, $n_{\text{tot}} = 2$ & 4 , independent of N . Both ground-states involve an equal superposition of a $\mu_d^z = 1/2$ state and a $\mu_d^z = 0$ state, such that the net impurity magnetisation in either of the renormalised polarised states is

$$m_d^z = \frac{1}{2} \left(0 + \frac{1}{2} \right) = \frac{1}{4}. \quad (16.5)$$

The reduced density matrix $\rho_{\text{imp}} = \text{Tr}' \left(|\psi_{\text{gs}}\rangle^{(n_{\text{tot}})} \langle \psi_{\text{gs}}|^{(n_{\text{tot}})} \right)$ (where Tr' indicates partial trace over all states apart from those of the impurity) for the impurity spin in the polarised states can now be immediately obtained. It is independent of n_{tot} , and actually turns out to be halfway between the maximally mixed density matrix $\rho_{\text{MM}} = \frac{1}{2}I$ and the non-entangled density matrix $\rho_{\text{NE}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$:

$$\rho_{\text{imp}} = \begin{pmatrix} 3/4 & 0 \\ 0 & 1/4 \end{pmatrix} = \frac{1}{2} (\rho_{\text{MM}} + \rho_{\text{NE}}), \quad S_{\text{EE}}(d) = -\text{Tr} (\rho_{\text{imp}} \ln \rho_{\text{imp}}) \simeq 0.81 \log 2. \quad (16.6)$$

This fractional value of the entropy in units of $\log 2$ is another signature of the non-Fermi liquid. We can express this in terms of an effective degeneracy $\tilde{g}$ of the impurity degree of freedom. We define this effective degeneracy as $\tilde{g} = 1 + 2|m_d^z|$, in order to recover the unique ground-state in the local Fermi liquid phase (that has $m_d^z = 0$, $\tilde{g} = 1$) and the magnetic doublet in the local moment phase (that has $m_d^z = 1/2$, $\tilde{g} = 2$). The proposed form also results in a non-integer impurity degeneracy for the NFL at the QCP, $\tilde{g}^* = 3/2$. Using the effective degeneracy $\tilde{g}$, one can also write down a general ground-state

$$|\psi_{\text{gs}}(g)\rangle = \frac{1}{\sqrt{(2-\tilde{g})^2 + (\tilde{g}-1)^2}} [(2-\tilde{g}) |n_{\text{tot}}, 0\rangle - (\tilde{g}-1) |n_{\text{tot}}, 1/2\rangle], \quad (16.7)$$

that interpolates between the strong-coupling ground-state for $r < r_{c2}$ (therefore $\tilde{g} = 1$) and the weak coupling ground-state $r > r_{c2}$ (therefore $\tilde{g} = 2$), also describing the QCP in between ($\tilde{g} = 3/2$).

The entanglement entropy, when written in terms of $\tilde{g}$, takes the form

$$\rho_{\text{imp}} = \frac{1}{2} \begin{pmatrix} 2 - \tilde{g} & 0 \\ 0 & \tilde{g} \end{pmatrix} = \frac{1}{2} + \begin{pmatrix} m_d^z & 0 \\ 0 & -m_d^z \end{pmatrix}, \quad (16.8)$$

$$S_{\text{EE}}(d) = - \left(1 - \frac{\tilde{g}}{2}\right) \ln \left(1 - \frac{\tilde{g}}{2}\right) - \frac{\tilde{g}}{2} \ln \frac{\tilde{g}}{2}. \quad (16.9)$$

The relation between the reduced density matrix ρ_{imp} and the impurity magnetisation m_d^z is consistent with that obtained by Kopp et al., [182].

The general form $|\psi_{\text{gs}}(g)\rangle$ of polarised interpolating ground-state also allows us to calculate the excess charge (n_{exc}) added, by the impurity, to the conduction bath Fermi surface (through gapless excitations). This then allows the scattering phase shift of the conduction electrons (obtained via the Friedel sum rule, [159–161]) to be linked with the impurity magnetisation m_d^z . From the definitions of the states $|n_{\text{tot}}, 0\rangle$ and $|n_{\text{tot}}, 1.2\rangle$, we note that the former state involves an impurity-bath singlet while the latter involves a decoupled local moment. The singlet state contributes an excess charge of unity to the conduction bath: $n_{\text{exc}}^{\text{SS}} = 1$, because of the impurity-bath entanglement within it. The local state moment does not contribute any excess charge, because the impurity spin is no longer hybridising with the bath in that state: $n_{\text{exc}}^{\text{LM}} = 0$. Putting these together, the net excess charge contributed by the state in eq. (16.7) is

$$n_{\text{exc}} = \frac{1}{(2 - \tilde{g})^2 + (\tilde{g} - 1)^2} \left[(2 - \tilde{g})^2 n_{\text{exc}}^{\text{SS}} + (\tilde{g} - 1)^2 n_{\text{exc}}^{\text{LM}} \right] = \left[1 + \left(\frac{2 - \tilde{g}}{\tilde{g} - 1} \right)^2 \right]^{-1}. \quad (16.10)$$

At the QCP, the excess charge acquires a fractional value of $1/2$, indicating once more that only half of the impurity remains coupled with the bath at the transition. The Friedel sum rule can be used to equate this excess charge with the phase shift δ^* suffered by conduction electrons as they scatter off the impurity, at the fixed point:

$$\delta^* = \pi n_{\text{exc}} = \left[1 + \left(\frac{2 - \tilde{g}}{\tilde{g} - 1} \right)^2 \right]^{-1} = \frac{4m_d^{z2}}{1 - 4|m_d^z| + 8m_d^{z2}}. \quad (16.11)$$

Equation (16.11) unifies the effect of several important quantities that lead to the frustration of the impurity and the destruction of the Kondo effect. The presence of U_b introduces local moment states into the spectrum and leads to states with non-vanishing impurity magnetisation m_d^z at the QCP; this non-zero m_d^z can be interpreted as an effective impurity degeneracy $\tilde{g}$ that is only partially screened, resulting in only half the excess charge n_{exc} being contributed to the conduction bath. This reduction in the excess charge acts as a shift in the boundary conditions felt by the conduction electrons and manifests as a phase shift that is less than the unitarity limit of π .

2.17 Appendix: Additional correlations near the ESQPT and the QCP

We have included some additional correlations for the extended SIAM here to highlight the features that are mentioned in the main manuscript. Left panel of Fig. (2.14) shows the impurity and zeroth site double occupancy. Both vanish at r_{c1} , indicating the emergence of the Kondo model. Very close to r_{c2} , the zeroth site doublon occupancy picks up (see inset), showing the destruction of the Kondo cloud. Right panel of Fig. (2.14) shows various measures of entanglement close to r_{c2} . The impurity-bath mutual information (blue) decreases as the transition is approached, showing the breakdown of impurity screening. The impurity and first site MI (violet), the zeroth site and first site MI (red) and the impurity, zeroth site and first site tripartite information (grey) increase, showing the emergence of long-ranged correlations near the transition. Fig. (2.15) shows the emergence of long-ranged intra-bath mutual information near r_{c1} and r_{c2} , signalling critical behaviour near the ESQPT and QCP respectively. Intra-bath CDW correlations and impurity-bath Ising correlations in Fig. (2.16) also show the previously-mentioned emergence of long-ranged correlations.

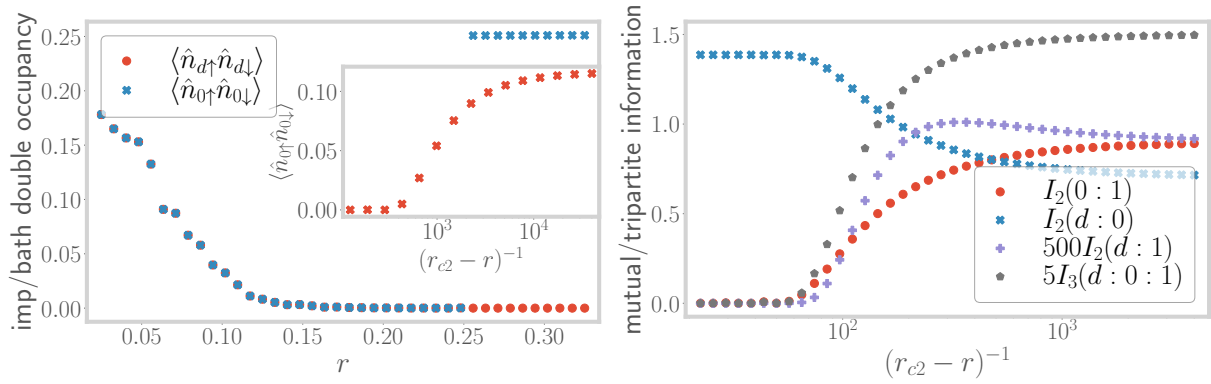


Figure 2.14: Left: Evolution of the double occupancy of impurity (red) and zeroth sites (blue) ground-state double occupancy, with r . Right: Evolution of various quantum information-theoretic measures, very close to the transition.

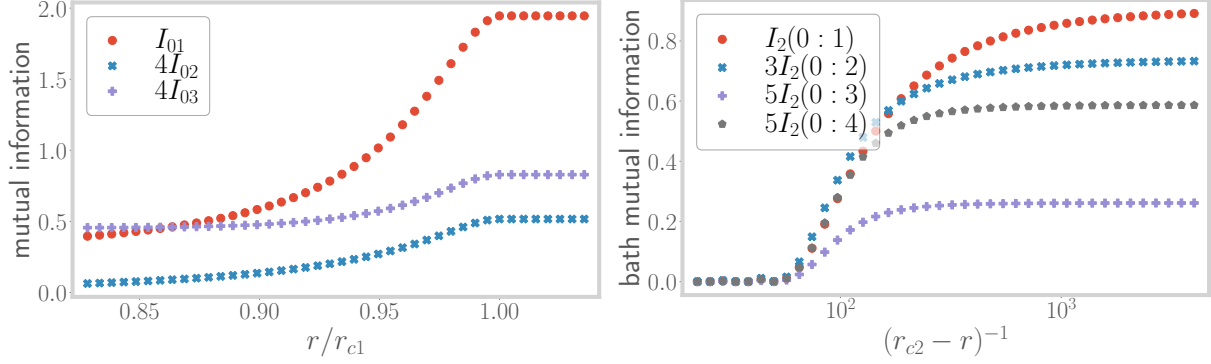


Figure 2.15: Variation of intra-bath mutual information between zeroth site and sites 1, 2 and 3, near r_{c1} (left) and r_{c2} (right).

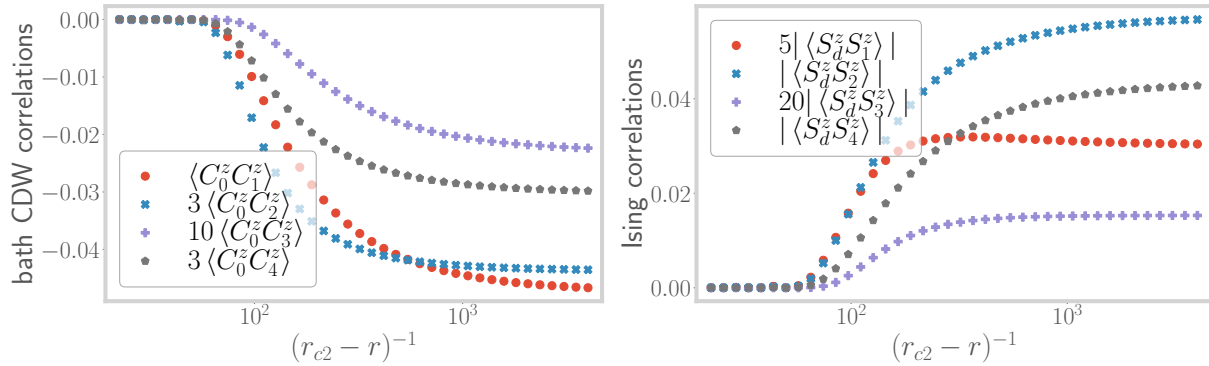


Figure 2.16: Left: Variation of charge isospin Ising correlations between the zeroth site and bath sites further down the chain. Right: Ising correlations between the impurity site and bath sites beyond the zeroth site. Both sets of correlations show an overall increase as $r \rightarrow r_{c2}$.

Chapter 3

Mapping From Auxiliary Model To Lattice: The Tiling Approach

3.1 The Philosophy of Auxiliary Models

In the previous chapter, we extended the Anderson impurity model to host a correlation-driven delocalisation-localisation transition on the impurity site, from a local Fermi liquid state to a local moment state. We demonstrated that this transition captures very well the phenomenology of the Hubbard dimension in infinite dimensions (as seen from, say, dynamical mean-field theory (DMFT)). In this chapter, we extend this approach by developing a formalism to study the more realistic case of finite-dimensional interacting models by using appropriately chosen impurity models as auxiliary models. The key to an *auxiliary model method* is to construct an appropriate auxiliary model that while being tractable captures some essential property of the full lattice model [183]. This involves separating the complete degrees of freedom into a system part (H_S) which one treats exactly and the rest of the system (H_R) that one might simplify in order to be able to solve the problem. The non-trivial nature of the problem arises of course from the hybridisation H_{SR} between the system and the bath:

$$H = \begin{bmatrix} H_R & H_{RS} \\ H_{RS}^* & H_S \end{bmatrix}. \quad (1.1)$$

This separation can always be done exactly, if one does not care about tractability. For example, one can take the Hubbard model that describes electrons hopping on a lattice and interacting when two electrons reside on the same lattice site:

$$H_{\text{hub}} = -t \sum_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (1.2)$$

and separate this into parts, the system being the local orbitals of a specific site l and the rest of the system being all the other sites:

$$H_S = U n_{l\uparrow} n_{l\downarrow}, \quad H_R = -t \sum'_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) + U \sum'_i n_{i\uparrow} n_{i\downarrow}, \quad H_{SR} + H_{RS} = -t \sum_{i,\sigma} (c_{l,\sigma}^\dagger c_{i,\sigma} + \text{h.c.}), \quad (1.3)$$

where the primed summation is carried out over all nearest-neighbour pairs that do not involve l and the summation in $H_{SR} + H_{RS}$ is over all sites adjacent to l .

The smaller system H_S (often called the *cluster*) is typically chosen such that its eigenstates are known exactly. Progress is then made by choosing a simpler form for H_R and its coupling H_{RS} with the smaller system. This combination of the cluster and the simpler bath is then called the *auxiliary system*. A tractable auxiliary system for the Hubbard model is the single-impurity Anderson model (SIAM); the correlated impurity site represents the set of local orbitals of any given site, while the conduction bath represents the rest of the lattice sites in an approximate fashion. Such a construction is shown in fig. 3.1.

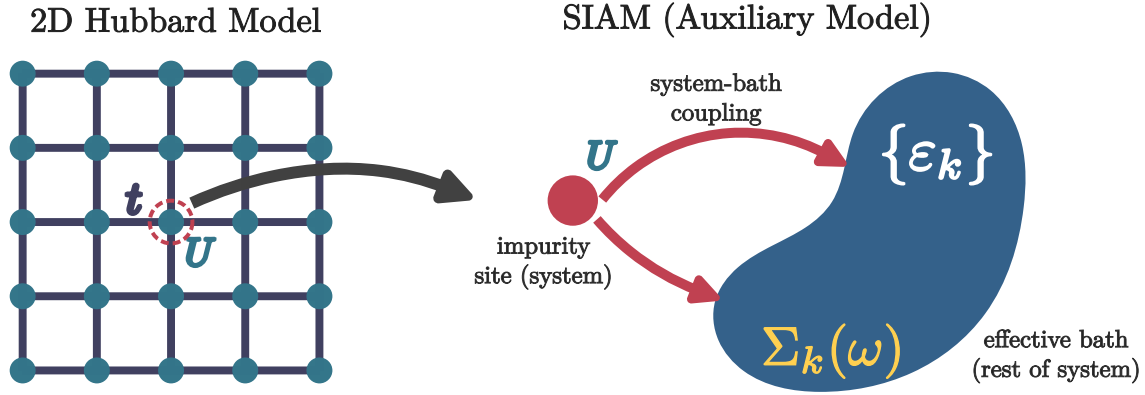


Figure 3.1: *Left*: Full Hubbard model lattice with onsite repulsion U^H on all sites and hopping between nearest neighbour sites with strength t^H . *Right*: Extraction of the auxiliary (cluster+bath) system from the full lattice. The central site on left becomes the impurity site (red) on the right (with an onsite repulsion ϵ_d), while the rest of the $N - 1$ sites on the left form a conduction bath (green circles) (with dispersion ϵ_k and correlation modelled by the self-energy $\Sigma_k(\omega)$) that hybridizes with the impurity through the coupling V .

It should be noted that any reasonable choice of the cluster and bath would break the translational symmetry of the full model: To allow computing quantities, one would need to make the bath (which is a much larger system) simpler than the cluster (which is a single site). This distinction breaks the translational symmetry of the Hubbard model. As a result, calculations that make use of the translation symmetry of the full model (such as k -space objects) will require a periodisation operation of an auxiliary model quantity.

The present work is aimed towards developing a new auxiliary model method to study correlated electronic systems. As described above, the first step is to identify the correct impurity model which can faithfully capture the physics of the lattice model we are trying to solve. The local behaviour of this impurity model should reflect the essential local physics of the lattice model. Typically, we will consider impurity model geometries where the real-space bath site connected directly to the impurity hosts some local interaction. We will henceforth refer to this site as the *zeroth site*. This model must be solved using an impurity solver. In the present work, we use the recently-developed unitary renormalisation group method [10, 18, 26, 122–124]. The leap to the bulk model is then made by applying **manybody lattice translation operators** on the auxiliary model. This process, referred

to as *tiling* here, allows us to reconstruct the lattice model Hamiltonian from that of the auxiliary model and restores the translation invariance of the full lattice model. Once the Hamiltonians can be mapped onto one another, it is possible to relate the eigenstates and correlations functions between the auxiliary model and lattice model, using a manybody version of **Bloch's theorem**.

3.2 What Is The Correct Auxiliary Model For Correlated Electrons on a 2D Square Lattice?

As discussed in more detail in the previous chapter, it was necessary to extend the single-impurity Anderson model (SIAM) in order to obtain a phase transition on the impurity site. Without any modification, the standard Anderson model (at particle-hole symmetry) leads to the emergence of an $\omega = 0$ Kondo resonance in the impurity ($d\sigma$) spectral function at any value of the impurity correlation U_d , at sufficiently low temperatures:

$$H_{\text{SIAM}} = \sum_{k,\sigma} \varepsilon_k n_{k,\sigma} - V \sum_{\sigma} (c_{0,\sigma}^\dagger c_{d,\sigma} + \text{h.c.}) + \varepsilon_d \sum_{\sigma} n_{d,\sigma} + U_d n_{d,\uparrow} n_{d,\downarrow}. \quad (2.1)$$

The operator $c_{0,\sigma}$ acts on the local conduction bath density that couples with the impurity site. The low-energy is one of local Fermi liquid for all parameter regimes. Increasing U_d only serves to reduce the width of the central peak at the cost of the appearance of side bands at energy scales of the order of U_d , but the resonance never disappears. The extended Anderson model that we developed augments the SIAM by allowing for local correlation in the local conduction bath charge density that couples with the impurity spin:

$$H_{\text{ESIAM}} = H_{\text{SIAM}} + J_K \mathbf{S}_d \cdot \mathbf{S}_0 - \frac{U_b}{2} (n_{0,\uparrow}^2 - n_{0,\downarrow}^2); \quad (2.2)$$

tuning this bath correlation to large attractive values (see [169] for some recent findings of non-local effective attractive interactions within the Hubbard model) *does* lead to a localisation transition on the impurity where the local Fermi liquid excitations are replaced by the physics of a gapped local moment on the impurity. We note that the generation of such correlations in the bath is something that occurs in DMFT as well – the conduction bath acquires a *non-trivial self-energy* during the convergence to self-consistency.

While this modified impurity model captures well the Mott metal-insulator transition of the Hubbard model in infinite dimensions, it *lacks any knowledge of the underlying lattice geometry*, and is therefore ill-suited to address the phenomenology of more realistic low-dimensional models. Examples of such phenomenology include the pseudogap phase of the cuprates (with nodal-antinodal dichotomy), strange metal (with T -linear resistivity) and high- T_c superconducting phases observed in several families of quantum materials as well as graphene. This leads us to propose the framework of *lattice-embedded quantum impurity models*. Upon embedding the impurity site within the underlying lattice, the coupling between it and its environment must be consistent with the lowered symmetry of the lattice (see Fig. 3.2). Impurity-bath correlations calculated within such an impurity model are sensitive to the lattice geometry, and are much better suited to address the phenomenology mentioned above.

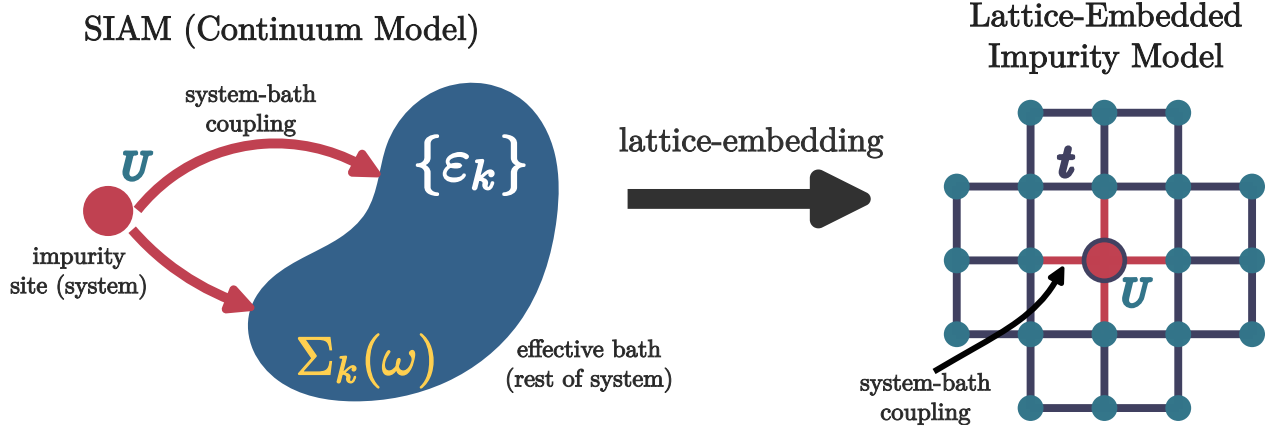


Figure 3.2

As a concrete example, we take the extended SIAM defined in eq. 2.2, and write down its lattice-embedded form on a 2D square lattice, with the impurity placed at the site r_d . This is done by allowing the impurity to hybridise with the four nearest-neighbours, through V and J_K . The same four sites also carry the local bath interaction. The complete auxiliary model is

$$\begin{aligned}
 H = & -t \sum_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) + \sum_{\langle i,r_d \rangle} \left[-\frac{V}{Z} \sum_{\sigma} (c_{r_d,\sigma}^\dagger c_{i,\sigma} + \text{h.c.}) + \frac{J_K}{Z} \mathbf{S}_{r_d} \cdot \mathbf{S}_i - \frac{W}{2Z} (n_{i,\uparrow}^2 - n_{i,\downarrow}^2) \right] \\
 & + \varepsilon_d \sum_{\sigma} n_{r_d,\sigma} + U_d n_{r_d,\uparrow} n_{r_d,\downarrow},
 \end{aligned} \tag{2.3}$$

where Z is the coordination number of the lattice, and the second summation $\sum_{\langle i,r_d \rangle}$ is carried out over all sites i that are nearest-neighbour to the site r_d . Note that the one-particle hybridisation V and the spin-exchange coupling J_K have their *symmetry lowered* from the s -wave spherical symmetry of the continuum impurity models to a C_4 -symmetry consistent with the underlying lattice. This feature is crucial in explaining the phenomenology of the various low-dimensional materials.

Various enhancements can be applied to such a model. The lattice can be modified to study different materials. More complicated geometries such as a two-layer model can be studied using a two-impurity auxiliary model, where each impurity couples with its own conduction bath as well as to each other.

3.3 Mapping Auxiliary Model To Lattice Model

In this subsection, we provide an explicit example of constructing a lattice model from a lattice-embedded impurity model. This will involve the use of manybody lattice translation operators, so we begin with a brief discussion of them.

3.3.1 Manybody Translation Operators

In order to create the bulk model, we need to translate the local auxiliary model over the entire lattice. For this, we define *many-particle global* translation operators $T(\mathbf{a})$ that translate all positions by a vector $\mathbf{a}$. In terms of manybody states and operators, their action is defined as

$$\begin{aligned} T^\dagger(\mathbf{a}) |\mathbf{r}_1, \mathbf{r}_2, \dots\rangle &= |\mathbf{r}_1 + \mathbf{a}\rangle \otimes |\mathbf{r}_2 + \mathbf{a}\rangle \dots \otimes |\mathbf{r}_n + \mathbf{a}\rangle \\ T^\dagger(\mathbf{a}) O(\mathbf{r}_1, \mathbf{r}_2, \dots) T(\mathbf{a}) &= O(\mathbf{r}_1 + \mathbf{a}, \mathbf{r}_2 + \mathbf{a}, \dots) , \end{aligned} \quad (3.1)$$

where $|\mathbf{r}_1, \mathbf{r}_2, \dots\rangle$ is a state in the manyparticle Fock-space basis with the particles localised at the specified positions. For example, for a local fermionic creation operator $c^\dagger(\mathbf{r})$, we have

$$T^\dagger(\mathbf{a}) c^\dagger(\mathbf{r}) T(\mathbf{a}) = c^\dagger(\mathbf{r} + \mathbf{a}) . \quad (3.2)$$

It acts similarly on the auxiliary model Hamiltonian:

$$T^\dagger(\mathbf{a}) H_{\text{aux}}(\mathbf{r}_d) T(\mathbf{a}) = H_{\text{aux}}(\mathbf{r}_d + \mathbf{a}) , \quad (3.3)$$

translating all sites by the vector $\mathbf{a}$. By introducing the Fourier transform to momentum space,

$$|\mathbf{r}_1, \mathbf{r}_2, \dots\rangle = \otimes_{j=1}^N \int d\mathbf{k}_j e^{i\mathbf{r}_j \cdot \mathbf{k}_j} |\mathbf{k}_j\rangle , \quad (3.4)$$

it is easy to see that the total momentum states are eigenstates of the global translation operators:

$$\begin{aligned} T^\dagger(\mathbf{a}) |\mathbf{k}_1, \mathbf{k}_2, \dots\rangle &= \otimes_{j=1}^N \int d\mathbf{r}_j e^{-i\mathbf{r}_j \cdot \mathbf{k}_j} T^\dagger(\mathbf{a}) |\mathbf{r}_j\rangle , \\ &= \otimes_{j=1}^N \int d\mathbf{r}_j e^{-i\mathbf{r}_j \cdot \mathbf{k}_j} |\mathbf{r}_j + \mathbf{a}\rangle \\ &= \otimes_{j=1}^N e^{i\mathbf{a} \cdot \mathbf{k}_j} \int d\mathbf{r}_j e^{-i\mathbf{r}_j \cdot \mathbf{k}_j} |\mathbf{r}_j\rangle \\ &= e^{i\mathbf{a} \cdot \mathbf{k}_{\text{tot}}} |\mathbf{k}_1, \mathbf{k}_2, \dots\rangle , \end{aligned} \quad (3.5)$$

where $\mathbf{k}_{\text{tot}} = \sum_j \mathbf{k}_j$ is the total momentum. This gives a compact form for the translation operator:

$$T^\dagger(\mathbf{a}) = e^{i\mathbf{a} \cdot \hat{\mathbf{k}}_{\text{tot}}} , \quad (3.6)$$

where $\hat{\mathbf{k}}_{\text{tot}}$ is the total momentum operator.

We end this section by calculating the effect of the translation operators on the k -space annihilation operators. We have

$$c(\mathbf{k}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{r}} e^{-i\mathbf{k} \cdot \mathbf{r}} c(\mathbf{r}) . \quad (3.7)$$

Applying the translation operator to this decomposition, we get

$$\begin{aligned} T(\vec{a}) c(\mathbf{k}) T^\dagger(\vec{a}) &= \frac{1}{\sqrt{N}} \sum_{\mathbf{r}} e^{-i\mathbf{k} \cdot \mathbf{r}} T(\vec{a}) c(\mathbf{r}) T^\dagger(\vec{a}) \\ &= \frac{1}{\sqrt{N}} \sum_{\mathbf{r}} e^{-i\mathbf{k} \cdot \mathbf{r}} c(\mathbf{r} - \vec{a}) \\ &= e^{-i\mathbf{k} \cdot \vec{a}} c(\mathbf{k}) . \end{aligned} \quad (3.8)$$

3.3.2 Periodisation of Auxiliary Model Into Extended Hubbard Model

We recall the lattice-embedded impurity model defined in eq. 2.3. As discussed above, the route to restoring translation symmetry and reconstructing the lattice model is to translate the auxiliary model using manybody translation operators. Accordingly, we identify the “unit cell” of translation. Since we expect the impurity site to capture most of the physics arising from the correlated electrons, a good choice for the unit cell is the impurity site and it’s surrounding correlated environment. In order to retain information of the lattice geometry, we convert the hybridisation and bath interaction terms from real space to k -space. The Hamiltonian for the unit cell is therefore

$$H_{\text{clust}}(\mathbf{r}_d) = - \sum_{\mathbf{k}, \sigma} e^{i\mathbf{k} \cdot \mathbf{r}_d} V(\mathbf{k}) (c_{\mathbf{r}_d, \sigma}^\dagger c_{\mathbf{k}, \sigma} + \text{h.c.}) + \sum_{\mathbf{k}_1, \mathbf{k}_2} e^{i(\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}_d} J_K(\mathbf{k}_1, \mathbf{k}_2) \mathbf{S}_{\mathbf{r}_d} \cdot \mathbf{S}_{\mathbf{k}_1, \mathbf{k}_2} + \varepsilon_d n_{\mathbf{r}_d} \\ + U_d n_{\mathbf{r}_d, \uparrow} n_{\mathbf{r}_d, \downarrow} - \frac{1}{2} \sum_{\{\mathbf{k}_i\}, \sigma, \sigma'} \sigma \sigma' e^{i(\mathbf{k}_1 - \mathbf{k}_2 + \mathbf{k}_3 - \mathbf{k}_4) \cdot \mathbf{r}_d} W(\{\mathbf{k}_i\}) c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma'}^\dagger c_{\mathbf{k}_4, \sigma'} , \quad (3.9)$$

where $\mathbf{S}_{\mathbf{k}_1, \mathbf{k}_2} = \sum_{\alpha, \beta} \sigma_{\alpha, \beta} c_{\mathbf{k}_1, \alpha}^\dagger c_{\mathbf{k}_2, \beta}$, and the Fourier transforms are defined by the following relations:

$$\sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_d} V(\mathbf{k}) c_{\mathbf{r}_d, \sigma}^\dagger c_{\mathbf{k}, \sigma} = \frac{V}{Z} \sum_{\mathbf{r} \rightarrow \text{NN of } \mathbf{r}_d} c_{\mathbf{r}_d, \sigma}^\dagger c_{\mathbf{r}, \sigma} , \\ \sum_{\mathbf{k}_1, \mathbf{k}_2} e^{i(\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}_d} J_K(\mathbf{k}_1, \mathbf{k}_2) \vec{S}_{\mathbf{r}_d} \cdot \vec{S}_{\mathbf{k}_1, \mathbf{k}_2} = \frac{J_K}{Z} \sum_{\mathbf{r} \rightarrow \text{NN of } \mathbf{r}_d} \mathbf{S}_{\mathbf{r}_d} \cdot \mathbf{S}_{\mathbf{r}} , \quad (3.10) \\ \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}_1, \mathbf{q}_2} e^{i(\mathbf{k}_1 - \mathbf{k}_2 + \mathbf{k}_3 - \mathbf{k}_4) \cdot \mathbf{r}_d} W(\{\mathbf{k}_i\}) c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma'}^\dagger c_{\mathbf{k}_4, \sigma'} = \frac{W}{2Z} \sum_{\mathbf{r} \rightarrow \text{NN of } \mathbf{r}_d} n_{\mathbf{r}, \sigma} n_{\mathbf{r}, \sigma'} ,$$

We now translate this unit cell to all sites of the lattice, and reconstruct a translationally-invariant correlated lattice model:

$$H_{\text{latt}} = \sum_{\mathbf{r}} T^\dagger(\mathbf{r} - \mathbf{r}_d) H_{\text{clust}}(\mathbf{r}_d) T(\mathbf{r} - \mathbf{r}_d) . \quad (3.11)$$

We consider the effect of the translation operations on each part of the Hamiltonian. The purely local terms translate directly into a chemical potential term and a Hubbard interaction term:

$$\sum_{\mathbf{r}} T^\dagger(\mathbf{r} - \mathbf{r}_d) (\varepsilon_d \sum_{\sigma} n_{\mathbf{r}_d, \sigma} + U_d n_{\mathbf{r}_d, \uparrow} n_{\mathbf{r}_d, \downarrow}) T(\mathbf{r} - \mathbf{r}_d) = \varepsilon_d \sum_{\mathbf{r}, \sigma} n_{\mathbf{r}, \sigma} + U_d \sum_{\mathbf{r}} n_{\mathbf{r}, \uparrow} n_{\mathbf{r}, \downarrow} . \quad (3.12)$$

The one-particle hybridisation term maps onto a tight-binding hopping term:

$$\sum_{\mathbf{r}} T^\dagger(\mathbf{r} - \mathbf{r}_d) \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_d} V(\mathbf{k}) c_{\mathbf{r}_d, \sigma}^\dagger c_{\mathbf{k}, \sigma} T(\mathbf{r} - \mathbf{r}_d) = \sum_{\mathbf{r}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} V(\mathbf{k}) c_{\mathbf{r}, \sigma}^\dagger c_{\mathbf{k}, \sigma} \\ = \frac{V}{Z} \sum_{\mathbf{r}} \sum_{\mathbf{r}' \rightarrow \text{NN of } \mathbf{r}} c_{\mathbf{r}, \sigma}^\dagger c_{\mathbf{r}', \sigma} \\ = \frac{2V}{Z} \sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} c_{\mathbf{r}, \sigma}^\dagger c_{\mathbf{r}', \sigma} , \quad (3.13)$$

where we used eq. 3.8 and the first equation from 3.10, and the fact that each nearest-neighbour pair $\mathbf{r}, \mathbf{r}'$ appears twice in the summation.

The Kondo terms map onto a nearest-neighbour spin-exchange term:

$$\begin{aligned} \sum_{\mathbf{r}, \mathbf{k}_1, \mathbf{k}_2} T^\dagger(\mathbf{r} - \mathbf{r}_d) e^{i(\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}_d} J_K(\mathbf{k}_1, \mathbf{k}_2) \vec{S}_{\mathbf{r}_d} \cdot \vec{S}_{\mathbf{k}_1, \mathbf{k}_2} T(\mathbf{r} - \mathbf{r}_d) &= \sum_{\mathbf{r}, \mathbf{k}_1, \mathbf{k}_2} e^{i(\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}} J_K(\mathbf{k}_1, \mathbf{k}_2) \vec{S}_{\mathbf{r}} \cdot \vec{S}_{\mathbf{k}_1, \mathbf{k}_2} \\ &= \frac{J_K}{Z} \sum_{\mathbf{r}} \sum_{\mathbf{r}' \rightarrow \text{NN of } \mathbf{r}} \vec{S}_{\mathbf{r}} \cdot \vec{S}_{\mathbf{r}'} , \\ &= \frac{2J_K}{Z} \sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} \vec{S}_{\mathbf{r}} \cdot \vec{S}_{\mathbf{r}'} , \end{aligned} \quad (3.14)$$

where we used the second equation from 3.10. Finally, the bath interaction term W maps onto another Hubbard term

$$\begin{aligned} &\sum_{\mathbf{r}, \mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} T^\dagger(\mathbf{r} - \mathbf{r}_d) e^{i(\mathbf{k}_1 - \mathbf{k}_2 + \mathbf{k}_3 - \mathbf{k}_4) \cdot \mathbf{r}_d} W(\{\mathbf{k}_i\}) c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma'}^\dagger c_{\mathbf{k}_4, \sigma'} T(\mathbf{r} - \mathbf{r}_d) \\ &= \sum_{\mathbf{r}, \mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} e^{i(\mathbf{k}_1 - \mathbf{k}_2 + \mathbf{k}_3 - \mathbf{k}_4) \cdot \mathbf{r}} W(\{\mathbf{k}_i\}) c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma'}^\dagger c_{\mathbf{k}_4, \sigma'} \\ &= \frac{W}{2Z} \sum_{\mathbf{r}} \sum_{\mathbf{r}' \rightarrow \text{NN of } \mathbf{r}} n_{\mathbf{r}', \sigma} n_{\mathbf{r}', \sigma'} \\ &= \frac{W}{2} \sum_{\mathbf{r}} n_{\mathbf{r}, \sigma} n_{\mathbf{r}, \sigma'} , \end{aligned} \quad (3.15)$$

where we used the fact that each point $\mathbf{r}'$ (which we relabel as $\mathbf{r}$) appears Z times in the summation.

The complete lattice Hamiltonian, after applying the translation operators, is

$$H_{\text{latt}} = \varepsilon_d \sum_{\mathbf{r}, \sigma} n_{\mathbf{r}, \sigma} + U_d \sum_{\mathbf{r}} n_{\mathbf{r}, \uparrow} n_{\mathbf{r}, \downarrow} - \frac{2V}{Z} \sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} c_{\mathbf{r}, \sigma}^\dagger c_{\mathbf{r}', \sigma} + \frac{2J_K}{Z} \sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} \vec{S}_{\mathbf{r}} \cdot \vec{S}_{\mathbf{r}'} - \frac{W}{2} \sum_{\mathbf{r}} (n_{\mathbf{r}, \uparrow} - n_{\mathbf{r}, \downarrow})^2 . \quad (3.16)$$

This is clearly an extended Hubbard model, with both 1-particle and spin-exchange nearest-neighbour hybridisation. The general form of the model is

$$H_{\text{ex-Hub}} = -t \sum_{\langle i, j \rangle, \sigma} (c_{i, \sigma}^\dagger c_{j, \sigma} + \text{h.c.}) + J \sum_{\langle i, j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + U \sum_i n_{i, \uparrow} n_{i, \downarrow} - \mu \sum_{i, \sigma} n_{i, \sigma} , \quad (3.17)$$

such that the effective couplings resulting from the tiling operation is

$$t = \frac{2V}{Z}, \quad J = \frac{2J_K}{Z}, \quad U = U_d - W, \quad \mu = -\varepsilon_d + \frac{W}{2} . \quad (3.18)$$

3.3.3 Form Of The Eigenstates: Bloch's Theorem

The tiled Hamiltonian defined in eq. 3.11 is manifestly symmetric under discrete translation, given that it's expressed in terms of the translation operators. The fact that the Hamiltonian H_{latt} commutes

with the manybody translation operator implies that the total crystal momentum is a conserved quantity. We will now use this to write down a “tiling” relation between the eigenstates of the auxiliary model and those of the lattice model.

In the tight-binding approach to lattice problems, the full Hamiltonian is obtained by translating the dynamics of the localised orbitals at each site over the entire lattice. The full eigenstate $|\Psi\rangle$ is obtained by constructing linear combinations of the eigenstates $|\psi(i)\rangle$ of the local Hamiltonians. According to Bloch’s theorem, $|\Psi\rangle$ is of the form $|\Psi_{n,K}\rangle = \sum_i e^{i\mathbf{K}\cdot\mathbf{r}_i} |\psi_n(i)\rangle$, where $\mathbf{K}$ is the single-particle crystal momentum, n is the band-index that labels various states within the local orbital, and $\mathbf{r}_i$ sums over the positions of the local Hamiltonians. Bloch’s theorem ensures that eigenstates satisfy the following relation under a translation operation by an arbitrary number of lattice spacings $n\mathbf{a}$:

$$T^\dagger(l\mathbf{a}) |\Psi_{n,K}\rangle = \sum_i e^{i\mathbf{K}\cdot\mathbf{r}_i} |\psi_n(i+l)\rangle = e^{-il\mathbf{K}\cdot\mathbf{a}} |\Psi_{n,K}\rangle . \quad (3.19)$$

As discussed earlier, the eigenstates $|\Psi_{n,K}\rangle$ of the lattice Hamiltonian obtained using eq. 3.11 enjoy the discrete translation symmetry of the lattice, and therefore enjoy a *many-body* Bloch’s theorem [184]. This means that the *local* eigenstates $|\psi_n(\mathbf{r}_d)\rangle$ (with the impurity located at an arbitrary position $\mathbf{r}_d$) of the unit cell auxiliary model Hamiltonian $H_{\text{clust}}(\mathbf{r}_d)$ defined in eq. 3.9 can be used to construct eigenstates of the lattice Hamiltonian. The index $n(= 0, 1, \dots)$ in the subscript indicates that it is the n^{th} eigenstate of the auxiliary model. The following combination of the auxiliary model eigenstates satisfies a many-particle equivalent of Bloch’s theorem [184]:

$$|\Psi_s\rangle \equiv |\Psi_{K,n}\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{r}_d} e^{i\mathbf{K}\cdot\mathbf{r}_d} |\psi_n(\mathbf{r}_d)\rangle , \quad (3.20)$$

where N is the total number of lattice sites and $\mathbf{r}_d$ is summed over all lattice spacings. The set of $n = 0$ states form the lowest band in the spectrum of the lattice, while higher values of n produce the more energetic bands.

3.4 Mapping Greens Functions from Auxiliary Model to Lattice Model

3.4.1 One-particle k -space Greens functions

In the previous part, we proposed a form for the eigenstates $|\Psi_s\rangle$ of the bulk Hamiltonian in terms of the ground-states $|\psi_n\rangle$ of the auxiliary models. In this section, we will relate one-particle Greens functions of the lattice model to those of the auxiliary model. We define the retarded time-domain lattice k -space Greens function at zero temperature as

$$G(k\sigma; t) = -i\theta(t) \langle \Psi_{\text{gs}} | \{c_{k\sigma}(t), c_{k\sigma}^\dagger\} | \Psi_{\text{gs}} \rangle . \quad (4.1)$$

where $|\Psi_{\text{gs}}\rangle$ is the lattice groundstate. Time evolution of the operators is governed by the lattice Hamiltonian H_{latt} :

$$c_{k\sigma}(t) = e^{itH_{\text{latt}}} c_{k\sigma} e^{-itH_{\text{latt}}} . \quad (4.2)$$

By using eq. 3.8 and the fact that the translation operators commute with H_{latt} , we have the relation

$$T(\vec{a}) c(\mathbf{k})(t) T^\dagger(\vec{a}) = e^{-i\mathbf{k} \cdot \vec{a}} c(\mathbf{k})(t), \quad (4.3)$$

which combined with eq. 3.8 gives

$$T(\mathbf{a}) c(\mathbf{k})(t) c^\dagger(\mathbf{k}) T^\dagger(\mathbf{b}) = e^{i\mathbf{k} \cdot (\mathbf{b} - \mathbf{a})} c(\mathbf{k})(t) T^\dagger(\mathbf{b} - \mathbf{a}) c^\dagger(\mathbf{k}). \quad (4.4)$$

In order to make use of the mapping between auxiliary model and lattice model Hamiltonians, we would like to express this lattice Greens function in terms of auxiliary model Greens functions – objects that can be computed purely from within a single auxiliary model. This will be a major simplification in our computations, because it is easier to work with a quantum impurity model, as compared to an interacting lattice model.

To make progress towards this, we express the lattice groundstate in terms of the auxiliary model groundstates (using eq. 3.20):

$$G(\mathbf{k}\sigma; t) = -i\theta(t) \frac{1}{N} \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_2) | \{c_{\mathbf{k}\sigma}(t), c_{\mathbf{k}\sigma}^\dagger\} | \psi_{\text{gs}}(\mathbf{r}_1) \rangle. \quad (4.5)$$

We take one of the terms in the anticommutator and proceed to simplify it:

$$\begin{aligned} \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_2) | c_{\mathbf{k}\sigma}(t) c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_1) \rangle &= \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_d) | T(\mathbf{r}_2 - \mathbf{r}_d) c_{\mathbf{k}\sigma}(t) c_{\mathbf{k}\sigma}^\dagger T^\dagger(\mathbf{r}_1 - \mathbf{r}_d) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &= \sum_{\mathbf{r}_1, \mathbf{r}_2} e^{i\mathbf{k} \cdot (\mathbf{r}_1 - \mathbf{r}_2)} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}(t) T^\dagger(\mathbf{r}_1 - \mathbf{r}_2) c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &= N \sum_{\mathbf{r}} e^{i\mathbf{k} \cdot \mathbf{r}} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}(t) T^\dagger(\mathbf{r}) c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle. \end{aligned} \quad (4.6)$$

where $\mathbf{r}_d$ is a reference lattice site, and we used eq. 4.4 in the second step. The other term in the anticommutator gives

$$\begin{aligned} \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_2) | c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma}(t) | \psi_{\text{gs}}(\mathbf{r}_1) \rangle &= \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_d) | T(\mathbf{r}_2 - \mathbf{r}_d) c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma}(t) T^\dagger(\mathbf{r}_1 - \mathbf{r}_d) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &= \sum_{\mathbf{r}_1, \mathbf{r}_2} e^{-i\mathbf{k} \cdot (\mathbf{r}_1 - \mathbf{r}_2)} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}^\dagger T^\dagger(\mathbf{r}_1 - \mathbf{r}_2) c_{\mathbf{k}\sigma}(t) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &= N \sum_{\mathbf{r}} e^{-i\mathbf{k} \cdot \mathbf{r}} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}^\dagger T^\dagger(\mathbf{r}) c_{\mathbf{k}\sigma}(t) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle. \end{aligned} \quad (4.7)$$

To resolve the time evolution of the operators, we insert $1 = \sum_m |\psi_m(\mathbf{r}_d)\rangle \langle \psi_m(\mathbf{r}_d)|$ into the equation. We again work with just the first term of the anticommutator for convenience:

$$\begin{aligned} &\sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_2) | c_{\mathbf{k}\sigma}(t) c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_1) \rangle \\ &= N \sum_{\mathbf{r}, n_1, n_2} e^{i\mathbf{k} \cdot \mathbf{r}} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}(t) | \psi_{n_1}(\mathbf{r}_d) \rangle \langle \psi_{n_1}(\mathbf{r}_d) | T^\dagger(\mathbf{r}) | \psi_{n_2}(\mathbf{r}_d) \rangle \langle \psi_{n_2}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &= N \sum_{\mathbf{r}, n_1, n_2} e^{i\mathbf{k} \cdot \mathbf{r}} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}(t) | \psi_{n_1}(\mathbf{r}_d) \rangle \langle \psi_{n_1}(\mathbf{r}_d) | \psi_{n_2}(\mathbf{r}_d + \mathbf{r}) \rangle \langle \psi_{n_2}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle. \end{aligned} \quad (4.8)$$

The overlap $\langle \psi_{n_1}(\mathbf{r}_d) | \psi_{n_2}(\mathbf{r}_d + \mathbf{r}) \rangle$ between the eigenstates of the auxiliary models at $\mathbf{r}_d$ and $\mathbf{r}_d + \mathbf{r}$ are appreciable only for $n_1 = n_2$. This is because, for small $\mathbf{r}$, the auxiliary models are very similar and the states are therefore almost orthogonal. For large $\mathbf{r}$, the overlap is again suppressed because the auxiliary model eigenstates are localised around the impurity. We therefore make the simplifying assumption: $\langle \psi_{n_1}(\mathbf{r}_d) | \psi_{n_2}(\mathbf{r}_d + \mathbf{r}) \rangle = \delta_{n_1, n_2} \langle \psi_{n_1}(\mathbf{r}_d) | \psi_{n_1}(\mathbf{r}_d + \mathbf{r}) \rangle = \delta_{n_1, n_2} F_{n_1}(\mathbf{r})$. Substituting this, we get

$$\sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_2) | c_{\mathbf{k}\sigma}(t) c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_1) \rangle = N \sum_n F_n(\mathbf{k}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle, \quad (4.9)$$

where we have defined the Fourier transform $F_n(\mathbf{k}) = \sum_{\mathbf{r}} e^{i\mathbf{k} \cdot \mathbf{r}} F_n(\mathbf{r})$ of the overlap function.

We now consider more carefully the transition operator $c_{\mathbf{k}\sigma}$ for the 1-particle excitation giving rise to the above Greens function. Within our auxiliary model approach, gapless excitations within the lattice model are represented by gapless excitations of the impurity site, specifically those that try to screen the impurity site. As a result, the uncoordinated scattering process $c_{\mathbf{k}\sigma}$ for the lattice model must be replaced by a coherent excitation $\mathcal{T}$ within the impurity model that captures those gapless excitations that occur in connection with the impurity, and projects out the uncorrelated excitations that take place even when the impurity site is decoupled from the bath.

In order to construct this auxiliary model $\mathcal{T}$ -matrix, we allow for all excitations of the impurity site in conjunction with those of the $\mathbf{k}$ -state:

$$\mathcal{T}_{\mathbf{k}, \sigma} = c_{\mathbf{r}_d \sigma} + \left[c_{\mathbf{r}_d \sigma}^\dagger + S_{\mathbf{r}_d}^\sigma + C_{\mathbf{r}_d}^+ \right] c_{\mathbf{k}\sigma}, \quad (4.10)$$

where $S_{\mathbf{r}_d}^+$ and $C_{\mathbf{r}_d}^+$ are spin and charge flip operators on the impurity site that cause transition from $|\downarrow\rangle \rightarrow |\uparrow\rangle$ and $|0\rangle \rightarrow |\uparrow\downarrow\rangle$ respectively. The first term encodes the local component of the $\mathbf{k}$ -space Greens function by creating an excitation on the impurity site (which is then translated across the lattice), and the rest of the terms encode the non-local components by tracking impurity-bath excitations.

Apart from modifying the excitation operator, we will now simplify the time-evolution as well. The first overlap on the right-hand side of eq. 4.9 expands to

$$\langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k}, \sigma}(t) | \psi_n(\mathbf{r}_d) \rangle = \langle \psi_{\text{gs}}(\mathbf{r}_d) | e^{i(H_{\text{aux}}(\mathbf{r}_d) + H_{\text{cav}}(\mathbf{r}_d))t} \mathcal{T}_{\mathbf{k}, \sigma} e^{-i(H_{\text{aux}}(\mathbf{r}_d) + H_{\text{cav}}(\mathbf{r}_d))t} | \psi_n(\mathbf{r}_d) \rangle, \quad (4.11)$$

where $H_{\text{aux}}(\mathbf{r}_d)$ (eq. 2.3) is the auxiliary model Hamiltonian consisting of the cluster Hamiltonian $H_{\text{clust}}(\mathbf{r}_d)$ along with all one-particle terms H_{1p} in the Hamiltonian (in a typical lattice model, this will be the tight-binding hopping terms), and $H_{\text{cav}}(\mathbf{r}_d)$ is the ‘‘cavity’’ Hamiltonian consisting of all interaction (2-particle and beyond) terms in the Hamiltonian. Together, they create the full lattice Hamiltonian: $H_{\text{latt}} = H_{\text{aux}}(\mathbf{r}_d) + H_{\text{cav}}(\mathbf{r}_d)$. Using the Zassenhaus form of the Baker-Campbell-Hausdorff formula, we have

$$e^{i(H_{\text{aux}}(\mathbf{r}_d) + H_{\text{cav}}(\mathbf{r}_d))t} \approx e^{iH_{\text{aux}}(\mathbf{r}_d)t} e^{iH_{\text{cav}}(\mathbf{r}_d)t} e^{[H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)]t^2/2}, \quad (4.12)$$

where we have dropped the higher-order terms because the commutators involve only the boundary degrees of freedom and become subsequently smaller as the order increases. One can think of the

operator $e^{iH_{\text{cav}}(\mathbf{r}_d)t} e^{[H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)]t^2/2}$ as renormalising the auxiliary model $\mathcal{T}$ -matrix through hybridisation with the rest of the system and formally leading to a new $\mathcal{T}$ -matrix:

$$\mathcal{T}_{\mathbf{k},\sigma}^* = e^{iH_{\text{cav}}(\mathbf{r}_d)t} e^{[H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)]t^2/2} \mathcal{T}_{\mathbf{k},\sigma} e^{-[H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)]t^2/2} e^{-iH_{\text{cav}}(\mathbf{r}_d)t} . \quad (4.13)$$

One can also absorb the modulating phase factor $F_n(\mathbf{k})$ (defined above and appearing in eq. 4.9) into this renormalisation:

$$\mathcal{T}_{\mathbf{k},\sigma}^* = e^{iH_{\text{cav}}(\mathbf{r}_d)t} e^{[H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)]t^2/2} \mathcal{T}_{\mathbf{k},\sigma} F_n(\mathbf{k}) \mathcal{P}_n(\mathbf{r}_d) e^{-[H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)]t^2/2} e^{-iH_{\text{cav}}(\mathbf{r}_d)t} , \quad (4.14)$$

where $\mathcal{P}_n(\mathbf{r}_d)$ projects onto the n^{th} eigenstate $|\psi_n(\mathbf{r}_d)\rangle$.

Since the overlap is calculated in the eigenstates of the auxiliary model at $\mathbf{r}_d$, we adopt the zeroth order approximation of ignoring this renormalisation. Later, we will specify a prescription of systematically improving our computation. The overlap expression becomes

$$\langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle = \langle \psi_{\text{gs}}(\mathbf{r}_d) | e^{i(H_{\text{clust}}(\mathbf{r}_d) + H_{1\text{p}})t} \mathcal{T}_{\mathbf{k},\sigma} e^{-i(H_{\text{clust}}(\mathbf{r}_d) + H_{1\text{p}})t} | \psi_n(\mathbf{r}_d) \rangle . \quad (4.15)$$

To make the problem tractable, we treat the effect of $H_{1\text{p}}$ at the level of an expectation value and note that the effect of the transition operator is to create a hole of momentum $\mathbf{k}$ (eq. 4.10). Let the Hamiltonian $H_{1\text{p}}$ be diagonal in certain modes $c_{\nu,\sigma}$ with energies ϵ_ν , and the operator $c_{\mathbf{k},\sigma}$ can be expressed in terms of these modes: $c_{\mathbf{k},\sigma} = \sum_\nu C_{\nu,\mathbf{k}} c_{\nu,\sigma}^\dagger$. A similar decomposition will hold for the $\mathcal{T}$ -matrix. Substituting it gives

$$\langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle = \sum_\nu C_{\nu,\mathbf{k}} \langle \psi_{\text{gs}}(\mathbf{r}_d) | e^{i(H_{\text{clust}}(\mathbf{r}_d) + H_{1\text{p}})t} \mathcal{T}_{\nu,\sigma} e^{-i(H_{\text{clust}}(\mathbf{r}_d) + H_{1\text{p}})t} | \psi_n(\mathbf{r}_d) \rangle . \quad (4.16)$$

Each excitation $\mathcal{T}_{\nu,\sigma}$ therefore leads to an energy cost $-\epsilon_\nu$. Moreover, $|\psi_n(\mathbf{r}_d)\rangle$ and $|\psi_{\text{gs}}(\mathbf{r}_d)\rangle$ are eigenstates of $H_{\text{clust}}(\mathbf{r}_d)$ with energies E_n and E_0 . This leads to the following simplified expression:

$$\langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle = \sum_\nu C_{\nu,\mathbf{k}} e^{-i(E_n - E_0 + \epsilon_\nu)t} \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\nu,\sigma} | \psi_n(\mathbf{r}_d) \rangle . \quad (4.17)$$

Treating the other overlap in eq. 4.9 in an identical fashion gives

$$\langle \psi_n(\mathbf{r}_d) | c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle = \sum_\nu C_{\nu,\mathbf{k}} \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{\nu,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle . \quad (4.18)$$

Substituting the expression back into 4.9 and then into 4.5, we get an expression for one half of the retarded Greens function:

$$\begin{aligned} & -i\theta(t) \langle \Psi_{\text{gs}} | c_{\mathbf{k}\sigma}(t) c_{\mathbf{k}\sigma}^\dagger | \Psi_{\text{gs}} \rangle \\ &= -i\theta(t) \sum_{\nu,n} C_{\nu,\mathbf{k}} e^{-i(E_n - E_0 + \epsilon_\nu)t} \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma} | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{\nu,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle . \end{aligned} \quad (4.19)$$

Working through the other term with the same strategy, we get the full expression:

$$\begin{aligned} & -i\theta(t) \langle \Psi_{\text{gs}} | \{c_{\mathbf{k}\sigma}(t), c_{\mathbf{k}\sigma}^\dagger\} | \Psi_{\text{gs}} \rangle \\ &= -i\theta(t) \sum_{\nu,n} C_{\nu,\mathbf{k}} e^{-i(E_n - E_0 + \epsilon_\nu)t} \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma} | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{\nu,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ & \quad - i\theta(t) \sum_{\nu,n} C_{\nu,\mathbf{k}} e^{i(E_n - E_0 - \epsilon_\nu)t} \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}^\dagger | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{\nu,\sigma} | \psi_{\text{gs}}(\mathbf{r}_d) \rangle . \end{aligned} \quad (4.20)$$

Changing from time to frequency domain using $g(\omega) = \int dt e^{i\omega T} f(t)$ gives

$$G(\mathbf{k}, \sigma; \omega) = \sum_{v,n} C_{v,k} \left[\frac{\langle \psi_{gs}(\mathbf{r}_d) | \mathcal{T}_{v,\sigma} | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{v,\sigma}^\dagger | \psi_{gs}(\mathbf{r}_d) \rangle}{\omega - (E_n - E_0 + \epsilon_v)} + \frac{\langle \psi_{gs}(\mathbf{r}_d) | \mathcal{T}_{v,\sigma}^\dagger | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{v,\sigma} | \psi_{gs}(\mathbf{r}_d) \rangle}{\omega + (E_n - E_0 - \epsilon_v)} \right]. \quad (4.21)$$

We recognise that the two parts of the lattice model Greens function are the greater and lesser Greens functions of the operator $\mathcal{T}_{\mathbf{k},\sigma}$, calculated within the auxiliary model at $\mathbf{r}_d$, but at shifted frequencies $\omega \mp \epsilon_k$:

$$G(\mathbf{k}, \sigma; \omega) = \sum_v C_{v,k} [\mathcal{G}^>(\mathcal{T}_{v,\sigma}; \omega - \epsilon_v) + \mathcal{G}^<(\mathcal{T}_{v,\sigma}; \omega + \epsilon_v)] . \quad (4.22)$$

We now discuss how to systematically improve this estimate. One way is to include the effect of other auxiliary models in the overlap function $F_n(\mathbf{r})$. This will of course require us to calculate an off-diagonal Greens function. As a concrete example, upon including the effects of the auxiliary models nearest to the reference point $\mathbf{r}_d$, the expression for the Greens function becomes

$$G(\mathbf{k}, \sigma; \omega) = \sum_{v,n} C_{v,k} \left[\frac{\langle \psi_{gs}(\mathbf{r}_d) | \mathcal{T}_{v,\sigma}^* | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{v,\sigma}^\dagger | \psi_{gs}(\mathbf{r}_d) \rangle}{\omega - (E_n - E_0 + \epsilon_v)} + \frac{\langle \psi_{gs}(\mathbf{r}_d) | \mathcal{T}_{v,\sigma}^\dagger | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{v,\sigma}^* | \psi_{gs}(\mathbf{r}_d) \rangle}{\omega + (E_n - E_0 - \epsilon_v)} \right], \quad (4.23)$$

with $\tau_{v,\sigma}^* = \mathcal{T}_{v,\sigma}(F_n(0) + \sum_a F_n(\mathbf{a})e^{i\mathbf{k}\cdot\mathbf{a}})$ where $\mathbf{a}$ sums over the primitive lattice vectors. Another way to include more non-local contributions is by considering the effects of the cavity Hamiltonian in eq. 4.14:

$$\mathcal{T}_{\mathbf{k},\sigma}^*(t) = e^{iH_{\text{cav}}(\mathbf{r}_d)t} (\mathcal{T}_{\mathbf{k},\sigma} + t^2/2 [H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)], \mathcal{T}_{\mathbf{k},\sigma}) e^{-iH_{\text{cav}}(\mathbf{r}_d)t} . \quad (4.24)$$

The time-domain Greens function associated with this operator will then need to be calculated, which can then be used to calculate frequency-domain functions.

3.4.2 One-particle real-space Greens functions

We now work with Greens functions for real-space excitations. The retarded Greens function in this case is

$$G(\mathbf{r}\sigma; t) = -i\theta(t) \langle \Psi_{gs} | \{c_{\mathbf{r}_d+\mathbf{r},\sigma}(t), c_{\mathbf{r}_d,\sigma}^\dagger\} | \Psi_{gs} \rangle . \quad (4.25)$$

Since our lattice model has discrete translation symmetry, we instead calculate the equivalent quantity

$$G(\mathbf{r}\sigma; t) = -i\theta(t) \sum_{\mathbf{r}_d} \langle \Psi_{gs} | \{c_{\mathbf{r}_d+\mathbf{r},\sigma}(t), c_{\mathbf{r}_d,\sigma}^\dagger\} | \Psi_{gs} \rangle . \quad (4.26)$$

Similar to the k -space Greens functions, express the groundstates as a superposition of auxiliary model Greens functions. We take the first term in the anticommutator and proceed to simplify it:

$$\begin{aligned} \sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d+\mathbf{r},\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \Psi_{\text{gs}} \rangle &= \frac{1}{N} \sum_{\mathbf{R}, \Delta, \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{R} + \Delta) | c_{\mathbf{r}_d+\mathbf{r},\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{R}) \rangle \\ &= \frac{1}{N} \sum_{\mathbf{R}, \Delta} \langle \psi_{\text{gs}}(-\mathbf{r}_d) | c_{\mathbf{r}-\mathbf{R}-\Delta,\sigma}(t) T(\Delta) c_{-\mathbf{R},\sigma}^\dagger | \psi_{\text{gs}}(-\mathbf{r}_d) \rangle . \end{aligned} \quad (4.27)$$

Local Greens function: $\mathbf{r} = 0$

We first consider the case of local Greens function, by setting $\mathbf{r} = 0$. We also change the dummy variable $\mathbf{r}_d$, $\mathbf{R}$ and Δ to $-\mathbf{r}_d$, $\mathbf{R}$ and $-\Delta$ to make the expressions tidier:

$$\sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d,\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \Psi_{\text{gs}} \rangle = \frac{1}{N} \sum_{\mathbf{R}, \Delta, \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{R}+\Delta,\sigma}(t) T^\dagger(\Delta) c_{\mathbf{R},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle . \quad (4.28)$$

We split the full summation into four parts, based on whether $\mathbf{R} = \mathbf{r}_d$ and whether $\Delta + \mathbf{R} = \mathbf{r}_d$:

$$\begin{aligned} \sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d,\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \Psi_{\text{gs}} \rangle &= \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d,\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &\quad + \sum_{\Delta \neq 0} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d+\Delta,\sigma}(t) T^\dagger(\Delta) c_{\mathbf{r}_d,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &\quad + \sum_{\mathbf{R} \neq \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d,\sigma}(t) T^\dagger(\mathbf{r}_d - \mathbf{R}) c_{\mathbf{R},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &\quad + \sum_{\mathbf{R} \neq \mathbf{r}_d, \Delta \neq \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\Delta,\sigma}(t) T^\dagger(\Delta - \mathbf{R}) c_{\mathbf{R},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle , \end{aligned} \quad (4.29)$$

where we have used the fact that the location of the auxiliary model does not affect the correlation calculated within the auxiliary model in order to cancel out the summation over $\mathbf{r}_d$ with the factor of $1/N$. The reason for splitting the expression into these parts is to explicitly write out which operators act on the impurity site $\mathbf{r}_d$ and which act on other real-space sites away from the impurity site. We will now replace the real-space excitation operators away from the impurity site with the associated T -matrices, following the discussion above eq. 4.10:

$$\mathcal{T}_{\mathbf{r},\sigma} = \left[\left(\sum_{\sigma'} c_{\mathbf{r}_d\sigma'}^\dagger + \text{h.c.} \right) + \left(S_{\mathbf{r}_d}^+ + \text{h.c.} \right) + \left(C_{\mathbf{r}_d}^+ + \text{h.c.} \right) \right] c_{\mathbf{r}\sigma} = \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} \mathcal{T}_{\mathbf{k},\sigma} . \quad (4.30)$$

We also insert a complete basis of states of the auxiliary model at $\mathbf{r}_d$, $1 = \sum_n |\psi_n(\mathbf{r}_d)\rangle \langle \psi_n(\mathbf{r}_d)|$, generating an overlap factor F_n as before:

$$\begin{aligned}
 & \sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d, \sigma}(t) c_{\mathbf{r}_d, \sigma}^\dagger | \Psi_{\text{gs}} \rangle \\
 &= \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d, \sigma}(t) c_{\mathbf{r}_d, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\
 &+ \sum_{\mathbf{k}, n} F'_n(\mathbf{k}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k}, \sigma}(t) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | c_{\mathbf{r}_d, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\
 &+ \sum_{\mathbf{k}, n} F'_n(\mathbf{k})^* \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d, \sigma}(t) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k}, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\
 &+ \sum_{\mathbf{k}_1, \mathbf{k}_2} F'_n(\mathbf{k}_1) F'_n(\mathbf{k}_2)^* \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k}_1, \sigma}(t) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k}_2, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle ,
 \end{aligned} \tag{4.31}$$

where $F'_n(\mathbf{k})$ is the truncated Fourier transform $\sum_{\mathbf{r} \neq 0} e^{i\mathbf{k} \cdot \mathbf{r}} F_n(\mathbf{r})$ of the overlap factor.

We next treat the time-evolution of the operators exactly as in the k -space Greens function. Without repeating the details, we state the final result here, for the frequency-domain real-space local Greens function:

$$\begin{aligned}
 & G_{\text{loc}}(\omega) \\
 &= \mathcal{G}(d\sigma; \omega) + \sum_{\mathbf{k}, \nu} C_{\nu, \mathbf{k}} \left[\mathcal{G}^>(\mathcal{T}_{\nu, \sigma}, d\sigma; \omega - \epsilon_\nu) + \mathcal{G}^<(\mathcal{T}_{\nu, \sigma}, d\sigma; \omega + \epsilon_\nu) \right] \\
 &+ \sum_{\mathbf{k}} \mathcal{G}(d\sigma, \mathcal{T}_{\mathbf{k}, \sigma}; \omega) + \sum_{\mathbf{k}, \nu, \mathbf{q}} C_{\nu, \mathbf{k}} \left[\mathcal{G}^>(\mathcal{T}_{\nu, \sigma}, \mathcal{T}_{\mathbf{q}, \sigma}; \omega - \epsilon_\nu) + \mathcal{G}^<(\mathcal{T}_{\nu, \sigma}, \mathcal{T}_{\mathbf{q}, \sigma}; \omega + \epsilon_\nu) \right] ,
 \end{aligned} \tag{4.32}$$

where $\mathcal{G}(d\sigma; \omega)$ is the retarded Greens function of the impurity site, $\mathcal{G}^>(\mathcal{T}_{\nu, \sigma}, d\sigma; \omega)$ is the off-diagonal retarded Greens function between the impurity site and the bath state $\nu\sigma$. $\mathcal{G}^>(\mathcal{T}_{\nu, \sigma}, \mathcal{T}_{\mathbf{q}, \sigma}; \omega)$ is the same but between multiple states in the bath.

Non-local Greens function: $\mathbf{r} \neq 0$

We next consider the case of non-local Greens function, where the excitations propagate a non-zero distance $\mathbf{r}$:

$$\sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d + \mathbf{r}, \sigma}(t) c_{\mathbf{r}_d, \sigma}^\dagger | \Psi_{\text{gs}} \rangle = \frac{1}{N} \sum_{\mathbf{R}, \Delta, \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{R} + \Delta + \mathbf{r}, \sigma}(t) T^\dagger(\Delta) c_{\mathbf{R}, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle . \tag{4.33}$$

To bring this expression closer to the one for the local Greens function, we redefine $\Delta + \mathbf{r} \rightarrow \Delta$:

$$\sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d + \mathbf{r}, \sigma}(t) c_{\mathbf{r}_d, \sigma}^\dagger | \Psi_{\text{gs}} \rangle = \frac{1}{N} \sum_{\mathbf{R}, \Delta, \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{R} + \Delta, \sigma}(t) T^\dagger(\Delta) T(\mathbf{r}) c_{\mathbf{R}, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle . \tag{4.34}$$

This allows to again split the full summation based on whether $\mathbf{R} = \mathbf{r}_d$ and whether $\Delta + \mathbf{R} = \mathbf{r}_d$:

$$\begin{aligned}
 \sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d + \mathbf{r}, \sigma}(t) c_{\mathbf{r}_d, \sigma}^\dagger | \Psi_{\text{gs}} \rangle &= \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d, \sigma}(t) T(\mathbf{r}) c_{\mathbf{r}_d, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\
 &+ \sum_{\Delta \neq 0} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d + \Delta, \sigma}(t) T^\dagger(\Delta - \mathbf{r}) c_{\mathbf{r}_d, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\
 &+ \sum_{\mathbf{R} \neq \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d, \sigma}(t) T^\dagger(\mathbf{r}_d - \mathbf{R} - \mathbf{r}) c_{\mathbf{R}, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\
 &+ \sum_{\mathbf{R} \neq \mathbf{r}_d, \Delta \neq \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\Delta, \sigma}(t) T^\dagger(\Delta - \mathbf{R} - \mathbf{r}) c_{\mathbf{R}, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle .
 \end{aligned} \tag{4.35}$$

Introducing Fourier transforms of the operators and inserting eigenstates of the auxiliary models leads to

$$\sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d + \mathbf{r}, \sigma}(t) c_{\mathbf{r}_d, \sigma}^\dagger | \Psi_{\text{gs}} \rangle = F_n^*(\mathbf{r}) \sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d, \sigma}(t) c_{\mathbf{r}_d, \sigma}^\dagger | \Psi_{\text{gs}} \rangle . \tag{4.36}$$

The net difference from the diagonal Greens function is the presence of the overlap factor $F_n^*(\mathbf{r})$.

3.4.3 Equal-Time Ground State Correlators: Real-space

We first consider a real-space operator $\mathcal{O}(\mathbf{r} + \Delta) \mathcal{O}^\dagger(\mathbf{r})$ that quantifies the presence of correlations over a distance Δ . The correlation function itself is given by the expectation value of this operator in the ground state:

$$C_O(\Delta) = \langle \Psi_{\text{gs}} | \mathcal{O}(\mathbf{r} + \Delta) \mathcal{O}^\dagger(\mathbf{r}) | \Psi_{\text{gs}} \rangle . \tag{4.37}$$

To obtain a tractable expression for this, we first replace the full ground state with its expression in terms of the auxiliary model ground states (eq. ??):

$$C_O(\Delta) = \frac{1}{N} \sum_{\mathbf{r}_1, \mathbf{r}'} \langle \psi_{\text{gs}}(\mathbf{r}_1 + \mathbf{r}' + \mathbf{r}_c + \mathbf{r}) | \mathcal{O}(\mathbf{r} + \Delta) \mathcal{O}^\dagger(\mathbf{r}) | \psi_{\text{gs}}(\mathbf{r}_1 + \mathbf{r}_c + \mathbf{r}) \rangle e^{-i\mathbf{k}_0 \cdot (\mathbf{r}_1 - \mathbf{r}_2)} , \tag{4.38}$$

where $\mathbf{r}_1$ and $\mathbf{r}_1 + \mathbf{r}'$ are the positions of the incoming and outgoing auxiliary model states, relative to $\mathbf{r}_c + \mathbf{r}$, and $\mathbf{k}_0$ is the crystal momentum of the ground state (which we will immediately set to zero). In order to convert the incoming and outgoing states into the same auxiliary mode at a reference location $\mathbf{r}_c$ (which would then allow computations purely within a single auxiliary model), we use the relation: $|\psi_{\text{gs}}(\mathbf{x})\rangle = T^\dagger(\mathbf{x} - \mathbf{r}_c) |\psi_{\text{gs}}(\mathbf{r}_c)\rangle$. Substituting this appropriately for both the auxiliary model states gives:

$$\begin{aligned}
 C_O(\Delta) &= \frac{1}{N} \sum_{\mathbf{r}_1, \mathbf{r}'} \langle \psi_{\text{gs}}(\mathbf{r}_c) | T(\mathbf{r}_1 + \mathbf{r}' + \mathbf{r}) \mathcal{O}(\mathbf{r} + \Delta) \mathcal{O}^\dagger(\mathbf{r}) T^\dagger(\mathbf{r}_1 + \mathbf{r}) | \psi_{\text{gs}}(\mathbf{r}_c) \rangle \\
 &= \frac{1}{N} \sum_{\mathbf{r}_1, \mathbf{r}'} \langle \psi_{\text{gs}}(\mathbf{r}_c) | T(\mathbf{r}') \mathcal{O}(\Delta - \mathbf{r}_1) \mathcal{O}^\dagger(-\mathbf{r}_1) | \psi_{\text{gs}}(\mathbf{r}_c) \rangle \\
 &= \frac{1}{N} \sum_{\mathbf{r}_1, \mathbf{r}'} \langle \psi_{\text{gs}}(\mathbf{r}_c + \mathbf{r}') | \mathcal{O}(\Delta - \mathbf{r}_1) \mathcal{O}^\dagger(-\mathbf{r}_1) | \psi_{\text{gs}}(\mathbf{r}_c) \rangle .
 \end{aligned} \tag{4.39}$$

As a final cosmetic change, we transform $\mathbf{r}_1 \rightarrow -\mathbf{r}$:

$$C_O(\Delta) = \frac{1}{N} \sum_{\mathbf{r}, \mathbf{r}'} \langle \psi_{\text{gs}}(\mathbf{r}_c + \mathbf{r}') | O(\Delta + \mathbf{r}) O^\dagger(\mathbf{r}) | \psi_{\text{gs}}(\mathbf{r}_c) \rangle . \quad (4.40)$$

Like in the Greens function calculation, we now consider the various kinds of contributions separately.

Intra-auxiliary model contributions: $\mathbf{r}' = 0$

These terms describe excitations that start and propagate within the same auxiliary model, upto a distance Δ :

$$C_O(\Delta) \rightarrow \frac{1}{N} \sum_{\mathbf{r}} \langle \psi_{\text{gs}}(\mathbf{r}_c) | O(\Delta + \mathbf{r}) O^\dagger(\mathbf{r}) | \psi_{\text{gs}}(\mathbf{r}_c) \rangle . \quad (4.41)$$

With an eye towards introducing the impurity operators into the correlations, we insert a complete basis defined by the eigenstates $\{|\psi_n(\mathbf{r}_c)\rangle\}$ of the auxiliary model into the expression:

$$C_O(\Delta) \rightarrow \frac{1}{N} \sum_{\mathbf{r}} \sum_n \langle \psi_{\text{gs}}(\mathbf{r}_c) | O(\Delta + \mathbf{r}) | \psi_n(\mathbf{r}_c) \rangle \langle \psi_n(\mathbf{r}_c) | O^\dagger(\mathbf{r}) | \psi_{\text{gs}}(\mathbf{r}_c) \rangle . \quad (4.42)$$

For the excitations that exist purely in the conduction bath ($\mathbf{r} \neq \mathbf{r}_c$), the excitation operators must be suitably modified (see the arguments around eq. 4.10) in order to incorporate Kondo screening. The modified excitation operators are generally defined as

$$\tilde{O}(\mathbf{r}) = O(\mathbf{r}) O^\dagger(d) \mathcal{P}_{\text{gs}} , \quad (4.43)$$

where $O^\dagger(d)$ is the hermitian conjugate of the correlation operator O , but applied on the impurity sites. This therefore constitutes a time-reversed scattering process on the impurity site relative to the process in the bath. The operator $\mathcal{P}_{\text{gs}}$ projects onto the ground state of the auxiliary model, since we are interested in ground state correlations. The operator multiplying the old correlation operator represents all possible excitations of the impurity site, and ensure that the bath and impurity excitation processes take place coherently. The projector ensures that only tripartite correlations between the impurity site and the two momentum states are captured by the correlation, which is what's desired within the tiling method.

With this modified operator, this class of correlation functions can be written as

$$C_O(\Delta) \rightarrow \frac{1}{N} \sum_{\mathbf{r}} \langle \psi_{\text{gs}}(\mathbf{r}_c) | \tilde{O}(\Delta + \mathbf{r}) | \psi_{\text{gs}}(\mathbf{r}_c) \rangle \langle \psi_{\text{gs}}(\mathbf{r}_c) | \tilde{O}^\dagger(\mathbf{r}) | \psi_{\text{gs}}(\mathbf{r}_c) \rangle . \quad (4.44)$$

This can be interpreted as the fact that within our auxiliary model formalism, the correlation between the sites $\mathbf{r}$ and $\Delta + \mathbf{r}$ can only occur through a transition process that connect one of the sites with the impurity site and then a return process that connects the impurity site with the other site.

Inter-auxiliary model contributions: $r' \neq 0$

These are the most non-local contributions, and involve excitations that connect different auxiliary models:

$$C_O(\Delta) \rightarrow \frac{1}{N} \sum_{r' \neq 0} \langle \psi_{\text{gs}}(r_c + r') | \tilde{O}(\Delta + r) | \psi_{\text{gs}}(r_c) \rangle \langle \psi_{\text{gs}}(r_c) | \tilde{O}^\dagger(r) | \psi_{\text{gs}}(r_c) \rangle . \quad (4.45)$$

3.4.4 Equal-Time Ground State Correlators: Momentum-space

We now consider momentum space correlations, through a general operator $O(k_2)O^\dagger(k_1)$:

$$C_O(k_1, k_2) = \langle \Psi_{\text{gs}} | O(k_2)O^\dagger(k_1) | \Psi_{\text{gs}} \rangle . \quad (4.46)$$

Note that O itself is a two-particle operator. To obtain a tractable expression for this, we first replace the full ground state with its expression in terms of the auxiliary model ground states (eq. ??):

$$C_O(k_1, k_2) = \frac{1}{N} \sum_{r_1, r_2} \langle \psi_{\text{gs}}(r_2) | O(k_2)O^\dagger(k_1) | \psi_{\text{gs}}(r_1) \rangle , \quad (4.47)$$

where r_1 and r_2 are the positions of the incoming and outgoing auxiliary model states. In order to convert the incoming and outgoing states into the same auxiliary mode at a reference location r_c (which would then allow computations purely within a single auxiliary model), we use the relation: $|\psi_{\text{gs}}(x)\rangle = T^\dagger(x - r_c) \psi_{\text{gs}} | \psi(r_c) \rangle$. Substituting this appropriately for both the auxiliary model states gives:

$$C_O(k_1, k_2) = \frac{1}{N} \sum_{r_1, r_2} \langle \psi_{\text{gs}}(r_c) | T(r_2 - r_c) O(k_2)O^\dagger(k_1) T^\dagger(r_1 - r_c) | \psi_{\text{gs}}(r_c) \rangle . \quad (4.48)$$

In order to simplify the translation operators, we can use eq. 3.8. For that, we would need to know whether the operator O involves a net transfer of momentum. If $O(k)$ commutes with the total number operator $n_k = \sum_\sigma n_{k,\sigma}$, there is no momentum transfer. Examples of such operators are spin operators, $S_{\alpha\beta}(k) \equiv c_{k\alpha}^\dagger c_{k\beta}$, and density operators $n_{k\alpha}$. If $O(k)$ does not commute with n_k , there is a net transfer of momentum, and one such operator would be the charge isospin operator $C^+(k) = c_{k\uparrow}^\dagger c_{k\downarrow}^\dagger$. For the first kind of operators, we have $T(a)O(k)T^\dagger(a) = O(k)$, while the for the latter, we get $T(a)O(k)T^\dagger(a) = e^{-a \cdot 2k} O(k)$, where $2k$ represents the momentum being transferred by the operator. We consider the two cases separately.

Momentum-conserving operators: $[O(k), n_k] = 0$

For these operators, the expression for the correlation gives

$$\begin{aligned} C_O(k_1, k_2) &= \frac{1}{N} \sum_{r_1, r_2} \langle \psi_{\text{gs}}(r_c) | T(r_2 - r_1) O(k_2)O^\dagger(k_1) | \psi_{\text{gs}}(r_c) \rangle \\ &= \sum_{\Delta} \langle \psi_{\text{gs}}(r_c + \Delta) | O(k_2)O^\dagger(k_1) | \psi_{\text{gs}}(r_c) \rangle . \end{aligned} \quad (4.49)$$

To obtain the last form, we defined $\Delta = \mathbf{r}_2 - \mathbf{r}_1$ as the distance between the incoming and outgoing auxiliary model states, and preformed the sum over the free variable $\mathbf{r}_1$ to cancel out the factor of $1/N$. Just like before, this expression can be decomposed into a term that involves a single auxiliary model and other terms that involve two distinct auxiliary models. By making the identification of the right transition operator $\tilde{O}(\mathbf{k}) = O(\mathbf{k})O^\dagger(d)\mathcal{P}_{\text{gs}}$, we get

$$C_O(\mathbf{k}_1, \mathbf{k}_2) = \sum_{\Delta} \langle \psi_{\text{gs}}(\mathbf{r}_c + \Delta) | \tilde{O}(\mathbf{k}_2) | \psi_{\text{gs}}(\mathbf{r}_c) \rangle \langle \psi_{\text{gs}}(\mathbf{r}_c) | \tilde{O}^\dagger(\mathbf{k}_1) | \psi_{\text{gs}}(\mathbf{r}_c) \rangle . \quad (4.50)$$

Non-momentum-conserving operators: $[O(\mathbf{k}), n_{\mathbf{k}}] \neq 0$

This class of operators incur an additional phase factor of $e^{-\mathbf{a} \cdot 2\mathbf{k}}$ when the translation operators are translated across them:

$$\begin{aligned} C_O(\mathbf{k}_1, \mathbf{k}_2) &= \frac{1}{N} \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_c) | T(\mathbf{r}_2 - \mathbf{r}_1) O(\mathbf{k}_2) O^\dagger(\mathbf{k}_1) | \psi_{\text{gs}}(\mathbf{r}_c) \rangle e^{2i(\mathbf{r}_1 - \mathbf{r}_c) \cdot (\mathbf{k}_1 - \mathbf{k}_2)} \\ &= \delta_{\mathbf{k}_1, \mathbf{k}_2} \sum_{\Delta} \langle \psi_{\text{gs}}(\mathbf{r}_c + \Delta) | O(\mathbf{k}_2) O^\dagger(\mathbf{k}_1) | \psi_{\text{gs}}(\mathbf{r}_c) \rangle . \end{aligned} \quad (4.51)$$

For this expression, we again defined Δ similar to before, and carried out the sum of $\mathbf{r}_1$ involving the exponential to obtain a factor of $N\delta_{\mathbf{k}_1, \mathbf{k}_2}$. The Kronecker delta factor is a manifestation of translational invariance and the associated total momentum conservation.

As it stands, operators like $O(\mathbf{k}_1)$ act purely on the conduction bath degrees of freedom. In order to incorporate impurity-bath correlation effects, we modify these operators using the appropriate T -matrices, by employing eq. 10.5. The final computations are carried out using these modified operators.

3.4.5 Entanglement Measures

We will now describe the prescription of calculating entanglement measures of the lattice model from within our auxiliary model treatment. In this section, we are interested mainly in two such measures, the entanglement entropy and the mutual information. Given a pure state $|\Psi\rangle$ describing the complete system, the entanglement entropy $S_{\text{EE}}(\nu)$ of a subsystem ν quantifies the entanglement of ν with the rest of the subsystem, and is defined as

$$S_{\text{EE}}(\nu) = -\text{Tr} [\rho(\nu) \log \rho(\nu)] , \quad \rho(\nu) = \text{Tr}_\nu [|\Psi\rangle \langle \Psi|] \quad (4.52)$$

where $\text{Tr} [\cdot]$ is the trace operator, and $\rho(\nu)$ is the reduced density matrix (RDM) for the subsystem ν obtained by taking the partial trace Tr_ν (over the states of ν) of the full density matrix $\rho = |\Psi\rangle \langle \Psi|$. If the subsystem ν describes local regions in real space (or states in k -space), we might be interested in the entanglement between two such subsystems ν_1 and ν_2 . The correct measure to quantify such entanglement is the mutual information:

$$I_2(\nu_1, \nu_2) = S_{\text{EE}}(\nu_1) + S_{\text{EE}}(\nu_2) - S_{\text{EE}}(\nu_1 \cup \nu_2) , \quad (4.53)$$

where $\nu_1 \cup \nu_2$ is a larger subsystem formed by combining ν_1 and ν_2 .

Real-space entanglement

Real-space entanglement measures can be used to probe delocalisation-localisation transitions. The simplest such measure is the entanglement of a local mode. Since the local entanglement entropy will be uniform at each lattice site for a system with translation invariance, it suffices to calculate the real-space averaged entanglement entropy $S_{EE}^{\text{loc}} = \frac{1}{N} \sum_{\mathbf{r}} S_{EE}(\mathbf{r})$. By visualising the lattice model as a superposition of auxiliary models placed at various sites (eq. ??), the real space average of the lattice model can be thought of as an average over sites of a particular impurity model, and then a second average over all the impurity models. But since all impurity models are equivalent to each other, the second average is redundant. Secondly, all correlations must derive from the impurity site, which can be formally encoded by subtracting, from this average, the corresponding contribution obtained in the absence of the impurity site. In total, the lattice-auxiliary model relation for the local entanglement entropy can be written as

$$S_{EE}^{\text{loc}} = \frac{1}{N} \sum_{\mathbf{r}} \left[S_{EE}(\mathbf{r}_d + \mathbf{r}) - S_{EE}^{(0)}(\mathbf{r}_d + \mathbf{r}) \right], \quad (4.54)$$

where $\mathbf{r}_d$ is the impurity site position, $\mathbf{r}$ is the distance of a conduction bath site from the impurity site (can be zero), and $S_{EE}(\mathbf{r}_d + \mathbf{r})$ is the entanglement entropy, calculated within the impurity model, at the location $\mathbf{r}_d + \mathbf{r}$. $S_{EE}^{(0)}(\mathbf{r}_d + \mathbf{r})$ is the same entanglement entropy, but calculated for an impurity model with vanishing impurity-bath hybridisation and bath interaction.

Momentum-space entanglement

Entanglement measures in k -space can provide valuable information regarding the Fermi surface structure and the nature of gapless excitations proximate to it. The most elementary measure is the entanglement entropy $S_{EE}(\mathbf{q})$ of a single excitation carrying momentum $\mathbf{q}$. Unlike S_{EE}^{loc} , this does not involve an average, and involves a single computation:

$$\tilde{S}_{EE}(\mathbf{q}) = S_{EE}(\mathbf{q}) - S_{EE}^{(0)}(\mathbf{q}). \quad (4.55)$$

It is possible to improve this by considering inter-auxiliary model contributions in the reduced density matrix $\rho(\mathbf{q})$. On the lattice model, $\rho(\mathbf{q})$ is defined as

$$\rho(\mathbf{q}) = \text{Tr}_{\mathbf{q}} \left[|\Psi_{\text{gs}}\rangle \langle \Psi_{\text{gs}}| \right], \quad (4.56)$$

where $\text{Tr}_{\mathbf{q}}[\cdot]$ is the partial trace over the Hilbert space of $\mathbf{q}$, and $|\Psi_{\text{gs}}\rangle$ is the tiled ground state. Using eq. ??, we can write the ground state in terms of those of the auxiliary model. These leads to two classes of terms, one purely within a single auxiliary model ground state $|\psi_{\text{gs}}\rangle$, and the other involving transitions across auxiliary models:

$$\rho(\mathbf{q}) \sim \text{Tr}_{\mathbf{q}} \left[|\psi_{\text{gs}}(\mathbf{r}_d)\rangle \langle \psi_{\text{gs}}(\mathbf{r}_d)| \right] + \frac{1}{N} \text{Tr}_{\mathbf{q}} \sum_{\mathbf{r}} \left[|\psi_{\text{gs}}(\mathbf{r}_d)\rangle \langle \psi_{\text{gs}}(\mathbf{r}_d + \mathbf{r})| \right], \quad (4.57)$$

where $|\psi_{\text{gs}}(\mathbf{r}_d)\rangle$ is a reference auxiliary model ground state, and $|\psi_{\text{gs}}(\mathbf{r}_d + \mathbf{r})\rangle$ sums over other auxiliary models at increasing distances from this reference model. The $\sim$ indicates that the RDM

needs to be normalised. The first term leads to the expression in eq. 4.56, while the second term can be used to improve this estimate. Away from any quantum critical points, the localised nature of the impurity-bath hybridisation ensures that impurity correlations decay exponentially away from the transition. This ensures that the second term is very small away from a critical point. We therefore restrict ourselves to just the first term in the present work.

3.5 Properties of the Bloch states

3.5.1 Translation invariance

It is easy to verify that $\Psi_{\vec{k}}(\{\mathbf{r}_k\})$ transforms like Bloch functions under translation by a displacement $\mathbf{r}$:

$$\Psi_{\vec{k}}(\{\mathbf{r}_k + \mathbf{r}\}) = \frac{1}{\lambda_{|\vec{k}\rangle} \mathcal{Z} N} \sum_{\mathbf{r}_i, \mathbf{a}} e^{i\vec{k} \cdot \vec{R}_i} \psi_{\text{aux}}(\mathbf{r}_i - \mathbf{r}, \mathbf{a}) = \frac{1}{\lambda_{|\vec{k}\rangle} \mathcal{Z} N} \sum_{\mathbf{r}_j, \mathbf{a}} e^{i\vec{k} \cdot (\vec{R}_j + \mathbf{r})} \psi_{\text{aux}}(\mathbf{r}_j, \mathbf{a}) = e^{i\vec{k} \cdot \vec{R}} \Psi_{\vec{k}}(\{\mathbf{r}_k\}) \quad (5.1)$$

In the last equation, we transformed $\mathbf{r}_i \rightarrow \mathbf{r}_j = \mathbf{r}_i - \mathbf{r}$. Note that the argument $\mathbf{a}$ does not change under a translation of the system, because that vector always represents the difference between the impurity lattice position and its nearest neighbours, irrespective of the absolute position of the impurity.

The wavefunction can be even brought into the familiar Bloch function form:

$$\begin{aligned} \Psi_{\vec{k}}^n(\{\mathbf{r}_k\}) &= \sum_{\mathbf{r}_i, \mathbf{a}} \frac{e^{i\vec{k} \cdot \vec{R}_i}}{\lambda_{|\vec{k}\rangle} \mathcal{Z} N} \psi_{\text{aux}}(\{\mathbf{r}_d - \mathbf{r}_i, \mathbf{r}_0 - \mathbf{r}_i - \mathbf{a}\}) = \frac{e^{i\vec{k} \cdot \frac{1}{N} \sum_k \vec{r}_k}}{\lambda_{|\vec{k}\rangle} \mathcal{Z} N} \sum_{\mathbf{r}_i, \mathbf{a}} e^{-i\vec{k} \cdot (\frac{1}{N} \sum_k \mathbf{r}_k - \vec{R}_i)} \psi_{\text{aux}}(\{\mathbf{r}_d - \mathbf{r}_i, \mathbf{r}_0 - \mathbf{r}_i - \mathbf{a}\}) \\ &= e^{i\vec{k} \cdot \vec{r}_{\text{COM}}} \eta_{\vec{k}}(\{\mathbf{r}_k\}) \end{aligned} \quad (5.2)$$

where $\mathbf{r}_{\text{COM}} = \frac{1}{N} \sum_k \mathbf{r}_k$ is the center-of-mass coordinate and $\eta_{\vec{k}}(\{\mathbf{r}_k\}) = \frac{1}{\lambda_{|\vec{k}\rangle} \mathcal{Z} N} \sum_{\mathbf{r}_i} e^{-i\vec{k} \cdot (\frac{1}{N} \sum_k \mathbf{r}_k - \vec{R}_i)} \psi_{\text{aux}}^n(\{\mathbf{r}_k - \mathbf{r}_i\})$ is the translation symmetric function. This form of the eigenstate allows the interpretation that tuning the Bloch momentum $\vec{k}$ corresponds to a translation of the center of mass of the system (or in this case, of the auxiliary models that comprise the system).

3.5.2 Orthonormality

It is straightforward to show that these states form an orthonormal basis. We start by writing down the inner product of two distinct such states:

$$\begin{aligned} \langle \Psi_{\vec{k}'} | \Psi_{\vec{k}} \rangle &= \frac{1}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{a}, \mathbf{a}'} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_j, \mathbf{a}') | \psi_{\text{aux}}(\mathbf{r}_i, \mathbf{a}) \rangle \\ &= \frac{1}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{a}, \mathbf{a}'} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') T^\dagger(\mathbf{r}_0 - \mathbf{r}_j) T(\mathbf{r}_0 - \mathbf{r}_i) | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \end{aligned} \quad (5.3)$$

At this point, we insert a complete basis of momentum eigenkets $1 = \sum_{\vec{q}} |\vec{q}\rangle \langle \vec{q}|$ to resolve the translation operators:

$$\begin{aligned}
 \langle \Psi_{\vec{k}} | \Psi_{\vec{k}} \rangle &= \frac{1}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{a}, \mathbf{a}', \vec{q}} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') T(\mathbf{r}_j - \mathbf{r}_i) |\vec{q}\rangle \langle \vec{q}| \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \frac{1}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{a}, \mathbf{a}', \vec{q}} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') e^{i\vec{q} \cdot (\mathbf{r}_j - \mathbf{r}_i)} |\vec{q}\rangle \langle \vec{q}| \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \frac{1}{|\lambda_{|\vec{k}\rangle}|^2 \mathcal{Z}^2} \delta_{\vec{k}, \vec{k}'} \sum_{\mathbf{a}, \mathbf{a}'} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') |\vec{k}\rangle \langle \vec{k}| \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle
 \end{aligned} \tag{5.4}$$

At the last step, we used the summation form of the Kronecker delta function: $\sum_{\mathbf{r}_i} e^{i\mathbf{r}_i \cdot (\vec{k} - \vec{q})} \sum_{\mathbf{r}_j} e^{i\mathbf{r}_j \cdot (\vec{q} - \vec{k}')} = N \delta_{\vec{k}, \vec{q}} N \delta_{\vec{k}', \vec{q}}$. The final remaining step is to identify that the inner products inside the summation are actually independent of the direction $\mathbf{a}, \mathbf{a}'$ of the connecting vector, and both are equal to the normalisation factor $\lambda_{|\vec{k}\rangle}$:

$$\langle \Psi_{\vec{k}'} | \Psi_{\vec{k}} \rangle = \frac{1}{|\lambda_{|\vec{k}\rangle}|^2 \mathcal{Z}^2} \delta_{\vec{k}, \vec{k}'} \sum_{\mathbf{a}, \mathbf{a}'} |\lambda_{|\vec{k}\rangle}|^2 = \frac{1}{|\lambda_{|\vec{k}\rangle}|^2 \mathcal{Z}^2} \delta_{\vec{k}, \vec{k}'} w^2 |\lambda_{|\vec{k}\rangle}|^2 = \delta_{\vec{k}, \vec{k}'} \tag{5.5}$$

This concludes the proof of orthonormality.

3.5.3 Eigenstates

To demonstrate that these are indeed eigenstates of the bulk Hamiltonian, we will calculate the matrix elements of the full Hamiltonian between these states:

$$\begin{aligned}
 \langle \Psi_{\vec{k}'} | H_{\text{H-H}} | \Psi_{\vec{k}} \rangle &= \frac{1}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k, \mathbf{a}, \mathbf{a}'} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_j, \mathbf{a}') | H_{\text{aux}}(\mathbf{r}_k) | \psi_{\text{aux}}(\mathbf{r}_i, \mathbf{a}) \rangle \\
 &= \frac{1}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k, \mathbf{a}, \mathbf{a}'} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | T_{\mathbf{r}_0 - \mathbf{r}_j}^\dagger T_{\mathbf{r}_0 - \mathbf{r}_k} H_{\text{aux}}(\mathbf{r}_0) T_{\mathbf{r}_0 - \mathbf{r}_k}^\dagger T_{\mathbf{r}_0 - \mathbf{r}_i} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \frac{1}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k, \mathbf{a}, \mathbf{a}'} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | T_{\mathbf{r}_k - \mathbf{r}_j}^\dagger H_{\text{aux}}(\mathbf{r}_0) T_{\mathbf{r}_k - \mathbf{r}_i} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle
 \end{aligned} \tag{5.6}$$

To resolve the translation operators, we will now insert the momentum eigenstates $|\vec{k}\rangle$ of the full model: $1 = \sum_{\vec{q}} |\vec{q}\rangle \langle \vec{q}|$ in between the operators.

$$\begin{aligned}
 \langle \Psi_{\vec{k}'} | H_{\text{H-H}} | \Psi_{\vec{k}} \rangle &= \sum_{\substack{r_i, r_j, r_k, \\ a, a', \vec{q}, \vec{q}'}} \frac{e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)}}{\lambda_{|\vec{k}'}^* \lambda_{|\vec{k}} \mathcal{Z}^2 N^2} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | \vec{q}' \rangle \langle \vec{q}' | T_{\mathbf{r}_k - \mathbf{r}_j}^\dagger H_{\text{aux}}(\mathbf{r}_0) T_{\mathbf{r}_k - \mathbf{r}_i} | \vec{q} \rangle \langle \vec{q} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \sum_{\substack{r_i, r_j, r_k, \\ a, a', \vec{q}, \vec{q}'}} \frac{e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)}}{\lambda_{|\vec{k}'}^* \lambda_{|\vec{k}} \mathcal{Z}^2 N^2} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | \vec{q}' \rangle e^{-i\vec{q}' \cdot (\mathbf{r}_k - \mathbf{r}_j)} e^{i\vec{q} \cdot (\mathbf{r}_k - \mathbf{r}_i)} \langle \vec{q}' | H_{\text{aux}}(\mathbf{r}_0) | \vec{q} \rangle \langle \vec{q} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \frac{N}{\lambda_{|\vec{k}'}^* \lambda_{|\vec{k}} \mathcal{Z}^2} \sum_{a, a', \vec{q}, \vec{q}'} \delta_{\vec{k}, \vec{q}} \delta_{\vec{q}, \vec{q}'} \delta_{\vec{q}', \vec{k}} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | \vec{q}' \rangle \langle \vec{q}' | H_{\text{aux}}(\mathbf{r}_0) | \vec{q} \rangle \langle \vec{q} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \frac{N}{|\lambda_{|\vec{k}}|^2 \mathcal{Z}^2} \delta_{kk'} \sum_{a, a'} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | \vec{k} \rangle \langle \vec{k} | H_{\text{aux}}(\mathbf{r}_0) | \vec{k} \rangle \langle \vec{k} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \frac{N}{|\lambda_{|\vec{k}}|^2 \mathcal{Z}^2} \delta_{kk'} \langle \vec{k} | H_{\text{aux}}(\mathbf{r}_0) | \vec{k} \rangle \underbrace{\sum_{a, a'} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | \vec{k} \rangle \langle \vec{k} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle}_{w^2 |\lambda_{|\vec{k}}|^2} \\
 &= N \delta_{kk'} \langle \vec{k} | H_{\text{aux}}(\mathbf{r}_0) | \vec{k} \rangle
 \end{aligned} \tag{5.7}$$

The momentum eigenket $|\vec{k}\rangle$ can be expressed as a sum over position eigenkets at varying distances from $\mathbf{r}_0$. This form allows a systematic method of improving the energy eigenvalue estimate, because the interacting in the impurity model is extremely localised, so the overlap between two auxiliary models decreases rapidly with distance. The majority of the contribution will come from the state at $\mathbf{r}_0$, and improvements are then made by considering auxiliary models at further distances.

Chapter 4

Mott Criticality as the Confinement Transition of a Pseudogap-Mott Metal: Insights from a lattice-embedded impurity model

4.1 Introduction

The Landau quasiparticle paradigm [185] underpins our understanding of metallic behaviour, where the screening of long-ranged repulsive interactions between delocalised electrons leads to well-defined excitations that carry the same quantum numbers as free electrons. This remarkable phenomenon - known as adiabatic continuity - is captured by excitations described by poles of the single-particle Greens function of the interacting system. But what defines metallicity when quasiparticles are no longer well-defined [?]? Can a surface of zeros of the Greens function [186], rather than poles forming a Fermi surface, define an entirely new kind of metal? Could such an exotic state serve as the precursor to Mott localisation driven purely by repulsion, and without invoking magnetic order [75]? These questions necessitate a radical departure in understanding conduction in correlated electron systems beyond the quasiparticle paradigm [?].

The anomalous pseudogap (PG) phase, observed in the cuprates, exhibits nodal-antinodal dichotomy with spectral weight concentrated on Fermi arcs around the nodal regions [?, ?, ?]. In the PG phase of such Mott systems [?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, 122, 187–193], the single-particle Green's function develops surfaces of zeros, also known as Luttinger surfaces [?, 186], that signals a fundamental obstruction to quasiparticle formation. As interaction strength increases, these zero surfaces proliferate, fragmenting the Fermi surface into nodal arcs [?, 187], violating the Luttinger sum rule by reconfiguring Fermi volume [15], and breaking an emergent $\mathbb{Z}_2$ symmetry associated with the conservation of spin currents in Fermi liquids [?, ?]. Unlike conventional instabilities driven by poles, this phenomenon marks a topological breakdown of Fermi liquid theory [?, ?]. We show that this breakdown arises from a strongly entangled, scale-invariant state of

quantum matter, one that serves as the parent metal to the Mott insulator in a half-filled lattice model of strongly correlated electrons. This offers a concrete, controlled framework to understand both the non-quasiparticle character of the pseudogap phase and its continuous evolution into a repulsion-driven insulator, in full alignment with Mott's original vision [75].

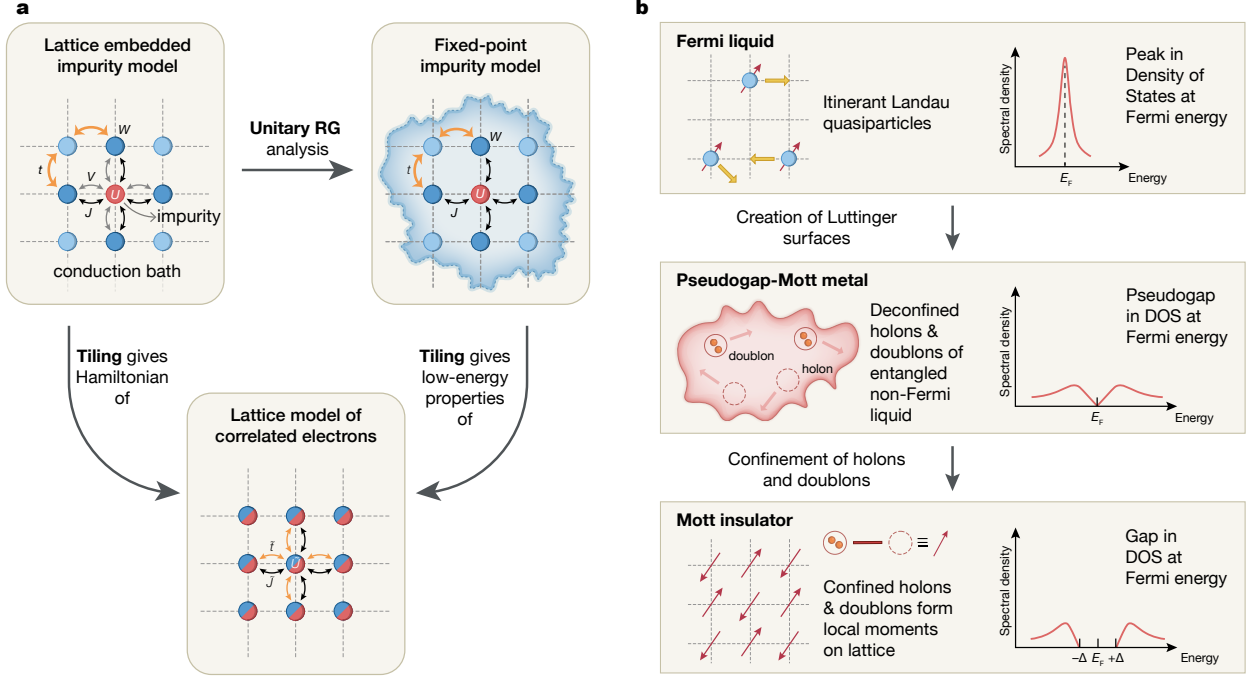


Figure 4.1: Schematic representation of (a) theoretical framework adopted by us (see main text for explanation of symbols), (b) phases obtained for the extended Hubbard model in two dimensions at $T = 0$, and passage between them.

Our approach involves developing a renormalization group (RG)-based impurity-to-lattice framework that reconstructs the low-energy physics of a strongly correlated system from the fixed-point structure of a quantum impurity model (Fig. 4.1(a)). Starting from a dynamical Anderson impurity embedded in a correlated bath, we solve the system using a unitary RG (URG) approach [18,26]. We then construct a translation-invariant lattice Hamiltonian through a many-body tiling procedure that preserves non-local correlations and inherits momentum-resolved self-energies and renormalized couplings from the impurity solution. The resulting model is an *extended*-Hubbard Hamiltonian on a two-dimensional square lattice at half-filling, with both on-site repulsion and nearest-neighbour spin exchange interactions. These correlations, derived directly from the impurity dynamics, enable tracking the evolution from the FL to the Mott insulator (MI) via an intermediate PG phase.

A central insight of our work is that the PG phase hosts a NFL state governed by Kondo frustration and doublon-holon deconfinement [194] (Fig. 4.1(b)). A correlation scale (W) in the conduction bath suppresses Kondo screening (with coupling J) beyond a critical ratio of W/J , triggering the emergence of Luttinger surfaces in the lattice model. This breakdown of quasiparticles begins at the antinodes and progressively engulfs the nodal regions, driving a continuous transformation into

the MI. The resulting NFL exhibits a pseudogapped density of states that vanishes as ω^2 as $\omega \rightarrow 0$, multipartite long-range entanglement, and a universal scattering rate of the form $\sim (a + b\omega^2)^{-1}$ that grows as $\omega \rightarrow 0$. At the critical endpoint of this phase, the URG flow identifies a singular NFL described by the Hatsugai–Kohmoto (HK) model [?, ?, ?], with fully deconfined holon–doublon excitations.

The emergence of Luttinger surfaces alters the anomaly structure associated with the generalized symmetry of the FS [?, ?, 17]. The corresponding topological charge is defined via the Luttinger–Ward functional. This charge is modified by Green’s function zeros, signalling a shift in the underlying anomaly. Within our framework, this reorganization grants topological protection to the RG flows terminating in the PG regime, ensuring the stability of its NFL excitations. These are adiabatically connected to the Hatsugai–Kohmoto model [?, ?], reinforcing the interpretation of the PG as a distinct quantum phase. We thus identify the NFL state as a new gapless, long-range entangled phase of correlated matter - the *Mott metal* - whose deconfined holon-doublon excitations are eventually confined across the Mott transition [75].

While we demonstrate our approach for an extended Hubbard model, the underlying framework is broadly applicable to correlated quantum systems. The impurity-plus-tiling construction accommodates multi-orbital degrees of freedom, frustrated geometries, and nontrivial band topology. By deriving lattice Hamiltonians directly from renormalized local dynamics - without relying on mean-field approximations - this method preserves nonlocal correlations and enhances momentum-space resolution. [Coupled with the unitary renormalisation group method as the impurity solver, this approach provides access to one-particle as well as two-particle correlations, entanglement measures and several dynamical quantities, making the method very versatile.](#) It thus provides a versatile and controlled route for engineering strongly correlated models, particularly in regimes where PG formation, NFL scaling, and Mott physics intersect.

We now describe the layout of the work. In section 4.2, we formulate the lattice-embedded auxiliary quantum impurity model and sketch the unitary renormalisation group (URG) method employed to analyse the impurity model. In section ??, we formulate the tiling procedure by which to map the impurity Hamiltonian, eigenstates and observables into a lattice model of strongly correlated electrons via the usage of many-body translation operators. Section ?? displays the destruction of the Fermi liquid from the breakdown of Kondo screening, leading to a pseudogapped non-Fermi liquid phase of matter. The novel properties of this phase are rigorously established in section 4.6. In section 4.7, we demonstrate that the continuous transition between the pseudogap and Mott insulating phases is described by the exactly solvable Hatsugai-Kohmoto model. We demonstrate in section 4.9 the universal organising principle that is responsible for the emergence of the pseudogap non-Fermi liquid phase. This also unveils the pseudogap as the strongly coupled long-range entangled parent metal of the Mott insulator. We conclude in section 4.10. Finally, we present appendices containing the derivation of the URG analysis of the impurity model, and the exactly solvable model at the critical point.

4.2 Auxiliary Model and URG analysis

To capture the interplay between Kondo physics and Mott localization, our approach relies on a two-dimensional auxiliary model consisting of an Anderson impurity embedded in a minimally-correlated bath, which we solve the system using a unitary RG (URG) approach [18, 26]. The impurity model and URG analysis are laid out in this section. In order to make contact with a lattice model, we construct a translation-invariant Hamiltonian from the impurity model through a many-body tiling procedure: this preserves non-local correlations, as well as allows for momentum-resolved self-energies and renormalized couplings generated from the impurity solution. For the impurity model chosen in this work, tiling results in an *extended*-Hubbard Hamiltonian on a two-dimensional square lattice at particle-hole symmetry (half-filling). The interaction terms within the lattice model are a result of analogous correlations present in the auxiliary model. This is laid out next in Section ??.

4.2.1 Lattice-embedded Impurity Model

The auxiliary model consists of a correlated impurity site (with a double occupancy cost U) embedded within a correlated conduction bath defined on a square lattice (see Fig. 4.1(a)). The impurity site hybridises with the conduction bath sites adjacent to it through one-particle hybridisation V as well as a spin-exchange interaction J arising from fluctuations in local spin densities of the electrons. The conduction bath has minimal correlations in the form of a local correlation term W on the sites that are adjacent to the impurity.

The Hamiltonian of the auxiliary model studied here is

$$\mathcal{H}_{\text{aux}} = H_{\text{imp}} + H_{\text{coup}} + H_{\text{cbath}}, \quad (2.1)$$

where the particle-hole symmetric impurity term is

$$H_{\text{imp}} = -\frac{U}{2} (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2, \quad (2.2)$$

describing on-site Coulomb interactions at the impurity. Here, $c_{d\sigma}^\dagger$ creates an electron with spin σ at the impurity site, and $\hat{n}_{d\sigma} = c_{d\sigma}^\dagger c_{d\sigma}$ denotes the corresponding number operator.

The impurity couples to four nearest-neighbour bath sites $c_{Z\sigma}$ through both hybridization and Kondo exchange:

$$H_{\text{coup}} = \frac{J}{2} \sum_{\sigma_1, \sigma_2, Z} \mathbf{S}_d \cdot c_{Z\sigma_1}^\dagger \boldsymbol{\tau}_{\sigma_1 \sigma_2} c_{Z\sigma_2} - V \sum_{\sigma, Z} (c_{Z\sigma}^\dagger c_{d\sigma} + \text{h.c.}), \quad (2.3)$$

where $\boldsymbol{\tau}$ are the Pauli matrices, and Z indexes the four neighbouring bath sites. The half-filled bath includes nearest neighbour hopping and on-site correlations on the site neighbouring the impurity:

$$H_{\text{cbath}} = \sum_{\mathbf{k}} \epsilon_{\mathbf{k}} c_{\mathbf{k},\sigma}^\dagger c_{\mathbf{k},\sigma} - \frac{W}{2} \sum_Z (\hat{n}_{Z\uparrow} - \hat{n}_{Z\downarrow})^2, \quad (2.4)$$

where $\epsilon_{\mathbf{k}} = -2t(\cos k_x + \cos k_y)$, and W parametrizes local repulsion in the bath that frustrates Kondo screening and drives the system towards local-moment formation. A crucial feature of this construction is that the Kondo coupling acquires a momentum structure upon Fourier transforming:

$$J_{\mathbf{k},\mathbf{k}'} = \frac{J}{2} [\cos(k_x - k'_x) + \cos(k_y - k'_y)] , \quad (2.5)$$

which respects the C_4 symmetry of the square lattice.

4.2.2 The unitary renormalisation group method

In order to obtain the various low-energy phases of our impurity model, we perform a scaling analysis of the associated Hamiltonian using the recently developed unitary renormalisation group (URG) method [19,39,40]. The method has been applied successfully on a wide variety of problems of correlated fermions [?, ?, 10, 106, 123, 124]. The method proceeds by resolving quantum fluctuations in high-energy degrees of freedom, leading to a low-energy Hamiltonian with renormalised couplings and new emergent degrees of freedom. Typically, for a system with Fermi energy ϵ_F and bandwidth E_N , the sequence of isoenergetic shells $\{E_{(j)}\}$, $E_j \in [E_0, E_N]$ define the states whose quantum fluctuations we sequentially resolve. The momentum states lying on shells E_N that are far away from the Fermi surface comprise the UV states, while those on shells near the Fermi surface comprise the IR states.

As a result of the URG transformations, the Hamiltonian $H_{(j)}$ at a given RG step j involves scattering processes between the k -states that have energies lower than $D_{(j+1)}$. The unitary transformation $U_{(j)}$ is then defined so as to remove the number fluctuations of the currently most energetic set of states $D_{(j)}$ [18, 26]:

$$H_{(j-1)} = U_{(j)} H_{(j)} U_{(j)}^\dagger , \text{ such that } [H_{(j-1)}, \hat{n}_j] = 0 . \quad (2.6)$$

The eigenvalue of $\hat{n}_j$ has, thus, been rendered an integral of motion (IOM) under the RG transformation.

The unitary transformations can be expressed in terms of a generator $\eta_{(j)}$ that has fermionic algebra [18, 26]:

$$U_{(j)} = \frac{1}{\sqrt{2}} \left(1 + \eta_{(j)} - \eta_{(j)}^\dagger \right) , \quad \left\{ \eta_{(j)}, \eta_{(j)}^\dagger \right\} = 1 , \quad (2.7)$$

where $\{\cdot\}$ is the anticommutator. The unitary operator $U_{(j)}$ that appears in Eq. (2.7) can be cast into the well-known general form $U = e^{\mathcal{S}}$, $\mathcal{S} = \frac{\pi}{4} \left(\eta_{(j)}^\dagger - \eta_{(j)} \right)$ that a unitary operator can take, defined by an anti-Hermitian operator $\mathcal{S}$. The generator $\eta_{(j)}$ is given by the expression [18, 26]

$$\eta_{(j)}^\dagger = \frac{1}{\hat{\omega}_{(j)} - \text{Tr}(H_{(j)} \hat{n}_j)} c_j^\dagger \text{Tr}(H_{(j)} c_j) . \quad (2.8)$$

The operators $\eta_{(j)}, \eta_{(j)}^\dagger$ behave as the many-particle analogues of the single-particle field operators $c_j, c_j^\dagger$ - they change the occupation number of the single-particle Fock space $|n_j\rangle$. The important

operator $\hat{\omega}_{(j)}$ originates from the quantum fluctuations that exist in the problem because of the non-commutation of the kinetic energy terms and the interaction terms in the Hamiltonian:

$$\hat{\omega}_{(j)} = H_{(j-1)} - H_{(j)}^i. \quad (2.9)$$

$H_{(j)}^i$ is the part of $H_{(j)}$ that commutes with $\hat{n}_j$ but does *not* commute with at least one $\hat{n}_l$ for $l < j$. The RG flow continues up to energy D^* , where a fixed point is reached from the vanishing of the RG function. Detailed comparisons of the URG with other methods (e.g., the functional RG, spectrum bifurcation RG etc.) can be found in Refs. [18, 123], and implementation on the present problem in Appendix A.

4.3 Derivation of unitary RG equations for the lattice-embedded impurity model

4.3.1 RG scheme

At any given step j of the RG procedure, we decouple the states $\{\mathbf{q}\}$ on the isoenergetic surface of energy ε_j . The diagonal Hamiltonian H_D for this step consists of all terms that do not change the occupancy of the states $\{\mathbf{q}\}$:

$$H_D^{(j)} = \varepsilon_j \sum_{q,\sigma} \tau_{q,\sigma} + \frac{1}{2} \sum_{\mathbf{q}} J_{\mathbf{q},\mathbf{q}} S_d^z (\hat{n}_{\mathbf{q},\uparrow} - \hat{n}_{\mathbf{q},\downarrow}) - \frac{1}{2} \sum_{\mathbf{q}} W_{\mathbf{q}} (\hat{n}_{\mathbf{q},\uparrow} - \hat{n}_{\mathbf{q},\downarrow})^2, \quad (3.1)$$

where $\tau = \hat{n} - 1/2$ and $W_{\mathbf{q}}$ is a shorthand for $W_{\mathbf{q},\mathbf{q},\mathbf{q}}$. The three terms, respectively, are the kinetic energy of the momentum states on the isoenergetic shell that we are decoupling, the spin-correlation energy between the impurity spin and the spins formed by these momentum states and, finally, the local correlation energy associated with these states arising from the W term. The off-diagonal part of the Hamiltonian on the other hand leads to scattering in the states $\{\mathbf{q}\}$. We now list these terms, classified by the coupling they originate from.

Arising from the Kondo spin-exchange term

$$\begin{aligned} T_{KZ1}^\dagger + T_{KZ1} &= \frac{1}{2} \sum_{\mathbf{k},\mathbf{q},\sigma} \sigma J_{\mathbf{k},\mathbf{q}} S_d^z \left[c_{\mathbf{q}\sigma}^\dagger c_{\mathbf{k},\sigma} + \text{h.c.} \right], \\ T_{KZ2}^\dagger + T_{KZ2} &= \frac{1}{2} \sum_{\mathbf{q},\sigma} \sigma J_{\mathbf{q},\bar{\mathbf{q}}} S_d^z \left[c_{\mathbf{q}\sigma}^\dagger c_{\bar{\mathbf{q}},\sigma} + \text{h.c.} \right], \\ T_{KT1}^\dagger + T_{KT1} &= \frac{1}{2} \sum_{\mathbf{k},\mathbf{q}} J_{\mathbf{k},\mathbf{q}} \left[S_d^+ \left(c_{\mathbf{q}\downarrow}^\dagger c_{\mathbf{k}\uparrow} + c_{\mathbf{k}\downarrow}^\dagger c_{\mathbf{q}\uparrow} \right) + \text{h.c.} \right], \\ T_{KT2}^\dagger + T_{KT2} &= \frac{1}{2} \sum_{\mathbf{q}} J_{\mathbf{q},\bar{\mathbf{q}}} \left[S_d^+ \left(c_{\mathbf{q}\downarrow}^\dagger c_{\bar{\mathbf{q}}\uparrow} + c_{\bar{\mathbf{q}}\downarrow}^\dagger c_{\mathbf{q}\uparrow} \right) + \text{h.c.} \right], \end{aligned} \quad (3.2)$$

Arising from spin-preserving scattering within conduction bath

$$\begin{aligned}
 T_{P1}^\dagger + T_{P1} &= - \sum_{\mathbf{q} \in \varepsilon_j} \sum_{\mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4 < \varepsilon_j} \sum_{\sigma} \left[W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma}^\dagger c_{\mathbf{k}_4, \sigma} + \text{h.c.} \right] \\
 T_{P2}^\dagger + T_{P3} &= - \sum_{\mathbf{q} \in \varepsilon_j} \sum_{\mathbf{k}_2 < \varepsilon_j} \sum_{\sigma} W_{\mathbf{q}, \mathbf{k}_2, \bar{\mathbf{q}}, \bar{\mathbf{q}}} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} n_{\bar{\mathbf{q}}, \sigma} - \sum_{\mathbf{q} \in \varepsilon_j} \sum_{\mathbf{k}_1 < \varepsilon_j} \sum_{\sigma} W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{q}, \sigma} n_{\mathbf{q}, \sigma} \\
 T_{P4} &= - \sum_{\mathbf{q} \in \varepsilon_j} \sum_{\mathbf{k}_2, \mathbf{k}_3 < \varepsilon_j} \sum_{\sigma} W_{\mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_3} c_{\mathbf{q}, \sigma}^\dagger c_{\bar{\mathbf{q}}, \sigma} c_{\mathbf{k}_2, \sigma}^\dagger c_{\mathbf{k}_3, \sigma} \\
 T_{P5} &= - \sum_{\mathbf{q} \in \varepsilon_j} \sum_{\mathbf{k}_2, \mathbf{k}_3 < \varepsilon_j} \sum_{\sigma} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \bar{\mathbf{q}}} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma}^\dagger c_{\bar{\mathbf{q}}, \sigma} \\
 &= + \sum_{\mathbf{q} \in \varepsilon_j} \sum_{\mathbf{k}_2, \mathbf{k}_3 < \varepsilon_j} \sum_{\sigma} W_{\mathbf{q}, \mathbf{k}_3, \mathbf{k}_2, \bar{\mathbf{q}}} c_{\mathbf{q}, \sigma}^\dagger c_{\bar{\mathbf{q}}, \sigma} c_{\mathbf{k}_2, \sigma}^\dagger c_{\mathbf{k}_3, \sigma} \\
 &= -T_{P4}
 \end{aligned} \tag{3.3}$$

Arising from spin-flip scattering within conduction bath

$$\begin{aligned}
 T_{F1}^\dagger + T_{F1} &= \sum_{\mathbf{q} \in \varepsilon_j} \sum_{\mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4 < \varepsilon_j} \sum_{\sigma} \left[W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} + \text{h.c.} \right] \\
 T_{F2} &= \sum_{\mathbf{q}, \mathbf{q}' \in \varepsilon_j} \sum_{\mathbf{k}_2, \mathbf{k}_3 < \varepsilon_j} \sum_{\sigma} W_{\mathbf{q}, \mathbf{q}', \mathbf{k}_2, \mathbf{k}_3} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{q}', \sigma} c_{\mathbf{k}_2, \bar{\sigma}}^\dagger c_{\mathbf{k}_3, \bar{\sigma}} \\
 T_{F3} &= \sum_{\mathbf{q}, \mathbf{q}' \in \varepsilon_j} \sum_{\mathbf{k}_2, \mathbf{k}_3 < \varepsilon_j} \sum_{\sigma} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{q}'} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{q}', \bar{\sigma}} \\
 T_{F4}^\dagger + T_{F4} &= \sum_{\mathbf{q}, \mathbf{q}' \in \varepsilon_j} \sum_{\mathbf{k}_1 < \varepsilon_j} \sum_{\sigma} \left[W_{\mathbf{q}, \mathbf{q}, \mathbf{q}', \mathbf{k}_1} n_{\mathbf{q}, \sigma} c_{\mathbf{q}', \bar{\sigma}}^\dagger c_{\mathbf{k}_1, \bar{\sigma}} + \text{h.c.} \right]
 \end{aligned} \tag{3.4}$$

In all of the terms $T_{P[i]}$ and $T_{F[i]}$, the factor of $1/2$ in front has been cancelled out by a factor of 2 coming from the multiple possibilities of arranging the momentum labels. We will henceforth ignore T_{P4} and T_{P5} because they cancel each other out.

The renormalisation of the Hamiltonian is constructed from the general expression

$$\Delta H^{(j)} = H_X \frac{1}{\omega - H_D} H_X. \tag{3.5}$$

The states on the isoenergetic shell $\pm|\varepsilon_j|$ come in particle-hole pairs $(\mathbf{q}, \bar{\mathbf{q}})$ with energies of opposite signs (relative to the Fermi energy). If $\mathbf{q}$ is defined as the hole state (unoccupied in the absence of quantum fluctuations), it will have positive energy, while the particle state $\bar{\mathbf{q}}$ will be of negative energy and hence below the Fermi surface. To be more specific, given a state $\mathbf{q}$ with energy $\pm|\varepsilon_j|$, we define its particle-hole transformed counterpart as the state $\bar{\mathbf{q}} = \pi + \mathbf{q}$, having energy $\mp|\varepsilon_j|$ and residing in the opposite quadrant of the Brillouin zone. Given this definition, we have the important property that

$$\begin{aligned}
 J_{\mathbf{k}, \bar{\mathbf{q}}} &= -J_{\mathbf{k}, \mathbf{q}}, \\
 W_{\{\mathbf{k}\}, \bar{\mathbf{q}}} &= -W_{\{\mathbf{k}\}, \mathbf{q}}.
 \end{aligned} \tag{3.6}$$

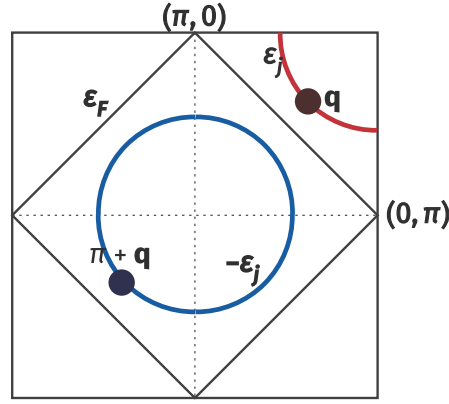


Figure 4.2: Particle and hole states.

4.3.2 Renormalisation of the bath correlation term W

The bath correlation term W can undergo renormalisation only via scattering processes arising from itself. Irrespective of whether the state $\mathbf{q}$ being decoupled is in a particle or hole configuration in the initial many-body state, the propagator $G = 1/(\omega - H_D)$ of the intermediate excited state is uniform, and equal to

$$G_W = 1/(\omega - |\varepsilon_j|/2 + W_{\mathbf{q}}/2) \quad , \quad (3.7)$$

where $W_{\mathbf{q}}$ is the same whether $\mathbf{q}$ is above or below the Fermi surface. The $|\varepsilon_j|/2$ in H_D arises from the excited nature of the state after the initial scattering process.

Scattering arising purely from spin-preserving processes

In this subsection, we calculate the renormalisation to W arising from the terms T_{P1} , T_{P2} and T_{P3} . The first term is

$$\begin{aligned} T_{P1}^\dagger G_W T_{P3} &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\mathbf{q}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma}^\dagger c_{\mathbf{k}_4, \sigma} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{q}, \sigma} n_{\mathbf{q}, \sigma} \\ &= - \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma}^\dagger c_{\mathbf{k}_4, \sigma} \sum_{\mathbf{q} \in \text{PS}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} \end{aligned} \quad (3.8)$$

The operators acting on the states being decoupled contract to form a number operator $n_{\mathbf{q}, \sigma}$ which projects the sum over $\mathbf{q}$ into the states that are initially occupied (particle sector, PS).

The second such contribution is obtained by flipping the sequence of scattering processes:

$$\begin{aligned} T_{P3} G_W T_{P1}^\dagger &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\mathbf{q}} W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{q}, \sigma} n_{\mathbf{q}, \sigma} G_W W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma}^\dagger c_{\mathbf{k}_4, \sigma} \\ &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma}^\dagger c_{\mathbf{k}_4, \sigma} \sum_{\mathbf{q} \in \text{HS}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} \end{aligned} \quad (3.9)$$

By virtue of eq. 3.6, the product of couplings $W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}}$ is the same irrespective of whether $\mathbf{q}$ belongs to the particle or hole sector. The two contributions therefore cancel each other.

Moreover, the remaining contributions $T_{P3}^\dagger G_W T_{P1}$ and $T_{P1} G_W T_{P2}^\dagger$ are effectively hermitian conjugates of the two contributions considered above, and therefore also cancel each other.

Scattering arising from spin-flip processes

We now come to the processes that involve spin-flips. Considering T_{F1} and T_{F4} first, we get

$$\begin{aligned}
 T_{F1}^\dagger G_W T_{F4} &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\mathbf{q}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{q}, \sigma} n_{\mathbf{q}, \bar{\sigma}} \\
 &= - \sum_{1,2,3,4} \sum_{\sigma} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} \sum_{\mathbf{q} \in \text{PS}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_4, \mathbf{k}_4} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} , \\
 T_{F4} G_W T_{F1}^\dagger &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\mathbf{q}} W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{q}, \sigma} n_{\mathbf{q}, \bar{\sigma}} G_W W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} \\
 &= \sum_{1,2,3,4} \sum_{\sigma} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} \sum_{\mathbf{q} \in \text{HS}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_4, \mathbf{k}_4} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} .
 \end{aligned} \tag{3.10}$$

By the same arguments as in the previous subsection, these terms cancel each other out. Their hermitian conjugate contributions $T_{F1} G_W T_{F4}^\dagger$ and $T_{F4}^\dagger G_W T_{F1}$ also cancel out. The other two terms are T_{F2} and T_{F3} , and their contributions also cancel out for the same reason:

$$\begin{aligned}
 T_{F2} G_W T_{F2} &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\mathbf{q}} W_{\mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\bar{\mathbf{q}}, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} G_W W_{\bar{\mathbf{q}}, \mathbf{q}, \mathbf{k}_1, \mathbf{k}_2} c_{\bar{\mathbf{q}}, \sigma}^\dagger c_{\mathbf{q}, \sigma} c_{\mathbf{k}_1, \bar{\sigma}}^\dagger c_{\mathbf{k}_2, \bar{\sigma}} \\
 &= \sum_{1,2,3,4} \sum_{\sigma} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} \sum_{\mathbf{q} \in \text{PS}} W_{\mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_3, \mathbf{k}_4} G_W W_{\bar{\mathbf{q}}, \mathbf{q}, \mathbf{k}_1, \mathbf{k}_2} , \\
 T_{F3} G_W T_{F3} &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\mathbf{q}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \bar{\mathbf{q}}} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\bar{\mathbf{q}}, \bar{\sigma}} G_W W_{\bar{\mathbf{q}}, \mathbf{k}_4, \mathbf{k}_1, \mathbf{q}} c_{\bar{\mathbf{q}}, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{q}, \sigma} \\
 &= - \sum_{1,2,3,4} \sum_{\sigma} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} \sum_{\mathbf{q} \in \text{PS}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \bar{\mathbf{q}}} G_W W_{\bar{\mathbf{q}}, \mathbf{k}_4, \mathbf{k}_1, \mathbf{q}} ,
 \end{aligned} \tag{3.11}$$

Scattering involving both spin-flip and spin-preserving processes

These processes involve the combination of terms like T_{P1} with T_{F4} , and T_{P2} with T_{F1} . These again cancel each other out for the same reasons as outline above.

Net renormalisation for the bath correlation term

Since all the contributions cancel out in pairs, the bath correlation term W is *marginal*.

4.3.3 Renormalisation of the Kondo scattering term J

We focus on the renormalisation of the spin-flip part of the Kondo interaction. For these processes, the intermediate many-body state always involves the impurity spin being anti-correlated with the conduction electron spin, such that the propagator for that state is $G_J = 1/(\omega - |\varepsilon_j|/2 + J_{\mathbf{q}}/4 + W_{\mathbf{q}}/2)$.

Impurity-mediated spin-flip scattering purely through Kondo-like processes

The following processes arising from the Kondo term renormalise the spin-flip interaction:

$$\begin{aligned}
 T_{KT1}^\dagger G_J (T_{KZ1} + T_{KZ1}^\dagger) &= \frac{1}{4} \sum_{\mathbf{k}_1, \mathbf{k}_1, \mathbf{q}} J_{\mathbf{q}, \mathbf{k}_2} S_d^+ \left[-c_{\mathbf{q}\downarrow}^\dagger c_{\mathbf{k}_2\uparrow} G_J c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{q}\downarrow} + c_{\mathbf{k}_2\downarrow}^\dagger c_{\mathbf{q}\uparrow} G_J c_{\mathbf{q}\uparrow}^\dagger c_{\mathbf{k}_1\uparrow} \right] J_{\mathbf{k}_1, \mathbf{q}} S_d^z \\
 &= -\frac{1}{8} \sum_{\mathbf{k}_1, \mathbf{k}_1, \mathbf{q}} J_{\mathbf{q}, \mathbf{k}_2} S_d^+ \left[c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{k}_2\uparrow} G_J n_{\mathbf{q}\downarrow} + c_{\mathbf{k}_2\downarrow}^\dagger c_{\mathbf{k}_1\uparrow} (1 - n_{\mathbf{q}\uparrow}) G_J \right] J_{\mathbf{k}_1, \mathbf{q}} \\
 &= -\frac{1}{8} \sum_{\mathbf{k}_1, \mathbf{k}_1} S_d^+ c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{k}_2\uparrow} \sum_{\mathbf{q} \in \text{PS}} [J_{\mathbf{q}, \mathbf{k}_2} J_{\mathbf{k}_1, \mathbf{q}} + J_{\bar{\mathbf{q}}, \mathbf{k}_1} J_{\mathbf{k}_2, \bar{\mathbf{q}}}] G_J .
 \end{aligned} \tag{3.12}$$

In getting the final expression, we used the sigma matrix relation $S_d^+ S_d^z = -\frac{1}{2} S_d^+$, and absorbed the projector $1 - n_{\mathbf{q}\uparrow}$ into the sum over the particle sector by replacing q with its particle-hole transformed counterpart $\bar{q}$. An identical contribution is obtained by switching the sequence of processes:

$$\begin{aligned}
 (T_{KZ1} + T_{KZ1}^\dagger) G_J T_{KT1}^\dagger &= \frac{1}{4} \sum_{\mathbf{k}_1, \mathbf{k}_1, \mathbf{q}} J_{\mathbf{k}_1, \mathbf{q}} S_d^z \left[-c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{q}\downarrow} G_J c_{\mathbf{q}\downarrow}^\dagger c_{\mathbf{k}_2\uparrow} + c_{\mathbf{q}\uparrow}^\dagger c_{\mathbf{k}_1\uparrow} G_J c_{\mathbf{k}_2\downarrow}^\dagger c_{\mathbf{q}\uparrow} \right] J_{\mathbf{q}, \mathbf{k}_2} S_d^+ \\
 &= -\frac{1}{8} \sum_{\mathbf{k}_1, \mathbf{k}_1} S_d^+ c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{k}_2\uparrow} \sum_{\mathbf{q} \in \text{PS}} [J_{\bar{\mathbf{q}}, \mathbf{k}_2} J_{\mathbf{k}_1, \bar{\mathbf{q}}} + J_{\mathbf{q}, \mathbf{k}_1} J_{\mathbf{k}_2, \mathbf{q}}] G_J .
 \end{aligned} \tag{3.13}$$

Scattering processes involving interplay between the Kondo interaction and conduction bath interaction

Looking at $T_{KT1}^\dagger$ first, we have

$$T_{KT1}^\dagger G_J (T_{F4} + T_{F4}^\dagger) = \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} J_{\mathbf{k}_2, \mathbf{q}} S_d^+ \left(c_{\mathbf{q}\downarrow}^\dagger c_{\mathbf{k}_2\uparrow} G_J W_{\mathbf{q}, \mathbf{q}, \mathbf{k}_1, \mathbf{q}} n_{\mathbf{q}\uparrow} c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{q}\downarrow} + c_{\mathbf{k}_2\downarrow}^\dagger c_{\mathbf{q}\uparrow} G_J W_{\bar{\mathbf{q}}, \bar{\mathbf{q}}, \mathbf{q}, \mathbf{k}_1} n_{\bar{\mathbf{q}}\downarrow} c_{\mathbf{q}\uparrow}^\dagger c_{\mathbf{k}_1\uparrow} \right) . \tag{3.14}$$

For either of the two choices of the functional form of W , it is easy to show that $W_{\mathbf{q}, \mathbf{q}, \mathbf{k}_1, \mathbf{q}} = W_{\bar{\mathbf{q}}, \bar{\mathbf{q}}, \mathbf{q}, \mathbf{k}_1}$.

$$T_{KT1}^\dagger G_J (T_{F4} + T_{F4}^\dagger) = \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} J_{\mathbf{k}_2, \mathbf{q}} W_{\mathbf{q}, \mathbf{q}, \mathbf{k}_1, \mathbf{q}} G_J S_d^+ \left[-c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{k}_2\uparrow} n_{\mathbf{q}\downarrow} n_{\mathbf{q}\uparrow} + c_{\mathbf{k}_2\downarrow}^\dagger c_{\mathbf{k}_1\uparrow} (1 - n_{\mathbf{q}\uparrow}) n_{\bar{\mathbf{q}}\downarrow} \right] . \tag{3.15}$$

Another contribution is obtained by switching the sequence of the scattering processes:

$$\begin{aligned}
 (T_{F4} + T_{F4}^\dagger) G_J T_{KT1}^\dagger &= \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} \left(W_{\mathbf{q}, \mathbf{q}, \mathbf{k}_1, \mathbf{q}} n_{\bar{\mathbf{q}}\uparrow} c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{q}\downarrow} G_J c_{\mathbf{q}\downarrow}^\dagger c_{\mathbf{k}_2\uparrow} + W_{\bar{\mathbf{q}}, \bar{\mathbf{q}}, \mathbf{q}, \mathbf{k}_1} n_{\mathbf{q}\downarrow} c_{\mathbf{q}\uparrow}^\dagger c_{\mathbf{k}_1\uparrow} G_J c_{\mathbf{k}_2\downarrow}^\dagger c_{\mathbf{q}\uparrow} \right) J_{\mathbf{k}_2, \mathbf{q}} S_d^+ \\
 &= \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} \left(c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{k}_2\uparrow} n_{\bar{\mathbf{q}}\uparrow} (1 - n_{\mathbf{q}\downarrow}) - c_{\mathbf{k}_2\downarrow}^\dagger c_{\mathbf{k}_1\uparrow} n_{\mathbf{q}\downarrow} n_{\mathbf{q}\uparrow} \right) W_{\mathbf{q}, \mathbf{q}, \mathbf{k}_1, \mathbf{q}} G_J J_{\mathbf{k}_2, \mathbf{q}} S_d^+
 \end{aligned} \tag{3.16}$$

The two contributions (eqs. 3.15 and 3.16) arising from T_{KT1} cancel each other.

We now consider the other spin-exchange process $T_{KT2}^\dagger$. One such contribution is

$$\begin{aligned}
 T_{KT2}^\dagger G_J T_{F3} &= \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} J_{\mathbf{q}, \bar{\mathbf{q}}} S_d^+ \left(c_{\mathbf{q}\downarrow}^\dagger c_{\bar{\mathbf{q}}\uparrow} G_J c_{\bar{\mathbf{q}}\uparrow}^\dagger c_{\mathbf{k}_2\uparrow} c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{q}\downarrow} + c_{\bar{\mathbf{q}}\downarrow}^\dagger c_{\mathbf{q}\uparrow} G_J c_{\mathbf{q}\uparrow}^\dagger c_{\mathbf{k}_2\uparrow} c_{\mathbf{k}_1\downarrow}^\dagger c_{\bar{\mathbf{q}}\downarrow} \right) W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}} \\
 &= -\frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} S_d^+ c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{k}_2\uparrow} \left[n_{\mathbf{q}\downarrow} (1 - n_{\bar{\mathbf{q}}\uparrow}) + n_{\bar{\mathbf{q}}\downarrow} (1 - n_{\mathbf{q}\uparrow}) \right] J_{\mathbf{q}, \bar{\mathbf{q}}} G_J W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}} \\
 &= -\frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2} S_d^+ c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{k}_2\uparrow} \sum_{\mathbf{q} \in \text{PS}} (J_{\mathbf{q}, \bar{\mathbf{q}}} W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}} + J_{\bar{\mathbf{q}}, \mathbf{q}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_1, \bar{\mathbf{q}}}) G_J.
 \end{aligned} \tag{3.17}$$

An identical contribution is obtained from the reversed term:

$$\begin{aligned}
 T_{F3} G_J T_{KT2}^\dagger &= \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}} \left(c_{\bar{\mathbf{q}}\uparrow}^\dagger c_{\mathbf{k}_2\uparrow} c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{q}\downarrow} G_J c_{\mathbf{q}\downarrow}^\dagger c_{\bar{\mathbf{q}}\uparrow} + c_{\mathbf{q}\uparrow}^\dagger c_{\mathbf{k}_2\uparrow} c_{\mathbf{k}_1\downarrow}^\dagger c_{\bar{\mathbf{q}}\downarrow} G_J c_{\bar{\mathbf{q}}\downarrow}^\dagger c_{\mathbf{q}\uparrow} \right) J_{\mathbf{q}, \bar{\mathbf{q}}} S_d^+ \\
 &= -\frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2} S_d^+ c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{k}_2\uparrow} \sum_{\mathbf{q} \in \text{PS}} (J_{\mathbf{q}, \bar{\mathbf{q}}} W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}} + J_{\bar{\mathbf{q}}, \mathbf{q}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_1, \bar{\mathbf{q}}}) G_J.
 \end{aligned} \tag{3.18}$$

Net renormalisation to the Kondo interaction

Combining the results from eqs. 3.12, 3.13, 3.17 and 3.18, as well as using the properties $J_{\bar{\mathbf{q}}, \mathbf{k}_1} J_{\mathbf{k}_2, \bar{\mathbf{q}}} = J_{\mathbf{q}, \mathbf{k}_2} J_{\mathbf{k}_1, \mathbf{q}} = J_{\mathbf{k}_2, \mathbf{q}} J_{\mathbf{q}, \mathbf{k}_1}$ and $J_{\mathbf{q}, \bar{\mathbf{q}}} W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}} = J_{\bar{\mathbf{q}}, \mathbf{q}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_1, \bar{\mathbf{q}}}$, the total renormalisation in the momentum-resolved Kondo coupling $J^{(j)}$ at the j^{th} step amounts to

$$\Delta J_{\mathbf{k}_1, \mathbf{k}_2}^{(j)} = - \sum_{\mathbf{q} \in \text{PS}} \frac{J_{\mathbf{k}_2, \mathbf{q}}^{(j)} J_{\mathbf{q}, \mathbf{k}_1}^{(j)} + 4 J_{\mathbf{q}, \bar{\mathbf{q}}}^{(j)} W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}}}{\omega - \frac{1}{2} |\mathcal{E}_j| + J_{\mathbf{q}}^{(j)} / 4 + W_{\mathbf{q}} / 2} \tag{3.19}$$

A representation of the RG phase diagram in the $-W/t$ vs. J/t plane of couplings has been provided in Fig. 4.4(a) from detailed numerical computations of eq.(??). We also provide a schematic representation of RG flows to various fixed point theories in the J vs. W plane of couplings in Fig. 4.3. This schematic figure shows the existence of three distinct phases (Fermi liquid metal, Mott Insulator, and Pseudogap-Mott metal) described by distinct fixed point theories of the RG flows. The Fermi liquid metal and Mott Insulator phases are distinguished from the Pseudogap-Mott metal by separatrices of the RG flows.

4.4 Reconstructing of lattice model from auxiliary model

To reconstruct a translationally invariant lattice Hamiltonian from the impurity model (with Hamiltonian $\mathcal{H}_{\text{aux}}$), we define a many-body tiling prescription that embeds the auxiliary impurity system uniformly across the two-dimensional square lattice. Thus, the tiling process systematically derives the full lattice dynamics from the renormalized fixed-point structure of the impurity. As described

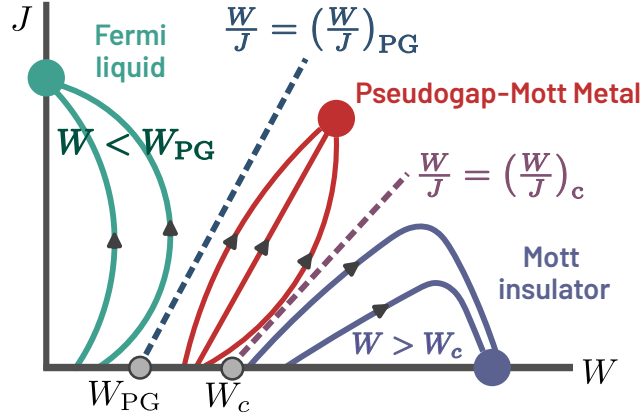


Figure 4.3: Renormalisation group (RG) flow diagram of the lattice model as obtained from a unitary RG analysis of the lattice-embedded impurity model. For $|W/J| < |W/J|_{PG}$, the flows lead to a Fermi liquid phase, while for $|W/J| < |W/J|_c$, they lead to a Mott insulator. For values in between, the system flows to a Mott metal phase that has a pseudogap in the density of states.

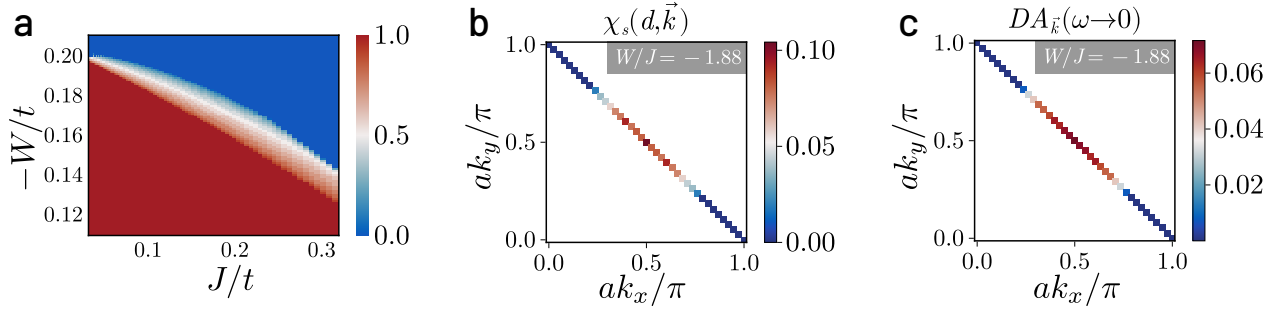


Figure 4.4: (a) Phase diagram of impurity model at strong coupling in U in terms of competing dimensionless Kondo (J/t) and bath correlation (W/t) couplings. Colourbar represents the fraction of states on the Fermi surface replaced by zeros of the Greens function (Luttinger surface). A pseudogap phase (shaded region) is observed between the local Fermi liquid (red region) and local moment (blue region) phases (b) k -space-resolved spin-spin correlation $\chi_s(d, \mathbf{k}) = \langle \mathbf{S}_d \cdot \mathbf{S}_{\mathbf{k}} \rangle$ in the pseudogap phase of the impurity model. Antinodal regions are observed to decouple from Kondo screening of the impurity. (c) Upon tiling, this leads to a k -space-resolved antinodal gap in the electronic density of states of the lattice model, corresponding to Luttinger surfaces of zeros.

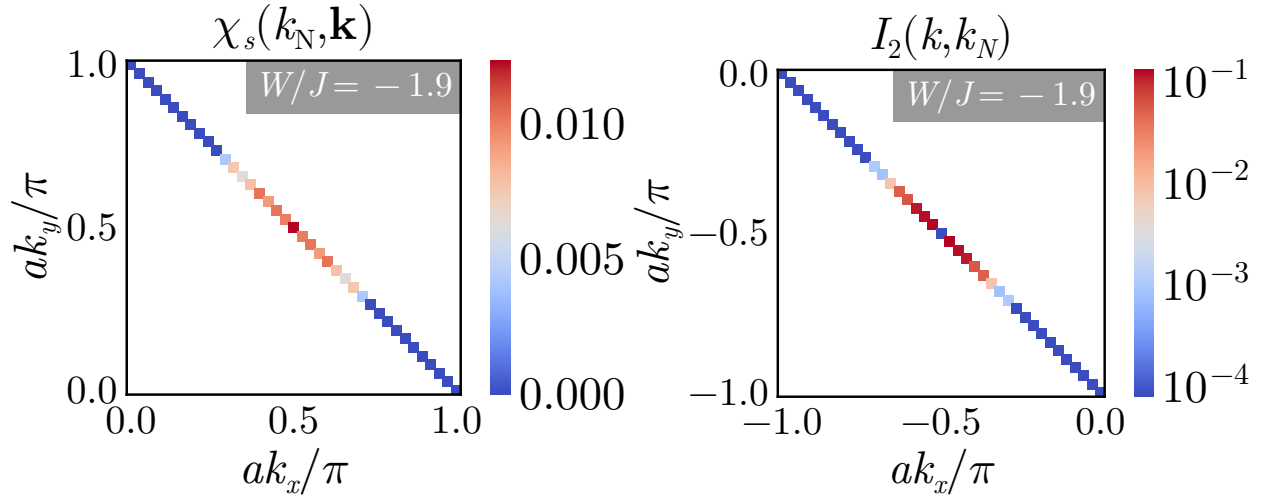


Figure 4.5: Left: Spin correlations χ_s for the *tiled model*, between the nodal point $k_N = (-\pi/2, -\pi/2)$ and an arbitrary k -point on the Fermi surface. In the PG, the spin-correlations near the antinode vanish, indicating that they have been removed from the metallic excitations, while the correlations near the nodal point become enhanced because of the increasingly correlated nature of the metal. Right: Mutual information I_2 between an arbitrary k -state and the nodal point. The nodal points remain strongly entangled with each other all the way through the PG.

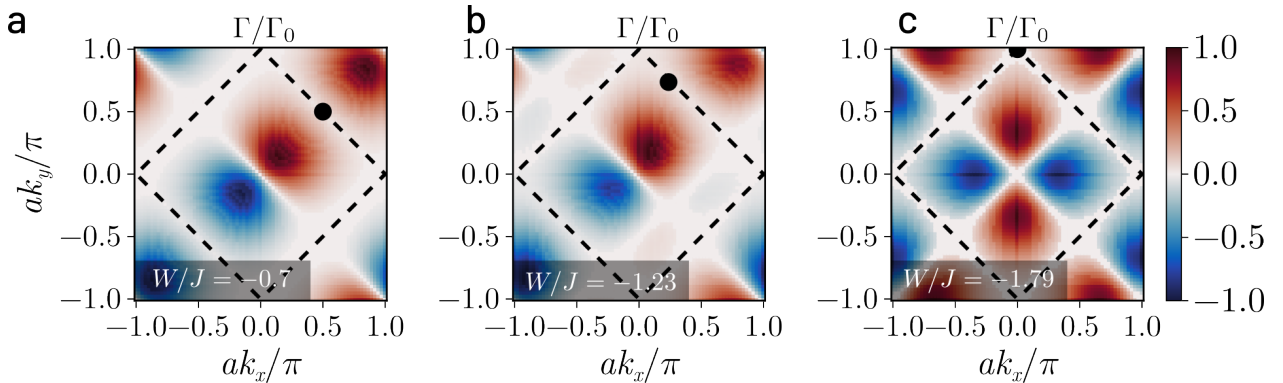


Figure 4.6: Initiation of the decoupling of $J_{\mathbf{k},\mathbf{k}'}$ (positive, negative and zeros shown in red, blue and white respectively), with $\mathbf{k}$ (black circle) at low-energies for $\mathbf{k}$ corresponding to the node (a), antinode (b) and a point mid-way between them (c) on the top right arm of the FS respectively with tuning W/J . As dictated by the symmetry of $J_{\mathbf{k},\mathbf{k}'}$, the decoupling for a given $\mathbf{k}$ proceeds via the appearance of zeros (white patches) of $J_{\mathbf{k},\mathbf{k}'}$ for $\mathbf{k}'$ initially on the nodal regions of adjacent arms, and progresses gradually towards the antinodes. The decoupling ends with the onset of the pseudogap.

in Section 4.2, the auxiliary model consists of a single impurity embedded in a correlated bath at the location $\mathbf{r}_d$:

$$H(\mathbf{r}_d) = H_{\text{imp}} + H_{\text{cbath}} + H_{\text{imp-cbath}}. \quad (4.1)$$

The form of various terms in eq.(4.1) have been defined in eqs.(2.2), (2.3) and (2.4) in Section 4.2.

The tiled lattice Hamiltonian is generated by symmetrically translating this impurity model:

$$\mathcal{H}_{\text{tiled}} = \mathcal{T}[H(\mathbf{r}_d)] = \sum_{\mathbf{r}} T^\dagger(\mathbf{r} - \mathbf{r}_d) H(\mathbf{r}_d) T(\mathbf{r} - \mathbf{r}_d), \quad (4.2)$$

where $T(\mathbf{a})$ are many-body translation operators [184]:

$$T^\dagger(\mathbf{a}) O(\{\mathbf{r}_i\}) T(\mathbf{a}) = O(\{\mathbf{r}_i + \mathbf{a}\}). \quad (4.3)$$

Applying this to each term yields:

$$\begin{aligned} \mathcal{T}[H_{\text{cbath}}] &= -\frac{(N-2)t}{\sqrt{\mathcal{Z}}} \sum_{\langle \mathbf{r}_i, \mathbf{r}_j \rangle; \sigma} \left(c_{\mathbf{r}_i \sigma}^\dagger c_{\mathbf{r}_j \sigma} + \text{h.c.} \right) \\ &\quad - \frac{1}{2} W \sum_{\mathbf{r}} (\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow})^2, \\ \mathcal{T}[H_{\text{imp}}] &= -\frac{U}{2} \sum_{\mathbf{r}} (\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow})^2, \\ \mathcal{T}[H_{\text{imp-cbath}}] &= \sum_{\langle \mathbf{r}_i, \mathbf{r}_j \rangle} \left[\frac{2J}{\mathcal{Z}} \mathbf{S}_{\mathbf{r}_i} \cdot \mathbf{S}_{\mathbf{r}_j} - \frac{2V}{\sqrt{\mathcal{Z}}} \sum_{\sigma} \left(c_{\mathbf{r}_i \sigma}^\dagger c_{\mathbf{r}_j \sigma} + \text{h.c.} \right) \right], \end{aligned} \quad (4.4)$$

with local spin operators defined as

$$\mathbf{S}_{\mathbf{r}} = \sum_{\sigma, \sigma'} c_{\mathbf{r} \sigma}^\dagger \boldsymbol{\tau}_{\sigma \sigma'} c_{\mathbf{r} \sigma'}. \quad (4.5)$$

To avoid overcounting of the non-interacting bath terms, we subtract extra copies of the kinetic energy term. The reconstructed extended-Hubbard model then reads:

$$\begin{aligned} \mathcal{H}_{\text{tiled}} &= -\frac{1}{\sqrt{\mathcal{Z}}} \tilde{t} \sum_{\langle \mathbf{r}_i, \mathbf{r}_j \rangle; \sigma} \left(c_{\mathbf{r}_i \sigma}^\dagger c_{\mathbf{r}_j \sigma} + \text{h.c.} \right) \\ &\quad + \frac{\tilde{J}}{\mathcal{Z}} \sum_{\langle \mathbf{r}_i, \mathbf{r}_j \rangle} \mathbf{S}_{\mathbf{r}_i} \cdot \mathbf{S}_{\mathbf{r}_j} - \frac{1}{2} \tilde{U} \sum_{\mathbf{r}} (\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow})^2, \end{aligned} \quad (4.6)$$

where the effective lattice couplings are given by:

$$\tilde{t} = t + 2V, \quad \tilde{U} = U + W, \quad \tilde{J} = 2J. \quad (4.7)$$

A non-zero chemical potential in the bath and/or the impurity can be included to study the effects of doping. Crucially, the eigenstates of $\mathcal{H}_{\text{tiled}}$ obey a many-body generalization of Bloch's theorem [184], which allows for the exact computation of momentum-resolved observables (presented here for a 77×77 $\mathbf{k}$ -space Brillouin zone grid). We shall demonstrate below that the lattice embedding and tiling procedures provide a controlled route to capture low-energy NFL behavior and Mott criticality directly from a quantum impurity model.

4.5 Effective action for the Hubbard-Heisenberg model and the eSIAM

In order to better understand the similarities and differences between the tiling approach outlined here and the approach adopted in dynamical mean-field approximations, we compute the effective action for the local impurity as well as the impurity-zeroth site system, in both the auxiliary and bulk models. This will be done in the limit of infinite coordination number, in order to create a tractable effective theory.

4.5.1 Local theory for the Hubbard-Heisenberg model

We will first work with the Hubbard-Heisenberg model. Let us recall that within the eSIAM, most of the dynamics is governed by the physics of the impurity site and the bath site coupled to the impurity. Accordingly, we will first obtain an effective action within the Hubbard-Heisenberg model for the small system consisting of a local site and all its nearest-neighbours.

We choose a certain local site that we call 0, whose nearest-neighbours will be denoted as $\{\bar{0}\}$. The action for the full Hubbard-Heisenberg model can be formally separated into three parts:

$$S_{\text{H-H}} = S_{0,\bar{0}} + S^{(0,\bar{0})} + S_{\text{int}}, \quad (5.1)$$

where $S_{0,\bar{0}}$ represents the part of the action that involves only the sites 0 and $\{\bar{0}\}$, $S^{(0,\bar{0})}$ represents the part that has all the sites apart from 0 and $\{\bar{0}\}$, while S_{int} contains all terms that connect these two parts. These three parts have the forms

$$\begin{aligned} S_{0,\bar{0}} = & \sum_{i=0,\bar{0}} \sum_{\sigma} \int_0^{\beta} d\tau \, c_{i\sigma}^{\dagger}(\tau) \partial_{\tau} c_{i\sigma}(\tau) - t \sum_{\bar{0},\sigma} \int_0^{\beta} d\tau \, \left[c_{0\sigma}^{\dagger}(\tau) c_{\bar{0}\sigma}(\tau) + \text{h.c.} \right] \\ & - \frac{U}{2} \sum_{i=0,\bar{0}} \int_0^{\beta} d\tau \, (n_{i\uparrow}(\tau) - n_{i\downarrow}(\tau))^2 + J \sum_{\bar{0}} \int_0^{\beta} d\tau \, \vec{S}_0(\tau) \cdot \vec{S}_{\bar{0}}(\tau), \end{aligned} \quad (5.2)$$

$$\begin{aligned} S^{(0,\bar{0})} = & \sum_{j \neq (0,\bar{0})} \sum_{\sigma} \int_0^{\beta} d\tau \, c_{j\sigma}^{\dagger}(\tau) \partial_{\tau} c_{j\sigma}(\tau) - t \sum_{\langle j,l \rangle \neq (0,\bar{0}),\sigma} \int_0^{\beta} d\tau \, \left[c_{j\sigma}^{\dagger}(\tau) c_{l\sigma}(\tau) + \text{h.c.} \right] \\ & - \frac{U}{2} \sum_{j \neq (0,\bar{0})} \int_0^{\beta} d\tau \, (n_{j\uparrow}(\tau) - n_{j\downarrow}(\tau))^2 + J \sum_{\langle j,l \rangle \neq (0,\bar{0})} \int_0^{\beta} d\tau \, \vec{S}_l(\tau) \cdot \vec{S}_j(\tau), \end{aligned} \quad (5.3)$$

$$S_{\text{int}} = -t \sum_{i \in \bar{0}} \sum_{j \in \text{NN of } i} \sum_{\sigma} \int_0^{\beta} d\tau \, \left(c_{i\sigma}^{\dagger}(\tau) c_{j\sigma}(\tau) + \text{h.c.} \right) + J \sum_{i \in \bar{0}} \sum_{j \in \text{NN of } i} \int_0^{\beta} d\tau \, \vec{S}_i(\tau) \cdot \vec{S}_j(\tau) = S_{\text{int}}^{\text{hop}} + S_{\text{int}}^{\text{spin}} \quad (5.4)$$

In order to obtain an effective theory $S_{\text{eff}}^{0,\bar{0}}$ purely for the local system $(0, \bar{0})$, we need to trace over all the other degrees of freedom. This partial trace will be carried out over the states of the so-called ‘‘cavity’’ system $S^{(0,\bar{0})}$. The effective action can be constructed by tracing over scattering processes

of all orders that leave the final action diagonal in the system $(0, \bar{0})$. The formal expression can be written as

$$S_{\text{eff}}^{0, \bar{0}} = S_{0, \bar{0}} + \sum_{n=1}^{\infty} \langle (S_{\text{int}}^{\text{hop}})^{2n} \rangle_{(0, \bar{0})} + \sum_{n=1}^{\infty} \langle (S_{\text{int}}^{\text{spin}})^{2n} \rangle_{(0, \bar{0})} \quad (5.5)$$

where, as mentioned before, the average is carried out in the cavity system $S^{(0, \bar{0})}$. Each hopping term in the expansion is of the form

$$\begin{aligned} (S_{\text{int}}^{\text{hop}})^{2n} &= t^{2n} \sum_{\sigma} \sum_{(i_1, i'_1), \dots, (i_n, i'_n)} \sum_{(j_1, j'_1), \dots, (j_n, j'_n)} \int_0^{\beta} d\tau_1 \dots d\tau_n d\tau'_1 \dots d\tau'_n \\ &\quad c_{i_1 \sigma}^{\dagger}(\tau_1) c_{i_2 \sigma}^{\dagger}(\tau_2) \dots c_{i_n \sigma}^{\dagger}(\tau_n) c_{i'_1 \sigma}(\tau'_1) c_{i'_2 \sigma}(\tau'_2) \dots c_{i'_n \sigma}(\tau'_n) \langle c_{j_1 \sigma}(\tau_1) c_{j_2 \sigma}(\tau_2) \dots c_{j_n \sigma} c_{j'_1 \sigma}^{\dagger}(\tau'_1) \dots c_{j'_n \sigma}^{\dagger}(\tau'_n) \rangle \\ &= t^{2n} \sum_{\sigma} \sum_{(i_1, i'_1), \dots, (i_n, i'_n)} \sum_{(j_1, j'_1), \dots, (j_n, j'_n)} \int_0^{\beta} d\tau_1 \dots d\tau'_n c_{i_1 \sigma}^{\dagger}(\tau_1) \dots c_{i_n \sigma}^{\dagger}(\tau_n) c_{i'_1 \sigma}(\tau'_1) \dots c_{i'_n \sigma}(\tau'_n) \\ &\quad G_n^{(0, \bar{0})}(\tau_1 \dots \tau_n; \tau'_1 \dots \tau'_n), \end{aligned} \quad (5.6)$$

where the indices i_1 through i_n and their primed counterparts run through $\bar{0}$, while j_l ($\in \{j_1, \dots, j'_n\}$) runs through the nearest-neighbours of the corresponding i_l index. The n -particle Greens function $G_n^{(0, \bar{0})}(\tau_1 \dots \tau_n; \tau'_1 \dots \tau'_n)$ for the cavity system is defined in the usual fashion

$$G_n^{(0, \bar{0})}(\tau_1 \dots \tau_n; \tau'_1 \dots \tau'_n) = \langle c_{j_1 \sigma}(\tau_1) c_{j_2 \sigma}(\tau_2) \dots c_{j_n \sigma} c_{j'_1 \sigma}^{\dagger}(\tau'_1) \dots c_{j'_n \sigma}^{\dagger}(\tau'_n) \rangle_{(0, \bar{0})} \quad (5.7)$$

In the limit of infinite coordination number, only the $n = 1$ term survives:

$$\begin{aligned} \lim_{Z \rightarrow \infty} \sum_{n=1}^{\infty} \langle (S_{\text{int}}^{\text{hop}})^{2n} \rangle_{(0, \bar{0})} &= t^2 \sum_{\sigma} \sum_{i, i'} \int_0^{\beta} d\tau d\tau' c_{i \sigma}^{\dagger}(\tau) c_{i' \sigma}(\tau') \sum_{j, j'} G_1^{(0, \bar{0})}(j\tau; j'\tau') \\ &= \sum_{\sigma} \sum_{i, i'} \int_0^{\beta} d\tau d\tau' \Delta_{ii', \sigma}(\tau - \tau') c_{i \sigma}^{\dagger}(\tau) c_{i' \sigma}(\tau'), \end{aligned} \quad (5.8)$$

where we have defined the bath hybridisation function $\Delta_{ii', \sigma}(\tau - \tau')$ as

$$\Delta_{ii', \sigma}(\tau - \tau') = t^2 \sum_{j \in \text{NN of } i} \sum_{j' \in \text{NN of } i'} G_1^{(0, \bar{0})}(j\sigma\tau; j'\sigma\tau'). \quad (5.9)$$

Under similar approximations, the spin term in the action reduces to

$$\lim_{Z \rightarrow \infty} \sum_{n=1}^{\infty} \langle (S_{\text{int}}^{\text{spin}})^{2n} \rangle_{(0, \bar{0})} = \sum_{i, i'} \int_0^{\beta} d\tau d\tau' \chi(\tau - \tau') \vec{S}_i(\tau) \cdot \vec{S}_{i'}(\tau'), \quad (5.10)$$

where the susceptibility $\chi_{ii'}(\tau - \tau')$ is defined as

$$\chi_{ii'}(\tau - \tau') = J^2 \sum_{j \in \text{NN of } i} \sum_{j' \in \text{NN of } i'} \vec{S}_j(\tau) \cdot \vec{S}_{j'}(\tau'). \quad (5.11)$$

The effective action for the $0, \bar{0}$ system, in the limit of large coordination number, simplifies to

$$\begin{aligned}
 S_{\text{eff}}^{0, \bar{0}} = & \sum_{i=0, \bar{0}} \sum_{\sigma} \int_0^{\beta} d\tau c_{i\sigma}^{\dagger}(\tau) \partial_{\tau} c_{i\sigma}(\tau) - t \sum_{\bar{0}, \sigma} \int_0^{\beta} d\tau \left[c_{0\sigma}^{\dagger}(\tau) c_{\bar{0}\sigma}(\tau) + \text{h.c.} \right] - \frac{U}{2} \sum_{i=0, \bar{0}} \int_0^{\beta} d\tau (n_{i\uparrow}(\tau) - n_{i\downarrow}(\tau))^2 \\
 & + J \sum_{\bar{0}} \int_0^{\beta} d\tau \vec{S}_0(\tau) \cdot \vec{S}_{\bar{0}}(\tau) + \sum_{i, i'} \int_0^{\beta} d\tau d\tau' \left[\sum_{\sigma} \Delta_{ii', \sigma}(\tau - \tau') c_{i\sigma}^{\dagger}(\tau) c_{i'\sigma}(\tau') + \chi_{ii'}(\tau - \tau') \vec{S}_i(\tau) \cdot \vec{S}_{i'}(\tau') \right]
 \end{aligned} \tag{5.12}$$

One can also write down an effective action purely for the local site 0. The sites $\bar{0}$ will now be a part of the environment action $S^{(0)}$ instead of the system action, but since the system is thermodynamically large, the cavity action can be assumed to remain the same, such the expectation values are calculated in the same system. As a result, the hybridisation and susceptibility functions remain unchanged. S_{int} is now made of terms that connect 0 and $\bar{0}$, so the transformation from $S^{(0, \bar{0})}$ can be made by replacing the set $\{j\}$ with the set $\bar{0}$, and the set $\bar{0}$ itself will be replaced by the local site 0. We quote the final form of the local effective action:

$$\begin{aligned}
 S_{\text{eff}}^0 = & \sum_{\sigma} \int_0^{\beta} d\tau c_{0\sigma}^{\dagger}(\tau) \partial_{\tau} c_{0\sigma}(\tau) - \frac{U}{2} \int_0^{\beta} d\tau (n_{0\uparrow}(\tau) - n_{0\downarrow}(\tau))^2 \\
 & + \int_0^{\beta} d\tau d\tau' \left[\sum_{\sigma} \Delta_{00, \sigma}(\tau - \tau') c_{0\sigma}^{\dagger}(\tau) c_{0\sigma}(\tau') + \chi_{00}(\tau - \tau') \vec{S}_0(\tau) \cdot \vec{S}_0(\tau') \right]
 \end{aligned} \tag{5.13}$$

4.5.2 Local theory for the extended SIAM

The Hamiltonian for the extended SIAM model is shown in eq. ???. We will denote the impurity with the label d and the bath sites with the label c . Among the bath sites, we will represent the site coupled to the impurity with z . The full action has the form

$$\begin{aligned}
 S_{\text{ES}} = & \sum_{d, c} \sum_{\sigma} \int_0^{\beta} d\tau c_{d\sigma}^{\dagger}(\tau) \partial_{\tau} c_{d\sigma}(\tau) - t \sum_{\langle i, j \rangle \in c, \sigma} \int_0^{\beta} d\tau \left[c_{i\sigma}^{\dagger}(\tau) c_{j\sigma}(\tau) + \text{h.c.} \right] - V \sum_{\sigma} \int_0^{\beta} d\tau \left[c_{d\sigma}^{\dagger}(\tau) c_{z\sigma}(\tau) + \text{h.c.} \right] \\
 & + J \int_0^{\beta} d\tau \vec{S}_d(\tau) \cdot \vec{S}_z(\tau) - \frac{U}{2} \int_0^{\beta} d\tau (n_{d\uparrow}(\tau) - n_{d\downarrow}(\tau))^2 - \frac{W}{2} \int_0^{\beta} d\tau (n_{z\uparrow}(\tau) - n_{z\downarrow}(\tau))^2
 \end{aligned} \tag{5.14}$$

As in the Hubbard-Heisenberg model, we first obtain an effective action for the pair of sites (d, z) . As in the previous calculation, this will again generate a hybridisation function $\Delta_{zz, \sigma}$ for the bath site nearest to the zeroth site. No susceptibility will be generated however, because there is no

spin-exchange coupling within the bath. The net result is

$$\begin{aligned}
 S_{\text{eff}}^{d,z} = & \sum_{d,c} \sum_{\sigma} \int_0^{\beta} d\tau c_{i\sigma}^{\dagger}(\tau) \partial_{\tau} c_{i\sigma}(\tau) - V \sum_{\sigma} \int_0^{\beta} d\tau \left[c_{d\sigma}^{\dagger}(\tau) c_{z\sigma}(\tau) + \text{h.c.} \right] + J \int_0^{\beta} d\tau \vec{S}_d(\tau) \cdot \vec{S}_z(\tau) \\
 & - \frac{U}{2} \int_0^{\beta} d\tau (n_{d\uparrow}(\tau) - n_{d\downarrow}(\tau))^2 - \frac{W}{2} \int_0^{\beta} d\tau (n_{z\uparrow}(\tau) - n_{z\downarrow}(\tau))^2 \\
 & + \int_0^{\beta} d\tau d\tau' \sum_{\sigma} \Delta_{zz,\sigma}(\tau - \tau') c_{z\sigma}^{\dagger}(\tau) c_{z\sigma}(\tau')
 \end{aligned} \tag{5.15}$$

This hybridisation $\Delta_{zz,\sigma}$ is calculated in the cavity model of the eSIAM, obtained by removing the impurity and the zeroth site from the full model; it is a completely non-interacting system.

One can go one step further and also remove the zeroth site in order to obtain a theory for the impurity site. With this choice, the cavity model now also includes the zeroth site, and hence contains the correlation term associated with W . The one-particle connection V will lead to a modified hybridisation $\mathcal{F}$ for this effective theory; the Kondo terms will generate a susceptibility. The resultant action is

$$\begin{aligned}
 S_{\text{eff}}^d = & \sum_{\sigma} \int_0^{\beta} d\tau c_{d\sigma}^{\dagger}(\tau) \partial_{\tau} c_{d\sigma}(\tau) - \frac{U}{2} \int_0^{\beta} d\tau (n_{d\uparrow}(\tau) - n_{d\downarrow}(\tau))^2 \\
 & + \int_0^{\beta} d\tau d\tau' \left[\sum_{\sigma} \mathcal{F}_{\sigma}(\tau - \tau') c_{i\sigma}^{\dagger}(\tau) c_{i'\sigma}(\tau') + \chi_d(\tau - \tau') \vec{S}_i(\tau) \cdot \vec{S}_{i'}(\tau') \right]
 \end{aligned} \tag{5.16}$$

where $\mathcal{F}$ is defined in terms of the local correlator of the zeroth site

$$\mathcal{F}_{\sigma}(\tau - \tau') = V^2 G_1^{(d)}(z\sigma\tau; z\sigma\tau') = V^2 \langle c_{z\sigma}(\tau) c_{z\sigma}^{\dagger}(\tau') \rangle_{(d)} \tag{5.17}$$

with the average computed in the interacting cavity model that has the impurity site removed. The interaction arises from the W -term. The susceptibility is also calculated in this interacting cavity model, and follows the same definition as in the bulk model:

$$\chi_d(\tau - \tau') = J^2 \vec{S}_z(\tau) \cdot \vec{S}_z(\tau') . \tag{5.18}$$

4.5.3 RG Phase Diagram

A detailed picture of the PG in the impurity model is obtained from a momentum-resolved breakdown of Kondo screening in the strong-coupling regime $U \gg t$ phase of H_{aux} . A unitary renormalisation group (URG) scaling analysis [18] obtains a flow equation of the Kondo coupling $J_{\mathbf{k}_1, \mathbf{k}_2}^{(j)}$ (see Appendix A for details)

$$\Delta J_{\mathbf{k}_1, \mathbf{k}_2}^{(j)} = - \sum_{\mathbf{q} \in \text{PS}} \frac{J_{\mathbf{k}_2, \mathbf{q}}^{(j)} J_{\mathbf{q}, \mathbf{k}_1}^{(j)} + 4 J_{\mathbf{q}, \bar{\mathbf{q}}}^{(j)} W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}}}{\omega - \frac{1}{2} |\varepsilon_j| + J_{\mathbf{q}}^{(j)} / 4 + W_{\mathbf{q}} / 2} , \tag{5.19}$$

where ε_j is the energy of the shell being decoupled at the j^{th} step, the sum is over all occupied momentum states $\mathbf{q}$ of the energy shell ε_j , and $\bar{\mathbf{q}} = \mathbf{q} + \boldsymbol{\pi}$ is the particle-hole transformed state associated with $\mathbf{q}$. The bath interaction coupling $W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}}$ is found to be marginal under these transformations. While we present a detailed numerical evaluation of the RG equation for $J_{\mathbf{k}_1, \mathbf{k}_2}$ (eq.(5.19)) below, it is clear that the frustration of Kondo screening due to charge fluctuations (for attractive bath interactions $W < 0$) leads to the Mott transition [195].

A detailed numerical computation of the URG equation in eq. (5.19) leads to the phase diagram shown in Fig. 4.4(a). Upon tuning the ratio of the bath and Kondo interactions (W/J) from zero to negative values, the following phases emerge in the impurity model from the competition between J and W in eq. (5.19): (i) for $W/J < (W/J)_{\text{PG}}$, an LFL phase (red region), where the entire FS participates in Kondo screening, (ii) for $\frac{W}{J} \in [(\frac{W}{J})_{\text{PG}}, (\frac{W}{J})_c]$ (shaded region), a local PG phase where disconnected parts of the FS around the node participate in Kondo screening, and (iii) a local moment phase for $\frac{W}{J} > (\frac{W}{J})_c$ (blue region), where the impurity remains unscreened at low-energies. These can be visualised from spin correlations, $\chi_s(d, \mathbf{k}) = \langle \mathbf{S}_d \cdot \mathbf{S}_{\mathbf{k}} \rangle$, as shown in Fig. 4.4(b) for the PG. The values $(W/J)_{\text{PG}}$ and $(W/J)_c$ are therefore the entry into and exit from the PG phase.

Mapping onto the lattice model via tiling, we observe that RG-induced nodal–antinodal dichotomy in $J_{\mathbf{k}, \mathbf{k}'}$ in the impurity model is the microscopic origin of the PG in the lattice model: the extinction of Kondo coherence translates into spectral zeros in the lattice Green’s function (i.e., Luttinger surfaces) at antinodal momenta (Fig. 4.4(c)). This establishes that the $T = 0$ Mott transition of the 2D *extended*-Hubbard model proceeds from FL to MI through an intervening PG phase. In Fig. 4.5, similar insights obtain from the spin-correlation $\chi_s(\mathbf{k}, k_N) = \langle \mathbf{S}_{\mathbf{k}} \cdot \mathbf{S}_{k_N} \rangle$ in k -space (where k_N is the nodal momentum) and mutual information $I_2(\mathbf{k} : k_N)$ between two momentum states: the PG displays Fermi arcs in the nodal regions along with vanishing correlation and entanglement in the antinodal regions.

4.5.4 Unravelling of Kondo Screening

The $\mathbf{k}$ -space anisotropy of Kondo breakdown can be visualized in terms of zeros of $J_{\mathbf{k}_N, \mathbf{k}}$, involving spin-flip scattering between the node $\mathbf{k}_N = (\pi/2, \pi/2)$ and a general wavevector $\mathbf{k}$. For any W/J , the C_4 lattice symmetry dictates that $J_{\mathbf{k}_N, \mathbf{k}}$ vanishes if $\mathbf{k}$ belongs to any of the antinodes or adjacent nodes. Tuning W/J towards $(W/J)_{\text{PG}}$ leads to an unravelling of the Kondo screening: $J_{\mathbf{k}_N, \mathbf{k}}$ for $\mathbf{k}$ close to the adjacent nodes turns RG-irrelevant first, and a patch of zeros subsequently appears in $J_{\mathbf{k}_N, \mathbf{k}}$ around this point (Fig. 4.6(a)). Tuning W/J further extends the patch of zeros towards the antinodes (Fig. 4.6(b) and 4.6(c)). Kondo screening thus unravels by a systematic decoupling of all $J_{\mathbf{k}_1, \mathbf{k}_2}$ that connect adjacent quadrants of the Brillouin zone.

Precisely at $W/J = (W/J)_{\text{PG}}$, the antinode joins this connected region of zeros in $J_{\mathbf{k}_1, \mathbf{k}_2}$, marking the decoupling of the antinodes from all other points in the neighbourhood of the FS. This is an interaction-driven Lifshitz transition of the FS, and marks the entry into a PG phase possessing Fermi arcs [?]. Importantly, it coincides with an emergent two-channel Kondo (2CK) impurity model, where each channel corresponds to a pair of Fermi arcs on opposite faces of the conduction bath FS (see Subection 4.5.5). The 2CK nature of the PG is guaranteed by the symmetry of $J_{\mathbf{k}, \mathbf{k}'}$:

$$J_{\mathbf{k}, \mathbf{k}'} = -J_{\mathbf{k}+\mathbf{Q}, \mathbf{k}'} = -J_{\mathbf{k}, \mathbf{k}'+\mathbf{Q}}, \quad \mathbf{Q} = (\pi, \pi). \quad (5.20)$$

The PG expands by shrinking these disconnected Fermi arcs towards the respective nodes, leading to nodal metals whose disappearance heralds the Mott transition.

4.5.5 Emergence of a two-channel Kondo Model

As the bath interaction W is tuned through the L-PG phase, the four nodal points in the Brillouin zone are the last to decouple from the impurity. This allows us to write down a simpler Kondo model near the transition, focusing on scattering processes involving the nodal region. Specifically, for the Kondo term $J_{\mathbf{k}_1, \mathbf{k}_2}$, we find that only the following classes of scattering processes survive close to the critical point:

- Both momenta in a neighbourhood around $\mathbf{k}_N$, where $\mathbf{k}_N$ can be any of the four nodal points,
- One momentum from the neighbourhood around $\mathbf{k}_N$, while the other from the neighbourhood around $\mathbf{k}_N + (\pi, \pi)$.

The second class is tied to the first through a symmetry of the Hamiltonian (eq. ??). The crucial feature of these processes is that each node only interacts with the other node directly opposite to it across the origin. That is, $(\pi/2, \pi/2)$ and $(-\pi/2, -\pi/2)$ only interact between themselves, while $(\pi/2, -\pi/2)$ and $(-\pi/2, \pi/2)$ form their own pair of connected regions.

Let $S^\pm$ be the set of momentum states in a small window around nodes along $k_x = \pm k_y$. As discussed immediately above, the Kondo spin-exchange term close to the transition does not lead to scattering of any k -state from S^+ to S^- . This is verified by calculating the ratio $\max \{J_{k_1^+, k_2^-}\} / \max \{J_{k_1^+, k_2^+}\}$, where the maximum in the numerator is calculated from all the low-energy (fixed point) Kondo scattering processes that connect the two sets $S^\pm$, while the denominator maximum is from within the set S^+ . This ratio vanishes (left panel of Fig. 4.9) within the local pseudogap regime, indicating that no direct Kondo scattering process persists between the two sectors $S^\pm$ at low energies, deep within the pseudogap regime. We call the value of W at which this ratio vanishes W^* .

The decoupling of the k -space regions $S^\pm$ is captured by the following simplified low-energy Hamiltonian:

$$H_{\text{eff}} = H(S^+) + H(S^-); \quad H(S) = \sum_{\mathbf{q} \in S, \sigma} \varepsilon_{\mathbf{q}} c_{\mathbf{q}\sigma}^\dagger c_{\mathbf{q}\sigma} + \sum_{\mathbf{q}_1 \in S, \mathbf{q}_2 \in S} \sum_{\alpha, \beta} J^*(\mathbf{q}_1, \mathbf{q}_2) \mathbf{S}_d \cdot \sigma_{\alpha\beta} c_{\mathbf{q}_1\alpha}^\dagger c_{\mathbf{q}_2\beta}, \quad (5.21)$$

where $H(S^\pm)$ represents the dynamics of momentum states residing in regions $S^\pm$. The only source of information transfer between the two regions is the fact that they interact with the same impurity local moment. Eq. 5.21 tells us that the low-energy dynamics of the embedded eSIAM close to the Kondo-breakdown transition is governed by an emergent **two-channel Kondo model**, with the conduction bath states in the sets $S^\pm$ making up the two channels.

4.5.6 Momentum-resolved Dynamical Spectral Weight Transfer

Passage through the PG phase is accompanied by a highly structured transfer of spectral weight across the FS. Strong charge fluctuations develop between the nodal and antinodal regions of the FS

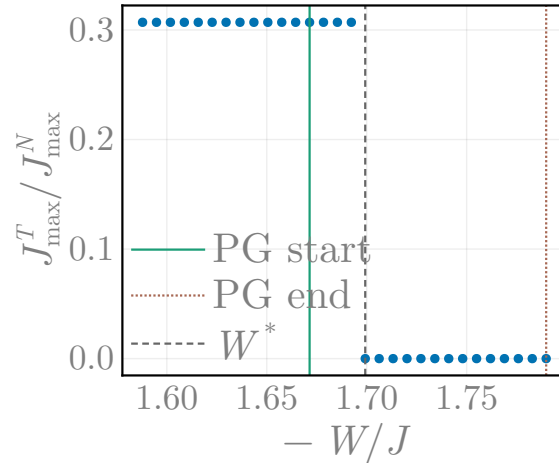


Figure 4.7: Ratio between the maximum values of the fixed point values of the Kondo couplings for scattering processes between and within the regions S^+ and S^- . The vanishing of the fixed point ratio indicates that the pseudogap regime features a singular low-energy effective theory, where the Fermi surface neighbourhood splits into two disconnected regions with no explicit term in the Hamiltonian connecting them directly.

in the PG regime of the impurity model (Fig. 4.8(a)), as captured by the correlator:

$$\chi_c(\mathbf{k}_1, \mathbf{k}_2) = \left\langle c_{\mathbf{k}_1\uparrow}^\dagger c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{k}_2\downarrow} c_{\mathbf{k}_2\uparrow} + \text{h.c.} \right\rangle. \quad (5.22)$$

These fluctuations dynamically redistribute low-energy spectral weight from the antinodes to higher energies, leading to selective gap formation. Accordingly, the Luttinger surfaces of the PG [186] coincides with the appearance of poles of the lattice model self-energy $\Sigma(\mathbf{k}, \omega = 0)$ near the antinodes; these poles approach the nodes on tuning towards the Mott transition (Fig. 4.8(b)). This mirrors the coalescing of finite-frequency poles of the self-energy poles towards zero frequency in the underlying impurity model (Fig. 4.8(c)).

4.6 Non-Fermi liquid excitations within the Pseudogap

4.6.1 Pseudogapped spectral function and singular self-energy

In the PG regime, the nature of gapless Fermi arcs changes dramatically. We have already argued that the low-energy dynamics of these gapless Fermi arcs are governed by an underlying two-channel Kondo (2CK) impurity model [196, 197]. This is consistent with the rapid fall of the impurity quasiparticle residue Z_{imp} (Fig. 4.9(a)) from finite values in the FL phase to vanishingly small values just before the onset of the PG. Inset shows the simultaneous emergence of increasingly uncompensated local magnetic moments upon traversing the PG phase. The accompanying impurity spectral function of the gapless arcs show a pseudogapped behaviour for $\omega \simeq 0$, with a rapid fall in the spectral weight at $\omega = 0$ upon traversing the PG (Fig. 4.9(c)). The collapse of the Kondo resonance into a pseudogapped spectral function is accompanied by the redistribution of spectral weight [186] in the

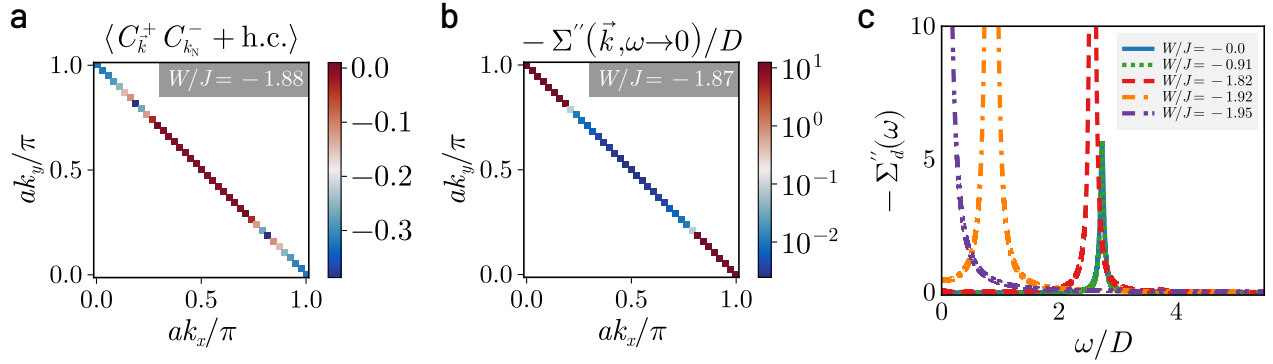


Figure 4.8: (a) Enhanced charge correlations $\chi_c(\mathbf{k}_1, \mathbf{k}_2) = \langle c_{\mathbf{k}_1\uparrow}^\dagger c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{k}_2\downarrow} c_{\mathbf{k}_2\uparrow} + \text{h.c.} \rangle$ between the nodal and antinodal regions, signalling Kondo breakdown in the pseudogap phase of the impurity model. (b) In turn, the breakdown leads to the gapping of the antinodal regions in the lattice model, seen from the appearance of poles in the imaginary part of the self-energy. (c) The imaginary part of the impurity self-energy $\Sigma''(\omega > 0)$ possesses a pole at non-zero ω in the PG that moves towards $\omega = 0$ as the Mott transition is approached.

impurity spectral function (Fig. 4.9(b)) from $\omega \sim 0$ to the Hubbard sidebands at finite frequencies $\omega \simeq \pm 3$ (in units of the bandwidth).

Concomitant with this is the emergence of a zero-frequency peak in the imaginary part of the self-energy of the NFL in the PG phase (Fig. 4.10(a)),

$$-1/\Sigma''(\omega) \sim 1/\Sigma''(0) + \omega^\beta. \quad (6.1)$$

Remarkably, we find that the exponent $\beta = 2$ characterises the NFL for the entire PG phase (see Fig. 4.10(b)), including the critical end-point. This is in stark contrast with the $\Sigma''(\omega) \sim \omega^2$ for the FL (Fig. 4.10(b)). A similar analysis reveals the power-law scaling for the low-frequency behaviour of the impurity spectral function in Fig. 4.11. We observe that $A_d(\omega) \sim A_d(0) + \omega^2$ for the pseudogap phase, while $1/A_d(\omega) \sim 1/A_d(0) + \omega^2$ for the Fermi liquid (inset of Fig. 4.11).

4.6.2 Finite-Frequency Scattering Rate and Optical Response Through the Pseudogap

All $\Sigma''(\omega)$ for the NFL are observed to lie above the Mott-Ioffe-Regel (MIR) bound [?, ?] (dashed blue line in Fig. 4.10(a)), while those for the FL lie below. The MIR bound is the maximum expected scattering rate in metals when the mean free path approaches the lattice spacing:

$$-2\Sigma''_{\text{MIR}} = 1/\tau_{\text{MIR}}, \quad \tau_{\text{MIR}} = l_{\text{min}}/v_F, \quad (6.2)$$

where l_{min} is the minimum mean free path in metals (equal to one lattice spacing) and τ_{MIR} is the associated lifetime of quasiparticles close to the FS (with Fermi velocity v_F). The height of the zero-frequency peak rises by almost 4 orders of magnitude from the start of the PG till its end at the Mott transition point (Fig. 4.10(c)); the dramatic growth of the peak height very near the Mott critical

point coincides with the coalescing of the finite-frequency poles of the self-energy into a single pole at zero-frequency, signalling the singular nodal NFL present at the Mott quantum critical point.

To further probe the breakdown of coherent transport in the pseudogap (PG) regime, we examine the product of the impurity spectral function $A(\omega)$ and the scattering lifetime $\tau(\omega)$, which approximates the optical conductivity $\sigma(\omega)$ via a Drude-type relation. As shown in the main panel of Fig. 4.12, this quantity exhibits a sharp Drude peak at $\omega = 0$ in the Fermi liquid (FL) regime (blue), consistent with long-lived quasiparticles. In stark contrast, the PG regime (green, red, orange) shows a pronounced suppression of low-frequency conductivity and the emergence of a finite-frequency peak centered near $\omega/D \sim 0.3$ – 0.4 (where D is the kinetic energy bandwidth).

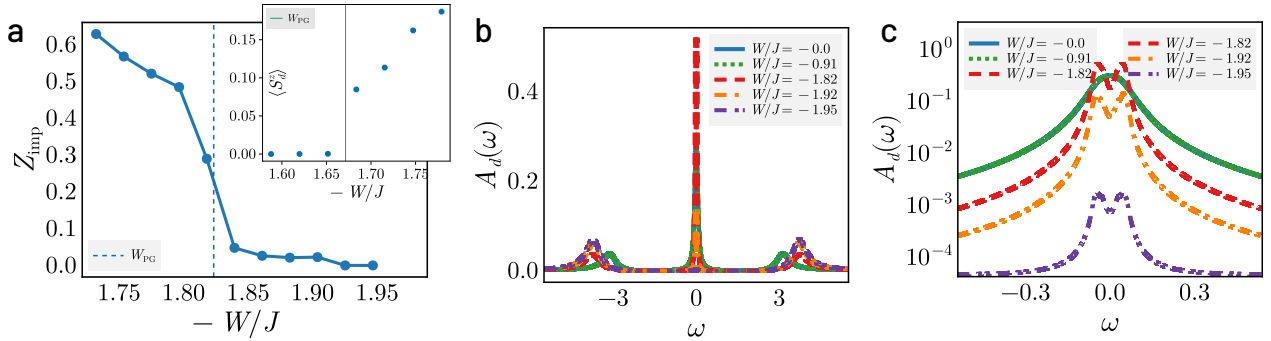


Figure 4.9: (a) Suppression of quasiparticle residue (Z_{imp}) as the impurity model is tuned towards the Mott transition. An initial drastic fall in Z_{imp} is observed for $W/J \lesssim (W/J)_{\text{PG}}$ from 0.3 to around 0.05, signalling the destruction of the FL with unravelling of Kondo screening. A steady decrease in Z_{imp} is observed in passage through the PG, and is vanishingly small close to the Mott transition due to a divergent self-energy. Inset shows the growth of unscreened impurity magnetic moment in the pseudogap phase, signalling the breakdown of Kondo screening. (b) Impurity Spectral function in the FL and PG phases. The evolution of the central peak from Kondo resonance to pseudogap is accompanied by dynamical spectral weight transfer to the Hubbard side bands at finite frequencies $\omega \simeq \pm 3$ (in units of the bandwidth). (c) Zoom-in of the central (Kondo) resonance of the impurity spectral function. While undisturbed in the Fermi liquid phase, it splits into a pseudogap in the PG phase, with its height diminishing rapidly as the Mott transition is approached.

Strikingly, this peak shifts very little across the PG regime (see inset), despite a deepening spectral gap and diverging self-energy. Its position lies well below the Mott gap ($\sim 2.5D$), suggesting that it represents an emergent mid-infrared (MIR) excitation intrinsic to the Mott metal characterising the PG. This feature closely mirrors the experimentally observed MIR peak in high- T_c cuprates [?, ?, ?], which has been variously attributed to excitations of electrons bound to vacancies [?], or the dressing of holes by spin excitations [?]. Here, its robust energy scale and insensitivity to increasing incoherence imply a universality persisting through the PG, likely rooted in short-range entanglement between deconfined doublons and holons excitations of the Mott metal. This is further evidence that the PG metal is not simply a degraded Fermi liquid, but a distinct phase whose optical response is driven by emergent degrees of freedom.

4.6.3 Non-local correlations in the pseudogap

In Fig. 4.13(a) and 4.13(b), the spin-flip correlations and mutual information between the impurity spin and conduction bath sites respectively are observed to undergo a crossover within the PG, from a short-ranged behaviour at its onset, to a long-ranged behaviour as the Mott transition approaches. The entanglement is also observed to be multipartite in nature: in Fig. 4.13(c), the quantum Fisher information (QFI) [133] computed for the ground state wavefunction using an operator corresponding to the sum of local spin-flip exchange processes shows a jump at the onset of the PG. Further, the FL is observed to possess bi-partite entanglement while the NFL of the PG phase displays pentapartite entanglement [?, ?].

These striking results imply that the Mott transition observed by us lies beyond the local quantum criticality scenario [156]. Instead, we observe the PG phase to be a novel state of strongly interacting quantum matter emergent from the breakdown of local Kondo screening. This state is described by a quantum critical Fermi surface [?] with NFL Fermi arcs that display increasingly critical behaviour, i.e., dynamics described by non-local quantum fluctuations, and excitations that become truly long-ranged close to the transition. It is tempting to conjecture that the long-range entangled critical state captured in Fig. 4.13(b) reflects the existence of a highly efficient scrambler [?, ?, ?], i.e., a state in which entanglement spreads among its constituents at the fastest rate permissible by quantum mechanics.

4.7 Exactly solvable nodal Non-Fermi liquid at Mott Criticality

Very close to the transition, the excitations of the nodal NFL correspond to those of a Hatsugai-Kohmoto model [?, ?]. This insight is obtained from a perturbation-theoretic treatment of the RG fixed point Hamiltonian of the impurity model for $W/J \lesssim (W/J)_{PG}$, by considering the effects of a small fixed point Kondo scattering probability J^* in the backdrop of a larger bath interaction parameter $|W|$. This yields the HK model [?, ?] as the singular part of the effective Hamiltonian arising from forward scattering processes.

As the bath interaction W is tuned through the L-PG phase, the nodal region is the last to decouple from the impurity. This allows us to write down a simpler Kondo model near the transition, where only the nodal region is hybridising with the impurity spin through Kondo interactions. This is done by retaining only those scattering processes $\mathbf{k}_1 \rightarrow \mathbf{k}_2$ that originate from and end at k -points within a small neighborhood of width $|\mathbf{q}|$ around the four nodal points: $\mathbf{k}_1, \mathbf{k}_2 \in \mathbf{N} + \mathbf{q}$, $|\mathbf{q}| \ll \pi$, where $\mathbf{N}$ can be any one of the four nodal points $\mathbf{N}_1 = (\pi/2, \pi/2)$, $\mathbf{N}_2 = (-\pi/2, \pi/2)$ and $\mathbf{N}_1 + \mathbf{Q}_1$ and $\mathbf{N}_2 + \mathbf{Q}_2$, where $\mathbf{Q}_1 = (-\pi, -\pi)$ and $\mathbf{Q}_2 = (\pi, -\pi)$ are the two nesting vectors. We assume that the window of $\mathbf{q}$ is small enough so that the fixed point Kondo coupling values $J^*(\mathbf{q}_1, \mathbf{q}_2)$ for the scattering processes involving $\mathbf{q}_1$ and $\mathbf{q}_2$ can be replaced by an average value J^* .

With these considerations, the simplified low-energy model near the transition describing the

Kondo scattering processes can be written as

$$\tilde{H}_{\text{imp-cbath}} = J^* \frac{1}{2} \sum_{l=1,2} \sum_{\mathbf{q}_1, \mathbf{q}_2} \sum_{\alpha, \beta} \mathbf{S}_d \cdot \boldsymbol{\sigma}_{\alpha\beta} \left(c_{\mathbf{N}_l + \mathbf{q}_1, \alpha}^\dagger c_{\mathbf{N}_l + \mathbf{q}_2, \beta} + c_{\mathbf{N}_l + \mathbf{Q}_l + \mathbf{q}_1, \alpha}^\dagger c_{\mathbf{N}_l + \mathbf{Q}_l + \mathbf{q}_2, \beta} - c_{\mathbf{N}_l + \mathbf{Q}_l + \mathbf{q}_1, \alpha}^\dagger c_{\mathbf{N}_l + \mathbf{q}_2, \beta} - c_{\mathbf{N}_l + \mathbf{q}_1, \alpha}^\dagger c_{\mathbf{N}_l + \mathbf{Q}_l + \mathbf{q}_2, \beta} \right). \quad (7.1)$$

The label l can take values 1 or 2, allowing us to consider both the decoupled channels in the PG (associated with $\mathbf{N}_1$ and $\mathbf{N}_2$). It also labels the nesting vectors $\mathbf{Q}_l$ associated with the two sets. α and β indicate spin indices, and $\mathbf{q}_1$ and $\mathbf{q}_2$ represent incoming and outgoing momenta in the scattering processes.

Because of the decoupling of the channel $l = 1$ and $l = 2$, we consider only the $l = 1$ channel for the rest of the calculations in this section. In order to simplify the Hamiltonian, we define new fermionic operators

$$\psi_{\mathbf{q}, \sigma} = \frac{1}{\sqrt{2}} (c_{\mathbf{N}_1 + \mathbf{q}, \sigma} + c_{\mathbf{N}_1 + \mathbf{Q}_1 - \mathbf{q}, \sigma}), \quad \phi_{\mathbf{q}, \sigma} = \frac{1}{\sqrt{2}} (c_{\mathbf{N}_1 + \mathbf{q}, \sigma} - c_{\mathbf{N}_1 + \mathbf{Q}_1 - \mathbf{q}, \sigma}), \quad (7.2)$$

The operator satisfy fermionic anticommutation relations. For convenience, we define new number operators for the sum and relative degrees of freedom:

$$s_{\mathbf{q}, \sigma} = \psi_{\mathbf{q}, \sigma}^\dagger \psi_{\mathbf{q}, \sigma}, \quad r_{\mathbf{q}, \sigma} = \phi_{\mathbf{q}, \sigma}^\dagger \phi_{\mathbf{q}, \sigma}. \quad (7.3)$$

We then have the following useful relation between the corresponding number operators $n = c^\dagger c$:

$$n_{\mathbf{N}_1 + \mathbf{q}, \sigma} + n_{\mathbf{N}_1 + \mathbf{Q}_1 - \mathbf{q}, \sigma} = s_{\mathbf{q}, \sigma} + r_{\mathbf{q}, \sigma}. \quad (7.4)$$

In terms of these new degrees of freedom, the auxiliary model Hamiltonian takes the form

$$\tilde{H} = -\frac{1}{2}W \sum_{\mathbf{q}, \sigma} r_{\mathbf{q}, \sigma} + W \sum_{\mathbf{q}_1, \mathbf{q}_2} r_{\mathbf{q}_1, \uparrow} r_{\mathbf{q}_2, \downarrow} + \sum_{\mathbf{q}_1, \mathbf{q}_2, \alpha, \beta} J^* \mathbf{S}_d \cdot \boldsymbol{\sigma}_{\alpha\beta} \phi_{\mathbf{q}_1 \alpha}^\dagger \phi_{\mathbf{q}_2 \beta} + \sum_{\mathbf{q}, \sigma} \varepsilon_{\mathbf{N}_1 + \mathbf{q}} \left(\psi_{\mathbf{q}, \sigma}^\dagger \phi_{\mathbf{q}, \sigma} + \phi_{\mathbf{q}, \sigma}^\dagger \psi_{\mathbf{q}, \sigma} \right), \quad (7.5)$$

where we have considered a simplified form of the bath interaction (for the channel $l = 1$), taking into account the density-density correlations in k -space.

For bath interaction strength close to the critical value ($W \lesssim W_c$), the fixed point coupling value J^* is much smaller than W . In order to obtain the gapless excitations of the system arising from the presence of the impurity site, we integrate out the impurity dynamics via a Schrieffer-Wolff transformation. The perturbation term $\mathcal{V}$ then consists of Hamiltonian terms that modify the impurity configuration,

$$\mathcal{V} = \sum_{\mathbf{q}, \sigma} \varepsilon_{\mathbf{N}_1 + \mathbf{q}} \left(\psi_{\mathbf{q}, \sigma}^\dagger \phi_{\mathbf{q}, \sigma} + \phi_{\mathbf{q}, \sigma}^\dagger \psi_{\mathbf{q}, \sigma} \right) + \sum_{\mathbf{q}_1, \mathbf{q}_2} J^* S_d^+ \phi_{\mathbf{q}_1 \downarrow}^\dagger \phi_{\mathbf{q}_2 \uparrow} + \text{h.c.}, \quad (7.6)$$

while the “non-interacting” Hamiltonian is

$$H_D = -\frac{1}{2}W \sum_{\mathbf{q}, \sigma} r_{\mathbf{q}, \sigma} + W \sum_{\mathbf{q}_1, \mathbf{q}_2} r_{\mathbf{q}_1, \uparrow} r_{\mathbf{q}_2, \downarrow}. \quad (7.7)$$

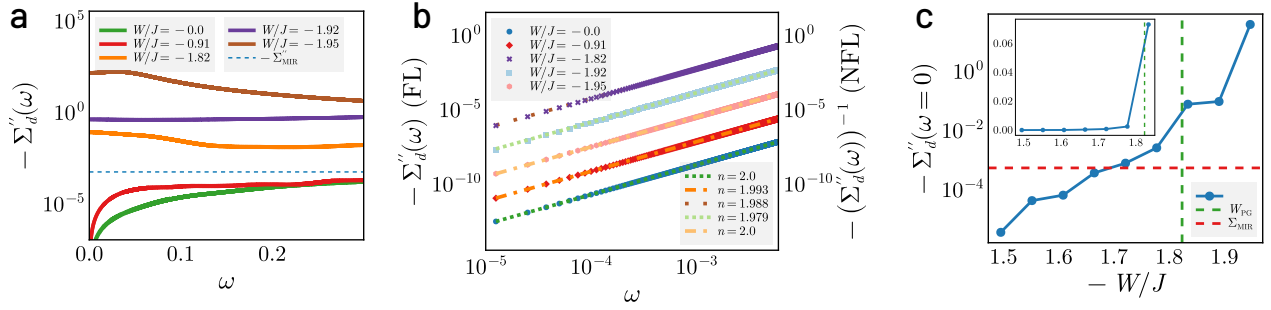


Figure 4.10: (a) Imaginary part of impurity self-energy $\Sigma''(\omega)$ as a function of frequency ω for Fermi liquid (FL, $|W/J| < 1.79$) and pseudogapped phases (NFL, $|W/J| \geq 1.79$). $\Sigma''(\omega)$ falls to zero as $\omega \rightarrow 0$ for the FL, while it attains a peak in the pseudogap. All $\Sigma''(\omega)$ for the NFL are observed to lie above the Mott-Ioffe-Regel (MIR) bound (dashed blue line), while those for the FL lie below. (b) Scaling of $\Sigma''(\omega)$ with frequency for Fermi liquid and pseudogapped phases. The FL self-energy fits to $\Sigma'' \sim \omega^\alpha$ with $\alpha \approx 2$, vanishing as $\omega \rightarrow 0$, while the NFL self-energy grows as $\Sigma'' \approx a - \omega^\beta$ for small ω , with $\beta \approx 2$. Remarkably, the NFL exponent remains mostly unchanged through the entirety of the PG phase. (c) Variation of the zero-frequency imaginary self-energy $-\Sigma''(\omega = 0)$ with ω in the FL and pseudogap phases (entry into the PG is marked by the vertical dashed line). The inset shows the same but in linear scale, in order to display the dramatic rise (by almost 30 times) on entering the PG.

For the present Hamiltonian, the low-energy state is the one that minimises the bath interaction term $\tilde{H}_{\text{cbath-int}}$. High-energy states are obtained by applying, on the state $|L\rangle$, the excitation operator $\phi_{\mathbf{q}_1, \sigma}^\dagger \phi_{\mathbf{q}_2, \bar{\sigma}}$ or its hermitian conjugate.

The complete second-order renormalised Hamiltonian is

$$\Delta \tilde{H} = \sum_{\mathbf{q}, \sigma} \epsilon_{\mathbf{q}} r_{\mathbf{q}, \sigma} + \mathcal{U} \sum_{\mathbf{q}, \sigma} r_{\mathbf{q}, \sigma} r_{\mathbf{q}, \bar{\sigma}} + \mathcal{U} \sum_{\mathbf{q}_1 \neq \mathbf{q}_2, \sigma} \left[r_{\mathbf{q}_1, \sigma} r_{\mathbf{q}_2, \bar{\sigma}} + \phi_{\mathbf{q}_1, \bar{\sigma}}^\dagger \phi_{\mathbf{q}_1, \sigma}^\dagger \phi_{\mathbf{q}_2, \sigma} \phi_{\mathbf{q}_2, \bar{\sigma}} \right]. \quad (7.8)$$

where $\epsilon_{\mathbf{q}} = \text{sign}(\epsilon_{\mathbf{N}_1 + \mathbf{q}}) \frac{\epsilon_{\mathbf{N}_1 + \mathbf{q}}^2}{-W}$ and $\mathcal{U} = \frac{J^{*2}}{4W}$.

The $\mathbf{q}_1 = \mathbf{q}_2$ component of the Hamiltonian shows the emergence of the exactly solvable Hatsugai-Kohmoto model [?, ?] at the critical point. The correlation term $\mathcal{U} \sum_{\mathbf{q}, \sigma} r_{\mathbf{q}, \sigma} r_{\mathbf{q}, \bar{\sigma}}$ leads to a transfer of spectral weight across the Fermi surface and separates the available states into three classes:

$$\langle n_{\mathbf{q}} \rangle = \begin{cases} 2, & \epsilon_{\mathbf{q}} < -|\mathcal{U}/2|, \\ 1, & |\mathcal{U}/2| > \epsilon_{\mathbf{q}} > -|\mathcal{U}/2|, \\ 0, & |\mathcal{U}/2| < \epsilon_{\mathbf{q}}. \end{cases} \quad (7.9)$$

In contrast to a Fermi liquid, there is a highly degenerate single-occupied region in the middle, and this gives rise to non-Fermi liquid excitations.

We now discuss the effects of the $\mathbf{q}_1 \neq \mathbf{q}_2$ component. The first term partially lifts the degeneracy of the central singly-occupied region and allows only zero magnetisation configurations. The second term creates gapped excitations involving the regions of zero and double occupancy; these represent

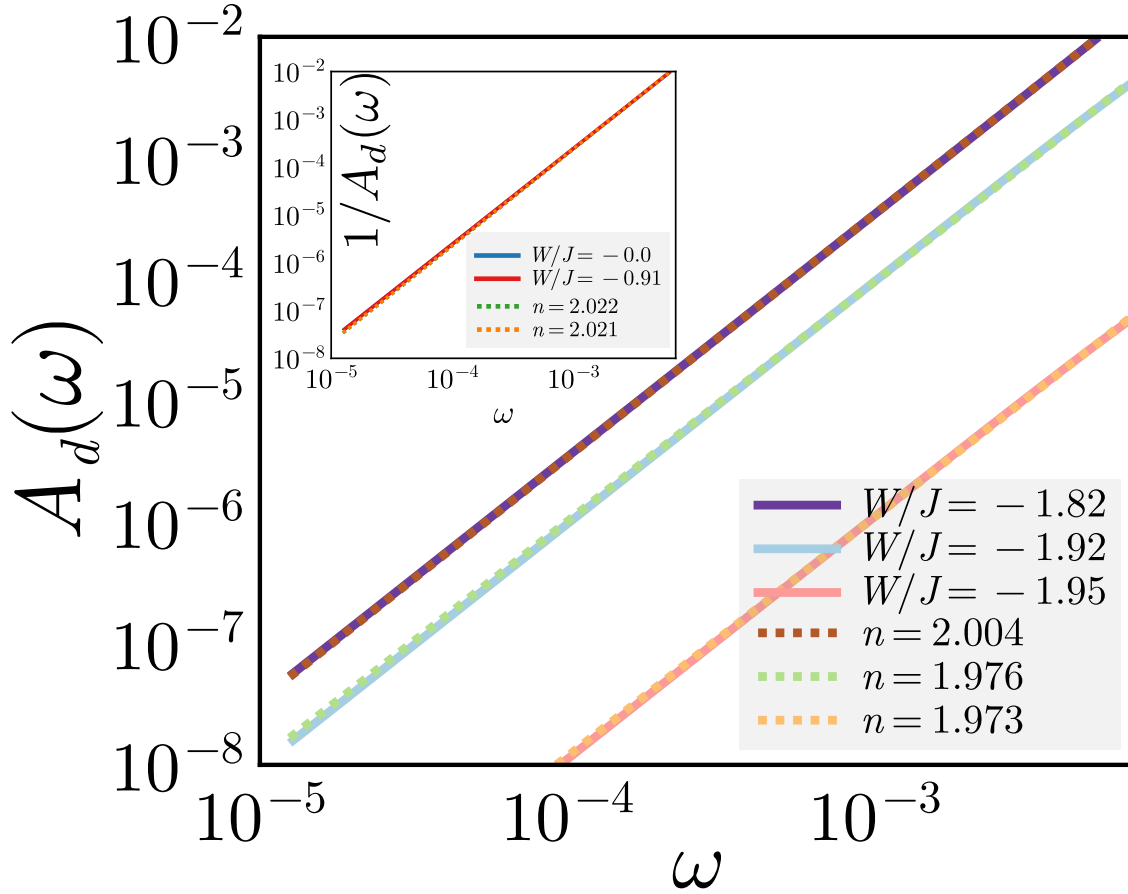


Figure 4.11: Impurity spectral function $A_d(\omega)$ in the local Fermi liquid (inset) and pseudogap (main). The PG spectral function is fit to a power law (with the $\omega = 0$ contribution subtracted, see Fig. 4.10(c) for the full spectral function). The exponent of the power law comes out to be $n \approx 2$ throughout the PG. The local FL spectral function (inset) is fit to a lorentzian (by fitting the inverse to a power law).

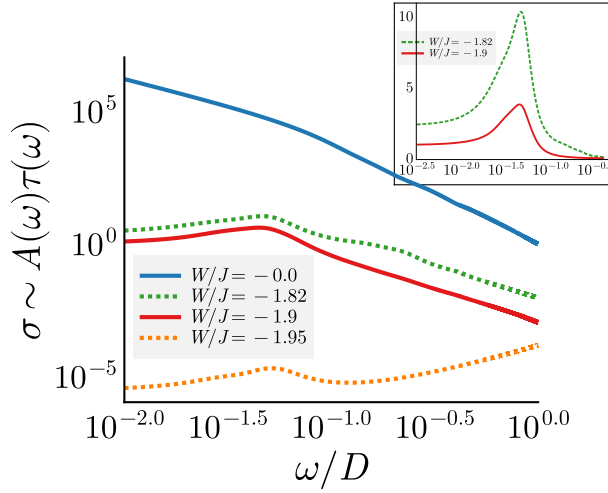


Figure 4.12: Optical conductivity $\sigma(\omega)$ for excitations of the impurity model, approximated through the product of the carrier density (taken from the density of states $A(\omega)$) and the lifetime $\tau(\omega)$ (obtained from the inverse scattering rate $-1/\Sigma''(\omega)$, where Σ'' is the imaginary part of the self-energy). The Fermi liquid (blue) shows a sharp Drude peak at $\omega = 0$ due to the presence of long-lived quasiparticles. The PG (green, red, orange) shows a shifted Drude peak at a non-zero frequency which is reminiscent of the mid-infrared (MIR) peak seen experimentally in the cuprates [?, ?, ?]. Inset shows the robustness of the position of the MIR peak through the PG, indicating it as a universal feature of the Mott metal.

subdominant pairing fluctuations of the nodal non-Fermi liquid. We recall that similar fluctuations across the Fermi surface were observed in our calculations of charge correlations in Fig. 4.8 (a), and are consistent with pairing fluctuations observed from quantum Monte Carlo simulations and found to give rise to quantum-critical behavior and PG physics [?].

Consequently, the resulting NFL metal of the lattice model involves long-lived excitations of multiple $\mathbf{k}$ -states, and manifest in the form of a divergent one-particle self-energy at the non-interacting FS [?]: $\Sigma_{\mathbf{q}}(\omega) = -\mathcal{U}^2/4(\omega - \epsilon_{\mathbf{q}})$, such that $\Sigma_{\epsilon_{\mathbf{q}}=0}(\omega \rightarrow 0) \rightarrow \infty$. This zero-frequency self-energy pole presages the transition into a Mott insulating phase, where it marks a hard gap in the spectral function for charge excitations. This is consistent with our findings for the lattice (Fig. 4.8(c)) and impurity self-energies (Fig. 4.10(c)). The nodal Mott metal thus comprises of a Greens function zero at the Fermi energy together with an anisotropic massless Dirac dispersion, leading to a non-zero Chern number [?, ?, ?]. This topological index survives into the Mott insulator.

For small but non-zero values of $\omega - \epsilon_{\mathbf{k}}$, we obtain a quasiparticle residue that vanishes with ω , $Z_{\text{imp}} \sim \omega^2/\mathcal{U}^2$. The scattering rate of this singular NFL possesses a sharp peak at the FS ($\omega = \epsilon_{\mathbf{k}_F}$):

$$\Gamma \sim \mathcal{U}^2 \delta(\omega - \epsilon_{\mathbf{k}}), \quad (7.10)$$

consistent with the sharply peaked Lorentzian $\Sigma''_{\mathbf{k}}(\omega) \sim \omega^{-2}$ captured in Fig. 4.10(a). This quantum critical NFL metal is an example of a strongly coupled scale-invariant form of quantum matter. The exact solution for eq.(?) reveals the presence of low-energy excitations comprised of holons and doublons [?]. These features point to the nodal NFL as a long-ranged and multipartite entan-

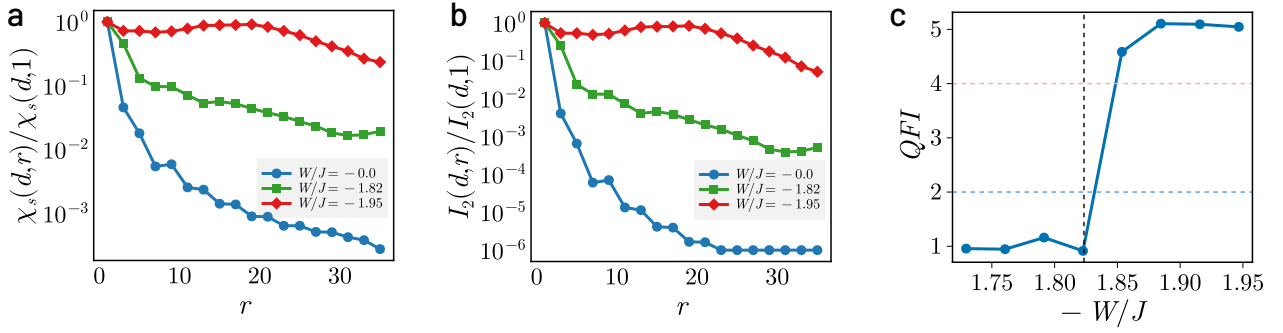


Figure 4.13: (a) Spin correlation $\langle \mathbf{S}_d \cdot \mathbf{S}_r \rangle$ and (b) mutual information $I_2(d, r)$ between the impurity spin and conduction bath local spin density as a function of the distance $\mathbf{r}$ between them, normalised against the value at $r = 1$. Both decay very quickly in the FL phase (blue), but show long-ranged behaviour in the NFL phase (green and red), extending to the edges of the system at the critical point (red). (c) Evolution of the Quantum Fisher Information F_Q for a nearest-neighbour spin-flip operator $\mathcal{O} = \sum_{i \in \text{odd}} (S_i^+ S_{i+1}^- + \text{h.c.})$ through the first Lifshitz transition and the pseudogap. The vertical dashed line marks the onset of the PG. To the left of it, the QFI in the Fermi liquid phase shows at most bipartite entanglement ($F_Q < 2$ (below blue dashed line)), while the PG shows the presence of multipartite entanglement upto 5 parties ($F_Q > 4$ (red dashed line)).

gled, strongly interacting scale-invariant state of quantum matter [?, ?, ?, ?] that are completely disconnected from the quasiparticles of the FL. Following the arguments laid out in [?], at finite temperatures, such a scale-invariant highly entangled non-Fermi liquid is likely associated with Planckian dissipation (i.e., $\Sigma''(T) \sim k_B T$) and a resistivity that varies linearly with temperature ($\rho \sim T$) [?, ?, 198]. This sits well with the possibility that this state is also a highly efficient scrambler [?, ?, ?]. Additionally, as mentioned above, we observe that the nodal metal possesses pairing fluctuations that can become dominant upon doping [?].

4.8 Heisenberg Model as a low-energy description of the tiled Mott insulator

For simplicity, we consider two impurity sites labelled 1 and 2 associated with two nearest-neighbour auxiliary models. The ground state subspace is four-fold degenerate:

$$|\Psi_L\rangle = \{|\sigma_1, \sigma_2\rangle\}, \quad \sigma_i = \pm 1, \quad (8.1)$$

where σ_i is the spin state of site i . This ground state is derived from the following "zeroth order" Hamiltonian that emerges in the local moment phase of the auxiliary models when all scattering processes between the impurity and conduction bath are RG-irrelevant:

$$H_0 = -\frac{U}{2} \sum_{i=1,2} (n_{i\uparrow} - n_{i\downarrow})^2; \quad (8.2)$$

the local correlation on the impurity site becomes the largest scale in the problem in this phase and pushes the $|n_i = 2\rangle$ and $|n_i = 0\rangle$ states to high energies. This then defines the high-energy subspace for our calculation:

$$|\Psi_H\rangle = |C_1, C_2\rangle, \quad (8.3)$$

where C_i can take values 0 or 2, indicating that the state i is either empty or full, respectively. Both the double and hole states exist at a charge gap of the order of $U/2$ above the low-energy singly-occupied subspace defined by the states $|\Psi_L\rangle$.

In order to allow virtual fluctuations that can lift the large ground state degeneracy and lower the energy, we consider (perturbatively) the effects of an irrelevant single-particle hybridisation that connects the nearest-neighbour sites. This perturbation Hamiltonian is therefore of the form

$$H_t = \sum_{\omega} V(\omega) \mathcal{P}(\omega) \sum_{\sigma} \left(c_{1\sigma}^{\dagger} c_{2\sigma} + \text{h.c.} \right), \quad (8.4)$$

where $V(\omega)$ only acts on states at the energy scale ω ; the renormalisation of V is encoded in the fact that $V(\omega)$ is largest for the excited states and vanishes at low-energies: $V(\omega \rightarrow 0) = 0$.

In order to obtain a low-energy effective Hamiltonian for the impurity sites arising from this hybridisation, we integrate out H_t via a Schrieffer-Wolff transformation. This leads to the following second-order Hamiltonian:

$$H_{\text{eff}} = \mathcal{P}_L H_t G \mathcal{H}_t \mathcal{P}_L. \quad (8.5)$$

The operator $\mathcal{P}_L$ projects onto the low-energy subspace $|\Psi_L\rangle$ - this ensures that we remain in the low-energy subspace at the beginning and at the end of the total process. The Greens function $G = (E_L - H_0)^{-1}$ incorporates the excitation energy to go from the low-energy subspace $|\Psi\rangle_L$ (of energy E_L) to the excited subspace $|\Psi\rangle_H$ of energy $E_L + U/2$. Substituting the form of the perturbation Hamiltonian and the excitation energy into the above expression gives

$$H_{\text{eff}} = \frac{V_H^2}{-U/2} \sum_{\sigma, \sigma'} \left[c_{1\sigma}^{\dagger} c_{2\sigma} c_{2\sigma'}^{\dagger} c_{1\sigma'} + c_{2\sigma}^{\dagger} c_{1\sigma} c_{1\sigma'}^{\dagger} c_{2\sigma'} \right]. \quad (8.6)$$

where $V_H \equiv V(\omega \rightarrow U/2)$ is the impurity-bath hybridisation at energy scales of the order of the Mott gap, in the sense of an RG flow. Terms with consecutive creation or annihilation operators on the same site are prohibited because each site is singly-occupied in the ground state. It is now easy to cast this Hamiltonian into a more recognizable form. For $\sigma' = \sigma$, we get

$$\sum_{\sigma} \delta_{\sigma, \sigma'} c_{1\sigma}^{\dagger} c_{2\sigma} c_{2\sigma'}^{\dagger} c_{1\sigma'} = \sum_{\sigma} (n_{1\sigma} - n_{1\sigma} n_{2\sigma}), \quad (8.7)$$

while $\sigma = -\sigma' = \pm 1$ gives

$$\sum_{\sigma} \delta_{\sigma, -\sigma'} c_{1\sigma}^{\dagger} c_{2\sigma} c_{2\sigma'}^{\dagger} c_{1\sigma'} = - (S_1^+ S_2^- + \text{h.c.}). \quad (8.8)$$

For the latter expression, we introduced the local spin-flip operators $S_i^\pm$. The expression above it can also be cast into spin variables, using the equations

$$\begin{aligned}\frac{1}{2} \sum_{\sigma} n_{i\sigma} &= \frac{1}{2}, \\ \frac{1}{2} \sum_{\sigma} \sigma n_{i\sigma} &= S_i^z,\end{aligned}\tag{8.9}$$

where the first equation is simply the condition of half-filling at each site, and the second equation is the definition of the local spin operator in z -direction. Adding and subtracting the equations gives $n_{i\sigma} = \frac{1}{2} + \sigma S_i^z$.

Substituting everything back into eq. 8.6 and dropping constant terms gives

$$H_{\text{eff}} = 2 \frac{V_H^2}{U/2} (2S_1^z S_2^z + S_1^+ S_2^- + S_1^- S_2^+) = J_{\text{eff}} \mathbf{S}_1 \cdot \mathbf{S}_2, \tag{8.10}$$

where the effective antiferromagnetic Heisenberg coupling is $J_{\text{eff}} = \frac{8V_H^2}{U}$.

4.9 Pseudogap as a strongly coupled phase of quantum matter

We now unveil an organising principle that leads to the remarkable properties observed above for the strongly interacting NFL of the PG phase. The existence of a sharp connected FS at $T = 0$ can be understood as the existence of a topologically protected manifold of gapless chiral excitations in $\mathbf{k}$ -space at the FS [17]. The FS is characterised by a topological index corresponding to an anomaly in the quantum many-body theory for electrons, and can be understood as a generalised symmetry of such a system [?, ?]. A theorem by Luttinger and Ward [14] shows that a count of the physical charge (known as Luttinger's volume) is identical to the topological index (a so-called homotopy charge known as Luttinger's count) even in the presence of electronic interactions that do not disturb the FS. We will now argue that the emergence of antinodal Luttinger surfaces involve a disconnection of the FS (into Fermi arcs) and that, by following La Nave et al. [?], the accompanying change in its topological properties leads to the existence of gapless NFL excitations that are non-local in nature.

The antinodal Luttinger surfaces arise from the splitting of double poles of the single-particle Greens function on the FS into poles lying on opposite complex half-planes, together with zeros that are pinned at the FS. These changes in the analytic structure of the single-particle Greens function have important consequences. First, the emergent zeros break a $\mathbb{Z}_2$ symmetry of the FS [?, ?, ?]. This symmetry is guaranteed within FL theory because the presence of quasiparticles vanishingly close to the FS allows the interchange of spin and charge degrees of freedom, promoting the separate $SU(2)$ symmetries of spin and charge to the larger symmetry of $O(4) = SU(2) \times SU(2) \times \mathbb{Z}_2$ [?]. The insertion of zeros of the Greens function at the Fermi surface in the PG, and the associated splitting of the Greens function, breaks this $\mathbb{Z}_2$ symmetry by placing the pole for the spin excitation in one half of the complex plane while placing that for the charge excitation in the other half [?, ?].

Second, they signal a divergent electronic self-energy as a function of the wavevector $\mathbf{k}$, render ill-behaved the Luttinger-Ward functional of the interacting electronic problem, and violate the generalised symmetry encoded within it. The changes in the pole structure change the Luttinger count

topological invariant, while the zeros give rise to an additional topological contribution (linked to the Adler-Bell-Jackiw-type chiral anomaly) [?, ?, 199, 200]. As a consequence of the half-filled particle-hole symmetric nature of the system at hand, Luttinger's volume is preserved upon taking into account topological contributions from both the Luttinger count *and* the zeros [15]. Importantly, La Nave et al. [?] argue, following recent developments in understanding generalised symmetries [?], that the additional anomaly arising from the Luttinger surfaces guarantees the existence of gapless NFL excitations that are non-local in nature.

Thus, the drastic change in nature of the real-space excitations - from locally well-defined Landau quasiparticles of the FL to the increasingly nonlocal excitations of the NFL Fermi arcs in the PG phase - appears to be dictated by a topological principle and, therefore, robust under renormalisation. This signals the NFL Fermi arcs of the PG as an emergent phase of strongly interacting quantum matter - which we dub the *Mott metal* - that is the parent metal of the MI. [Such an identification of the pseudogap as a distinct phase is backed up by angle-resolved photoemission spectroscopy \(ARPES\) experiments \[?\]; the anomalous momentum, temperature and doping dependence of the gap function observed in the cuprates suggests that the pseudogap maybe described by an order parameter distinct from that of the superconductor \[?, ?\].](#) The same topological principle connects the nonlocal unparticle-like gapless excitations [?, ?, ?, ?] of the scale-invariant nodal NFL of the HK model observed precisely at the Mott critical point to those of the rest of the PG phase, e.g., a universal scaling of $\Sigma''_{\mathbf{k}}(\omega) \sim (a + \omega^2)^{-1}$ (Fig. 4.10(b)) and local density of states $A_d(\omega) \sim \omega^2$ (Fig. 4.11) of the NFL throughout the PG phase, such that the value of $\Sigma''_{\mathbf{k}}(\omega = 0)$ value continues to grow upon violating the MIR bound (Fig. 4.10(c)).

4.10 Conclusions

Our analysis reveals that the Mott transition proceeds via a continuous evolution through a PG regime characterized by a singular NFL metal - the Mott metal - with deconfined holon-doublon excitations confined to nodal Fermi arcs, and a scaling of it's spectral features with universal power-law exponents. As the system approaches criticality, this metallic phase exhibits increasingly non-local correlations and a divergent self-energy, signalling the breakdown of Landau quasiparticles, formation of local moments and the onset of long-range quantum entanglement. Anchored in two-channel Kondo dynamics at intermediate scales and governed by Hatsugai-Kohmoto physics near the critical endpoint, the Mott metal provides a unified framework for understanding anomalous metallicity in strongly correlated systems. It's fate under finite doping presents a compelling direction for future investigation. [More complicated models can be studied by suitably extending the tiling approach and choosing an appropriate auxiliary model. Electron-phonon coupling can be taken into account by studying an Anderson-Holstein model that couples the impurity site to local lattice modes \[?\]. The effects of disorder can be studied by averaging quantities over various disorder configurations of the conduction bath. The versatility of the unitary RG solver and the impurity model-based construction makes it convenient to explore a large variety of lattice models.](#)

Chapter 5

Kondo Breakdown in a Heavy-Fermion Model

5.1 Bilayer extended Hubbard model: Impurity model

[195] We approach the heavy-fermion problem by starting from a bilayer extended Hubbard model, consisting of two layers (f and c). Towards studying this lattice model, we adopt a two-layer impurity problem that hosts a correlated impurity site in each layer (S_f and S_d):

$$H_{\text{aux}} = H_{\text{lp}} + H_f + H_d + H_{fd} , \quad (1.1)$$

where H_{lp} is the Hamiltonian for the non-interacting itinerant electrons of either layer,

$$H_{\text{lp}} = -t \sum_{\sigma, \alpha} \sum_{\langle i, j \rangle} \left(c_{i, \sigma, \alpha}^\dagger c_{j, \sigma, \alpha} + \text{h.c.} \right) - \sum_{i, \sigma, \alpha} \mu_\alpha n_{i, \sigma, \alpha} , \quad (1.2)$$

such that α sums over the two layers f and d . H_f and H_d describe the dynamics of the correlated impurity sites (and their local neighbourhood) in each layer:

$$\begin{aligned} H_f = & -\frac{U_f}{2} (n_{f, \uparrow} - n_{f, \downarrow})^2 + \sum_{Z \in \text{NN}} \left[-V \sum_{\sigma} \left(c_{f, \sigma}^\dagger c_{Z, \sigma, f} + \text{h.c.} \right) + \frac{J}{2} \sum_{\alpha, \beta} \mathbf{S}_f \cdot \boldsymbol{\sigma}_{\alpha\beta} c_{Z, \alpha, f}^\dagger c_{Z, \beta, f} \right. \\ & \left. - \frac{W_f}{2} (n_{Z, \uparrow, f} - n_{Z, \downarrow, f})^2 \right] , \\ H_d = & -\eta_d n_d - \frac{U_d}{2} (n_{d, \uparrow} - n_{d, \downarrow})^2 + \sum_{Z \in \text{NN}} \left[-V \sum_{\sigma} \left(c_{d, \sigma}^\dagger c_{Z, \sigma, d} + \text{h.c.} \right) + \frac{J}{2} \sum_{\alpha, \beta} \mathbf{S}_d \cdot \boldsymbol{\sigma}_{\alpha\beta} c_{Z, \alpha, d}^\dagger c_{Z, \beta, d} \right] , \end{aligned} \quad (1.3)$$

where $c_{v, \sigma, \alpha}^\dagger$ can refer to creation operator for either the f -layer. Finally, H_{fd} represents the inter-layer hybridisation:

$$H_{fc} = -V_\perp \sum_{\sigma} \left(c_{d, \sigma}^\dagger c_{f, \sigma} + \text{h.c.} \right) . \quad (1.4)$$

5.2 Tiling Into Bilayer Extended Hubbard Model

Tiling the impurity model leads to a bilayer extended Hubbard model on the 2D square lattice. In order to tile our auxiliary model to restore translation invariance, we identify the cluster which acts as the “unit” cell of translation. As discussed in the text surrounding eq. 3.9, a good choice for the unit cell is the impurity site and its surrounding correlated environment. Placing the impurity sites at $\mathbf{r}_d$ in each layer, we get

$$H_{\text{clust}}(\mathbf{r}_d) = H_f(\mathbf{r}_d) + H_d(\mathbf{r}_d) + H_{fc}(\mathbf{r}_d) . \quad (2.1)$$

The details of tiling a cluster Hamiltonian were already shown in Sec. 3.3.2; the general idea is to use manybody translation operators (details in Sec. 3.3.1) to center the cluster Hamiltonian at various sites of the lattice, such that combining all such terms leads to a periodised Hamiltonian. we write down the final result here:

$$\begin{aligned} H_{\text{latt}} &= \sum_{\mathbf{r}} T^\dagger(\mathbf{r}) H_{\text{clust}}(\mathbf{r}_d) T(\mathbf{r}) \\ &= -\frac{2V}{Z} \sum_{\langle i,j \rangle, \sigma} \left(c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.} \right) - \sum_i \left[\eta_d n_{i,d} + \frac{U_d}{2} (n_{i,\uparrow,d} - n_{i,\downarrow,d})^2 \right] + \frac{2J}{Z} \sum_{\langle i,j \rangle} \mathbf{S}_{i,d} \cdot \mathbf{S}_{j,d} \\ &\quad - \frac{2V}{Z} \sum_{\langle i,j \rangle, \sigma} \left(f_{i,\sigma}^\dagger f_{j,\sigma} + \text{h.c.} \right) - \frac{U_f - W_f}{2} \sum_i (n_{i,\uparrow,f} - n_{i,\downarrow,f})^2 + \frac{2J}{Z} \sum_{\langle i,j \rangle} \mathbf{S}_{i,f} \cdot \mathbf{S}_{j,f} \\ &\quad - V_\perp \sum_{i,\sigma} (f_{i,\sigma}^\dagger c_{i,\sigma} + \text{h.c.}) \end{aligned} \quad (2.2)$$

Tiling the impurity model therefore leads to a bilayer extended Hubbard model, with effective couplings that depend on the auxiliary model couplings. The value of η_d defines the effective chemical potential of the d -layer; a non-zero value moves the layer away from half-filling. The f -layer is pinned at half-filling.

Following Sec. 3.4.1, the retarded k -space Greens function in either layer can be expressed in terms of auxiliary model Greens functions:

$$G(\mathbf{k}, \sigma, \alpha; \omega) = \sum_{\nu} C_{\nu, \mathbf{k} \alpha} \left[\mathcal{G}^>(\mathcal{T}_{\nu, \sigma, \alpha}; \omega - \mathcal{E}_\nu(\mathbf{k})) - \mathcal{G}^<(\mathcal{T}_{\nu, \sigma, \alpha}; \omega + \mathcal{E}_\nu(\mathbf{k})) \right] , \quad (2.3)$$

where $\alpha = f, d$ denotes the layer in which the Greens function is being calculated, the modes ν indicate the operators which diagonalise the Hamiltonian H_{1p} , with energies $\mathcal{E}_\nu(\mathbf{k})$:

$$H_{1p} = -\frac{2V}{Z} \sum_{\langle i,j \rangle, \sigma} \left(c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.} + f_{i,\sigma}^\dagger f_{j,\sigma} + \text{h.c.} \right) - V_\perp \sum_{i,\sigma} (f_{i,\sigma}^\dagger c_{i,\sigma} + \text{h.c.}) - \sum_i \eta_d n_{i,d} , \quad (2.4)$$

and the $\mathcal{T}$ -matrices are defined in eq. 4.10:

$$\mathcal{T}_{\nu, \sigma, \alpha} = \left[\left(\sum_{\sigma'} c_{\alpha \sigma'}^\dagger + \text{h.c.} \right) + (S_\alpha^+ + \text{h.c.}) + (C_\alpha^+ + \text{h.c.}) \right] c_{\nu, \sigma} , \quad (2.5)$$

In order to calculate $\mathcal{E}_v(\mathbf{k})$, we introduce the k -space modes via a Fourier transform:

$$c_{i,\sigma}^\dagger = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{r}_i} c_{\mathbf{k},\sigma}^\dagger, \quad f_{i,\sigma}^\dagger = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{r}_i} f_{\mathbf{k},\sigma}^\dagger. \quad (2.6)$$

In terms of these modes, the 1-particle Hamiltonian decouples in k -space:

$$H_{1p} = \sum_{\mathbf{k},\sigma} \left[(\epsilon_{\mathbf{k}} - \eta_d)(n_{\mathbf{k},\sigma,d} + n_{\mathbf{k},\sigma,f}) - V_\perp (c_{\mathbf{k},\sigma}^\dagger f_{\mathbf{k},\sigma} + \text{h.c.}) \right]. \quad (2.7)$$

The diagonal modes are given by

$$v_{\mathbf{k},\sigma,\pm}^\dagger = C_{\pm,k} c_{\mathbf{k},\sigma}^\dagger + \sqrt{1 - C_{\pm,k}^2} f_{\mathbf{k},\sigma}^\dagger, \quad (2.8)$$

the coefficients obtained from diagonalising the matrix

$$H_{1p}(\mathbf{k}) = \begin{pmatrix} \epsilon_{\mathbf{k}} & -V_\perp \\ -V_\perp & \epsilon_{\mathbf{k}} - \eta_d \end{pmatrix}. \quad (2.9)$$

The energies are given by $\mathcal{E}_\pm(\mathbf{k}) = \epsilon_{\mathbf{k}} - \eta_d/2 \pm \Delta$, with $\epsilon_{\mathbf{k}}$ being $-\frac{4V}{Z}(\cos k_x + \cos k_y)$ for a 2D square lattice and $\Delta = \sqrt{V_\perp^2 + \eta_d^2}/4$. Substituting into the expression for the lattice Greens function, we get

$$G(\mathbf{k}, d; \omega) = \sum_{\alpha=\pm} C_{\alpha,k} \left[\mathcal{G}^>(\mathcal{T}_{\mathbf{k},\alpha,d}; \omega - \epsilon_{\mathbf{k}} + \eta_d/2 + \alpha\Delta) + \mathcal{G}^<(\mathcal{T}_{\mathbf{k},\alpha,d}; \omega + \epsilon_{\mathbf{k}} - \eta_d/2 - \alpha\Delta) \right]. \quad (2.10)$$

Following eq. 4.32, we can similarly write down an expression the local lattice Greens function, in real-space:

$$\begin{aligned} G_{\text{loc}}(\omega) &= \mathcal{G}(d\sigma; \omega) \\ &+ \sum_{\mathbf{k},\alpha=\pm} C_{\alpha,k} \left[\mathcal{G}^>(\mathcal{T}_{\mathbf{k},\alpha,d}, d\sigma; \omega - \epsilon_{\mathbf{k}} + \eta_d/2 + \alpha\Delta) + \mathcal{G}^<(\mathcal{T}_{\mathbf{k},\alpha,d}, d\sigma; \omega + \epsilon_{\mathbf{k}} - \eta_d/2 - \alpha\Delta) \right] \\ &+ \sum_{\mathbf{k}} \mathcal{G}(d\sigma, \mathcal{T}_{\mathbf{k},\sigma}; \omega) \\ &+ \sum_{\mathbf{k},\mathbf{q},\alpha=\pm} C_{\alpha,k} \left[\mathcal{G}^>(\mathcal{T}_{\mathbf{k},\alpha,d}, \mathcal{T}_{\mathbf{q},\sigma,d}; \omega - \epsilon_{\mathbf{k}} + \eta_d/2 + \alpha\Delta) + \mathcal{G}^<(\mathcal{T}_{\mathbf{k},\alpha,d}, \mathcal{T}_{\mathbf{q},\sigma,d}; \omega + \epsilon_{\mathbf{k}} - \eta_d/2 - \alpha\Delta) \right] \end{aligned} \quad (2.11)$$

5.3 Unitary RG Analysis of The Bilayer Lattice-Embedded ESIAM

In the limit of large U_α , we carry out a Schrieffer-Wolff transformation and work with the following low-energy Hamiltonian:

$$H_{\text{aux}} = \sum_{\mathbf{k}, \sigma, \alpha} \epsilon_{\mathbf{k}, \alpha} n_{\mathbf{k}, \sigma, \alpha} + \sum_{\alpha} \sum_{Z \in \text{NN}} \left[\frac{1}{2} J_{\alpha} \sum_{\sigma, \sigma'} \mathbf{S}_{\alpha} \cdot \boldsymbol{\sigma}_{\alpha \beta} c_{Z, \sigma, \alpha}^{\dagger} c_{Z, \sigma', \alpha} - \frac{W_{\alpha}}{2} (n_{Z, \uparrow, \alpha} - n_{Z, \downarrow, \alpha})^2 \right] + J \mathbf{S}_f \cdot \mathbf{S}_d. \quad (3.1)$$

In order to study the low-energy physics of the impurity model, we iteratively integrate out high-energy degrees of freedom using the unitary RG method. Let the momentum states being decoupled from the two layers be $|\mathbf{q}, \sigma, f\rangle$ and $|\mathbf{q}, \sigma, d\rangle$ if they are above the Fermi energy (unoccupied in the low-energy configuration) and $|\bar{\mathbf{q}}, \sigma, f\rangle$ and $|\bar{\mathbf{q}}, \sigma, d\rangle$ if they are below the Fermi energy (occupied in the low-energy configuration). For every shell of conduction bath states of width ΔD integrated out, the Hamiltonian couplings renormalise by folding in the effects of virtual excitations to these states. We already have the renormalisation group equations for the couplings J_{α} in the case of $J = 0$:

$$\frac{\Delta J_f(\mathbf{k}_1, \mathbf{k}_2)}{\Delta D} = \int_{UV} d\mathbf{q} \rho(\mathbf{q}) [J_f(\mathbf{k}_1, \mathbf{q}) J_f(\mathbf{q}, \mathbf{k}_2) + J_f(\mathbf{q}, \bar{\mathbf{q}}) W_f(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)] G_f^{(0)}(\mathbf{q}), \quad (3.2)$$

where $\mathbf{q}$ sums over the UV states being integrated out, $\rho(\mathbf{q})$ is the density of states for the momentum states being integrated out and $G_f^{(0)}$ is the propagator for the excited intermediate state (for $J = 0$):

$$G_f^{(0)}(\mathbf{q}) = \frac{1}{\omega - |\epsilon_f(\mathbf{q})|/2 + J_f(\mathbf{q})/4 + W_f(\mathbf{q})/2}. \quad (3.3)$$

An identical equations holds for ΔJ_d , with all f -quantities replaced with d -quantities. Switching on J modifies the excitation energy in the propagator:

$$G_f(\mathbf{q}) = \frac{1}{1/G_f^{(0)} - J \vec{S}_f \cdot \vec{S}_d}. \quad (3.4)$$

This expression can be simplified by noting that the operator in the denominator has only two eigenvalues, $-3/4$ in the singlet sector and $1/4$ in the triplet sector. Defining the projectors $\mathcal{P}_0$ and $\mathcal{P}_1$ for the two sectors, we have

$$\vec{S}_f \cdot \vec{S}_d = \frac{-3}{4} \mathcal{P}_0 + \frac{1}{4} \mathcal{P}_1, \quad \mathcal{P}_0 + \mathcal{P}_1 = \mathbb{I}. \quad (3.5)$$

Using these relations, we can write

$$\begin{aligned} G_f(\mathbf{q}) &= \frac{1}{1/G_f^{(0)} + 3J/4} \mathcal{P}_0 + \frac{1}{1/G_f^{(0)} - J/4} \mathcal{P}_1 \\ &= \left(\frac{1/4}{1/G_f^{(0)} + 3J/4} + \frac{3/4}{1/G_f^{(0)} - J/4} \right) \mathbb{I} + \left(\frac{1}{1/G_f^{(0)} + 3J/4} - \frac{1}{1/G_f^{(0)} - J/4} \right) \vec{S}_f \cdot \vec{S}_d. \end{aligned} \quad (3.6)$$

Noting that only the first operator renormalises the Kondo interaction, we get the following modified RG equation for the Kondo coupling J_α :

$$\frac{\Delta J_\alpha}{\Delta D} = \int_{UV} d\mathbf{q} \rho(\mathbf{q}) [J_\alpha(\mathbf{k}_1, \mathbf{q}) J_\alpha(\mathbf{q}, \mathbf{k}_2) + J_\alpha(\mathbf{q}, \bar{\mathbf{q}}) W_\alpha(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)] \left(\frac{1}{4} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) + 3J/4} + \frac{3}{4} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) - J/4} \right) \quad (3.7)$$

We now turn to the renormalisation of J , arising from hybridisation of f - and d -layers. In order to capture coherent scattering processes between the two layers, we project onto the following rotated basis of excited states:

$$|H\rangle_\pm = \frac{1}{\sqrt{2}} \left(|H(\mathbf{q})\rangle_f \pm |H(\mathbf{q})\rangle_d \right), \quad (3.8)$$

$$\mathcal{P}(\mathbf{q})_H = |H\rangle_+ \langle H|_+ + |H\rangle_- \langle H|_- ,$$

where $|H(\mathbf{q})\rangle_\alpha$ is the state obtained upon exciting both states in the layer α : $|H(\mathbf{q})\rangle_\alpha = c_{\mathbf{q},\alpha}^\dagger c_{\bar{\mathbf{q}},\alpha} |L\rangle_f |L\rangle_d$, and $\mathcal{P}(\mathbf{q})_H$ projects onto this rotated excited basis. We focus on the spin-flip component $JS_f^+ S_d^-$ of the Hamiltonian. There are two kinds of processes that renormalise this coupling. We first consider one that involves a spin-flip of the d -layer followed by a spin-flip of the f -layer:

$$\Delta H = \frac{1}{N^2} \sum_{\mathbf{q} \in UV} \langle L | \frac{1}{2} J_f(\bar{\mathbf{q}}, \mathbf{q}) S_f^+ c_{\mathbf{q}-\downarrow, f}^\dagger c_{\mathbf{q}+\uparrow, f} \mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H \frac{1}{2} J_d(\mathbf{q}, \bar{\mathbf{q}}) S_d^- c_{\mathbf{q}\uparrow, d}^\dagger c_{\bar{\mathbf{q}}\downarrow, d} | L \rangle, \quad (3.9)$$

where $|L\rangle = |L\rangle_f |L\rangle_d$ is the initial configuration for both layers. G is the propagator for the excited state:

$$G = \frac{1}{\omega - H_D}, \quad (3.10)$$

where H_D is the diagonal part of the Hamiltonian corresponding to the excited(intermediate) state. The diagonal part consists of the kinetic energy and the Ising part of the Kondo interaction:

$$H_D = \sum_{\mathbf{q}, \sigma \in \mathbf{q}_\pm, \alpha} \epsilon_\alpha(\mathbf{q}) \tau_{\mathbf{q}, \sigma} + \frac{1}{2} \sum_{\alpha} J_\alpha S_\alpha^z \sum_{\mathbf{q}, \sigma} \sigma n_{\mathbf{q}, \sigma, \alpha} + J \vec{S}_f \cdot \vec{S}_d. \quad (3.11)$$

Calculating the matrix elements of the propagator in the rotated basis gives

$$\mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H = \frac{1}{2} (G_f + G_d) (|H\rangle_+ \langle H|_+ + |H\rangle_- \langle H|_-) + \frac{1}{2} (G_f - G_d) (|H\rangle_+ \langle H|_- + |H\rangle_- \langle H|_+), \quad (3.12)$$

where

$$G_\alpha(\mathbf{q}) = \frac{1}{\omega - \frac{1}{2} [\epsilon_\alpha(\mathbf{q}) - \epsilon_\alpha(\bar{\mathbf{q}})] + \frac{1}{2} J_\alpha(\mathbf{q}) + W_\alpha(\mathbf{q}) - \frac{1}{4} J}. \quad (3.13)$$

In eq. 3.9, since the first process is an excitation into a d -state and the second process is a de-excitation from an f -state, the expression can be simplified into

$$\langle L | S_f^+ c_{\mathbf{q}-\downarrow, f}^\dagger c_{\mathbf{q}+\uparrow, f} \mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H S_d^- c_{\mathbf{q}\uparrow, d}^\dagger c_{\bar{\mathbf{q}}\downarrow, d} | L \rangle = S_f^+ S_d^- \langle H_f | \mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H | H_d \rangle = \frac{1}{2} (G_f + G_d) S_f^+ S_d^-. \quad (3.14)$$

The net Hamiltonian renormalisation is finally a sum over the contribution from each mode:

$$\Delta H = \frac{1}{8N^2} \sum_{\mathbf{q} \in UV} J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) (G_f(\mathbf{q}) + G_d(\mathbf{q})) S_f^+ S_d^- . \quad (3.15)$$

Converting the sums to integrals (assuming a density of states $\rho(\mathbf{q})$) gives the final expression

$$\frac{\Delta H}{\Delta D} = \frac{1}{8N} S_f^+ S_d^- \int_{UV} d\mathbf{q} \rho(\mathbf{q}) J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) (G_f(\mathbf{q}) + G_d(\mathbf{q})) . \quad (3.16)$$

The time-reversed of this process can be obtained simply by exchanging the d -interaction with the f -interaction. Since the renormalisation itself is symmetric, the contribution is equal to what we obtained here. The total renormalisation in the coupling for the term $S_f^+ S_d^-$ is therefore

$$\begin{aligned} \frac{\Delta J}{\Delta D} &= \frac{1}{2} J_f J_d \int_{UV} d\mathbf{q} \rho(\mathbf{q}) J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) (G_f(\mathbf{q}) + G_d(\mathbf{q})) \\ &= \frac{1}{2} \int_{UV} d\mathbf{q} \rho(\mathbf{q}) \left[\frac{J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}})}{\omega - \frac{1}{2} \bar{\epsilon}_f(\mathbf{q}) + \frac{1}{2} J_f(\mathbf{q}) + W_f(\mathbf{q}) - \frac{1}{4} J} + \frac{J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}})}{\omega - \frac{1}{2} \bar{\epsilon}_d(\mathbf{q}) + \frac{1}{2} J_d(\mathbf{q}) + W_d(\mathbf{q}) - \frac{1}{4} J} \right] , \end{aligned} \quad (3.17)$$

where $\bar{\epsilon}_\alpha(\mathbf{q}) = \epsilon_f(\mathbf{q}) - \epsilon_f(\bar{\mathbf{q}})$ is the total excitation energy for the pair of states $\mathbf{q}, \bar{\mathbf{q}}$.

5.4 Symmetries preserved under renormalisation

By Fourier transforming the real-space forms of the Kondo coupling J_f and bath interaction W , we get their k -space forms:

$$\begin{aligned} J_f(\mathbf{k}, \mathbf{q}) &= \frac{J_f}{2} [\cos(k_x - q_x) + \cos(k_y - q_y)] , \\ W(\mathbf{k}, \mathbf{q}, \mathbf{k}', \mathbf{q}') &= \frac{W}{2} [\cos(k_x - q_x + k'_x - q'_x) + \cos(k_y - q_y + k'_y - q'_y)] . \end{aligned} \quad (4.1)$$

These are of course the unrenormalised forms; the Kondo coupling k -space dependence can evolve during the RG flow. The k -space sensitive form of the Kondo coupling and conduction bath interactions are invariant under symmetry transformations in the Brillouin zone.

Translation by a nesting vector into opposite quadrant

Define the reciprocal lattice vectors (RLVs) $\mathbf{Q}_1 = (\pi, \pi)$ and $\mathbf{Q}_2 = (\pi, -\pi)$. The bare Kondo coupling in eq. 4.1 is (anti)symmetric under translation of (one)both momentum by either of the two RLVs:

$$\begin{aligned} J_f(\mathbf{k} + \mathbf{Q}_i, \mathbf{q}) &= J_f(\mathbf{k}, \mathbf{q} + \mathbf{Q}_i) = -J_f(\mathbf{k}, \mathbf{q}) ; i = 1, 2 ; \\ J_f(\mathbf{k} + \mathbf{Q}_i, \mathbf{q} + \mathbf{Q}_j) &= J_f(\mathbf{k}, \mathbf{q}) ; i = 1, 2 \quad j = 1, 2 . \end{aligned} \quad (4.2)$$

These symmetries survive under the renormalisation group transformations. For the first transformation (under which the Kondo coupling is antisymmetric), note that each of the terms on the right

hand side of the RG equation for J_f (eq. ??) are antisymmetric as well - the third term because it's the Kondo coupling itself which automatically has the symmetry, the second term (involving W) because W is also antisymmetric under transformation of one momentum, and the first term because only one of the two J_f in the product will transform (to obtain a minus sign). This ensures that the entirety of the renormalisation transforms antisymmetrically. A very similar argument shows that the symmetry under transformation of both momenta also survives under renormalisation.

Translation into adjacent quadrant

We will make use of another symmetry. Consider two momenta $\mathbf{k}$ and $\mathbf{q}$ in the first and second quadrant, with the same y -component but opposite x -components:

$$\mathbf{k}_y = \mathbf{q}_y, \mathbf{k}_x = -\mathbf{q}_x. \quad (4.3)$$

We refer to $\mathbf{q}$ as $\bar{\mathbf{k}}$ to signal the fact the above relation between the two momenta. We first consider the bare interaction, where we have the symmetry

$$\begin{aligned} J_f(\mathbf{k}, \bar{\mathbf{k}}') &= J_f(\bar{\mathbf{k}}, \mathbf{k}'), \\ J_f(\bar{\mathbf{k}}, \bar{\mathbf{k}}') &= J_f(\mathbf{k}, \mathbf{k}'). \end{aligned} \quad (4.4)$$

We now argue that these symmetries are preserved during the RG flow. Using the properties $W_{\bar{\mathbf{q}}, \mathbf{k}_2, \bar{\mathbf{k}}_1, \mathbf{q}} = W_{\bar{\mathbf{q}}, \bar{\mathbf{k}}_2, \mathbf{k}_1, \mathbf{q}}$ and $J_f^{(j)}(\mathbf{k}_2, \mathbf{q}) J_f^{(j)}(\mathbf{q}, \bar{\mathbf{k}}_1) = J_f^{(j)}(\bar{\mathbf{k}}_2, \bar{\mathbf{q}}) J_f^{(j)}(\bar{\mathbf{q}}, \mathbf{k}_1)$ and the fact that $\bar{\mathbf{q}}$ lies on the same isoenergy shell as $\mathbf{q}$ and is already part of the summation over PS in eq. ??, we can see that $\Delta J_f^{(j)}(\bar{\mathbf{k}}_1, \mathbf{k}_2) = \Delta J_f^{(j)}(\mathbf{k}_1, \bar{\mathbf{k}}_2)$. A similar line of argument shows that $\Delta J_f^{(j)}(\bar{\mathbf{k}}_1, \bar{\mathbf{k}}_2) = \Delta J_f^{(j)}(\mathbf{k}_1, \mathbf{k}_2)$.

5.5 Reduction to Truncated 1D Representation

At the renormalisation group fixed point, we have a renormalised theory for the interaction of the impurity spin S_f with the f -layer Fermi surface:

$$H^* = \sum_{\alpha, \beta} \mathbf{S}_f \cdot \boldsymbol{\sigma}_{\alpha, \beta} \sum_{\mathbf{k}, \mathbf{q}} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k}, \alpha}^\dagger f_{\mathbf{q}, \beta}. \quad (5.1)$$

We will now obtain a more minimal representation of this interaction. Each momentum label is summed over all four quadrants Q_1 through Q_4 ; eq. 4.2 relates Q_1 with Q_3 and Q_2 with Q_4 :

$$\begin{aligned} \sum_{\mathbf{k}, \mathbf{q}} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k}, \alpha}^\dagger f_{\mathbf{q}, \beta} &= \sum_{\mathbf{q}} \left[\sum_{\mathbf{k} \in Q_1, Q_2} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k}, \alpha}^\dagger + \sum_{\mathbf{k} \in Q_3, Q_4} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k}, \alpha}^\dagger \right] f_{\mathbf{q}, \beta} \\ &= \sum_{\mathbf{q}} \left[\sum_{\mathbf{k} \in Q_1, Q_2} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k}, \alpha}^\dagger + \sum_{\mathbf{k} \in Q_1} J_f^*(\mathbf{k} - \mathbf{Q}_1, \mathbf{q}) f_{\mathbf{k} - \mathbf{Q}_1, \alpha}^\dagger + \sum_{\mathbf{k} \in Q_2} J_f^*(\mathbf{k} + \mathbf{Q}_2, \mathbf{q}) f_{\mathbf{k} + \mathbf{Q}_2, \alpha}^\dagger \right] f_{\mathbf{q}, \beta} \\ &= \sum_{\mathbf{q}} \left[\sum_{\mathbf{k} \in Q_1} J_f^*(\mathbf{k}, \mathbf{q}) (f_{\mathbf{k}, \alpha}^\dagger - f_{\mathbf{k} - \mathbf{Q}_1, \alpha}^\dagger) + \sum_{\mathbf{k} \in Q_2} J_f^*(\mathbf{k}, \mathbf{q}) (f_{\mathbf{k}, \alpha}^\dagger - f_{\mathbf{k} + \mathbf{Q}_2, \alpha}^\dagger) \right] f_{\mathbf{q}, \beta}. \end{aligned} \quad (5.2)$$

For ease of notation, we define new fermionic operators $A_{\mathbf{k},\sigma}$:

$$A_{\mathbf{k},\sigma,\pm} = \begin{cases} \frac{1}{\sqrt{2}} (f_{\mathbf{k},\sigma} \pm f_{\mathbf{k}-\mathbf{Q}_1,\sigma}) , & \mathbf{k} \in Q_1 , \\ \frac{1}{\sqrt{2}} (f_{\mathbf{k},\sigma} \pm f_{\mathbf{k}+\mathbf{Q}_2,\sigma}) , & \mathbf{k} \in Q_2 , \end{cases} \quad (5.3)$$

which satisfy the appropriate algebra: $\{A_{\mathbf{k},\sigma,p}, A_{\mathbf{k}',\sigma',p'}\} = \delta_{\mathbf{k},\mathbf{k}'} \delta_{\sigma,\sigma'} \delta p, p'$, with $p = \pm$ denoting the flavours of the A fields. Note that only the $p = -1$ flavour enters the Hamiltonian. Henceforth, we drop the label $\pm$ and it is implied that A refers to the $p = -1$ variant.

Decomposing the sum over $\mathbf{q}$ in a similar fashion, we get

$$\sum_{\mathbf{k},\mathbf{q}} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k},\alpha}^\dagger f_{\mathbf{q},\beta} = 2 \sum_{\mathbf{k},\mathbf{q} \in Q_1, Q_2} J_f^*(\mathbf{k}, \mathbf{q}) A_{\mathbf{k},\alpha}^\dagger A_{\mathbf{q},\beta} . \quad (5.4)$$

5.6 Correlations in truncated representation

In order to calculate equal-time correlations (such as $\langle S_d^+ f_{\mathbf{k}\downarrow}^\dagger f_{\mathbf{k}'\uparrow} \rangle$), we use eq. 5.3 to express the bare fields $f_{\mathbf{k},\sigma}$ in terms of the new fields $A_{\mathbf{k},\sigma,\pm}$ and $B_{\mathbf{k},\sigma,\pm}$:

$$\begin{aligned} f_{\mathbf{k},\sigma} &= \frac{1}{\sqrt{2}} (A_{\mathbf{k},\sigma,+} + A_{\mathbf{k},\sigma,-}) , & \mathbf{k} \in Q_1, Q_2 \\ f_{\mathbf{k}-\mathbf{Q}_1,\sigma} &= \frac{1}{\sqrt{2}} (A_{\mathbf{k},\sigma,+} - A_{\mathbf{k},\sigma,-}) , & \mathbf{k} \in Q_1 \\ f_{\mathbf{k}+\mathbf{Q}_2,\sigma} &= \frac{1}{\sqrt{2}} (A_{\mathbf{k},\sigma,+} - A_{\mathbf{k},\sigma,-}) , & \mathbf{k} \in Q_2 \end{aligned} \quad (6.1)$$

where the four relations act on operators in four quadrants. These relations allow us to express correlations $F_{k_i,k_j} = \langle S_d^+ c_{k_i,\uparrow}^\dagger c_{k_j,\downarrow} \rangle$ for the bare fields f in terms of the rotated fields. Using the fact that the Hamiltonian does not couple the fields B_+ and A_+ , we have

$$\begin{aligned} F_{k_1,k_1} &= F_{k_3,k_3} = -F_{k_1,k_3} = \frac{1}{2} \langle S_d^+ A_{k_1,\downarrow,-}^\dagger A_{k_1,\uparrow,-} \rangle , \\ F_{k_2,k_2} &= F_{k_4,k_4} = -F_{k_2,k_4} = \frac{1}{2} \langle S_d^+ A_{k_2,\downarrow,-}^\dagger A_{k_2,\uparrow,-} \rangle , \\ F_{k_1,k_2} &= F_{k_3,k_4} = -F_{k_1,k_4} = -F_{k_2,k_3} = \frac{1}{2} \langle S_d^+ A_{k_1,\downarrow,-}^\dagger A_{k_1,\uparrow,-} \rangle , \end{aligned} \quad (6.2)$$

where the momentum state k_i belongs to the quadrant Q_i .

5.7 Effective Star Graph Representation of Kondo Interaction

To make computation less intensive, we want to make the Hamiltonian in eq. 5.4 sparser, by going to a star graph representation where each node of the graph couples only with the impurity spin

(central node) and not to the other nodes of the graph. We therefore consider a rotated basis of excitations

$$\gamma_{i,\sigma}^\dagger = \sum_k \Lambda_{i,k} A_{k,\sigma}^\dagger, \text{ where } \sum_k \Lambda_{i,k}^* \Lambda_{j,k} = \delta_{i,j}. \quad (7.1)$$

These fermions of course satisfy the appropriate algebra:

$$\{\gamma_{i,\sigma}, \gamma_{j,\sigma'}^\dagger\} = \sum_{k,q} \Lambda_{i,k}^* \Lambda_{j,q} \{A_{k,\sigma}, A_{q,\sigma'}^\dagger\} = \delta_{i,j} \delta_{\sigma,\sigma'}. \quad (7.2)$$

In order to calculate the coefficients $\Lambda_{i,k}$, we impose the star graph simplification condition described above:

$$\sum_{\mathbf{k}, \mathbf{q}} J_f^*(\mathbf{k}, \mathbf{q}) A_{\mathbf{k},\alpha}^\dagger A_{\mathbf{q},\beta} = \sum_i K_i \gamma_{i,\alpha}^\dagger \gamma_{i,\beta}, \quad (7.3)$$

where K_i are the eigenvalues of the Kondo coupling matrix $J_f^*(\mathbf{k}, \mathbf{q})$. Substituting the definition of the γ fields (eq. 7.1) leads to the equation

$$J_f^*(\mathbf{k}, \mathbf{q}) = \sum_i \Lambda_{k,i} K_i \Lambda_{i,q}^\dagger \implies \mathcal{J} = \Lambda K \Lambda^\dagger, \quad (7.4)$$

where $\mathcal{J}$ is the Kondo coupling matrix with matrix elements $J_f^*(\mathbf{k}, \mathbf{q})$, K is a diagonal matrix with the eigenvalues (K_i) of $\mathcal{J}$ along its diagonal, and Λ is the matrix $\Lambda_{i,k}$ (the rotated indices i along the row, and the earlier momentum indices along the column). From eq. 7.4, it is clear that since $\mathcal{J}$ is Hermitian, the matrix Λ is simply the unitary matrix which diagonalises the Kondo matrix $\mathcal{J}$:

$$\Lambda^\dagger \mathcal{J} \Lambda = K. \quad (7.5)$$

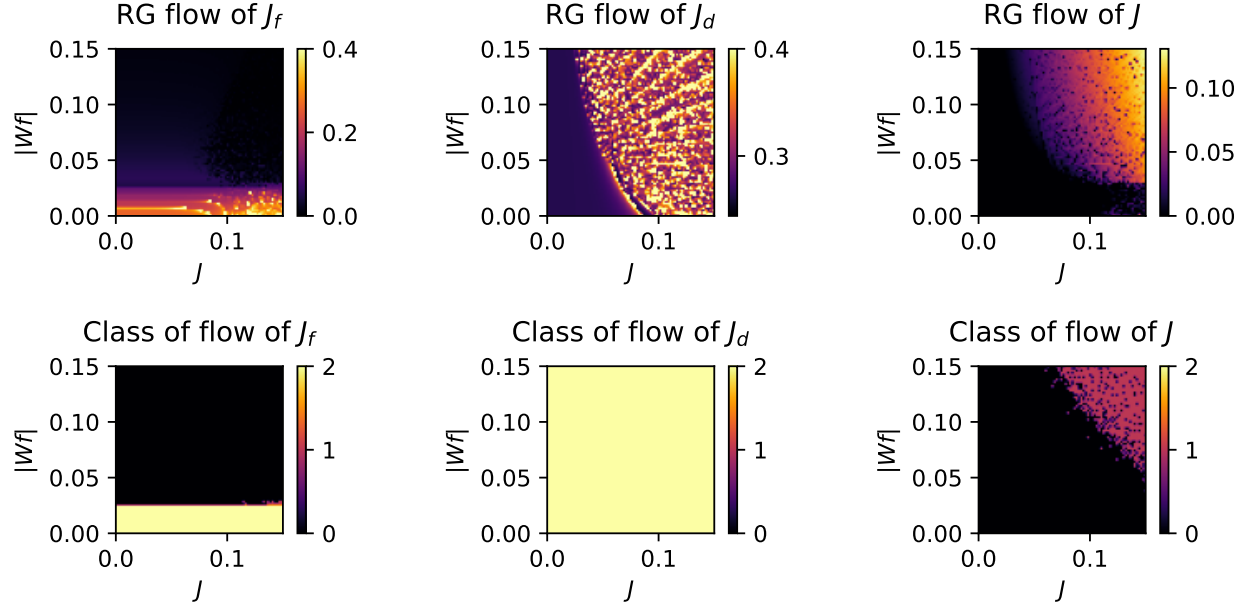
The prescription is therefore as follows. We compute the unitary transformation Λ that diagonalises $\mathcal{J}$ and obtain the eigenvalues $\{K_i\}$. Using the eigenvalues, we work with the sparser Hamiltonian

$$\vec{S}_f \cdot \sum_{\alpha,\beta} \vec{\sigma}_{\alpha,\beta} \sum_i K_i \gamma_{i,\alpha}^\dagger \gamma_{i,\beta}. \quad (7.6)$$

If we have to calculate a certain correlation $C_{k,q}$, we instead calculate the correlation matrix $\tilde{C}_{i,j}$, and then transform back using the unitary transformation:

$$C = \Lambda \tilde{C} \Lambda^\dagger. \quad (7.7)$$

5.8 RG flow simulation on the $d = \infty$ Bethe lattice at half-filling



5.9 RG flow simulation on the 2D square lattice

Kondo coupling and bath interaction acquire k -dependence:

$$J(\mathbf{k}, \mathbf{q}) = \frac{J}{2} [\cos(\mathbf{k}_x - \mathbf{q}_x) + \cos(\mathbf{k}_y - \mathbf{q}_y)] , \quad (9.1)$$

$$W(\mathbf{k}, \mathbf{q}, \mathbf{k}', \mathbf{q}') = \frac{W}{2} [\cos(\mathbf{k}_x - \mathbf{q}_x + \mathbf{k}'_x - \mathbf{q}'_x) + \cos(\mathbf{k}_y - \mathbf{q}_y + \mathbf{k}'_y - \mathbf{q}'_y)] .$$

The particle-hole transformed state $\bar{\mathbf{q}}$ associated with $\mathbf{q}$ is defined as $\bar{\mathbf{q}} = \mathbf{q} + (\pi, \pi)$ if $\mathbf{q}$ is in the first or third quadrant and $\mathbf{q} + (\pi, -\pi)$ is in the second or fourth quadrant.

We first introduce a non-zero chemical potential in the conduction bath of the two layer:

$$H_{\text{aux}} = \sum_{\mathbf{k}, \sigma, \alpha} (\epsilon_{\mathbf{k}, \alpha} - \mu) n_{\mathbf{k}, \sigma, \alpha} + \sum_{\alpha} \sum_{Z \in \text{NN}} \left[\frac{1}{2} J_{\alpha} \sum_{\sigma, \sigma'} \mathbf{S}_{\alpha} \cdot \boldsymbol{\sigma}_{\alpha\beta} c_{Z, \sigma, \alpha}^{\dagger} c_{Z, \sigma', \alpha} - \frac{W_{\alpha}}{2} (n_{Z, \uparrow, \alpha} - n_{Z, \downarrow, \alpha})^2 \right] + J \mathbf{S}_f \cdot \mathbf{S}_d . \quad (9.2)$$

The effect of this is to split the RG equation into contributions from particle and hole sectors:

$$\begin{aligned} \frac{\Delta J_\alpha}{\Delta D} = & \frac{1}{2} \int_{PS} d\mathbf{q} \rho(\mathbf{q}) [J_\alpha(\mathbf{k}_1, \mathbf{q}) J_\alpha(\mathbf{q}, \mathbf{k}_2) + J_\alpha(\mathbf{q}, \bar{\mathbf{q}}) W_\alpha(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)] \left(\frac{1/4}{1/G_\alpha^{(0)}(\mathbf{q}) + \mu/2 + 3J/4} + \frac{3/4}{1/G_\alpha^{(0)}(\mathbf{q}) + \mu/2 - J/4} \right) \\ & + \frac{1}{2} \int_{HS} d\mathbf{q} \rho(\mathbf{q}) [J_\alpha(\mathbf{k}_1, \mathbf{q}) J_\alpha(\mathbf{q}, \mathbf{k}_2) + J_\alpha(\mathbf{q}, \bar{\mathbf{q}}) W_\alpha(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)] \left(\frac{1/4}{1/G_\alpha^{(0)}(\mathbf{q}) - \mu/2 + 3J/4} + \frac{3/4}{1/G_\alpha^{(0)}(\mathbf{q}) - \mu/2 - J/4} \right) \end{aligned} \quad (9.3)$$

where PS and HS integrate over the particle and hole sectors of the excited states (where the excited states are occupied and unoccupied respectively).

Since the bath interaction doesn't renormalise, its functional form remains unchanged and can be substituted into the RG equations. In fact, we can write the expressions in a compact matrix form for efficient computation:

$$\frac{\Delta J_\alpha}{\Delta D} = J_\alpha \mathcal{G}_\alpha J_\alpha^\dagger + \mathcal{W}_\alpha \text{Tr} [\mathcal{G}'_\alpha], \quad (9.4)$$

where all objects are matrices. J_α is the Kondo coupling matrix. $\mathcal{G}_\alpha$ is a diagonal matrix with matrix elements

$$\mathcal{G}_\alpha[q, q] = \begin{cases} \rho(q) \left(\frac{1}{8} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) + \mu/2 + 3J/4} + \frac{3}{8} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) + \mu/2 - J/4} \right), & \mathbf{q} \in PS, \\ \rho(q) \left(\frac{1}{8} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) - \mu/2 + 3J/4} + \frac{3}{8} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) - \mu/2 - J/4} \right), & \mathbf{q} \in HS, \end{cases} \quad (9.5)$$

if q is in the UV subspace and zero otherwise. $\mathcal{W}_\alpha$ is the bath interaction matrix whose matrix elements are given by $\mathcal{W}_\alpha[k, k'] = \frac{W}{2} [\cos(k_x - k'_x) + \cos(k_y - k'_y)]$. Finally, $\mathcal{G}'_\alpha$ is another diagonal matrix whose matrix elements are defined as $\mathcal{G}'_\alpha[q, q] = \mathcal{G}[q, q] J_\alpha[q, \bar{q}]$.

The RG equation for the inter-layer coupling J can be written as

$$\frac{\Delta J}{\Delta D} = \frac{1}{2} \mathbf{J} \cdot \mathbf{G}, \quad (9.6)$$

where $\mathbf{J}$ and $\mathbf{G}$ are vectors defined by the elements $\mathbf{J}[q] = J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}})$ and

$$\mathbf{G}[q] = \rho(\mathbf{q}) \left[\frac{1}{\omega - \frac{1}{2} \bar{\epsilon}_f(\mathbf{q}) + \frac{1}{2} J_f(\mathbf{q}) + W_f(\mathbf{q}) - \frac{1}{4} J} + \frac{1}{\omega - \frac{1}{2} \bar{\epsilon}_d(\mathbf{q}) + \frac{1}{2} J_d(\mathbf{q}) + W_d(\mathbf{q}) - \frac{1}{4} J} \right]. \quad (9.7)$$

5.10 Mapping Correlations from Auxiliary Model to Lattice Model

We follow a similar procedure as the one laid out in the earlier work. Using Bloch's theorem, the eigenstates for the tiled lattice model can be expressed as a superposition of the "local" eigenstates of the auxiliary model. Let $|\psi_{\text{gs}}(\mathbf{r}_d)\rangle$ be the groundstate of the bilayer impurity model with the f -

and d -impurity sites located at $\mathbf{r}_d$ (only tracking the horizontal position). The low-lying eigenstates of the tiled model for the total crystal momentum $\mathbf{k}$ are then given by

$$|\Psi_{\mathbf{k}}\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{r}_d} e^{i\mathbf{k}\cdot\mathbf{r}_d} |\psi_{\text{gs}}(\mathbf{r}_d)\rangle . \quad (10.1)$$

We first consider intra-layer equal-time correlators, of the form

$$C_{O_\alpha}(\Delta) = \langle \Psi_{\mathbf{k}} | O_\alpha(\mathbf{r} + \Delta) O_\alpha^\dagger(\mathbf{r}) | \Psi_{\mathbf{k}} \rangle , \quad (10.2)$$

where α can be f or d and indicates the layer in which the correlation is being calculated (or equivalently, the layer in which the operators act).

Following arguments identical to the ones laid out in the previous work, we substitute eq. 10.1 into eq. 10.2 and use global translation operators to transfer the sum over the local eigenstates into a sum over the operators, allowing us to compute correlations within a single auxiliary model:

$$C_{O_\alpha}(\Delta) = \frac{1}{N} \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_2 + \mathbf{r}_c) | O_\alpha(\Delta + \mathbf{r}_1) O_\alpha^\dagger(\mathbf{r}_1) | \psi_{\text{gs}}(\mathbf{r}_c) \rangle e^{-i\mathbf{k}\cdot\mathbf{r}_2} . \quad (10.3)$$

Using the treatment laid out in the previous work, we can extract an intra-auxiliary model component from the full expression (by setting $\mathbf{r}_2 = 0$):

$$\frac{1}{N} \sum_{\mathbf{r}_1} \langle \psi_{\text{gs}}(\mathbf{r}_c) | \tilde{O}_\alpha(\Delta + \mathbf{r}_1) | \psi_{\text{gs}}(\mathbf{r}_c) \rangle \langle \psi_{\text{gs}}(\mathbf{r}_c) | \tilde{O}_\alpha^\dagger(\mathbf{r}_1) | \psi_{\text{gs}}(\mathbf{r}_c) \rangle . \quad (10.4)$$

The modified excitation operators are generally defined as

$$\tilde{O}_\alpha(\mathbf{r}) = O_\alpha(\mathbf{r}) O^\dagger(\alpha) \mathcal{P}_{\text{gs}} , \quad (10.5)$$

where $O^\dagger(d)$ is the hermitian conjugate of the correlation operator O , but applied on the impurity sites. This therefore constitutes a time-reversed scattering process on the impurity site relative to the process in the bath. The operator $\mathcal{P}_{\text{gs}}$ projects onto the ground state of the auxiliary model, since we are interested in ground state correlations. The operator multiplying the old correlation operator represents excitation of the impurity site, and ensures that the bath and impurity excitation processes take place coherently. The projector ensures that only tripartite correlations between the impurity site and the two momentum states are captured by the correlation, which is what's desired within the tiling method.

For inter-layer correlations, we have

$$C_O(\Delta) = \langle \Psi_{\mathbf{k}} | O_{\bar{\alpha}}(\mathbf{r} + \Delta) O_\alpha^\dagger(\mathbf{r}) | \Psi_{\mathbf{k}} \rangle , \quad (10.6)$$

where $\bar{\alpha}$ represents the layer opposite to α . An identical treatment leads to the following intra-auxiliary model contribution

$$\frac{1}{N} \sum_{\mathbf{r}_1} \langle \psi_{\text{gs}}(\mathbf{r}_c) | \tilde{O}_{\bar{\alpha}}(\Delta + \mathbf{r}_1) | \psi_{\text{gs}}(\mathbf{r}_c) \rangle \langle \psi_{\text{gs}}(\mathbf{r}_c) | \tilde{O}_\alpha^\dagger(\mathbf{r}_1) | \psi_{\text{gs}}(\mathbf{r}_c) \rangle . \quad (10.7)$$

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